Think, Prove, Write
A Project-Based Approach to Proofwriting

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Preface

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Part I

Building Blocks

Chapter 1

Writing a Proof

You are now at one of the most exciting points in your mathematical journey: transitioning from a focus on procedures and computation to a focus on abstraction and proof. Up to this point, perhaps mathematics has felt mostly formulaic; you can become rather good at algebra and calculus by memorizing a small number of problem templates and formulas and then recognizing when to apply them. Most higher-level mathematics does not work this way. (Indeed, if it did, few people would be motivated to become mathematicians!) While calculation and computation may remain a part of your mathematical practice, increasingly you will be asked to contend with open-ended and theoretical questions that cannot be solved through computational procedures alone. The purpose of this book is to help you explore what such questions may look like, how to approach their solutions, and how to communicate those solutions to other mathematicians.

You have probably already encountered, or at least wondered about, some questions of this more open-ended nature. For example, why do certain procedures (like the division algorithm, the Chain Rule, or the matrix inversion formula) always work? Is there a largest prime number, or are there infinitely many primes – and either way, how could we know for sure without checking infinitely many numbers? If two friends playing Tic-Tac-Toe each want to win (or, if winning is impossible, draw), what is the best strategy for each to adopt? Abstract mathematics seeks to explore, understand, and write good answers to questions like these.

At this point, this process probably seems mysterious. If these questions cannot be solved by applying some known procedure, then how do their solutions come about? If a question is open-ended, does it still have an objectively correct answer, or might two different mathematicians disagree as to its solution? How can you tell if a proposed solution is right, or good? What happens if you are exploring a question and get stuck, unable to figure out the answer?

Approaching abstract questions can be more difficult than executing a procedure, and the questions above give some sense as to why. Fortunately, there are objectively true and false statements in mathematics – but mathematical truths can be subtle and hard to find. If you simply assert that a statement is true because it "seems to work" or "feels true", other mathematicians will likely not be convinced. To actually verify that a statement is true and convince other mathematicians of its truth, you need a logical argument, called a *proof*. Here is the single most important idea we can impart about constructing and writing proofs: *a*

proof is just a clear explanation of why something is true. Beyond this guiding idea, there is a great diversity among proofs. There is no set language or template that all proofs must follow (although there are some general conventions we will discuss as we progress). You might immediately see why a certain statement is true, or it might take you a large amount of study to discover a fact. Some facts are most easily explained by applying specific proof techniques that we will learn as we go; in other cases, there is no "technique" to a proof beyond thinking logically and systematically. Often, there are *multiple* valid proofs of a single statement – distinct ways that you can logically verify its truth. And until you start exploring a problem, you may have no idea what the process of its proof will look like!

This state of affairs may seem mysterious and even a bit intimidating – but hopefully also exciting. The process of finding and writing a proof is as much an art as a science and can be as creative and satisfying as any other artistic pursuit. Moreover, because proofs do not all fit a certain "template", we have good news: you can start writing some of them right away, with no specialized terminology or notation required. So, before we talk about specific techniques, let's just jump in and see what this proof-writing business is about. The problems in this chapter should require no mathematical prerequisites beyond algebraic fluency and the willingness to think carefully. If you can clearly explain why some statement in this chapter is true, then, congratulations, you have written your first proof.

1.1 Getting Your Feet Wet

Consistent with the philosophy that proof-writing is best learned by doing, we suggest that you start this section by trying to think about and explain a solution to one of the following questions. As you do so, think about the following questions: Does your solution just state the correct answer, or does it also explain why it is correct? Is your work clearly organized and easy to follow? Could your classmates or professors understand your solution with no other context?

After you have tried your hand at writing a solution, we'll model the proof-writing process with a similar question, so you can see how we approach the task and what we try to achieve and avoid in our finished product. We'll then invite you to revisit your first attempt and see if there is anything you want to modify. But it is important that you try on your own first! Remember, the purpose is not to mimic what we do – it is for you to start developing your own exploration skills and to think about what elements are needed to clearly explain your discoveries.

Exercise 1.1.1. Think about at least one of the following questions until you believe you understand its answer. Then, write a careful explanation of your solution.

(a) In a classroom of 20 people, every person shakes hands exactly once with every other person. How many total handshakes take place?

(b) Three students sit in a circle, and the teacher gives each a hat. Each hat is either white or black. Students can see the colors of their classmates' hats, but not their own. The teacher tells each student to raise their hand if they see at least one black hat on a classmate, and all students raise a hand. She then tells the students to speak up if

they are certain of their own hat color on the basis of this information. If no student speaks up, what can you conclude about the students' hat colors?

(c) The game of Poison is a 2-player game played with 10 pennies in a row. On each player's turn, the player may remove either 1 or 2 pennies from the row. The object is to leave one's opponent with the last penny, which is "poison". Is there a strategy by which Player 1 can always win? If so, describe it. If not, describe a strategy by which Player 2 can win.

Hopefully, you feel that you understand one of the above questions. (And hopefully, figuring out its solution was kind of fun!) Before we reflect on your proof, let's go through the proof-writing process together with a different problem to help you gather your thoughts about the attempt you just made. This next problem is about a game called Soap.

The game of Soap is played with a rectangular chocolate bar divided into rows and columns of squares. For this example, suppose our bar has 3 rows of 4 squares per row. One corner of this chocolate bar has been replaced by a square of soap (which is not palatable). Soap is a 2-player game, in which players take turns breaking the chocolate along a row or column and eating whatever they break off. Each player's goal is to leave their opponent with nothing but the soap square. Does Player 1 have a strategy whereby they can always win? If not, how can Player 2 stop Player 1 from winning?

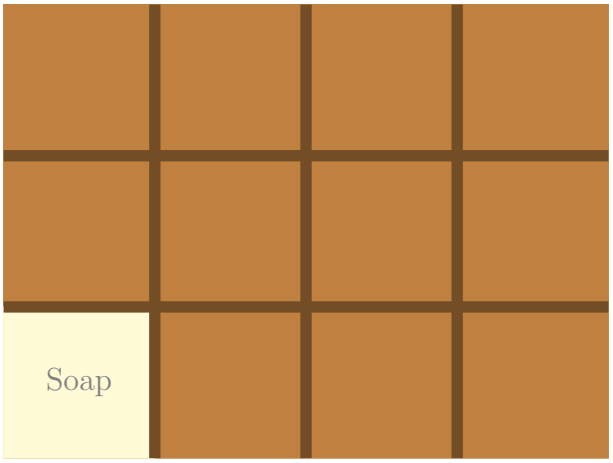


Figure 1.1: The bar of chocolate in Soap. Player 1 may break this bar along any row or column on their first turn.

Before we assert an answer and start drafting a proof, we need to know what the answer is! Indeed, the first part of the proof-writing process is always the proof-discovery process. To start exploring this problem, we might recruit a friend and play a couple of games of Soap. If we take this "experimental" approach, we might notice that there are actually many different choices that Players 1 and 2 can make, even on the small bar of chocolate we've described. Indeed, just on Player 1's first turn, they can break the bar along the first, second, or third column, or along the first or second row. This is five different choices, and Player 2 has multiple responses to each choice. It seems clear that listing all possible trajectories for the game is not an efficient strategy for understanding it.

One natural way to explore the game is to think of starting with an even smaller bar. Often, the mathematical exploration process starts by considering some small examples; understanding a few examples may not yield a general argument, but it can really help build intuition. In the simplest case, who wins a game of Soap played with a 1×1 bar? Well, Player 1 has to eat the soap! So Player 2 wins. On the other hand, with a 1×2 bar (or even a $1 \times k$ bar, for some integer $k > 1$), we can quickly see that Player 1 is always able to win: Player 1 can choose to break off all but the soap square from a $1 \times k$ bar, leaving Player 2 with only soap.

Note that this analysis does not itself address the situation we are asked to analyze, but it helps us make progress in understanding how the game works. The next smallest example we could try is a 2×2 chocolate bar. In this case, Player 1 must break off a 1×2 (or 2×1) piece of chocolate and hand Player 2 a 1×2 (or 2×1) rectangle. But we know that if a player is handed a 1×2 rectangle, that player wins: they break off everything except the soap and hand soap back to their opponent. So starting with a 2×2 bar, Player 1 always loses.

You might be starting to see that we could continue this type of analysis for progressively larger bars. Indeed, one valid way to discover and prove a winning strategy is to go through just such an exhaustive analysis. In the proof below, we model such an argument.

Claim 1. *In the game of Soap played with a 3×4 bar of chocolate, Player 1 is always able to win.*

Proof 1 of Claim 1. We describe what happens in the game of Soap if a player is handed a bar with dimensions $m \times k$, where the smaller dimension is at most 3. We order these cases by the smaller dimension.

Firstly, if a player is handed a 1×1 bar, then that bar contains nothing but a soap square, and so that player loses.

Next, if a player is handed a $1 \times k$ bar for $k > 1$, then that player can win by breaking off all but the soap square and handing only the soap square to their opponent.

Next, if a player is handed a 2×2 bar, then that player must break off either a 2×1 bar or a 1×2 bar and hand their opponent a 2×1 or 1×2 bar. In either case, their opponent can break off the chocolate square in this bar and hand back only soap. So, if a player is handed a 2×2 bar, they will lose.

Next, if a player is handed a $2 \times k$ bar for $k > 2$, they may break off $k - 2$ columns and hand their opponent a 2×2 bar. As we described above, any player who is handed a 2×2 bar will lose. So, if a player is handed a $2 \times k$ bar for $k > 2$, their strategy is to break off all but a 2×2 square, which they hand to their opponent, thereby guaranteeing a win.

Next, if a player is handed a 3×3 bar, they must break off either a 1×3 or 2×3 or 3×1 or 3×2 bar, handing their opponent a 2×3 or 1×3 or 3×2 or 3×1 bar. In any of these cases, we have described above a strategy for the opponent to win. (Specifically, if the opponent receives a bar with one dimension equal to 1, they can break off all but the soap and hand this square back to the first player. If the opponent receives a bar with both dimensions at least 2, then they will hand back a 2×2 square.) In any case, the player who is handed the 3×3 bar loses.

Finally, we are ready to argue that Player 1 can always win Soap. On turn 1, Player 1 is handed a 3×4 bar, which they can choose to break so that Player 2 is handed a 3×3 bar.

Hence, as described above, Player 2 loses (as long as Player 1 chooses to play optimally). □

What do you think of the proof above? Does it seem clear? Does it seem like the best way to explain how Soap works? Some readers may think this proof is great; some may find it lacking. Sometimes, the evaluation of a proof is subjective – different readers may disagree as to whether a proof is "good" or "bad". One thing that is not subjective is that the above proof is *correct*. It completely describes how Player 1 can ensure that they win Soap; it covers all the possible scenarios without making any logical leaps or unjustified assertions. A reader who understands the rules of Soap but has not analyzed the problem could use the above proof alone to fully understand how Soap works on a 3×4 bar.

However, some aspects of the above proof may be unsatisfying. The argument is somewhat lengthy and repetitive; similar analysis is performed many times. Do we need to go through all of these cases separately, or is there some more general reason why they all work? A related consideration is one of generalization. How easy would it be to modify the above argument for a 14×17 bar? Could we analyze Soap played on a much larger bar without going through similar analysis? Actually, we could – and the process of writing our first proof may illuminate an idea that lets us write a second, shorter, much more generalizable proof. As you gain experience with abstract math, you will often experience this phenomenon: once you understand one proof well enough, it may indicate a faster or more general proof that you could not have seen initially.

Let's use the ideas in the "first draft" proof above to give another, shorter proof. If we think about the pattern that emerged in our first argument, it seems like any time a player is handed a square bar (whether the bar is 1×1, 2×2, or 3×3), this is bad news! Now, perhaps this is not a "general" pattern; doing a few examples and noticing a pattern does not guarantee that the pattern will continue. We cannot just assert that a player who is handed a square bar will always lose. However, now that we have some evidence in that direction, it is worth thinking a little more abstractly about the following conjecture about a bar with any dimensions.

Conjecture 1.1.1. *In the game of Soap, if a player is handed a square bar at any point, then that player will lose the game as long as their opponent plays optimally.*

A *conjecture* in math is just a statement that you think is true based on some intuition or evidence but have not fully justified yet. We can turn this conjecture into an established fact, or *proposition*, by writing a full proof. You may already see that proving this conjecture would be very relevant to explaining how the game of Soap works in general. So let's do so! This next argument may seem a bit more abstract than our first proof, and you may wonder where the ideas came from. But if you look back at our initial analysis of 3×4 Soap, hopefully you will see that the ideas in this proof are very similar to the ideas in our first proof. Now that we have intuition about 3×4 Soap, we can *generalize* our argument significantly.

Proof of Conjecture 1.1.1. Suppose a player is handed a square bar at some point in the game of Soap. If this bar is a 1×1 bar, then (since a player never voluntarily eats soap before the end of the game), the bar must consist of just the soap square. Being handed the soap square means the player has lost the game.

If the player is handed a $k \times k$ bar for some $k > 1$, then they have not yet lost the game. However, the player must break off some amount of the bar before handing it back to their opponent. No matter how the player breaks the bar, they must hand their opponent a non-square rectangle, say with dimensions $k \times m$, where m is smaller than k. Note that if the opponent is handed a non-square rectangle, then they have not yet lost the game either, because the losing condition is to be handed a 1×1 square.

Now, upon being handed a $k \times m$ bar, the opponent can always break the bar so as to hand back an $m \times m$ square. Thus, as soon as a player is handed a square bar, they cannot prevent themselves from being handed back a smaller square bar on their next turn. Thus, as long as the opponent always hands back a square bar, the player who is "stuck with the square" will eventually be handed a 1×1 bar and lose the game. □

We have now proved Conjecture 1.1.1, raising its status from a conjecture to a proposition. We have *not* yet accomplished our goal of giving a second analysis of the 3×4 game of Soap. However, we can *use* Conjecture 1.1.1 to completely settle the matter.

Proof 2 of Claim 1. When Soap is played with a 3×4 bar, Player 1 is not handed a square. On their first turn, Player 1 may break off a 3×1 bar and hand a 3×3 square bar to Player 2. By the previous proposition, because Player 2 is handed a square bar, Player 2 will lose Soap as long as Player 1 plays correctly. Player 1 can play optimally by always breaking the bar so as to hand a square to Player 2. □

In fact, it may occur to you that this argument doesn't rely any more on the fact that the game starts with a 3×4 bar. See if you can use the ideas we've generated here to explain the general strategy for Soap.

Exercise 1.1.2. Suppose the game of Soap is played with an $n \times m$ bar, where n and m are both positive integers. Can Player 1 always find a strategy to win Soap if $n < m$? What if $n = m$? Write a proof justifying your answers.

Now that we have modeled proof-writing via the game of Soap, you might have a clearer feeling about which features are important in a proof. Before we move on, let's reflect.

Exercise 1.1.3. Re-read the proof you wrote for Exercise 1.1.1. What similarities or differences do you observe between your proof-writing process and ours? Are there any things you wish to modify or expand on in your proof? Edit your proof accordingly.

1.2 When is a Proof Complete?

You've now tried your hand at writing some proofs and seen us go through the proof-discovery and proof-writing process. Likely, you still have some questions. The biggest may be: How do we know when a proof is "good enough"? We said that a proof is just a clear, complete explanation for why some fact is true. But how do we quantify clarity? What if the explanation has the right ideas, but leaves out some details? What makes an argument convincing to a reader? Is it possible for mathematicians to disagree about whether a proof is complete? What if we come across a convincing-sounding "proof" justifying a statement that actually *isn't* true? How can we identify such a "false proof"?

The questions of correctness and completeness are extremely important, and we want to start thinking about them early. Good proofs don't necessarily follow specific templates, but they do all have some features in common. What do we need to look out for when we're trying to write a good proof?

1.2.1 In search of mathematical rigor

Mathematical rigor is a shorthand term that encapsulates all of the ideas we've been discussing. As an everyday word, *rigor* can be a synonym for strictness, stiffness, or precision. Hopefully, you see how these meanings could fit in with the ideal of a complete and clear proof. Indeed, these connotations are part of the reason why we use the word *rigor* instead of some other word when we talk about a proof's completeness. However, one thing you'll notice as you study math is that usually, mathematical terms have more precise definitions than everyday words. When math shares a word with common English, we want to make sure that we understand the math sense of the word, not just the conversational sense. For example, in modern, casual English, many people use the word *unique* to suggest "unusual and remarkable", as in "I met this cool girl in class yesterday! She has a pretty unique fashion sense." But in math, the word *unique* has a very precise and inflexible meaning; an object is *unique* if it is the one and only object satisfying some property. For example, when we say that -3 is the *unique* solution to the equation $x + 3 = 0$, we do not mean "solutions to the equation $x + 3 = 0$ are rare" or "-3 is a particularly interesting solution to the equation $x + 3 = 0$". We are making a very specific claim that -3 is the only number such that $-3 + 3 = 0$. In order to understand and justify this claim, it is important that we understand the mathematical definition of all its words.

Similarly, in the mathematical sense, when we say a proof is *rigorous*, we don't just mean to communicate a vague sense that "This explanation seems pretty careful and complete to me". There are specific things that mathematicians look for to decide if we call a proof rigorous or not. These include:

1. Is every statement in this proof true?
2. Are all necessary details shown to explain *why* each statement in the proof is true?
3. Does every sentence in this proof follow logically from the previous sentences?
4. Does the writer make any hidden or unjustified assumptions in the proof?
5. Is the conclusion of the argument the same as the stated goal?

These are questions that working mathematicians are constantly asking themselves. We hope that you will start asking yourself these questions, too. In fact, to help you develop this habit, the next subsection is entirely devoted to the art of *reading* proofs – looking at an argument that someone else has written and asking yourself whether these tests of rigor check out.

However, we first give a word of caution. Mathematicians *do* like to be as precise and objective as possible, and the questions above *will* help you think about the conditions under

which a proof can be called rigorous. But determining rigor does not just mean going through a checklist to make sure each box is checked. It means thinking critically about arguments as they are presented.

Because the quality of rigor is tied to the reader's experience of a proof, there is even a sense in which it is subjective. Mathematical claims are objectively true or false, but the mathematical community subjectively decides how much detail is needed to justify a claim. When we say that a rigorous proof must contain all necessary details, we are implicitly suggesting that *some* details may not be necessary. When we say that a proof cannot contain unjustified assumptions, this suggests that *some* assumptions are justifiable. In other words, the standard for "completeness" *can* vary based on the audience.

We can see examples of this phenomenon by looking into the history of mathematics. Many historical proofs – often written by extremely talented and famous mathematicians – were considered completely rigorous at the time but are not considered complete today. This does not mean that the writers of these proofs were bad mathematicians or that their ideas were wrong. It means that as mathematics has evolved, we have come to see that some statements that were once considered "obvious" are either quite subtle or constitute assumptions that must be stated up front.

As a historical example, consider one of the most famous proof-based works, Euclid's *Elements*. The *Elements* is among the most historically and culturally important texts in mathematics – the earliest large-scale instance of math on an axiomatic foundation. (While historians note that portions of the *Elements* were known to mathematicians before Euclid and were just collected by him into a common text, these earlier sources are now lost to history. Euclid still gets the credit for presenting the body of results in a unified way and for filling in the gaps to make the collection as rigorous as he could. Historians also believe that some theorems in the *Elements* are Euclid's original results.) Euclid begins the *Elements* with a list of definitions and axioms that form the basis for all that follows. Starting with this foundation is necessary for grounded mathematics; we must know precisely what the words we use mean so that we have a common understanding of what statements we are trying to prove. Euclid's axioms are basic assumptions about how geometry and algebra work – for instance, that all right angles are in fact congruent and the transitive property of equality holds (that is, if $a = b$ and $b = c$, then $a = c$). Euclid then endeavors to prove all subsequent statements assuming nothing but these axioms. With such an approach, one can in theory trace a direct logical line from the small set of initial assumptions to any more complicated claim. This approach is still used by mathematicians today.

The *Elements* was considered to be the best example of mathematical rigor for more than two millennia. Written around 300 B.C.E., its text was extensively copied, translated, and studied across multiple periods and cultures. It was only in the 19th century that some mathematicians began to believe that the *Elements* was not, in fact, completely rigorous. As mathematics entered the modern era and new sub-fields were being systematically explored, mathematicians noticed that some of Euclid's proofs required facts or assumptions that he never states explicitly. Euclid sometimes assumes that certain diagrams represent generally true statements; for instance, he draws two circles with a shared line segment as their radius and assumes on the basis of this figure that two circles drawn in such a way must intersect at two points. While his conclusion is true (and may seem obvious), it actually takes a fair amount of work to prove. A variety of other additional assumptions appear throughout

Euclid's work, and modern editions of the *Elements* give expanded lists of axioms to fill in the gaps. (In Project 1.5.4, you will have the opportunity to learn more about Euclid's work or about some other famous examples of historical "gaps" in proofs.)

Hopefully, this example leaves you with the feeling that the question of rigor is a bit thorny. In our proofs, we want to show enough detail, check that we don't make unwarranted assumptions, and avoid logical leaps. On the other hand, if we have to spend pages of work justifying that "$0 \cdot 3 = 0$" and "The algorithm for multiplying multi-digit numbers works the way I learned in third grade", we'll never get anywhere. In theory, a completely rigorous proof should explain every detail, no matter how small, but in practice, there is an art to understanding which details in an argument are important, which details really *need* to be checked, and which details we can safely skip. When mathematicians get into trouble for leaving out details, it's typically because the details they left out were big, difficult, or central to the argument. If the work needed to "fill in" a proof is as hard as the work required for the rest of the proof, hopefully you'll agree that this detail is likely important. In fact, the harder a detail is to understand, the more important it is that we carefully explain it in our proof.

Here is the most important take-away about rigor as it pertains to practical proof-writing: A human writing a proof will basically never show every single detail. Don't get bogged down justifying every line of algebra or arithmetic. From now on, you are allowed to know that the rules of arithmetic and high-school algebra work the way you were taught, and unless we are specifically asking you to prove a fact about calculus, you can assume that the various derivative and integration rules work the way you were taught. We will not try to trick you by introducing very subtle issues like the ones that plagued Euclid (although you can take some lessons from our discussion of Euclid; for instance, if you are trying to justify a conclusion with a picture, you need to verify that your picture is general and accurate). Most of the math that you currently know is built on strong foundations, and the point of this book is not to re-litigate all of that from the start.

What you *should* do as you move forward is think critically about the statement you are trying to prove and the audience who will read your proof. These are the things that inform what details are important and what details can be left out. When you are deciding whether you have explained all necessary details, ask yourself questions like the following:

Question 1: *If I make a claim that I do not fully justify, do I at least understand why it is true? Is this claim noticeably easier to understand than the actual result I'm trying to prove?* A good rule is to never make a claim that you couldn't personally justify if pressed. Students sometimes say "It's clear that this is true" when they really mean "This seems plausible; I'll assert it and see if the reader will buy it". You can get into trouble this way! Some statements "feel" true but aren't. Some true facts "feel" like they should be easy to verify but actually take a shocking amount of effort to justify. So please do not say that something is clearly true unless you know you *could* explain it. That said, it's fine to sometimes leave out some details. When you do, make sure the details you leave out really are minor – they should, at least, be much easier to understand than whatever statement you are ultimately trying to justify. For example, if your goal is actually to prove that $0 \cdot 3 = 0$, it might make sense to belabor things about arithmetic and the meaning and properties of the number 0. There is a time and a place for such things! But if your goal is to prove a much more advanced fact, like the Fundamental Theorem of Calculus, you probably don't need to spend

time in your proof justifying basic rules of algebra or arithmetic.

Question 2: *If a member of my intended audience read my proof and noticed that I claim something without justification, would they accept my claim and understand why it's true?* When mathematicians write for different audiences, they often adjust how much detail they include. A mathematician writing a calculus textbook might spend an entire page going through a calculation to justify a statement like

$$\int_0^1 \sqrt{1+x^2}\,dx = \frac{\sqrt{2}+\ln(1+\sqrt{2})}{2}.$$

The same mathematician writing a research paper in which the above integral arose as part of a larger argument might simply assert this result with a comment that it follows from standard techniques of integration. The difference is that in the first case, the point of the mathematical writing is to show how to evaluate the integral, and the intended audience is a group of students who probably have never thought about such an integral before and would have a hard time figuring it out for themselves. In the second case, the point of the mathematical writing is presumably not just to evaluate the integral, but to use its value in a much larger argument. Now the intended audience is probably a group of mathematicians who have many years of experience calculating such integrals. Hopefully you see the difference between the two situations!

In short, when you are deciding whether a detail is important, think about both the centrality of the detail to your argument and the audience for your writing. This discussion leads to another obvious question.

Question 3: *Who is my audience?* Sometimes, there is a very specific and obvious answer. For instance, if you are a professor writing a transition-to-proof textbook for an undergraduate course, then your audience is undergraduate students in a transition-to-proof class. But often, we don't write to such a well-defined group. If you are working on a proof for a homework set, or an independent study project, or a senior capstone paper, your document could be read by both students and professors in your department. Our advice in such situations is to imagine you are writing to a *peer* – a student at about your level of mathematical experience. Don't leave out a ton of details just because you assume your professor will be able to fill them in. Write so that another student in your class would be able to follow your argument. This perspective will not only help you communicate with your peer group, but it will also naturally help you make sure that *you yourself* understand all the important steps in your argument.

Now that we know the basic things to look for in a rigorous proof, let's gain some practical experience. In the next subsection, we'll practice asking all the questions that help us decide whether a proof is rigorous – not by writing our own proofs, but by *reading* proofs written by others. As you'll discover, the better you are at detecting rigor (or its lack) in an argument you read, the easier it will be to write rigorous arguments of your own.

1.2.2 Reading others' proofs critically

When we talk about reading proofs "critically", we aren't talking about reading with the intent to tear others down. Instead, "critical" reading is active reading – it is reading in

such a way that with each assertion, we ask "Does it make sense?" or "Does it follow?" or "Is it true?" Especially with the prevalence of AI, there's no shortage of "proofs" out there that *sound* good – that are written in the style of correct proofs, in an authoritative voice. But we don't evaluate proofs according to these criteria. A proof is correct if all its assertions are true, fully justified, and in the correct logical order. There are no short-cuts to identifying correct proofs; we must read and understand the whole thing before we can certify that it is complete. In this subsection, we consider some elementary propositions that you either already know to be true or at least will readily believe. In each case, we suggest some possible arguments and evaluate them to determine whether they meet the standards of proofs.

We first consider a simple proposition about arithmetic with integers. We recall two important definitions.

Definition 1.2.1. An integer n is **even** if it can be written as $n = 2k$ for some integer k. An integer n is **odd** if it can be written as $n = 2k + 1$ for some integer k.

The following is a true proposition. Think about it on your own until you believe it.

Proposition 1.2.2. *The sum of two odd integers is an even integer.*

We will give several arguments, some better, some worse, and read them critically. Start by reading each of these arguments yourself and identify any issues you see.

First Argument. The first few odd positive integers are 1, 3, 5, and 7. Taking various sums, we find that $1 + 1 = 2$, $3 + 5 = 8$, $3 + 7 = 10$, and $5 + 7 = 12$. All of these sums are even. A similar pattern holds if we take other sums of odd integers, even if they are very large. For example, 1001 and 2973 are both odd and their sum is 3974, which is even. Because the statement works for any example we choose, the proposition is true.

Second Argument. Suppose we are given two odd integers. By definition, both can be written as $2k + 1$ for another integer k. When we add them, their sum is

$$(2k + 1) + (2k + 1) = 4k + 2 = 2(2k + 1),$$

which is even. Therefore the proposition holds.

Third Argument. Let n and m be odd integers. Then we can write $n = 2j+1$ and $m = 2k+1$ for integers j and k. Adding them gives

$$n + m = (2j + 1) + (2k + 1) = 2j + 2k + 2,$$

which is the sum of three even integers and so is even. Therefore the proposition is true.

Although the above arguments differ in completeness, in fact each of them is lacking, so we take the time to discuss them in turn.

The first argument is probably the most problematic. In fact, it isn't an argument at all, but a report of some explorations of the situation. There are infinitely many pairs of odd integers that could be added, so it is impossible to make an exhaustive list of all the sums to check that in each case the answer is an even integer. *Whenever a proposition makes an assertion about infinitely many things, no finite list of examples counts as a a valid argument.*

That said, you should not shy away from doing examples. Good general advice is that you shouldn't try to prove a proposition until you believe it. Although examples don't prove general statements, almost everyone's intuition for what is true is based on working with examples. Doing enough examples carefully can also often help you start to understand the deeper reasons why something is true. Every once in a while, examples will also show you that things aren't as you thought they were, that something you thought was always true in fact isn't. Sometimes the failure is massive – the proposition you are trying to prove could simply be false. Other times, it is *mostly* true and all you need to do is modify the statement to add an additional assumption or exclude a handful of exceptions. (We'll see specific examples of this phenomenon below.) So, yes, explore mathematical situations by working specific examples. Just realize that typically, the examples are *evidence* for the truthfulness of a proposition and not the proof itself.

What about the second argument above? In many ways, it is better than the first argument. One of its big strengths is that it uses the definitions of odd and even integers; it starts by recalling the form an odd integer has – that it can be written in the form $2k + 1$ – and it aims to show that the sum is even by showing that it can be written as two times an integer. These are great features. But the argument has an error. The proposition states that the sum of any two odd integers is even, but the proof actually assumes that the two odd integers are the same. Yes, any odd integer can be written in the form $2k + 1$ for some integer k, but if we have two possibly different odd integers, we also have two possibly different integers k.

The third argument above shows how to get around the issue we noticed with the second argument. It starts off by naming the two odd integers and stating that each of them can be written as two times an integer plus 1, but not necessarily for the same integer. Thus we write the first as $2j + 1$ and the second as $2k + 1$. Things then seem to be going well – we add them together. But then the next line of the argument just asserts that the sum "is the sum of three even integers and so is even." Is this statement true? Well, actually it is. Was it justified in the argument? No. In fact one could reasonably argue that the statement "The sum of two odd integers is even" is roughly the same level of difficulty as "The sum of three even integers is even." This third argument can, however, be turned into a complete proof by taking it a couple steps further. Here is a complete proof.

Proof. Let n and m be odd integers. Then we can write $n = 2j + 1$ and $m = 2k + 1$ for integers j and k. Adding n and m gives

$$\begin{aligned} n + m &= (2j + 1) + (2k + 1) \\ &= 2j + 2k + 2 \\ &= 2(j + k + 1). \end{aligned}$$

Because we have displayed $n + m$ as two times an integer, we may conclude that $n + m$ is indeed even. □

Exercise 1.2.1. The above proof would be considered to be complete by the standards of most transition-to-proof courses. However, as we previously discussed, almost all proofs include steps that are not fully justified – things that the author assumes the reader will know and accept. The proof above is no exception. Re-read it and identify any steps or

assertions that are technically not justified. Do you believe that the unjustified steps are valid?

Now we'll give you a chance to read a proof completely on your own. Here is another statement about integers that you probably believe. We give three arguments to justify the statement. Read them all and make notes of the strengths and weaknesses you perceive.

Proposition 1.2.3. *No integer is both even and odd.*

Like we said, hopefully you believe this is true! Here are the proposed arguments. See what you think.

First Argument. We know that the even integers are $\ldots, -4, -2, 0, 2, 4, \ldots$ and the odd integers are $\ldots, -5, -3, -1, 1, 3, 5, \ldots$. Looking at these lists, clearly there is no overlap. We also know that every integer will be on one of these lists. So every integer is either even or odd, but can't be both.

Second Argument. Let n be an integer. Our goal is to argue that n is not both even and odd. If n is not even, then n is certainly not both even and odd. In this case, there is nothing to argue. So for the rest of the proof, we can assume that n *is* even. Now, our goal is to show that n is not also odd.

Because we assume that n is even, we can write $n = 2k$ for some integer k. We want to show that n cannot also be odd. If n *were* also odd, then we could write $n = 2j + 1$ for some integer j. It would follow that

$$2k = 2j + 1,$$

which we can rearrange to say

$$1 = 2k - 2j = 2(k - j).$$

We can solve the above equation for $k - j$:

$$\frac{1}{2} = \frac{2(k-j)}{2} = k - j.$$

So we conclude that if n is both even and odd, then there are some integers j and k such that $k - j = \frac{1}{2}$. Since the difference of two integers is an integer, this would mean that $\frac{1}{2}$ is an integer. We know that $\frac{1}{2}$ is *not* an integer, so we conclude that n cannot be both even and odd.

Third Argument. We want to show that if n is an integer, then n can't be both even and odd. So let's suppose that some integer *is* both even and odd, and show that this supposition leads to some nonsensical conclusion. If we go through valid reasoning from a premise and reach a nonsensical conclusion, then that means our original premise is also nonsensical.

We'll assume that n is positive; the argument for if n is negative or zero would be very similar. If n is both even and odd, then we can write $2k = n = 2j + 1$ for some integers j and k. Now, let's consider $n - 1$. We can write

$$n - 1 = 2k - 1 = 2(k - 1) + 1$$

and

$$n - 1 = 2j + 1 - 1 = 2j,$$

which shows that if n is both even and odd, then $n - 1$ is also both even and odd. Applying the same argument to $n - 1$, this means that $n - 2$ is both even and odd, and applying the same argument to $n - 2$, this means that $n - 3$ is both even and odd. Proceeding in this way forever, if n is both even and odd, then we can conclude that every integer smaller than n is both even and odd. This means that 0 is both even and odd, since we assume n is an integer bigger than 0. But clearly 0 is not both even and odd, since we cannot write $0 = 2\ell + 1$ for any integer ℓ. So, we have arrived at our nonsensical conclusion and we conclude that it is impossible for an integer n to be both even and odd.

Exercise 1.2.2. Write a brief reflection on the three arguments above. Which, if any of them, do you find convincing? Do you think that each rises to the level of a full proof? If not, what weaknesses or errors do you perceive? If you are in a group setting, discuss your thoughts with the group. Does everyone agree about which arguments are convincing and complete?

As you practice reading proofs critically, it will get easier[1]. Our advice to help you develop this skill is: Don't rush yourself. Every time you read a claim, pause and ask yourself "Is this claim true? What could go wrong here?" It is good to approach each statement in a proof with a degree of skepticism. Again, we are not coming from a place of disrespect for the proof's author; we are actively engaging with each statement in the proof and personally affirming that we understand it before we move on to the next. The next exercise is intended to get you thinking about things that could go wrong and about how a plausible-sounding statement might not always hold. If you read one of the following statements in a proof, would you stop and query it?

Exercise 1.2.3. Each of the following statements is "almost" true, meaning that it is true except for a small number of special cases where it fails. Describe those exceptions and then replace the statement with one that is true. For example, if the statement were "Every prime number is odd," the statement would be false because 2 is prime but 2 is not odd. We could replace the false statement with the true statement "Every prime greater than 2 is odd."

(a) For every real number x, x^2 is positive.

(b) Let m and b be real numbers. The line with equation $y = mx + b$ has an x-intercept at $x = -b/m$.

(c) For every integer x, the fraction $\frac{1}{1+x}$ is less than or equal to 1.

[1]In fact, if you do it for too long, your critical proof reading skills will start to bleed over into the "non-math" world. Hence the joke: a mathematician, a physicist, and an engineer ride a train through Scotland. The engineer looks out the window and sees a white sheep. He says, "Ah, sheep in Scotland are white." The physicist says, "Well, we can say that some sheep in Scotland are white." The mathematician objects: "No, no, on the available data, the most we can claim is that one sheep in Scotland is white on its right-hand side."

(d) For every real number a, the polynomial $p(x) = x^2 - a^2$ has two distinct roots, at $x = a$ and $x = -a$.

Exercise 1.2.4. Build on Exercise 1.2.3 by writing proofs of each of your "corrected" statements. After you write a proof for a "corrected" statement, think about the exception to the original statement that you found in Exercise 1.2.3. Do you see why your proof is not valid for this exceptional case?

1.3 Making (and Refuting) Conjectures

Often in classes and textbooks, we are told true statements and asked to prove them. But in the real world, mathematical facts don't just fall out of the sky. In fact, every true statement in mathematics was once unknown and had to be discovered. Even now, new mathematical truths are being discovered every day; the job of mathematical research is to find and prove interesting new results. So how do mathematicians keep discovering new facts? The answer leads to one of the most fun parts of doing math: exploring problems and conjecturing.

As we said above, a *conjecture* is just a mathematical statement whose truth has not yet been determined one way or another. You might have heard the word applied to some famous problems in math, but you don't have to be a famous mathematician in order to make a conjecture. In fact, we touched on the conjecturing process in Section 1.1 when we explored the game of Soap. Anyone who is investigating a mathematical question can play with it, do examples, and make guesses about what is happening. This is how conjectures typically arise. There are some very long-standing conjectures with their own special names – statements that the mathematical community may largely believe to be true, but that nobody has yet been able to prove. There are also many conjectures made every day of which we are unaware. And every time you sit down to work on an open-ended homework question or project, you make conjectures, even if you don't formally state them. Sometimes these conjectures are *true* – they represent a correct guess about the mathematical situation that can later be justified with a proof. Sometimes conjectures are *false* – they make an incorrect guess. This situation is common and part of the process of exploring and learning. But it is important to remember that any conjecture *may or may not* be true. When we make a conjecture, no matter how confident we feel or how extensive the evidence, we shouldn't just go on assuming it is true without justification, and we should be prepared for the possibility that it is false.

To show that a conjecture is true, we need to prove it. Finding a proof may not be easy, but we are developing a general idea of how the process goes: we explore the statement, try to develop some intuition until we have a general idea about why it is true, and carefully write down this idea to make a complete logical argument. Depending on the complexity of the statement we are trying to prove, we might be able to "see" the idea of its proof and write it down pretty quickly, or we might need several iterations of the process where we explore the question, write ideas of an argument, discover that some details of our argument are complicated or hard to justify, and go back to explore these ideas more carefully. The proof-writing process is not always straightforward! But at least we know that there is a process.

However, we can only prove that a statement is true if it *is* true. How do we show that a conjecture is false? Just as we need to justify all of our assertions when we write a proof of a true statement, when we claim that a conjecture is false, we need to offer justification. So, to show that a conjecture is false, we are technically still writing a proof. However, instead of writing a proof of the conjecture, we are writing a proof to justify the claim that the conjecture is false. Such a proof is often faster and simpler than proving a general mathematical claim. For example, consider the following (false) conjecture.

Conjecture 1.3.1. *Let k be an integer. Then $k^2 - 1$ is divisible by 4.*

Conjecture 1.3.1 makes a claim about *every single integer* k. If we were trying to prove that Conjecture 1.3.1 is true (which, again, it is not), then we would need to make an argument that applies to *every single integer* k. On the other hand, what does it mean to say that Conjecture 1.3.1 is false? We aren't necessarily claiming that there is *no* integer k such that $k^2 - 1$ is divisible by 4 – we are saying that it is not *generally* true. Because the conjecture says that *every single integer* k has the property that $k^2 - 1$ is divisible by 4, we can refute the conjecture by finding *just one* integer k that fails to have the desired property. This single example demonstrates that Conjecture 1.3.1 is not a true statement (even if the conclusion is true for some specific integers).

Claim 2. *Conjecture 1.3.1 is false.*

Proof. Consider the integer $k = 2$. We see that $2^2 - 1 = 4 - 1 = 3$, which is not divisible by 4. Thus, it is not true that for every integer k, the integer $k^2 - 1$ is divisible by 4. □

In this situation, $k = 2$ is what we call a *counterexample*. It is a specific instance that satisfies the hypotheses of Conjecture 1.3.1 (it is an integer) but does not satisfy the conclusion ($2^2 - 1$ is not divisible by 4). We can disprove false conjectures about *all* objects of a certain type by finding one counterexamples. If you can exhibit even one specific case where a conjecture fails, you have done all you need to disprove the conjecture.

How can this discussion be consistent with our earlier statement that doing one specific example does *not* comprise a valid proof of a general statement? For example, in Section 1.2, we proved that the sum of two odd integers is always an even integer. At the time, we discussed that we *cannot* prove this statement by taking a few examples of odd integer pairs and showing that their sums are even. It is important to understand why the two scenarios are different. When is showing one example enough to justify a statement?

The answer is: *It depends on what type of statement we are trying to show!* If we make a claim about infinitely many things – for example, the claim that every pair of odd integers has an even sum – then showing a finite list of examples is not good enough, because a finite list cannot cover every one of our infinitely many cases. However, if we make a claim about just one thing – for example, the existence of some integer k such that $k^2 - 1$ is not divisible by 4 – then a single example *is* good enough. In the second case, we are not making a claim about infinitely many integers; rather, we are saying that there exists *at least one* integer with the claimed property. To show that at least one such integer exists, it suffices to give an example of one such integer.

Let's also note the relationship between the form of Conjecture 1.3.1 and the claim that refutes it – its *negation*. Conjecture 1.3.1 makes a claim that *all* integers satisfy some

property, so to refute this claim, we need to show *there exists* an integer that does *not* satisfy the property. This example illustrates a general phenomenon: When we claim that a certain property is satisfied by *all* objects of some type, then that claim can be refuted by exhibiting *just one* object of the given type that does not have the given property. (If the relationship between this type of claim and its negation seems interesting, read on; we will talk in much more depth about such quantified statements in Chapter 3.) Perhaps the most important thing to note is that the way we refuted Conjecture 1.3.1 depended on the type of claim that Conjecture 1.3.1 was making. A single counterexample can refute a false conjecture *if* the conjecture makes a claim about all objects of a certain type. However, some conjectures can not be refuted by a single example. For instance, here's another conjecture that we know to be false.

Conjecture 1.3.2. *There exist two odd integers j and k such that $j + k$ is odd.*

Do you see the difference? We could give many examples that seem to go against Conjecture 1.3.2 – but Conjecture 1.3.2 is not claiming that *every* pair of odd integers has an odd sum. It just says that *at least one* pair of odd integers has an odd sum. In order to refute Conjecture 1.3.2, we would need a general argument for why there is *no* pair of odd integers with an odd sum – that is, an argument showing that the sum of two odd integers is always even. We gave such an argument in Section 1.2; that entire argument is required to refute Conjecture 1.3.2.

If we disprove a conjecture, it doesn't always mean that the conjecture was worthless. Conjecture 1.3.2 happens to be entirely meritless – there is no experimental evidence that you could gather to support its statement. But while it is possible to make such a baseless conjecture, in practice, we make conjectures based on some intuition – examples we have worked or similar statements that we know to be true. Even if this type of exploration does not lead us to a true conjecture, it can lead us in the direction of truth. This is the case with Conjecture 1.3.1. Again, Conjecture 1.3.1 is false, but it is a conjecture that you might make based on real experimental evidence. Suppose you are asked to investigate the divisibility properties of $k^2 - 1$ when k is an integer. Very naturally, you might start with a few examples, and you notice a pattern:

$$k = 1: \quad 1^2 - 1 = 1 - 1 = 0 = 0 \cdot 4$$

$$k = 11: \quad 11^2 - 1 = 121 - 1 = 120 = 30 \cdot 4$$

$$k = -23: \quad (-23)^2 - 1 = 529 - 1 = 528 = 132 \cdot 4$$

$$k = 1025: \quad 1025^2 - 1 = 1050625 - 1 = 1050624 = 262656 \cdot 4$$

It certainly seems like many integers have the property that $k^2 - 1$ is divisible by 4. Of course, if you included the $k = 2$ case in your exploration, you would immediately realize that not *every* integer has the property that $k^2 - 1$ is divisible by 4. Perhaps based on these examples, you already have an idea of which integers have this divisibility property.

Exercise 1.3.1. Modify Conjecture 1.3.1 to make a new conjecture describing which integers k have the property that $k^2 - 1$ is divisible by 4. Prove your conjecture.

Making false conjectures is a part of doing math; we shouldn't feel bad when we make a conjecture and later discover that it is wrong. However, we would probably prefer to make true conjectures. When we are conjecturing, it is good to spend enough time in the evidence-gathering step that we are at least tentatively convinced that there is something to our conjecture. Doing only two or three examples may suggest a false pattern or may provide insufficient data to reveal a true one. So, do many examples! Try to do examples with different "flavors". For instance, if you are exploring a property of integers, do examples with different kinds of integers: big, small, even, odd, prime, composite, positive, negative. It is surprisingly easy to fall into a false conjecture based on limited data.

Example 1.3.3. For a positive integer n, the number $n!$ (called n *factorial*) is defined as the product $n \cdot (n-1) \cdot \cdots \cdot 2 \cdot 1$. How many divisors does $n!$ have?

We might not have any great intuition for this problem. Is it simple or complex? How might we prove something about the situation? But we can start the conjecturing process. Let's make a table for small n.

n	$n!$	Divisors	Number of Divisors
1	1	1	1
2	2	1, 2	2
3	6	1, 2, 3, 6	4
4	24	1, 2, 3, 4, 6, 8, 12, 24	8
5	120	1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120	16

On the basis of this information, you might make a completely reasonable conjecture: It really looks like the number of divisors of $n!$ is 2^{n-1}. If true, it would be a very interesting property of factorials! When we see a sequence of numbers like this one, it is natural to believe that the pattern is "real" – how often does the sequence $1, 2, 4, 8, 16$ come up coincidentally? And of course, we are motivated to believe our conjecture because it would be a simple and beautiful result.

You have likely guessed that, unfortunately, the purpose of this example is to illustrate the pitfalls of limited data[2]. While the first five factorials have this lovely pattern of divisors, the pattern does *not* hold in general. The next factorial is $6! = 720$, which has 30 divisors, not 32.

Although the pattern we hoped for is not true, we *can* explain why we see these numbers of divisors for our factorials. We might start by looking at the lists of divisors themselves and ask how each list is related to the corresponding factorial. We might recall that each integer greater than 1 has a *prime factorization* – a (unique) way of writing it as a product of primes. For example,

$$4! = 24 = 2^3 \cdot 3.$$

Notice that every divisor of 24 has its own prime factorization, which is related to the prime factorization of 24. In fact, using our experience of how prime factors work, we might guess that the divisors of 24 are exactly the numbers that can be obtained by multiplying together

[2]For some more examples like this presented through the medium of song, see "How They Fool Ya" as performed by Grant Sanderson.

some (or all) of its prime factors. A general statement to this effect statement requires some justification, but it is true. If we believe that we can generate divisors of 24 by selecting some pieces of its prime factorization, then the problem becomes tractable. We can think of selecting this group of prime factors as equivalent to selecting an exponent for 2 and an exponent for 3, where the exponent for 2 is between 0 (corresponding to no factors of 2 in our divisor) and 3 (corresponding to the maximum number of factors of 2 in our divisor) and the exponent for 3 is either 0 or 1. So our divisors of 24 are

$$1 = 2^0 \cdot 3^0, \quad 2 = 2^1 \cdot 3^0, \quad 4 = 2^2 \cdot 3^0, \quad 8 = 2^3 \cdot 3^0$$

$$3 = 2^0 \cdot 3^1, \quad 6 = 2^1 \cdot 3^1, \quad 12 = 2^2 \cdot 3^1, \quad 24 = 2^3 \cdot 3^1$$

In total, how many ways can we make a divisor of 24? Well, in our divisor, the exponent on 2 will be either 0, 1, 2, or 3 (so we have 4 choices) and the exponent on 3 will be 0 or 1 (so we have 2 choices). Thus, we have $4 \cdot 2 = 8$ ways to make a divisor of 24.

We can use the same type of argument to understand the number of divisors of any positive integer. We start by finding the number's prime factorization, and then count how many ways we can choose some of its prime factors to make a divisor. If our number factors as

$$p_1^{a_1} \cdot p_2^{a_2} \cdot \ldots \cdot p_k^{a_k},$$

then to make a divisor, we have $a_1 + 1$ choices for how many factors of p_1 we select, $a_2 + 1$ choices for how many factors of p_2, and so on. So our total number of divisors should be

$$(a_1 + 1)(a_2 + 1) \cdots (a_k + 1).$$

(Such arguments get us into a branch of math called number theory; see Chapter 7 for much more on this type of thing. For more details on problems that involve counting interesting objects, see Chapter 4.) Once we have this insight into the problem, we understand how to quickly compute the number of divisors for other factorials. We can test 6! and discover that the number of divisors of 6! is not 32.

n	$n!$	Prime Factorization	Number of Divisors
1	1	1	1
2	2	2^1	$(1+1) = 2$
3	6	$2^1 \cdot 3^1$	$(1+1)(1+1) = 4$
4	24	$2^3 \cdot 3^1$	$(3+1)(1+1) = 8$
5	120	$2^3 \cdot 3^1 \cdot 5^1$	$(3+1)(1+1)(1+1) = 16$
6	720	$2^4 \cdot 3^2 \cdot 5^1$	$(4+1)(2+1)(1+1) = 30$

Looking at the table, we see that we get the prime factorization for $n!$ by taking the prime factorization for $(n-1)!$ and multiplying by the prime factorization for n. (If you think about how factorials relate to each other, hopefully this makes sense.) There is a relationship between the number of divisors of $(n-1)!$ and of $n!$, but this relationship is determined by how many divisors n has and whether these divisors divide $(n-1)!$ or not. In the first few cases, it happens to work out that the number of divisors increases by a factor of 2, but this rule only applies in special cases.

This discussion makes a number of big assertions about how divisors and prime numbers work; it should not be considered fully rigorous. To rigorously discuss this problem, we probably need more background. We need to carefully define what it means for one number to divide another and for a number to be prime. We also need to know that every positive integer has a unique prime factorization. We will deal with formalizing such notions in Chapter 7. However, you probably have considerable experience finding prime factorizations of integers going back to elementary school, so you probably believe our assertions and are comfortable using these ideas to explore related problems for yourself. One great thing about conjecturing is that it is less formal than the process of writing a full proof. While it is good to be careful when gathering evidence to make conjectures, you shouldn't feel paralyzed by the need to formalize all of your exploration right away.

Exercise 1.3.2. Using the method above, count the divisors of $n!$ for $n = 7, 8, 9, 10, 11$. Then make a conjecture about a condition under which $n!$ has twice as many divisors as $(n-1)!$. Without giving a rigorous proof, give a few sentences justifying your conjecture.

We spent so much time on this last example partially to illustrate that incorrect conjectures sometimes seem very reasonable – indeed, many experienced mathematicians would initially make the same conjecture when presented with this problem. However, we also want to illustrate that the conjecturing process is self-correcting; if you start out with a false conjecture, you are not stuck forever. As you continue to think about your problem, you will come to understand more deeply where the patterns you observe come from. If this understanding eventually leads you to conclude that the conjecture you made is not true, that doesn't mean you wasted your time – it means you are closer to formulating and proving a new, true conjecture.

Just like other aspects of the proof-discovery and proof-writing process, there is not a magic formula for making good conjectures. Sometimes you confidently make a conjecture and later discover a counterexample. Sometimes you explore a question for a long time and have trouble formulating any conjecture – depending on the problem, it may be difficult to detect a pattern or otherwise decide what is happening. As with anything, you will get better and more confident at making conjectures the more you do it. To get started, here are a few situations that are not too hard to explore. See if you can make a conjecture in each case. (Depending on the case, the best tools to *prove* these conjectures may or may not be known to you. In Chapter 2, we will meet our first big proof technique, which will help with some of these examples.)

Exercise 1.3.3. For each of the problems below, explore the problem and make a conjecture. If you believe you have an argument that could be used to justify your conjecture, feel free to write it. But the main purpose of this exercise is to make the conjectures, not prove them.

(a) Consider the integers of the form $7^n - 4$, where n is a positive integer. Make a conjecture about the divisibility properties of such integers.

(b) Let $f(x) = 2x + 1$. We denote by $f^n(x)$ the composition $f \circ f \circ \cdots \circ f$ of f with itself n times. So for instance, $f^2(x) = f(f(x)) = 2(2x + 1) + 1$. What form will $f^n(x)$ have?

(c) More generally, let $f(x) = ax + b$ for some real numbers a and b. What form will $f^n(x)$ have?

(d) A certain fast food restaurant sells chicken nuggets in three quantities: a small box of 3 nuggets, a medium box of 8 nuggets, or a large box of 12 nuggets. If you want to buy a number of nuggets that is not 3, 8, or 12, you may buy multiple boxes, but you cannot buy "partial" orders – so, for instance, it is possible to buy 11 nuggets (since $11 = 3 + 8$) but it is impossible to buy 5 nuggets. Is there a largest number of nuggets that it is impossible to buy?

(e) Consider the 2×2 matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Let n be a positive integer. Is there a simple way to describe the form of A^n?

(f) Let $O(n)$ be the sum of the first n odd integers:

$$O(n) = \sum_{i=1}^{n} (2i - 1).$$

Is there a simpler way to express the value of $O(n)$?

1.4 Problems

Problem sections in this book include the kinds of questions that make up typical problem sets in a first proofs course. Such questions might be more involved than the exercises embedded in the text but they are still smaller than the projects proposed at the end of each chapter.

1. Let n be an integer that can be written in the form $3k + 2$ for some other integer k.

   (a) List at least 5 integers n having the form described above.

   (b) Suppose, as above, that n can be written in the form $3k + 2$ for some integer k. If you square n, does the resulting number also have this form? If so, prove the claim. If not, is there some other way that n^2 is predictably related to multiples of 3? Formulate and prove such a proposition.

2. The purpose of this problem is to explore the divisibility properties of $k^3 - k$ for k an integer.

   (a) Explore $k^3 - k$ for different values of k and make a conjecture about a factor they all share.

   (b) Prove your conjecture.

3. Consider the 3×3 matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Find A^2, A^3, A^4. Make and prove a conjecture about A^n for any positive integer n.

4. Consider the 3×3 matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Find A^2, A^3, A^4. Make and prove a conjecture about A^n for any positive integer n.

5. Consider the 2×2 matrix

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Find A^2, A^3, A^4. Make and prove a conjecture about A^n for any positive integer n.

Typically, we think of proofs as formal mathematical writing, but remember, a proof is just a clear explanation for why something is true. You can write proofs to justify logical statements that you might not think of as mathematical. Here are some logic puzzles. For each, give a careful explanation for the solution.

6. On a certain ship, captives may choose to be taken prisoner by one of two pirates. One pirate is a seasoned sailor who always lies, and one is a novice who always tells the truth. When captives board the ship, they may ask one of the two pirates exactly one question. The captives do not know which pirate always lies. The captives' goal is to select the captor who always tells the truth, because he is known to be a merciful sort and will release his prisoners if they ask nicely.

   A group of captives is unsure of how to proceed. A wise mathematician among them says: "Simply ask either one of the pirates which of them the other would say is the novice." Explain the mathematician's answer. How can you use this question to ensure that you are taken prisoner by the novice?

7. A famous pirate captain has hidden his treasure in three chests. One chest contains gold coins, one chest contains counterfeit coins, and one chest contains a mixture of gold and counterfeit coins. Each chest is labeled to indicate its contents: either G for gold, C for counterfeit, or B for both. Because this pirate captain is very clever and dishonest, none of the labels match the actual contents of the chest. You cannot visually distinguish a gold coin from a counterfeit one, but you can perform a lengthy chemical test to tell the two apart. What is the minimum number of coins you need to test to determine the contents of all three chests, and what strategy should you use?

8. According to the Pirate Code, when a group of pirates is dividing treasure, they must proceed as follows. Firstly, the pirates rank themselves by order of seniority (with no "ties" allowed). Once they rank themselves, the most senior pirate in the group proposes a way to divide the treasure. All pirates, including the proposer, then vote whether they wish to accept this proposed division. If a majority of pirates votes yes, the division is adopted. In the case of a tie vote, the proposer breaks the tie. If a majority of pirates votes no, then the proposer must walk the plank. Among the remaining pirates, the next most senior then proposes a new division of treasure, and the voting cycle resumes. Pirates vote according to the following considerations.

(a) Every pirate would prefer to not walk the plank.

(b) Given that they will survive a vote, every pirate wants to ensure they receive as much treasure as possible.

(c) Because pirates are scoundrels, all other things being equal, they prefer to throw their crew mates overboard. So, a pirate will not vote yes on a proposed division if he knows that he will be able to do equally well by voting no.

(d) All pirates are perfectly rational. Subject to the above considerations, they will always act in their own self-interest.

Suppose three pirates are dividing a chest of 100 gold coins. Coins cannot be cut in half; each pirate must receive a non-negative integer number of coins in the division. If all pirates propose and vote based on the above considerations, how will the gold be divided?

Next, suppose four pirates are dividing a chest of 100 gold coins. How will the gold be divided? What about five pirates? Based on these examples, do you have a conjecture about how 1000 pirates will divide 5000 coins?

In Exercise 1.1.1, we posed some concrete questions for you to explore. All of these questions can be asked more generally. The next exercises return to these questions and expand upon them.

9. In Exercise 1.1.1 part (a), we asked you about a special case of a general fact known as the *Handshaking Lemma*. We posed the question for a group of 20 people, but we can think about the same question for any number of people.

   (a) If you solved part (a) when you did Exercise 1.1.1, summarize your solution. If not, do part (a) of Exercise 1.1.1 now.

   (b) Let n be an integer bigger than 1. Suppose n people gather in a room and every person shakes hands exactly once with every other person. State a formula for the total number of handshakes that take place in this room. Prove that your formula is correct.

10. In Exercise 1.1.1 part (b), we asked you to reason about a problem involving three children wearing hats. We could pose many similar "hat deciding" problems; the following is one way to generalize this exercise.

    (a) If you solved part (b) when you did Exercise 1.1.1, summarize your solution. If not, do part (b) of Exercise 1.1.1 now.

    (b) Suppose four students sit in a circle and the teacher gives each a hat. Each hat is either black or white; students can see the colors of their classmates' hats but not their own. The teacher tells each student to raise their hand if they see at least one black hat on a classmate, and all students raise their hands. She asks the students to speak up if they are certain of their own hat color on the basis of this information, but no student does so. What is the minimum number of black hats the teacher must have distributed?

(c) Let n be an integer greater than 2. Suppose n students sit in a circle. Their teacher distributes hats and asks questions as above. All students raise their hands, but none are immediately sure of their own hat color. What is the minimum number of black hats the teacher must have distributed?

(d) Suppose the game from part (b) continues. After it becomes clear that no students can immediately decide their own hat color, the teacher notes that everyone is unsure. She tells her students to take note of the fact that everyone is unsure, and see if this extra information can help them decide. Still, no student can immediately decide their own hat color. What hat colors must the students have?

11. In Exercise 1.1.1 part (c), we introduced the game of Poison. In that exercise, we specified that Poison is played with 10 pennies in a row, but we can in fact play the game with any number of pennies. Refer to Exercise 1.1.1 for the rules of Poison, and then answer the following questions.

(a) If you solved part (c) when you did Exercise 1.1.1, summarize your solution. If not, do part (c) of Exercise 1.1.1 now.

(b) Suppose the game of Poison is played with 11 pennies instead of 10. Is there a strategy by which Player 1 can always win? If so, describe it. If not, describe a strategy that Player 2 can adopt to stop Player 1 from winning.

(c) Perform a similar analysis for Poison played with 12 pennies, and with 13 pennies.

(d) Based on your exploration, formulate a conjecture about Poison played with n pennies. Does the value of n influence which player will win? Try to prove your conjecture.

1.5 Suggested Projects

Projects are a key component of this book, so we take a moment to discuss the rationale and how they work. As you have no doubt noticed, our perspective is that you will best learn how to do math by actually doing it. In many courses on proof-writing or in later proof-based courses, you are told what is true and asked to write a proof, often using a proof of a similar proposition as a model. Such an approach can help you to learn content quickly and to develop a kind of confidence in proof-writing, because you can feel that you have mastered some standard proof templates. However, such an approach takes out the biggest part of the mathematical process, which involves having a question whose answer you don't know and exploring it until you understand it. In this process, discovering what's true and discovering why it's true are intertwined, and the final proof comes about not by following your nose on a template but often by writing successively more precise arguments for what you have discovered.

Projects are designed to be more open-ended and more challenging than questions posed in the embedded exercises or problem sections of this text. We will pose a problem or set-up for you to explore and suggest the format that your report might take, but it is up to you how you approach the project. Not all successful projects will look the same – some

projects will have multiple directions to explore, and even when two students take the same direction, there may be multiple valid approaches. So we suggest that, as much as possible, you approach projects on your own and don't worry too much about what solutions others might write. You *should not* try to look up answers or hints to these problems; doing so defeats the entire purpose of completing them. The idea is for *you* to come to terms with the problem and write a solution as precisely as you can.

Projects are ranked by expected time investment (denoted by the ◷ symbol), from 1◷ to 3◷. This does not necessarily mean that 1◷ projects are easier than 2◷ projects; projects of all designations are designed to push your thinking and writing skills in different ways. But a 1◷ project will typically focus on a single problem or idea, and typically will not require any knowledge or research beyond the contents of this text. 2 and 3◷ projects might ask you to explore several related questions or explore one problem in multiple ways, and might require you to consult some outside sources for additional background. You could think of 1◷ projects as analogous to a single homework assignment, requiring perhaps a week of work. A 2◷ project might be more like a midterm essay or final project in a course. 3◷ projects are still more expansive; most of these could be the jumping-off points for a whole semester of independent study or a senior capstone paper. Some 3◷ projects even include the potential for original research. Sometimes, a project might fit into several categories depending on how deeply you choose to explore it. In these cases, we will indicate a range of ratings to reflect the minimum and maximum investment you would need to engage with the project.

1.5.1 The Prisoners' Hat Problem

(1◷) A pirate captain has captured 100 prisoners, but will offer them the opportunity to win their freedom. The prisoners are told that they will be assigned numbers 1 through 100 and lined up accordingly. Prisoner 1 will be at the front of the line and unable to see any other prisoners. Prisoner 2 will be second in line, able to see Prisoner 1 but no other prisoners, and so on. The pirate captain will then go down the line and give each prisoner either a red hat or a blue hat. A prisoner cannot see her own hat color, but can see the hats of all prisoners ahead of her in line. Prisoners do not know anything about the total number of blue and red hats distributed.

Starting with Prisoner 100 and proceeding in reverse numerical order, the pirate captain will ask each prisoner to guess her hat color. Prisoners who guess correctly will be released. Any prisoners who guess incorrectly will remain captive. When a prisoner is asked to guess her hat color, she may only say "blue" or "red"; she may not communicate information to her fellow prisoners in any other way. As soon as a prisoner guesses, she and all other prisoners immediately learn whether this guess was correct. All of the prisoners are friends, and their shared goal is to ensure that as many prisoners as possible go free. They are allowed to confer before the hats are distributed to decide upon their strategy.

Find a strategy which will guarantee that the maximum possible number of prisoners go free. Write a careful description of this strategy. Then write a discussion of this strategy, including

1. A complete description of its possible outcomes. How many prisoners are guaranteed to

go free under your strategy? Will your strategy always save exactly the same number of prisoners, or is its outcome variable?

2. A proof that your strategy will always succeed in saving the number of prisoners you claim that it will.

3. A proof that the prisoners could not adopt any other strategy guaranteed to save more prisoners than your strategy does.

4. A reflection on how you might generalize your strategy. Does it matter that there are exactly 100 prisoners in this scenario?

1.5.2 Scorpions and Diamonds

$(1-2\clock)$ In a certain doom-filled temple, 100 small boxes are arranged in a line. The boxes are numbered sequentially 1 to 100. Initially, each box contains a deadly scorpion. However, the contents of the boxes will be altered by the following procedure. 100 attendants work in the temple. The attendants are also numbered 1 through 100. Starting with attendant 1 and proceeding in order, the attendants walk down the line of boxes. Each time attendant k reaches a box whose number is a multiple of k, that attendant will check if the box contains a scorpion. If so, the attendant will remove the scorpion and replace it with a large diamond. If the box contains a diamond already, the attendant will remove the diamond and add a scorpion. The process ends once attendant 100 has finished going down the line.

A certain amateur archaeologist is visiting this temple and has heard that diamonds and scorpions are distributed according to the above procedure. His goal is to take all diamonds from the boxes. He has no antivenom, so must not open any box containing a scorpion. He is also in a hurry, so he does not have time to write out all 100 steps of the procedure and compute by brute force which boxes contain diamonds. Find a simple criterion by which the archaeologist can determine whether a box contains a diamond or a scorpion. (This criterion cannot involve opening the box to check directly!) Write a careful explanation for why this criterion always works. If the procedure is performed with n boxes, where n is any positive integer, which boxes will contain diamonds when the procedure ends?

For a more involved project, in addition to addressing the question above by hand, write a program which will simulate the problem for 100 boxes. Ensure that your program runs and that its conclusion matches the conclusion of your work above! In the presentation of your project, include both your proofs addressing the questions above and your code, with sufficient documentation that it is clear how your code works. After presenting both approaches, include a reflection on the following questions. Does the program you wrote constitute a valid proof of the outcome in the case when 100 boxes are present? Does it constitute a proof of what will happen in general when n boxes are present? Why or why not? If you do not think that your code constitutes a valid proof in either or both cases, is there utility in approaching this problem computationally?

1.5.3 The Josephus Problem

(2◷) The Josephus Problem is a famous mathematical puzzle originating with an account written by an ancient Jewish historian named Josephus. The problem has many variations, many of which in fact correspond to very dark story problems. The curious reader will be able to find much on the history of the problem and its sometimes upsetting variants. For a modern audience, we elect to give a version of the story which involves less violence than usual.

On a certain pirate ship, the captain has just been thrown overboard by his mutinous crew.[3] The crew must now decide upon a way to elect a new captain. Being mathematically minded pirates, they come up with the following scheme. They will stand in a large circle, with their positions numbered 1 through n (with n being the total number of starting pirates). They will gradually (non-violently) eliminate pirates from the circle, according to a pattern which we will describe below. The last pirate left standing in the circle will be the new captain.

The pirate standing at position 1 is the "starting pirate". When the election begins, Pirate 1 shoves the pirate to his right (that is, Pirate 2) out of the circle. The action then proceeds to the next pirate left in the circle, Pirate 3, who shoves the pirate to his right (that is, Pirate 4) out of of the circle. We continue around the circle, with each pirate on his turn able to shove his right neighbor out. The process continues around the circle as many times as needed until only one pirate is left.

The goal of this project is to model and make conjectures about this problem, with the hope of coming to a firm conjecture about what will happen in an n-pirate election. We will lay out some suggestions for how you might explore and gather data about this problem. The idea is for you to go through the same type of work and thought that we went through when we illustrated the conjecture and solution process. (See for example the discussions of Soap, or the process for understanding the number of divisors of $n!$. The Josephus Problem does not necessarily need exactly the same ideas as those examples, but our narration should give you a sense of how much exploration you might want to do to come to grips with the question.) It may be that you don't feel able to prove your conjecture in all possible cases, so if you get to a strong conjecture but don't yet know how to fully justify it, don't worry too much. You can always come back to this conjecture after working through more of this book and flesh out a full proof. We will indicate at least one special case where you *should* be able to find a proof for your conjecture already.

Because this is primarily a conjecturing project, we suggest that your final product for this project should be a report describing your conjecturing process, showing representative examples that you did to help you understand the problem, and clearly explaining the conjecture that you come to. After presenting this data and your conjecture, you can also include any proofs that you come up with justifying some or all of your conjecture. While there is not necessarily one single correct way to work up to a conjecture, we suggest you at least include the following in your work process.

1. Model the problem by hand to make sure you understand the election procedure and see who wins in small elections. Make sure you at least understand how the problem

[3] It is shockingly difficult to motivate the Josephus Problem without at least a little bit of violence.

works for every number n between 1 and 10.

2. Understanding the problem for a few more values higher than 10 is going to help in your conjecturing. If you want to do all your work by hand, we suggest you model the problem for all n up through 16. If you have a little bit of programming experience, try to write a program which will automate the problem, and generate data for as many values of n as you wish (we'd suggest from $n = 1$ to 40 if you'd like a large and useful but still manageable sample).

3. Clearly organize your data and see if you notice any patterns. Based on what you see, can you extend the pattern and guess what position will win in an n-pirate election?

As we said, you might not currently have all the tools to write a full proof of your general conjecture. It is still valuable to think about whether there is a way you could justify some or all of your conjecture. As perhaps a small hint, you should have noticed something special about powers of 2. See if you can at least prove your conjecture for the case when n is a power of 2.

1.5.4 History of Proof

$(2 - 3\clock)$ Mathematical facts are sometimes proved multiple times. Sometimes, a fact is initially discovered, but its original proof is later determined to be somehow lacking; when this happens, mathematicians will try to "correct" or "complete" the original proof in order to justify continued acceptance of the fact. Sometimes, even when the original proof of a fact is complete and correct, other mathematicians will still try to "re-prove" the fact by finding different arguments that lead to the same conclusion.

For this project, choose an example of either a fact whose initial proof was later found lacking and had to be completed, or of a fact which has been proved multiple times in significantly different ways. We list here some examples and ideas to get you started:

1. The Pythagorean Theorem is a basic result from geometry that we probably all know. It is also perhaps the most re-proved theorem in mathematics, with hundreds of distinct known proofs recorded. See if you can learn at least three meaningfully distinct proofs of this famous fact.

2. Gauss's Sum is a famous formula which you may have seen in calculus; it is a nice closed form formula for the sum of the first n positive integers. In Chapter 2, we will discuss Gauss's Sum and show one way to prove the formula using a technique called induction. However, there are other "elementary" proofs of Gauss's Sum which do not require any fancy tools. Learn about some such proofs. As a starting point, the formula is called Gauss's Sum because supposedly, the mathematician Carl Friedrich Gauss found a simple justification for the formula as a child. There are also some "picture proofs" of this result which demonstrate the formula in a visual way. For much more on "picture proofs" (of Gauss's Sum and many other results), find a book or article on *proofs without words*.

3. We mentioned in Section 1.2 that Euclid's *Elements* was considered the standard for mathematical rigor for a long time, but modern mathematicians have identified hidden assumptions or gaps in some of its proofs. Learn about some of the specific arguments which were found lacking, including an explanation of the specific assumptions or holes in reasoning which modern mathematicians have since filled in.

You do not need to confine yourself to this list; feel free to choose another example that you know of, or ask a professor to suggest more examples. But make sure that you choose an example for which you can understand the proofs (or for which you are willing to self-study in order to understand the proofs).

This is a research and writing project: instead of finding a proof for yourself, you will read primary sources and write a paper which could teach your peers about the historical context, mathematical background, and proof details necessary to understand the example you have selected. The goal is to produce a substantial piece of writing, perhaps including sections of formal mathematics, which will both teach a reader about the problem you have selected and reflect upon the process of its proof. Make sure that your essay communicates, at a minimum, all of the following things:

1. the fact whose proofs you have chosen to highlight
2. the history of this fact's discovery
3. the ideas that go into its proofs, including the major differences between the different proofs
4. a reflection on the history of your chosen problem

To research this essay, you will need to find sources containing the original and new proofs of your fact, and potentially other sources discussing the mathematical background and history necessary to understand your example. The idea with this essay is not just to copy down proofs that you find for some fact, but to engage with them. Make sure that you write or summarize the fact and its proofs in your own words, with enough background and explanation that one of your peers could follow the main ideas based on your writing alone. A successful project should not just contain correct mathematical ideas, but help a reader understand how the math in your example works, why it is interesting, and how its discovery helps us understand the process of mathematical research. Does the history of this problem influence your thoughts about how math works? Are there ways in which the new proof(s) of your result change or deepen your understanding of the problem? What potential value do you see in re-visiting historic mathematics?

Chapter 2

Induction

We hope that you are starting to appreciate the proof-writing process and the many ways it can look. Finding and writing a mathematical proof is a creative and complex process. It cannot and should not be reduced to a recipe. That said, there are some ideas and techniques that are frequently useful in proof writing. Learning to use these tools will help you to think through arguments more precisely and expand your ability to write solid, convincing proofs. And, while they will not eliminate the need for creativity and experimentation, having these tools at your fingertips can help you to get going when looking for a proof idea. If you know that certain techniques often work, you can start your exploration by trying to apply one and seeing if it leads somewhere useful.

The first major technique we discuss is called the *principle of mathematical induction*. It is a powerful tool that has applications to problems in all areas of math. Every mathematician is fluent with this technique, and for some, it is literally used on a daily basis. It is particularly useful for establishing that apparent patterns in numerical sequences are, in fact, present. For example, the technique could be used to establish many conjectures along these lines that you made in the exercises or problems in Chapter 1.

2.1 The Principle of Induction

2.1.1 The Idea of Induction

Let's motivate our discussion by first thinking about a concrete question.

Exercise 2.1.1. Consider the integers of the form $11^n - 6$, where n is a positive integer. Make a conjecture about a common divisor of all such integers. Can you prove your conjecture? Either give the most complete proof you can, or explain where the difficulty lies in writing a proof.

Think about the problem on your own before reading on; there is a lot of value in trying it by yourself and seeing where you might get stuck.

There are several different approaches to this problem, and you may or may not have found an argument that works. If you came up with an idea different from what we discuss below, that's a *good* thing, and one of the reasons why math is such a beautiful subject. But

let us show you one standard way to think about the problem. It will introduce the main logical principle of the rest of this chapter.

To start exploring this problem, let's gather a little data. We want to base our conjectures on actual evidence. Here's a table with the first few integers of the given form.

n	$11^n - 6$	Divisors
1	5	$1, 5$
2	115	$1, 5, 23, 115$
3	1325	$1, 5, 25, 53, 265, 1325$
4	14635	$1, 5, 2927, 14635$

There's one obvious pattern in these lists of divisors: for every n we have tested, $11^n - 6$ is divisible by 5. Let's recall that an integer m is divisible by 5 if we can find some integer k such that $m = 5k$. So, we conjecture that for any positive integer n, $11^n - 6$ is divisible by 5. Or, in other words, that there is an integer k such that $11^n - 6 = 5k$.

However, it may be unclear how we can proceed with this conjecture. When we have specific values of n, like in the table above, we can directly test. For each number of the form $11^n - 6$ in our table, we can divide the number by 5 explicitly. But we cannot generate infinitely many of these numbers and test them all. In fact, even for $n = 4$, $11^n - 6$ is large. Even testing specific cases for large n will become prohibitively difficult because $11^n - 6$ will have too many digits for our calculator.

As in some of the examples in Chapter 1, we are in a situation where we are trying to prove infinitely many statements, and a finite list of examples will not be sufficient. We can do larger and larger examples to further support our conjecture, but that experimentation alone seems unlikely to produce deeper understanding. We need a general *reason* why these numbers are all divisible by 5.

Here's an idea. Instead of considering integers of the form $11^n - 6$ (which seem somewhat contrived), let's start by looking at integers of the form 11^n. Even if we have never though about such integers before, we probably looked at some while generating data for our conjecture. Here are the first few powers of 11:

n	11^n	Ends With
1	11	1
2	121	1
3	1331	1
4	14641	1

Aren't these surprisingly pretty numbers? One interesting feature is that they all end with a 1. Of course, we can not just take this statement as given, even if we have heard previously that it is true. But maybe it seems that it wouldn't be too hard to how that for any positive integer n, the number 11^n ends in a 1.

If we show that 11^n ends in 1, then here is how we could proceed. A number that ends in 1 has the form $10m + 1$ for some integer m. So we could write $11^n - 6 = 10m + 1 - 6$ for some integer m. Subtracting and factoring out a 5, we have

$$11^n - 6 = 10m - 5 = 5(2m - 1),$$

which shows that $11^n - 6$ is divisible by 5.

We have made progress, but we still need to show that 11^n ends in 1 for any positive integer n. We have verified the claim for the first few n by direct computation. But what about a general argument? The key is to exploit how successive powers of 11 are related. For example, we have not computed 11^5 for our table above, but we could get it by taking 11^4 and multiplying this by 11. We see that 11^4 ends in 1, as does 11. So, we can write

$$11^5 = 11^4 \cdot 11 = (1464 \cdot 10 + 1)(10 + 1).$$

Multiplying this out, we see that every term in the product is a multiple of 10 except for the $1 \cdot 1$ term. Thus, 11^5 also has the form $10m + 1$ for some m (which we could explicitly determine by multiplying things above). Now, let's not explicitly compute 11^5, but try to do the same thing again. If we know that 11^5 ends in 1, then we can write $11^5 = 10m + 1$ for some m, and

$$11^6 = 11^5 \cdot 11 = (10m + 1)(10 + 1) = 100m + 10m + 10 + 1,$$

which again results in a sum where all terms are divisible by 10 except the final 1. Notice that at this point, it didn't actually matter that we started with 11^5 and were trying to understand 11^6. Exactly this same argument would show that *if* 11^k ends in 1, then 11^{k+1} also does.

It might seem to you that we actually can conclude from the above discussion that 11^n always ends in 1. We know that the first few powers of 11 end in 1, because we computed them. We have shown that, when one power of 11 ends in 1, then the next higher power of 11 also ends in 1. Shouldn't this mean that all positive integer powers of 11 actually end in 1, because the property of ending in 1 holds for the first few and "passes" from one power to the next? *The answer is yes,* and the mathematical principle underlying an argument like this is the main idea of this chapter. It is called the principle of mathematical induction.

Every mathematician has one or more metaphors to describe the principle of induction. Some describe it as knocking over dominoes; some talk about climbing rungs of a ladder. We like to think of induction as a train. To understand induction via train cars, start by writing a careful answer to the following question.

Exercise 2.1.2. A train has 5 cars. You know the first car is red. You also know that this train obeys the following rule: Any time one train car is red, the next car in the train is also red. What color is the fifth car?

Hopefully, you concluded that the fifth car must be red! Since the first car is red, the second car is red too (because the second car comes after the first, and any time one car is red, the car that comes after is red too). Then, because the second car is red, and the third car follows the second, the third car must be red too. Because the third car is red, and the fourth car follows the third, the fourth car must be red. Finally, because the fourth car is red, and the fifth car follows the fourth, the fifth car must be red.

It is important to note the "step by step" nature of this logical argument. The train's rule does *not* say that "whenever one car is red, then all cars are red". We can only know that the fifth car is red because we were first able to argue that the fourth car must be red. However, we *can* ultimately conclude that all 5 cars are red, because the first car is red and we can go down the line one by one. Perhaps you see where we're going.

Exercise 2.1.3. A train has 500 cars. You know the first car is red. You also know that this train obeys the following rule: Any time one train car is red, the next car in the train is also red. What color do you think the 500th car is? Could you explicitly verify this claim?

The next exercise extends the idea even further. Before we pose it, we note that we use the name *natural numbers* to describe the collection of all positive integers $1, 2, 3, \ldots$.

Exercise 2.1.4. A train has infinitely many cars, one for each natural number. You know the first car is red. You also know that this train obeys the following rule: Any time one train car is red, the next car in the train is also red. Let $n > 1$ be a natural number. What color do you think the n-th car is? Could you explicitly verify this claim?

In some sense, Exercises 2.1.2, 2.1.3, and 2.1.4 all seem to be the same question. If you understand the answer to Exercise 2.1.2, you probably already think you know the answers to the other exercises. But while we were able to write a repetitive paragraph explicitly answering Exercise 2.1.2, it would not be so easy to approach Exercises 2.1.3 and 2.1.4 in this way. In the case of Exercise 2.1.3, we *could* write out 499 separate steps; because car 1 is red, so is car 2. Because car 2 is red, so is car 3. Because car 3 is red, so is car 4, and so on. It would clearly be absurd, but eventually we would be able to get through each individual logical step and directly conclude that car 500 is red, because in the previous step we concluded that car 499 is red. The proof would be incredibly annoying to write, it would not be illuminating, and it would be terrible for anyone to read.

The case of Exercise 2.1.4 is even worse, because here we are asked about the n-th car, where n is not a specific number. There is not a set number of statements that we would have to write to get from car 1 to car n. If n were 3, we would need to write 2 statements (car 1 being red implies that car 2 is red, car 2 being red implies that car 3 is red). If n were 1 million, we would need $999,999$ statements. For any specific instance of n, there is a finite number of statements that we could write down to directly show that car n is red. But because there are infinitely many numbers, there is no finite list of statements that we could write down to explicitly and separately determine the colors of all cars in our infinite train.

However, *we still know that in all of these exercises, all of the cars should be red.* We have an ordered list of objects (train cars), and we know the first object in the list has a certain property (being red). We'll call this property "P" for short. We know that any time one object in our ordered list has property P, then the next object in the list has property P also. Logically, it should follow that *every* object in the list has property P. For any fixed number n, we *could* write down a fixed number $(n-1)$ of statements moving from object 1 to object n with the conclusion that each object between 1 and n has property P. We do not explicitly need to write down these statements separately, as long as we know that object 1 has property P and that property P "moves down the line" in the correct way.

Now that you know the idea of induction, perhaps you see why the other metaphors we mentioned are apt. Induction is like knocking down dominoes; if you knock down the first domino in a line, and any time one domino is knocked down, it knocks down the next in line, then knocking down the first domino means all dominoes get knocked down. Induction is also like a ladder; if you can climb to the first rung on a ladder, and any time you've climbed

to one rung you can climb to the next higher rung, then you can climb to any rung on the ladder.

Formally, we state the principle of induction as follows.

Principle 2.1.1 (The Principle of Mathematical Induction)**.** *Consider a series of statements $P(1), P(2), P(3), \ldots$ indexed by the natural numbers. If*

1. *$P(1)$ is a true statement*

   and

2. *whenever $P(k)$ is a true statement, then $P(k+1)$ is also true*

then $P(n)$ is a true statement for all natural numbers n.

Note that our decision to index the statements from 1 is sensible but not necessary. If you have a list of statements $P(0), P(1), \ldots$, you can use the same idea to conclude that all statements in the list are true provided that $P(0)$ holds and that whenever $P(k)$ holds, then $P(k+1)$ holds. You could even index your list starting from 2, or 3, or some other natural number. The important thing is that the statements are indexed in a sequence and that this sequence has a starting point – the "first car" in the train or the "first rung" on the ladder.

2.1.2 Using the Principle of Induction

Induction has a big strength as a proof idea: it is a way that we can rigorously justify infinitely many statements at once[1]. However, it may not be so clear how to use the principle of induction in practice. If we are handed all the set-up, we know what the principle of induction says, but most problems in the world are not phrased in the language of Principle 2.1.1. In Exercise 2.1.1, we saw a problem which seems to relate to the idea of Principle 2.1.1, but we only discovered that relationship after some experimentation with the problem. If we meet a problem in the wild, how do we know if we can approach it using Principle 2.1.1 – and, when we *can* apply Principle 2.1.1 to a problem, how should we formally structure the resulting argument? Let's look at another example to start.

Exercise 2.1.5. Consider the following proposition: Let $x \geq -1$ be a fixed real number. For any natural number n,

$$(1+x)^n \geq 1 + nx.$$

(a) We can interpret the proposition as a series of statements $P(n)$ indexed by the natural numbers. How so?

(b) Carefully write down $P(1)$, the statement corresponding to the index $n = 1$. Show that $P(1)$ is true.

[1]Note that induction is not the only way that we could justify infinitely many statements at once. For example, in Chapter 1, we proved some statements about the properties of even and odd integers. These were not induction proofs; we proceeded directly from the definitions of even and odd integers. So we don't *always* need induction to prove that infinitely many things are true. Induction is still an extremely powerful and appealing idea.

(c) Next write down the statement $P(2)$ and explicitly prove that it is true.

We will prove this proposition by induction below.

Statements made about *all natural numbers* n naturally correspond to ordered lists of the kind described in Principle 2.1.1. For every natural number, we have a different instance of the statement! Once you identify such a list, induction can potentially help to show that every statement on the list is true. But the principle of induction require more than a list of statements indexed by the natural numbers. We need to be able to do two important steps: We need to show that the first statement in our list is true, and we need to show that *if* statement k is true, then statement $k+1$ is also true.

Proving that the first statement on our list is true is called the *base case*, or *basis step*, of our argument. Often (but not always!) the base case is straightforward to verify. In the second part of Exercise 2.1.5, you hopefully already got the experience of proving a base case on your own. There is no single technique for proving a base case – you just have one statement that you are trying to prove, and you can do so by whatever argument you can find.

The second half of using the principle of induction is called the *inductive step*. The goal is to prove that, for *any* natural number k, if the k-th statement on our list is true, then so is the $(k+1)$-st statement. Here, we need a general argument. It is *not* enough to show that if the first statement on the list is true, then so is the second. We need an argument that gets us between any pair of statements that are "next to" each other on the list, no matter where on the list they are.

To do this, we start by fixing a natural number k – but we do not specify what number k is. It could correspond to any statement on our list (including the first). We *suppose* that the k-th statement is true and, using this assumption, try to show that the $(k+1)$-st must hold too.

If this is all making sense, then we encourage you to go back to Exercise 2.1.5 and try the inductive step for yourself. Fix some $k \geq 1$ and suppose the k-th statement on your list is true. Can you use this supposition to argue that the $(k+1)$-st statement on your list is also true? (In a moment, we will give a full proof by induction of the proposition in Exercise 2.1.5. So, you will have a chance to compare your argument to ours.)

Before we give our proof, let's note that induction is *not* guaranteed to work, even when your goal is to prove a list of statements indexed by the natural numbers. In order for induction to be a useful proof technique, several things have to happen with your list of statements:

1. Every statement is actually true.
2. You can directly prove that the first statement is true.
3. The statements are related in a manner that knowing the truth of statement k has a direct bearing on the truth of statement $k+1$.

If the first thing does not happen, then somewhere in your induction argument you will be trying to prove a false statement – so of course you will not be able to arrive at a correct proof. If the second thing does not happen, then you will have nowhere for the induction to

start. Trying to induct without a base case is like saying "If one car in this train is red, then the next car on the train is red" without actually establishing the color of the first train car. What if the first train car *isn't* red? Maybe all cars are green, or some are green and some are pink, or maybe the first few train cars are green and eventually they switch to being red. There is no conclusion you can draw. Finally, if the third thing doesn't happen, then you will have no hope of extending your base case to imply infinitely many things. In the train analogy, we just have an arbitrary rule and we are told it always applies: If one car is red, then the next is red. But when we actually apply induction, we need to find a *reason* why the $k-$th statement on our list implies the $(k+1)$-st.

Fortunately, we often *do* have situations where true statements are related and build on one another. Let's write out a full proof of the statement from Exercise 2.1.5.

Proposition 2.1.2. *Let $x \geq -1$ be a fixed real number. For any natural number n,*

$$(1+x)^n \geq 1 + nx.$$

Proof. The proof is by induction on n. Note that the proposition corresponds to a list of statements indexed by the natural numbers, where the n-th statement on our list is $P(n)$: For the natural number n, $(1+x)^n \geq 1 + nx$.

For the base case, consider $n = 1$. We must show that $(1+x)^1 \geq 1 + 1 \cdot x$. Observe,

$$(1+x)^1 = 1 + x = 1 + 1 \cdot x,$$

so the desired inequality is clearly true. Our base case holds.

For the inductive step, we must show that if our k-th statement holds, then our $(k+1)$-st statement also holds. Fix $k \geq 1$ and suppose that the k-th statement holds, that is, that

$$(1+x)^k \geq 1 + kx.$$

Consider the case for $k+1$. Note that, because $x \geq -1$, $(1+x) \geq 0$. Thus, we can multiply the inequality above by $(1+x)$ to obtain another true inequality:

$$(1+x) \cdot (1+x)^k \geq (1+x)(1+kx).$$

Now, observe that $(1+x) \cdot (1+x)^k = (1+x)^{k+1}$ and

$$(1+x)(1+kx) = 1 + x + kx + kx^2 = 1 + (k+1)x + kx^2.$$

So we have

$$(1+x)^{k+1} \geq 1 + (k+1)x + kx^2 \geq 1 + (k+1)x$$

because $k \geq 1$ and x^2 is non-negative, and so $kx^2 \geq 0$. Hence, we have shown that for any fixed natural number k, if our k-th statement holds, then

$$(1+x)^{k+1} \geq 1 + (k+1)x,$$

which is our $(k+1)$–st statement. The inductive step is satisfied. By the principle of mathematical induction, we conclude that all statements on our list are true. So, for any natural number n, we have $(1+x)^n \geq 1 + nx$. □

Because this was our first example of proof by induction, we wrote out the details of the logic very carefully. Usually, mathematicians are not so explicit when writing induction arguments. For instance, they don't usually describe how the statement they are proving is really a list of statements indexed by the natural numbers. (If writing these things out explicitly helps you keep the logic straight, however, feel free to do so!) For comparison, here is a somewhat shorter version of the above proof that is still considered complete and more closely resembles what you would see in practice.

Shorter proof of Proposition 2.1.2. We proceed by induction on n.

Base Case: Consider $n = 1$. Observe,

$$(1 + x)^1 = 1 + x = 1 + 1 \cdot x,$$

and so the base case holds.

Inductive Step: Fix $k \geq 1$ and suppose that

$$(1 + x)^k \geq 1 + kx.$$

Consider the case for $k + 1$. Observe, $(1 + x)^{k+1} = (1 + x)(1 + x)^k$. Furthermore, because $x \geq -1$, we have $(1 + x) \geq 0$. Thus, multiplying the above inequality by $(1 + x)$ gives

$$(1 + x)^{k+1} = (1 + x)(1 + x)^k \geq (1 + x)(1 + kx).$$

Simplifying the right-hand side yields

$$(1 + x)^{k+1} \geq 1 + (k + 1)x + kx^2 \geq 1 + (k + 1)x,$$

where the last inequality follows from the fact that $k \geq 1$ and x^2 is always non-negative, so $kx^2 \geq 0$. Thus, the inductive step succeeds.

By the principle of mathematical induction induction, $(1 + x)^n \geq 1 + nx$ for all natural numbers n. □

The remainder of this section is devoted to more exercises, examples, and discussion about the technique of induction. The best way for you to become comfortable with the idea is use it yourself. As you do so, we *strongly* suggest that you think of induction as a creative tool to use, not a template to memorize. Induction arguments tend to all have the same "shape"; they always require a base case and then an inductive step. But this does not mean that induction is a boring technique that always works exactly the same way. Even if you have a sense that induction may be useful, it is not always obvious *how* to make it work. Sometimes, you have to be very creative to prove either the base case or the inductive step. Sometimes, induction seems like a great idea but it actually *doesn't* work – typically, because it is not possible to show that one statement on your list implies the next. So, yes, practice induction, and notice how different arguments by induction are similar. But please also note all the different styles of argument and different flavors of problem that induction can cover.

One well-known problem type for which induction helps is proving *sum formulas*. Suppose you add up the first n integers with some special property. Does this sum have a predictable

form? Well, it probably depends upon what special property they share – but, oftentimes, there are beautiful and compact formulas for such specially-constructed sums. Let's work through a famous example together known as *Gauss's Sum*, after the mathematician Carl Friedrich Gauss.

Example 2.1.3. Let $S(n) = \sum_{j=1}^{n} j$ be the sum of the first n positive integers. Is there a simpler formula for $S(n)$?

To answer this question, we either need some special insight or some data upon which to base a conjecture. Let's find the first few sums of this type and see if we notice any interesting patterns.

n	Sum We Compute	Value of $S(n)$
1	1	1
2	$1 + 2$	3
3	$1 + 2 + 3$	6
4	$1 + 2 + 3 + 4$	10
5	$1 + 2 + 3 + 4 + 5$	15
6	$1 + 2 + 3 + 4 + 5 + 6$	21

Looking at this table, it may not be immediately clear what is going on. Our goal is to understand $S(n)$ as a "simple" function of n. At the very least, we would like a formula for $S(n)$ that does not involve a sum. Before we can prove such a formula, we need a guess for what said formula is! How do n and $S(n)$ seem to be related?

Perhaps we just start writing down some observations to see if anything jumps out at us. One way to gain insight into $S(n)$ is to compare it to $S(n-1)$. How much does the function increase when n increases by 1? (If, for example, $S(n)$ were a linear function of n, then the difference between $S(n)$ and $S(n-1)$ would be constant, and equal to the slope of this linear function.) Looking at the data, we see that

$$S(2) - S(1) = 2$$

$$S(3) - S(2) = 3$$

$$S(4) - S(3) = 4$$

$$S(5) - S(4) = 5$$

$$S(6) - S(5) = 6$$

which probably makes it clear what the general pattern is. $S(n)$ is obtained by adding n to $S(n-1)$. You may have noticed this general relationship even without doing these calculations. Indeed, in general,

$$S(n) = \sum_{j=1}^{n} j = n + \sum_{j=1}^{n-1} j = n + S(n-1).$$

This relationship is useful, especially if we ultimately want to write an induction argument, but it does not immediately give us a "simple" formula for $S(n)$ because we have expressed $S(n)$ in terms of the same sum function we are trying to understand.

We now know that $S(n)$ is not a linear function of n – the bigger n is, the more distance we see between $S(n-1)$ and $S(n)$. At this point, there are a number of sensible next steps. We might guess that $S(n)$ is a quadratic function of n. Quadratic functions are nice to work with and exhibit the type of growth behavior we observed. We could fit a quadratic to the data we have so far and then try to prove that the formula holds for all n by induction. Or perhaps we decide to just look at our data for longer to see if we notice any other nice relationship between n and $S(n)$. Either approach might lead to the question: What do we multiply n by to obtain $S(n)$? If $S(n)$ is a polynomial in n, then we would have $S(n) = n \cdot p(n) + c$ for some polynomial $p(n)$ and constant c; if $c = 0$, then we could perhaps guess $p(n)$ by looking at our data, specifically at the ratio $S(n)/n$:

n	Value of $S(n)$	$S(n)/n$
1	1	1
2	3	3/2
3	6	2
4	10	5/2
5	15	3
6	21	7/2

This pattern is fairly convincing. It looks like we have

$$S(n) = n \cdot (S(n)/n) = n \cdot \frac{n+1}{2}$$

for all natural numbers n.

So, after a bunch of exploration, we have a *conjecture* for $S(n)$. Note that, as in previous examples, our conjecture can really be expressed as a list of claims indexed by the natural numbers. The n-th claim on our list is that, for the specific natural number n, we have $S(n) = \frac{n(n+1)}{2}$. Let's prove this claim using induction.

Proposition 2.1.4. *For any natural number n, $S(n) = \sum_{j=1}^{n} j = \frac{n(n+1)}{2}$*

Proof. We induct on n. Our base case is $n = 1$. We have

$$S(1) = 1 = \frac{1(1+1)}{2},$$

so the formula holds for $n = 1$.

For the inductive step, fix $k \geq 1$ and suppose that the result holds for k, so that $S(k) = \frac{k(k+1)}{2}$. We must show that the result holds for $k+1$. We have seen that

$$S(k+1) = (k+1) + S(k).$$

By the inductive hypothesis, $S(k) = \frac{k(k+1)}{2}$, so

$$\begin{aligned} S(k+1) &= (k+1) + \frac{k(k+1)}{2} \\ &= \frac{2(k+1) + k(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}, \end{aligned}$$

which is the formula we desire for $S(k+1)$. Thus, the inductive step succeeds. By induction, we conclude that $S(n) = \frac{n(n+1)}{2}$ for all natural numbers n. □

Let's note that, once we *knew* what the formula for $S(n)$ should be, the induction proof was actually not too long. The bulk of the work went into experimenting with the problem in order to conjecture the correct formula. If you practice many induction problems, writing an argument like this one may seem fairly formulaic; if you are told that the formula is $S(n) = \frac{n(n+1)}{2}$ and told to prove it by induction, then the solution is pretty fast. However, to solve the problem from scratch, we are *not* just applying a formula. We require experimentation, patience, and insight.

Exercise 2.1.6. Gauss's Sum is named after the mathematician Carl Friedrich Gauss because, supposedly, Gauss discovered the formula $S(n) = \frac{n(n+1)}{2}$ as a child. According to the story, Gauss was asked by a teacher to add the first 100 positive integers and he astounded his teacher by doing so almost immediately, without explicitly adding them all. When asked by his teacher how he did it, he stated the formula for $S(100)$ and proved that it is valid. However, he did not use induction. Can you prove Gauss's Sum formula without using induction or any other "fancy" tools?

There are many other sum formulas. They are very frequently proved by induction (even when they can also be proved by other means) because they so naturally fit into the induction framework. Here's another sum for you to think about; see if you can discover its formula on your own.

Exercise 2.1.7. Conjecture a formula for $\sum_{j=1}^{n}(2j-1)$, the sum of the first n odd integers. Prove your conjecture.

We have now seen a few examples where induction can be used. While these were not all exactly the same type of problem, they all seemed naturally "set up" for induction to work. In each example, we had a proposition about some function of natural numbers that very naturally gave us an indexed list of claims. Induction is often useful in such situations. But it can be used in many other situations. Here is an example where induction can be applied, but it requires some creativity to see how.

Example 2.1.5. You are given a standard chess board (so, an 8×8 board divided into 64 squares) and a bag of L-shaped tiles, each built to exactly cover 3 squares of the chess board. You also receive one special square tile that covers exactly one square of the chess board. Your goal is to perfectly cover the chess board with these tiles, so that every square on the board is covered by a tile, all tiles sit completely on the board, and no tiles overlap.

(a) Show that it is possible to tile the board with the tiles provided.

(b) Show that, in fact, there are many such tilings; you can choose any square on the board to place the special square tile and find a perfect tiling which has the square tile in this chosen position.

(c) Show that, for any integer $n \geq 0$, you can perfectly tile a $2^n \times 2^n$ chess board with a bag of L-shaped tiles and exactly one square tile, and moreover, you can find such a tiling where the square tile sits on any specified square.

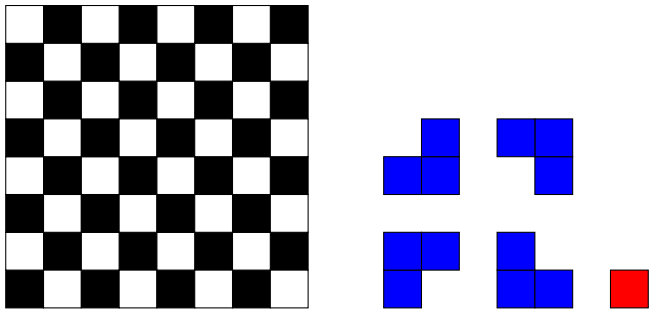


Figure 2.1: An 8×8 chess board and the available tiles. We have as many L-shaped tiles as we wish, but only one square. Different rotations of the L-shaped tiles are shown; note that when we are covering our board, we can rotate tiles however we wish.

We illustrate the board and available tiles in Figure 2.1.

This problem probably seems very different from the type of problem we've been doing in this chapter so far. It is! There is a very wide range of problems to which induction can be applied. Even once you feel comfortable with the technique, you may still need to think hard about when and how to use it. This chess board example is a problem that you might not immediately think has anything to do with induction. We *can* use induction on this problem, but it is not so obviously "set up" for the technique. To understand how we will apply induction here, we first need to have some insight into the problem, which can only come by trying it.

The most concrete of the tasks above is to show that we *can* perfectly tile this board with the available tiles. You could just experiment with the picture and probably find a tiling that works. But looking ahead, we are also asked to show that *many* tilings work. In fact, apparently we can first place our single square tile on *any* square of this board and fill in L-shaped tiles around it. Because we will eventually need to understand how to tile this board in a lot of detail, let's try to do our experimentation a little more systematically.

Here's an idea that might seem related to eventually finding an inductive argument, but is also just a generally good idea. Tiling an 8×8 board is maybe too big of an example to start with. What can we say about tiling smaller boards? Often, doing smaller examples helps us build intuition about how a problem works. What smaller examples can we try?

Well, the smallest possible square chess board is 1×1. Clearly, we can tile a 1×1 chess board with the available tiles: we have exactly one square tile that perfectly covers the single square on the 1×1 board. The next example we could try is 2×2, and again, this is easy to understand – a 2×2 board can be tiled with a single L-shaped tile and the square tile.

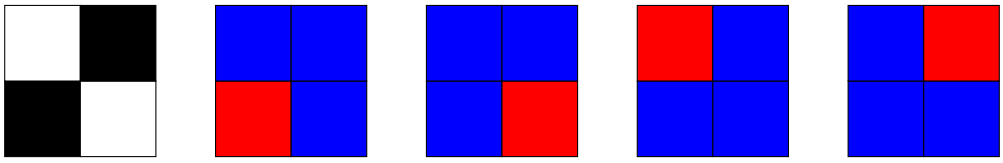


Figure 2.2: A 2×2 chess board and the possible tilings. Note that we can place the square tile wherever we wish.

Looking at these small examples, perhaps we feel a little inspiration. We are able to

easily understand tiling a 2×2 chess board. As we look at the 2×2 chess board, we might also notice that it is made up of four 1×1 chess boards put next to each other. Can we use our understanding of tiling a 2×2 board to understand bigger boards? We could make a 4×4 board by putting four 2×2 boards together. If we had *four* special square tiles, we would definitely know how to tile a 4×4 chess board. In each of the four "quadrants" of the board, we could put down a square tile, and we see how the L-shaped tiles would then fill in each quadrant. However, we only have one square tile. But, because we have these lovely small pictures, we can maybe see how to get around this issue. Suppose for a moment we ignore the rules of this problem and tile our 4×4 board with four square tiles, not just one. In each quadrant, we can choose any position we want for the special square tile. What if we place three special square tiles "next to" each other so as to make an L shape?

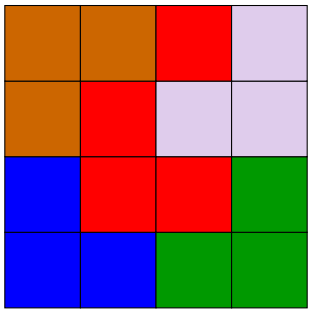


Figure 2.3: A 4×4 chess board tiled with L-shaped tiles and four squares. The square tiles are colored red. L-shaped tiles are given different colors for visual distinction.

We built this tiling by using four different red square tiles, so the three red squares in the center are not really an L-shaped tile. But of course, we could *replace* these with an L-shaped tile because they form exactly the right shape. By doing so, we would turn our "illegal" tiling using four square tiles into a perfectly good tiling using exactly one square tile. Note also that because we can freely choose the position of the single square tile in any quadrant, we could use the same strategy to make a tiling of the 4×4 board where the single square tile occurs in *any* position. We first place a square tile in the desired location. For the three quadrants that do not receive the "designated" square tile, we tile each using an L-shaped tile and a square, placing the square tile in the corner of the quadrant nearest the board's center. When we tile all four quadrants in this way, we will be able to replace three of the four square tiles with an L-shaped tile, and only the "designated" square tile will be left.

Now, we still have not actually exhibited a tiling of the 8×8 board, much less shown all the other things we are asked to show. But our work basically shows us how to understand *all parts* of Example 2.1.5, and indicates how induction may be helpful. This trick of breaking up a board into four smaller boards can be generalized. The following claim is the third part of Example 2.1.5, but it implies the first and second parts of the example. It is actually *easier* to just prove the general statement than to work out the specific large cases by hand. This is often the case in inductive problems – if you understand how to build up smaller cases to do one specific big-ish case, you often understand how to do the problem in general and may as well just write the general solution.

Proposition 2.1.6. *Let $n \geq 0$ be an integer and consider a $2^n \times 2^n$ chess board. Given any designated square on this board, it is possible to tile the board with L-shaped tiles and exactly one square tile so that the square tile sits on the designated square.*

Proof. The proof is by induction on n.. In this case, our base case is $n = 0$, because the list of claims is indexed from 0, not 1.

Base Case: $n = 0$. The claim in this case is that given a $2^0 \times 2^0$ board with one designated square, it is possible to tile this board with some number of L-shaped tiles and a single square tile so that the square tile sits on the designated square. Because a $2^0 \times 2^0$ board consists of a single square, there is only one way to designate a square and only one way to tile the board. The unique acceptable tiling is obtained by placing the single square tile on the single square of the board.

Inductive Step: Fix $k \geq 0$ and suppose that given a $2^k \times 2^k$ board and any designated square on this board, it is possible to find a tiling of this board such that the square tile sits on the designated square. We must show that this implies that, given a $2^{k+1} \times 2^{k+1}$ board and any designated square, it is possible to find a tiling of this board so that the square tile sits on the designated square.

We label four "quadrants" of our $2^{k+1} \times 2^{k+1}$ board. The first quadrant contains squares in the 2^k topmost rows and 2^k rightmost columns of the board (so visually, the top right quarter of the whole board). The second quadrant contains squares in the 2^k topmost rows and 2^k leftmost columns of the board (visually, the top left quarter). The third quadrant contains squares in the 2^k bottommost rows and 2^k leftmost columns (the bottom left quarter) and the fourth quadrant contains squares in the 2^k bottommost rows and 2^k rightmost columns (the bottom right quarter). Note that every square is in exactly one quadrant, so there is a unique quadrant containing the designated square. For concreteness, suppose that the designated square sits in the first quadrant. (If it sits in another quadrant, an analogous argument works.)

Each quadrant is a $2^k \times 2^k$ board, so by the inductive hypothesis, we can tile each quadrant using L-shaped tiles and exactly one special square tile, which may be placed on any chosen square in the quadrant. First, we use the inductive hypothesis to find a tiling of the first quadrant in which the square tile sits on the designated square. Then, we choose a tiling of the second quadrant in which the square tile sits in the bottom right corner of the second quadrant. We choose a tiling of the third quadrant in which the square tile sits in the top right corner of the third quadrant, and a tiling of the fourth quadrant in which the square tile sits in the top left corner of the fourth quadrant. Combining these four tilings, we have a tiling of the full $2^{k+1} \times 2^{k+1}$ board using L-shaped tiles and four square tiles. One square tile sits on the designated square (in the first quadrant) and the other three square tiles form the shape of an L-shaped tile. We replace these three squares by a single L-shaped tile, thereby creating a valid tiling of the $2^{k+1} \times 2^{k+1}$ board. The inductive step succeeds, and by the principle of mathematical induction, the proposition follows. □

Exercise 2.1.8. Even though the above proof is complete and correct, you can solidify your understanding of the inductive step by writing out explicitly how one specific small case is built from the previous. Consider the 4×4 board and suppose the "designated square" is to

go in the upper right corner. The inductive step above tells us how to tile it by first dividing it into four 2×2 boards. Draw the valid tiling given by this argument.

Because this example was a different "flavor" (and was perhaps more complex) than our other examples, let's briefly reflect upon the proof. If you were not told that this problem yields to induction, you might never have tried this approach. Perhaps you would have tried to find "rules" for constructing valid tilings of the chessboard – ways to start with the square tile in a specified spot and to place tiles around it to "fill in" a valid tiling. Or perhaps you would have tried some totally different proof strategy. Induction is not a magic bullet – there are many problems that cannot be approached with induction. But it is *sometimes* useful even for problems that do not seem naturally "set up" for the induction process.

Another interesting feature of our solution is the way in which we answered the parts of Example 2.1.5. The first part of this example just asked us to find one valid tiling of the 8×8 board, which we never explicitly did. Instead, we gave a "recipe" for doing so, which also provides a much more general and flexible understanding of the problem. If our only goal were to justify that there exists some valid tiling of the 8×8 board, then Proposition 2.1.6 seems to be much stronger than required. Proposition 2.1.6 doesn't just say that we *can* find a valid tiling of a $2^n \times 2^n$ board, but that we can find such valid tilings where the square tile sits anywhere we like. Is the extra strength of this statement really necessary?

Actually, *yes*. Suppose the only question you cared about was the following: for $n \geq 0$, is there a valid tiling of the $2^n \times 2^n$ board? You could make the following claim, which seems weaker than Proposition 2.1.6.

Claim 3. *Let $n \geq 0$ be an integer. There exists a tiling of the $2^n \times 2^n$ chessboard using L-shaped tiles and exactly one square tile.*

Given the work we've already done, we know that Claim 3 is true – it follows from Proposition 2.1.6. It might seem like Claim 3 should be easier to prove than Proposition 2.1.6 – after all, we are trying to show that just one small part of Proposition 2.1.6 holds. But it is *not* easier to prove Claim 3. This is because there is no clear way to prove Claim 3 by induction. Indeed, let's think about what would happen if we tried.

Our base case would be the same as in the proof of Proposition 2.1.6. For the inductive step, we would fix $k \geq 0$ and suppose that it is possible to tile a $2^k \times 2^k$ board using L-shaped tiles and exactly one square. Our goal would then be to show that we can so tile a $2^{k+1} \times 2^{k+1}$ board. Here is where we run into trouble. We can break up this board into four quadrants as before, but this is no longer so helpful. Each quadrant is a $2^k \times 2^k$ board, so the inductive hypothesis says we can separately tile it using L-shaped tiles and exactly one square. We obtain a tiling of the $2^{k+1} \times 2^{k+1}$ board using L-shaped tiles and four square tiles. In the proof of Proposition 2.1.6, there was no problem – we guaranteed that three of these square tiles sat next to each other and could be replaced by an L-shaped tile. But in this "simpler" case, we have no control over where these four square tiles appear, because our inductive hypothesis just says there is *some* way to tile a $2^k \times 2^k$ board using exactly one square. There is no reason to expect that three of our four square tiles sit next to each other when we take these tilings of the $2^k \times 2^k$ board and "combine" them.

This illustrates an interesting phenomenon: Sometimes, it is easier to inductively prove a stronger claim than a weaker claim! This may seem counterintuitive. If a strong claim

implies a weaker claim, then surely there is "more" to show when proving the strong claim. But think about how induction works: We have some claim that we prove is true for one small case. Then we get to *assume* our claim for some fixed case and use this assumption to argue that the next case holds. It is true that if we start with a strong claim, we might have to do a bit more work to check that the claim holds for the base case. But then we get a benefit – we get to assume a stronger claim holds in the inductive step, and we can use all the strength of this claim to argue for the next case.

We hope these few examples convince you that induction is both a powerful idea and a versatile one. Once we have induction in our mathematical toolbox, many problems will open up to us. Keep in mind that, while this chapter focuses exclusively on induction, there may *also* be inductive arguments in future chapters and exercises. The principle comes up in almost every area of math – sometimes in the least expected places.

2.2 Strong Induction

Before we finish our discussion of induction, we describe a common variation on the idea. Let's motivate this variation with an example. Think about it before reading on.

Exercise 2.2.1. A group of students is ordering egg rolls for an end-of-semester party. The restaurant sells egg rolls in two order sizes – a "single person" order of 3 or a "group sized" order of 8. The restaurant will not sell partial orders of egg rolls, so if the students want to have a specific number of egg rolls, they must achieve this by purchasing some number of 3-roll orders and some number of 8-roll orders. There are 41 students attending the party, and the group wants to have exactly one egg roll for each. Is it possible for the students to buy exactly 41 egg rolls? In general, what numbers of egg rolls can the students order?

Hopefully you found that an order of 41 egg rolls *is* possible – for example, you can buy 11 orders of 3 and one order of 8 for a total of $3(11) + 8 = 41$ egg rolls. But doing this one specific case does not necessarily give us insight into the problem more generally. 41 seems like a fairly random number. It is not "trivially" achievable based on the set up (for example, it is not a multiple of 3 or a multiple of 8), but we were still able to find a way to order 41 egg rolls. We might guess that it is possible to buy orders of most sizes.

On the other hand, we certainly see that some order sizes are *not* possible. For instance, the students cannot order exactly 1 egg roll or exactly 2 egg rolls, because the minimum order size is 3. If the students wanted exactly 10 egg rolls, there is no way to order this either; 10 is not a multiple of 3, so to order 10 egg rolls, the students cannot just buy some number of 3-roll orders. Since 10 is also smaller than 16, they also cannot buy more than one 8-roll order if they want 10 rolls. So they must buy exactly one 8-roll order – but then they are stuck. To get exactly 10 rolls, they would need to add 2 rolls to their 8-roll order, and the restaurant will not sell a 2 egg roll order.

So some numbers of rolls can be ordered and some cannot. How can we tell which numbers can and can't be achieved? Well, as usual, we can start by generating some data. Here are the first 16 cases.

Number	Possible?	Number (cont.)	Possible?
1	No	9	Yes (3×3)
2	No	10	No
3	Yes (1×3)	11	Yes $(3 + 8)$
4	No	12	Yes (4×3)
5	No	13	No
6	Yes (2×3)	14	Yes $((2 \times 3) + 8)$
7	No	15	Yes (5×3)
8	Yes (1×8)	16	Yes (2×8)

Now, what do we think about this data? Should we continue collecting, or do we have enough to go on? There may not be an obvious pattern to the "yes" and "no" answers here. Some small numbers are possible and others are not – it just depends on whether there is some way to write the number as $3j + 8k$ for two non-negative integers j and k. To verify that a number is not possible, we have to go through all possible combinations of j and k and argue that none work.

However, we might notice one way that we can immediately verify that a number *is* possible. For instance, 11 is possible because 8 is possible – we get an order of 11 by first making an order of 8 and then adding an order of 3. Similarly, 14 is possible because 11 is possible – we order 14 by first ordering 11 and then ordering an extra 3. In general, if we can order n egg rolls by some combination of size 8 and size 3 orders, then we can also order $n + 3$ (and $n + 8$, in case that's relevant) by ordering n in a valid way and then ordering an extra 3 (or 8).

Now, notice the last 3 data points in our table. It is possible to order 14, 15, and 16 egg rolls. What does this mean about larger orders? Well, it is possible to order 17, because 17 is $14 + 3$. It is possible to order 18, because 18 is $15 + 3$. And it is possible to order 19, because 19 is $16 + 3$. Then, we see that it is possible to order 20, 21, and 22, because these are 3 more than 17, 18, and 19 respectively, and so on. The conclusion here should be that it is possible to order any number larger than 13, because we can always "build up" to such a number by adding some number of 3-roll orders to 14, 15, or 16.

We could explicitly write out instructions for making such an order, but we also notice an induction-style thought process here. Is there a way to do this problem quickly with induction? Let's try. Here is the claim we are trying to show.

Proposition 2.2.1. *For any number $n \geq 14$, it is possible to order exactly n egg rolls.*

We can think of this as either a list of statements indexed $14, 15, 16, \ldots$ (in which case our base case would be $n = 14$) or we can slightly rephrase the statement so as to naturally index from 1. Let's see what that would look like. Statement 1 is supposed to say "It is possible to order exactly 14 egg rolls", statement 2 should say "It is possible to order exactly 15 egg rolls", and so on. How about we rephrase statement 1 to read "It is possible to order $13 + 1$ egg rolls", statement 2 to read "It is possible to order $13 + 2$ egg rolls" and so on. Then our i-th statement will read "It is possible to order $13 + i$ egg rolls".

Now, whichever indexing convention we choose, our base case is the statement "It is possible to order 14 egg rolls". We know this statement is true. For the inductive step, we fix $k \geq 1$ and suppose the statement "It is possible to order $13 + k$ egg rolls" is true. Our

goal is to show that it is also possible to order $13+k+1$ egg rolls. But here we see a problem. The fact that we can order $13+k$ egg rolls does *not* imply that we can order $13+k+1$. It implies that we can order $13+k+3$ egg rolls, because we can order $13+k$ egg rolls and then add a 3-roll order.

We can get around this problem. We just need a flavor of induction that is able to "look back" to more previous cases. Let's start by establishing *three* small cases; we calculated directly that we can order $13+i$ egg rolls when $i = 1, 2$, or 3. Now, let's think about the following statement: If $k \geq 3$ is an integer and for all integers i with $1 \leq i \leq k$, it is possible to order $13+i$ egg rolls, then it is also possible to order $13+k+1$ egg rolls.

Based on our discussion, this statement certainly seems true. If $k \geq 3$, then $k-2 \geq 1$, meaning that $k-2$ is an integer between 1 and k. So, by assumption, it is possible to order $13+k-2$ egg rolls. So it is also possible to order $13+(k-2)+3 = 13+k+1$ egg rolls.

Let's combine our three small cases with the statement we've just added to summarize where we are. We want to show that for any natural number n, it is possible to order $13+n$ egg rolls. We know

1. The first three statements ($n = 1, 2, 3$) hold

2. If $k \geq 3$ is an integer and the first k statements hold, then so does the $(k+1)$-st statement.

These two steps together imply that all statements hold. The first three together imply the fourth, the first four together imply the fifth, and so on.

This strategy may seem different from the induction arguments we've done so far. Instead of assuming that one case holds and using it to prove the next, we assume that *all previous* cases hold and use these together to prove the next. But is this idea really different from the basic induction idea? Are we *actually* assuming more than we ordinarily do? After all, when we apply "standard" induction, we think of "moving down the train" one car at a time. We start with the base case (the "first car") and then our logic goes as follows. We show that if one case holds (one car is red), then the next holds too (is also red) – so first, if the first car is red, then the second is red. Then, because the second is red, the third is red, and so on. When we get to car k, we use the fact that if the k-th car is red, then the $(k+1)$-st is red. But to *get* to the point where we know car k is red, we implicitly "went down the line" and concluded that every car between car 1 and car k *is also* red. Thought of from this angle, assuming that all statements between 1 and k hold is just assuming something we can get from the usual inductive hypothesis.

The idea we are discussing is called "strong" induction. Here is its formal statement:

Principle 2.2.2 (Strong Induction)**.** *Suppose a series of statements $P(1), P(2), P(3), \ldots$ is indexed by the natural numbers. If*

1. *for some natural number b, $P(1), P(2), \ldots, P(b)$ are true*

   and

2. *for $k \geq b$, whenever all of $P(1), P(2), \ldots, P(k)$ are true, then $P(k+1)$ is true*

then $P(n)$ is true for all natural numbers n.

Let's see how we can phrase the solution to Exercise 2.2.1 in terms of this principle.

Proof of Proposition 2.2.1. We proceed by strong induction on n. We have a list of statements indexed by the natural numbers: Statement $P(n)$ is "It is possible to order exactly $13 + n$ egg rolls via a combination of 3-roll and 8-roll orders".

Base Case: We show directly that when $n = 1, 2$, or 3, it is possible to order exactly $13 + n$ egg rolls. Indeed, we can order $13 + 1 = 14$ by taking 2 3-roll orders and 1 8-roll order, we can order $13 + 2 = 15$ by taking 5 3-roll orders, and we can order $13 + 3 = 16$ by taking 2 8-roll orders.

Inductive Step: Fix $k \geq 3$ and suppose that the first k statements hold – that is, for any integer i with $1 \leq i \leq k$, it is possible to order exactly $13 + i$ egg rolls via a combination of 3-roll and 8-roll orders. Consider $13 + k + 1$. Because $k \geq 3$, $k - 2 \geq 1$. So by our strong inductive hypothesis, it is possible to order exactly $13 + k - 2$ egg rolls via a combination of 3-roll and 8-roll orders. By adding one 3-roll order to such a combination, we get a total order of $13 + k - 2 + 3 = 13 + k + 1$ egg rolls. So it is possible to order $13 + k + 1$ egg rolls via a combination of 3-roll and 8-roll orders. By the principle of strong induction, all of our statements are true. □

Let's note one point that arose when we applied the "strong" form of induction. It might seem small but is important. As we have phrased this principle, there is not just one base case. There is some number b of statements that we first check by hand, and the value of this b is not given in the statement of Principle 2.2.2. In our proof above, $b = 3$. Why do we now need to check multiple cases by hand, and how did we know to write the proof above with $b = 3$? In our previous applications of induction, we only ever checked one base case.

The answer is that our inductive step needed to "look back" 3 cases. If $k \geq 3$, then statement $P(k - 2)$ is an actual statement on our list, and we can use its truth to imply the truth of statement $P(k + 1)$. But for $P(1), P(2)$, and $P(3)$, there is no statement on our list to look back to. They are not part of the inductive step because they do not "build on" previous statements – they all just happen to be true, and we need individual arguments for all of them. If our inductive step required us to "look back" some other number of cases, we would need that number of base cases.

If the importance of multiple base cases still seems a bit unclear, this next exercise will be illustrative. It sounds quite similar to Example 2.2.1. Can it be approached the same way? Think about it before moving on.

Exercise 2.2.2. The student group from Exercise 2.2.1 is also buying fried dumplings for their party. The restaurant will sell "single person" orders of 4 dumplings or "group sized" orders of 12. The restaurant will not sell partial orders. Is it possible for the students to buy exactly 41 dumplings? In general what numbers of dumplings can they purchase?

In this case, the answer is *no*, 41 dumplings cannot be purchased. In fact, we can see very quickly that many numbers of dumplings are impossible. Why is this situation so different from Exercise 2.2.1? Well, the possible order sizes are 4 and 12, and 12 is a multiple of 4. The students can buy dumplings in orders of size $4j + 12k$, where j and k are non-negative

integers. But we can write $4j + 12k = 4(j + 3k)$. So every possible order size is a multiple of 4. Of course, for any integer $m > 0$, it *is* possible to buy $4m$ dumplings – the students just buy m 4-dumpling orders. So, possible dumpling order sizes are exactly the positive integer multiples of 4.

We didn't need induction to understand this problem. But *could* we understand it with strong induction? Hopefully, we cannot use strong induction to prove that all sufficiently large orders of dumplings are possible, because we know this isn't true! So what goes wrong?

Exercise 2.2.3. Suppose you did not spend time experimenting with this problem and made the false conjecture that for any natural number n, it is possible to buy $11 + n$ dumplings via some combination of 4-dumpling orders and 12-dumpling orders. Suppose further that, because you have some experience with similar problems, you believe you can prove your false conjecture using strong induction. But again, you are not taking a lot of time to do examples and build intuition, so you just compute a single base case ($n = 1$) and skip right to the inductive step. What does your inductive step look like? Does it symbolically look like the inductive step should succeed? For what specific values of k does the inductive hypothesis *not* imply the desired conclusion?

Hopefully these introductory examples have given you some time to digest the idea of strong induction. As with our previous examples of induction, we'll take some time now to discuss a few more problems to which the idea can be applied. As with "standard" induction, there are many and varied applications of strong induction. Let's think about some problems beyond egg rolls and dumplings where this style of argument works.

This next application concerns a property of numbers that you may already have heard about. The *binary representation* of a positive integer n is the (unique) way of writing n as a sum of distinct integer powers of 2. For example, $11 = 8 + 2 + 1 = 2^3 + 2^1 + 2^0$. In computer science (and elsewhere), this type of representation is so common that there is some shorthand notation for it, the *binary string*. A binary string looks like a list of 1's and 0's, where each "digit" in the string corresponds to some power of 2 and the digit's value (0 or 1) corresponds to how many copies of that power of 2 appear in the representation. So the binary string corresponding to the (base 10) number 11 is 1011. Here, 1011 is not the number one thousand and eleven, but a compact way of saying "Take one copy of 2^3, zero copies of 2^2, one copy of 2^1, and one copy of 2^0". In fact, this binary string representation works analogously to our base 10 number system, but whereas in our base 10 number system, each digit is between 0 and 9, the possible digits in a binary string are only 0 or 1.

Exercise 2.2.4. Write the binary string that corresponds to your current age and the binary string that corresponds to the current year.

You have probably seen binary strings written somewhere before (if not in a math or computer science context, then in a movie or TV show about hackers). Perhaps you have even worked with binary representations of numbers. But there is a specific claim about positive integers that we implicitly make when we work with binary representations, which is that each positive integer actually has one. How do we know that it is possible to write every positive integer as a sum of distinct powers of 2? This sounds like a job for strong induction. We give a proof first, and then some commentary.

Example 2.2.3. Let n be a positive integer. Then n can be written as a sum of distinct powers of 2.

Proof. The proof is by strong induction on n.
Base Case: $n = 1$. Observe, $1 = 2^0$ is a representation of 1 as a sum of distinct powers of 2, so the base case holds.

Inductive Step: Fix $k \geq 1$ and suppose that for all integers i with $1 \leq i \leq k$, we can write i as a sum of distinct powers of 2. Consider $k + 1$.

First, observe that if $k + 1$ is itself a power of 2, say $k + 1 = 2^m$ for some integer m, then we have trivially displayed a representation of $k + 1$ as a sum of distinct powers of 2 (in this case, just a single power of 2). So in this case, we are done. Thus, we may assume that $k + 1$ is not a power of 2. Thus, because $k + 1 > 1$, there is some integer m such that $2^m < k + 1 < 2^{m+1}$. So we can write $k + 1$ as

$$k + 1 = 2^m + (k + 1 - 2^m).$$

Note that $k + 1 - 2^m < 2^m$, because $k + 1 < 2^{m+1} = 2 \cdot 2^m$. Now, using the inductive hypothesis, because $1 \leq k + 1 - 2^m \leq k$, there is a way to write $k + 1 - 2^m$ as a sum of distinct powers of 2. Moreover, because the value of this entire sum is less than 2^m, we know that 2^m does not appear in this representation. If S is the sum representing $k+1-2^m$, then $S + 2^m$ is a representation of $k + 1$ as a sum of distinct powers of 2.

Thus, the inductive step succeeds, and we conclude that all natural numbers n have a representation as a sum of distinct powers of 2. □

You may notice some differences between this argument and our first example of strong induction. Let's highlight those. First, note that we only do one base case. We spent a fair amount of time talking about how more than one base case is necessary for the egg roll example. Why do we only need one here? Our inductive step also feels different. We have the expected inductive hypothesis, but the way we use it is not as straightforward as in our first example. And sometimes we don't seem to need the inductive hypothesis – if $k + 1$ is a power of 2, then we don't use any previous case to show that $k + 1$ has the correct representation. The only bearing that the inductive hypothesis has is that we have fixed an integer k such that $k + 1 = 2^m$. When $k + 1$ is not a power of 2, we do clearly use the inductive hypothesis, but unlike with the egg roll problem, there is not a set number of cases that we "look back" in our inductive step. We apply the inductive hypothesis to the number $k+1-2^m$, which could be anywhere between 1 and $2^m - 1$, depending on where $k + 1$ falls between 2^m and 2^{m+1}.

These two differences actually go hand-in-hand. Again, the reason why multiple base cases may be necessary is that in some strong induction problems, the first few small cases do not "build on" one another. We can only start the inductive step at the point where we know that every successive case is related to a previous case. So in the egg roll problem, because we always needed to look three cases back, we needed three base cases – there are no real cases "before" case 1 that we can build upon to establish case 2 and case 3. In the binary string problem, however, we know that for any $k > 1$, case $k + 1$ either immediately works (if $k+1 = 2^m$) or relies on case $k+1-2^m$ (if $2^m < k+1 < 2^{m+1}$). Because $k+1-2^m$

is a positive integer smaller than $k+1$, we are guaranteed that case $k+1-2^m$ is an actual previous case that we can build upon.

Our binary string example is definitely still a strong induction problem. The differences in structure between these examples emphasize (again) that induction problems don't follow a single template. We need to critically think about how our statements build on each other to determine exactly how our argument should go.

2.3 Problems

This chapter is different from all others in the book; whereas the others are held together by a common topic or area of math, this one has addressed a specific proof technique. We have sought to show how versatile this technique is by applying it to problems from different areas of math. This problem section will be similarly unusual; whereas in other chapters' problem sections we try not to tell you what proof technique to use, for this one, it will be no surprise that all of the problems can be handled with induction. That said, many of the problems can be solved in other ways too, and you should not force yourself to use induction if you see another approach that you can make equally rigorous. As with all our problem sections, we try not to steal your fun by telling you what to prove but rather, whenever possible, ask you first to explore the situation and make a conjecture.

1. Find and prove a simple formula for the sum of the first $n+1$ powers of 2 beginning with 2^0, that is, for
$$2^0 + 2^1 + 2^2 + \ldots + 2^n = \sum_{j=0}^{n} 2^j.$$

2. Let a and r be real numbers. A *finite geometric series* is a sum of the form
$$a + ar + ar^2 + \ldots + ar^n = \sum_{j=0}^{n} ar^j.$$
You may have encountered such series in calculus along the way to understanding infinite geometric series. It is an important and useful fact that, if $r \neq 1$, then
$$\sum_{j=0}^{n} ar^j = \frac{a(1-r^{n+1})}{1-r}.$$
Prove this formula.

3. Make and prove a conjecture about a common divisor shared by all integers of the form $7^n - 4$ for n a non-negative integer.

4. Let a and b be real constants and let
$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}.$$
Make and prove a conjecture about A^n.

5. Let
$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$
Make and prove a conjecture about A^n.

6. Some inequalities about integers can be proved using induction arguments. Each of the following inequalities is true for all integers n beyond some threshold. Through exploration, find this threshold and then give a proof of each inequality for all n beyond the threshold you identified.
   (a) $2^n \geq n^2$
   (b) $2^n \geq 2n + 1$
   (c) $n! \geq 2^{n-1}$.

You may remember from calculus that a sequence (denoted $\{a_n\}$) is an ordered list of numbers $a_1, a_2, \ldots$. Typically, we understand sequences via a formula for the n-th term a_n, which might either be an *explicit* function of n or be given *recursively* in terms of previous numbers in the sequence. (We will talk much more about sequences in Chapter 10.) The next few problems ask you to think about different sequences and some of their properties.

7. The Fibonacci numbers are an infinite sequence $\{F_n\}$ defined recursively by
$$F_1 = F_2 = 1 \quad \text{and} \quad F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geq 3.$$
The first few Fibonacci numbers are therefore
$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144.$$
In Chapter 10, we will see how to obtain an explicit formula for the n-th term of the sequence. The formula is almost certainly impossible for you to guess, even with abundant experimentation.
   (a) Using any method, find the first 12 terms of the sequence $\{O_n\}$ of Fibonacci numbers with odd index and the first 12 terms of the sequence $\{E_n\}$ of Fibonacci numbers with even index.
   (b) Find and prove a proposition about the sum of the first k Fibonacci numbers with odd index, that is, for $\sum_{j=1}^{k} F_{2j-1}$.
   (c) Now find and prove a proposition about the sum of the first k Fibonacci numbers of even index, that is, for $\sum_{j=1}^{k} F_{2j}$.

8. Show that the n-th Fibonacci number F_n satisfies $F_n < 2^n$.

9. There are many beautiful patterns and identities relating to the Fibonacci numbers. One such beautiful identity describes the value of the expression
$$F_{n+1} \cdot F_{n-1} - F_n^2.$$
Find the value of this expression for several values of n. What do you conjecture about this expression? Prove your conjecture.

10. The *Lucas numbers* $\{L_n\}$ are a sequence very similar to the Fibonacci numbers. We define $L_1 = 2, L_2 = 1$, and for $n > 2$, we have $L_n = L_{n-1} + L_{n-2}$. Like the Fibonacci numbers, we will talk much more about the Lucas numbers in Chapter 10.

    (a) Generate the first 10 Lucas numbers. Are these the same as the first 10 Fibonacci numbers?

    (b) Explore the value of the expression

    $$L_{n+1}L_{n-1} - L_n^2.$$

    Recall that we discussed the "Fibonacci" version of this expression in a previous exercise. Conjecture and prove a formula for the value of this expression.

    (c) Lucas and Fibonacci numbers are closely connected in many ways. Compare the expressions L_n and $F_{n-1} + F_{n+1}$ for several values of n. How do you conjecture these expressions to be related? Prove your conjecture.

11. Define the sequence $\{a_n\}$ by $a_1 = 2$ and, for $n > 1$, $a_n = \frac{a_{n-1}+1}{2}$. Conjecture and prove a simple, explicit formula for a_n (that is, a formula expressing a_n as a function of n, not in terms of any previous terms in the sequence).

12. A sequence $\{a_n\}$ is said to be *bounded above* by B if $a_n \leq B$ for all n. We similarly define what it means for $\{a_n\}$ to be bounded below. Explore the sequence $\{a_n\}$ defined by $a_1 = 1$ and $a_n = \sqrt{2a_{n-1} + 3}$ for $n > 1$. Conjecture the best upper and lower bounds you can for this sequence, and prove your conjectures.

The next problems require you to know calculus. You can skip them if you feel it is logically suspect to use facts and techniques from calculus that we have not proved in this book. We pose the problems despite this fact because it opens up a wide variety of new problems to explore and because most calculus books do contain proofs of the basic rules. The missing ingredient is really only the precise definition of a limit (which some introductory texts include but many skip).

13. A function is said to be *infinitely differentiable* if its derivatives of all orders exist and are differentiable. Each of the following following functions is infinitely differentiable on the stated set. For each, find a formula for $f^{(n)}$, its n-th derivative, and prove your formula by induction, using the standard differentiation rules.

    (a) $f(x) = \dfrac{1}{1-x}$ for x real, $x \neq 1$.

    (b) $f(x) = e^{-ax}$ for x real and a a fixed real constant.

    (c) $f(x) = x^{-\frac{1}{2}}$ for positive real x.

    (d) $f(x) = \frac{e^x - e^{-x}}{2}$ for x real.

14. Let f be an infinitely-differentiable function on $\mathbb{R}$. Let a be any real constant and define a new infinitely-differentiable function by $g(x) = f(ax)$. Find and prove a formula for $g^{(n)}$ in terms of $f^{(n)}$.

15. The function given by $f(x) = e^{x^2}$ for x real is also infinitely differentiable. Although writing down an explicit formula with all the correct coefficients for the n-th derivative $f^{(n)}$ is somewhat difficult, it should not surprise you that it has the form $f^{(n)}(x) = e^{x^2}p_n(x)$ for some function p_n. By finding p_n for several small n, make as precise of a conjecture as you can about the properties of p_n. Prove it by induction.

The next few problems ask you to think about function composition. Let $f^n(x)$ denote the n-fold composition of $f(x)$ with itself n times, so that $f^2(x) = f(f(x))$, $f^3(x) = f(f(f(x)))$, etc. You may recall that we posed some of these questions in Section 1.3; we now have the tools to prove such properties.

16. If $f(x) = 2x + 1$, find and prove a simple formula for $f^n(x)$.

17. Let a and b be fixed (non-zero) real numbers. If $f(x) = ax + b$, find and prove a simple formula for $f^n(x)$. (Note that this generalizes the previous exercise.)

18. If $f(x) = (\sqrt{x} + 1)^2$, find and prove a simple formula for $f^n(x)$.

19. If $f(x) = \frac{1}{x}$, find and prove a simple formula for $f^n(x)$.

20. Bertrand's Postulate (also known as the Bertrand-Chebyshev Theorem) is a famous fact about the distribution of prime numbers in the integers. Originally proved by Chebyshev in 1850, the fact has been re-proved several times via successively simpler arguments. The mathematician N. J. Fine summarized both the substance and history of Bertrand's Postulate in the couplet:

    Chebyshev said it, but I'll say it again:

    There is always a prime between n and $2n$.

    While adding this precision would throw off the meter of the couplet, we should clarify that here, n is an integer greater than 1, and Bertrand's Postulate officially states that there is always a prime *strictly* between n and $2n$ (so, a prime number p with $n < p < 2n$).

    (a) Test out Bertrand's Postulate for the following specific positive integers n:

    $$2, 3, 4, 5, 10, 20, 100$$

    Does the statement hold in these cases? (You don't need to attempt a general proof!)

    (b) You may assume that Bertrand's Postulate is true. Using this, show that any positive integer $n \geq 7$ can be written as a sum of distinct prime numbers. Why do we require $n \geq 7$ here? (Hint: it is actually easier here to prove the following *stronger* statement. Any positive integer $n \geq 7$ can be written as a sum of distinct primes, the largest of which is at most the maximum of 11 and $n - 7$.)

    (c) What does Bertrand's Postulate imply about the total number of primes?

21. (a) Let n be a positive integer which is not a Fibonacci number, and say F_k is the biggest Fibonacci number which is less than n. Explain why $n - F_k < F_{k-1}$.

    (b) Use strong induction, along with part (a), to prove the following claim: every positive integer is either a Fibonacci number or can be written as a sum of two or more distinct Fibonacci numbers.

22. In a certain carnival game, players are asked to throw darts at a board. Each dart will score a certain number of points based on where it hits the board. Players may purchase as many darts to throw as they want, and a player's total score is the sum of their scores for each dart they throw.

    (a) Suppose a dart will score 3 points if it hits the board. If the dart hits a smaller "bullseye" region, the dart will score 7 points. What total scores can a player achieve?

    (b) Suppose the board has two "bullseye" regions marked, with one smaller than the other. A dart will score 3 points if it hits the board, 7 points if it hits the larger "bullseye" region, and 10 points if it hits the smaller "bullseye" region. What total scores can a player achieve? Is this different from part (a)? Why or why not?

    (c) Suppose the board has two "bullseye" regions marked, with one smaller than the other. A dart will score 3 points if it hits the board, 7 points if it hits the larger "bullseye" region, and 11 points if it hits the smaller "bullseye" region. What total scores can a player achieve? Is this different from part (a)? Why or why not?

2.4 Suggested Projects

2.4.1 The Arithmetic-Geometric Mean Inequality and Its Generalizations

This project concerns a famous inequality. At first glance, the project may seem to have little to do with the topic of this chapter. However, after you have had a chance to explore and prove the basic inequality, we will suggest how you can obtain generalizations. The crowning problem of this project involves giving an induction proof of a family of generalizations. Your product for this project should be an expository report similar to a section of this textbook in which you carefully define the objects of interest (the arithmetic and geometric mean), discuss their applications, and state and prove the inequalities relaing them. The exercises below are to help you structure your exploration. You should not just produce a list of solutions to the problems.

The inequality you will investigate is called the *arithmetic-geometric mean inequality*, or *AGM inequality*, for short. In its simplest form, it relates two different *means*, or averages. One of them is undoubtedly familiar to you, but the second is probably not. We give the two definitions.

Definition 2.4.1. Let a and b be real numbers. Their *arithmetic mean* is given by $\frac{a+b}{2}$.

Definition 2.4.2. Let a and b be non-negative real numbers. Their *geometric mean* is given by $\sqrt{ab}$.

Note that the arithmetic mean makes sense for both positive and negative numbers. For example, if the temperature in Kirksville, Missouri was $4°F$ on January 13 and $-10°F$ on January 14, you can conclude that the average temperature for the two days was $\frac{4+(-10)}{2} = -3°F$ and can begin planning your move to San Diego. On the other hand, the geometric mean doesn't make sense if the expression under the square root is negative, and so we only consider the geometric mean for non-negative numbers. But why is $\sqrt{ab}$ called the geometric mean of a and b? Imagine a rectangle with side lengths a and b. Then the geometric mean $\sqrt{ab}$ is the side length of a square whose area is the same as the area of this rectangle.

The first set of problems gives you an opportunity to explore these two types of averages of two numbers.

1. Suppose you invest \$10,000. In the first year, your investment goes up by 10% and in the second year, your investment goes down by 10%.

   (a) How much money do you have at the end of two years? Explain carefully why the answer is not \$10,000. This calculation shows that the effective annual interest rate is *not* the arithmetic mean of the annual interest rates (which is 0).

   (b) On the other hand, we can compute the effective annual interest rate by taking $\sqrt{1.1 \cdot 0.9}$. Find this effective annual interest rate and verify that it gives you the same answer as you obtained above. Explain why this procedure works. Note that this is an application of the geometric mean.

2. Explore the relationship between the arithmetic and geometric mean of two positive real numbers by computing both for at least 10 pairs (a, b). Present your data through a table or a graph and summarize your observations.

You should have observed that the geometric mean is always less than or equal to the arithmetic mean. This proposition is called the *arithmetic-geometric mean (AGM) inequality*.

Proposition 2.4.3. *Let a and b be non-negative real numbers. Then*

$$\sqrt{ab} \leq \frac{a+b}{2}. \tag{2.1}$$

The next few exercises guide you through a proof. Just to be clear, we are not yet at the point in the project where induction is useful.

3. Show that for any real numbers A and B (not assumed non-negative),

$$2AB \leq A^2 + B^2.$$

4. Now prove the AGM inequality for non-negative real numbers a and b.

5. Determine the precise conditions under which there is equality in (2.1).

Next, we seek to generalize the AGM inequality to products of collections of more than two non-negative real numbers. One would expect the generalization to say that for N non-negative real numbers $a_1, \ldots, a_N$,

$$(a_1 a_2 \cdots a_N)^{\frac{1}{N}} \leq \frac{a_1 + a_2 + \ldots + a_N}{N}.$$

This inequality is harder to prove for some N than others. It turns out that it is not hard to prove when N is a power of 2[2], that is when $N = 2^n$ for some natural number n, and in fact an induction proof is possible. We would like to guide you towards giving this proof.

If we were to just tell you "Go to it – prove the generalized AGM inequality for $N = 2^n$ by induction on n," you'd start off well, noting that the base case is just the AGM inequality for two numbers, but you would likely get tangled up in notation at the inductive step. In many cases, it is instructive to do one or two more small cases first. These cases don't become part of your final proof, but it is quite likely that without doing them, you won't be able to write the general proof. Never shy away from doing more examples. It is a principle of doing mathematics that if you can work out a sufficiently complicated case, you can probably prove the general result.

6. Prove the generalized AGM inequality for $N = 4 = 2^2$ by somehow using the AGM inequality for $N = 2 = 2^1$.

7. Now prove the generalized AGM inequality for $N = 8 = 2^3$ using (at most) the inequalities for $N = 2$ and $N = 4$.

8. Now try to prove the generalized AGM inequality for $N = 2^n$ by induction on n.

2.4.2 Binomial Coefficients, Part I

(1 ⏲) "He is a man of good birth and excellent education, endowed by nature with a phenomenal mathematical faculty. At the age of twenty-one he wrote a treatise upon the Binomial Theorem, which has had a European vogue." (*The Final Problem*, Sir Arthur Conan Doyle)

This description of the evil genius James Moriarty strikes many mathematicians as humorous because, even in the late 1800's, the Binomial Theorem would hardly have been considered cutting edge mathematics capable of earning one a reputation throughout Europe. Still, this famous theorem has many useful applications and connections that we explore in this project. This project focuses on those facts that can be established using a combination of basic algebra and, at times, induction. If you enjoy this project, there is a continuation in Chapter 4 that connects the special *binomial coefficients* that appear in the Binomial Theorem to problems in *combinatorics*, which is the field of mathematics devoted to the counting of finite sets. (It's harder but also more interesting and beautiful than it sounds.)

The Binomial Theorem seeks to answer the question: What polynomial results if I raise a binomial to a positive integer power? In other words, if a and b are real numbers and n is a natural number, what is $(a + b)^n$? The following table shows the answer for $n = 1, 2, 3, 4, 5$.

[2]Note the similarity with the chessboard tiling problem from earlier in the chapter.

n	$(a+b)^n$
1	$a+b$
2	$a^2+2ab+b^2$
3	$a^3+3a^22b+3ab^2+b^3$
4	$a^4+4a^3b+6a^2b^2+4ab^3+b^4$
5	$a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$

Some patterns in the table are immediately apparent; in the n-th row, the polynomial has $n+1$ terms, starting with a^n, and each term has a power in a that is one less than the previous and a power in b that is one more. The coefficients themselves also seem to follow some patterns; the first and last coefficients always seem to be 1, and the second and second last always seem to be n. However, it would be rather difficult for us to motivate a conjecture about a general formula for the coefficients. Instead, we will tell you the formula that people have discovered, and one of the goals of the project will be to prove that it works.

Definition 2.4.4. Let n and k be non-negative integers with $0 \leq k \leq n$. We define $\binom{n}{k}$, read "n choose k", by the formula

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}. \tag{2.2}$$

Numbers of the form $\binom{n}{k}$ are called binomial coefficients. This terminology reflects their central role in the Binomial Theorem, which we are now ready to state.

Theorem 2.4.5. *Let n be a positive integer. Let a and b be real numbers. Then*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k. \tag{2.3}$$

The problems in this project will guide you through a proof and show you a number of interesting applications. The product from this project should be a report on the binomial theorem and the binomial coefficients that includes their definition, the statement and proof of the Binomial Theorem, and some interesting applications of the theorem and facts about the coefficients. In preparation you should do these problems, but your report should not just be a list of solutions to them.

1. To start to understand the binomial coefficients, make a table whose rows correspond to values of n between 1 and 6. In each row, list, in order, the binomial coefficients $\binom{n}{k}$ for $0 \leq k \leq n$. So, for example, in the row for $n = 3$, you will compute the four binomial coefficients $\binom{3}{0}$, $\binom{3}{1}$, $\binom{3}{2}$, and $\binom{3}{3}$. Compare these numbers with the table we made above.

2. The binomial coefficients in the n-th row of your table are related in a very interesting way to those in the previous row. In particular, if $n \geq 1$ and $1 \leq k \leq n-1$, then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}. \tag{2.4}$$

This formula is called *Pascal's identity.*

(a) Illustrate Pascal's identity by verifying that the binomial coefficients $\binom{5}{1}$, $\binom{5}{2}$, $\binom{5}{3}$ and $\binom{5}{4}$ are related to the binomial coefficients in the fourth row according to the equation (2.4).

(b) Although we will see other elegant arguments later, one can prove Pascal's identity algebraically using the definition (2.2). Prove Pascal's identity using this method.

3. Now prove the Binomial Theorem, Theorem 2.4.5, by induction on n.

4. Use the Binomial Theorem to evaluate each sum.

(a) $\displaystyle\sum_{k=0}^{n} \binom{n}{k}$

(b) $\displaystyle\sum_{k=0}^{n} (-1)^k \binom{n}{k}$

(c) $\displaystyle\sum_{0\leq k\leq n,\ k \text{ even}} \binom{n}{k}$

(d) $\displaystyle\sum_{0\leq k\leq n,\ k \text{ even}} \binom{n}{k} a^k$ for a any non-zero real number.

(e) Aren't these fun?

5. Let f and g be infinitely-differentiable functions on $\mathbb{R}$. Find and prove a formula for $(fg)^{(n)}$, the n-th derivative of their product.

2.4.3 Fractals

(2◷) You may have heard of fractals before – beautiful and sometimes paradoxical geometric objects which are often defined by recursive procedures. Fractals are characterized by *self-similarity*: if you "zoom in" on part of a fractal, you will see the fractal's shape reproduced at a smaller scale. In fact, no matter how much you "zoom in" on a fractal, you will see the same shape at smaller and smaller scales. That is, fractals look *infinitely detailed*.

It is not possible for us to actually draw an infinitely detailed, infinitely self-similar object. But we can often describe how such an object arises, and draw steps "towards" such an object, by defining a recursive building procedure. The purpose of this project is to explore the recursive procedures which define three famous fractals: the Cantor Set, the Sierpinski Triangle, and the Koch Snowflake. Because these fractals are built up recursively – that is, by applying some procedure over and over, one step at a time – they lend themselves naturally to inductive arguments. We caution readers that, in order to describe and work with fractals, some knowledge of concepts relating to the infinite (limits, series) will be necessary. The introduction you receive to these topics in your first year of calculus should be sufficient; however, if you do not remember these topics well or do not feel that you have studied them on rigorous enough footing to justify their use in a proof-writing project, we suggest choosing another project.

Chapter 3

Logic

In the first chapter of this book, we described a proof as a clear, complete argument in which each statement follows logically from the previous. And it certainly seems as if we have given many good proofs in the last two chapters. So why do we need an entire chapter now on logic? Were we doing it wrong? Should we have started here?

Of course this chapter is where we intended it to be. We have taken an immersion approach to teaching proof in which we have begun by focusing on the idea that proof serves a purpose – to communicate one's understanding of a mathematical situation to another – and it is good to lead with that purpose. You can learn a lot about what makes a proof good by simply reading some, thinking about them, trying to write your own, and engaging with others in a critical analysis of these proofs. But there is a time and place for learning all the rules of logic, and, no surprise, it is now.

Perhaps an analogy will help. Parents don't say to each other one day "Phoebe is 6 months old. Time to teach her our native language. Get out the grammar books." Instead, they start with the idea that learning one's native language starts with a desire to communicate – "I'm hungry", "I'm cold", "I love you", "I don't like peas." Others model the use of (more or less) correct language, the child tries to understand and to imitate, and with time, communication improves. Children become fluent in their native language this way. However, at some point, their ability to communicate is improved by studying the rules, and so we start to teach grammar, proper usage, spelling, and punctuation. All this additional instruction helps the student move from fluent to literate, articulate,and even eloquent.

This chapter is the beginning of your training on the "grammar" of proofs. We have already used informally many of the rules of logic we will discuss in this chapter, just as parents and teachers model many of the rules of grammar before starting formal instruction. You will find that you already have some familiarity and facility with many concepts in this chapter because you often think in logical patterns, even if you do not assign formal words and symbols to them. So, while you will definitely learn some new vocabulary in this chapter, we hope that you will notice that you already understand most of the basic principles. But one reason to study formal logic carefully is that, although 15 years of education have helped you internalize some basic logical principles, as the statements you want to prove become increasingly complex, it will become increasingly important to think explicitly and carefully about their logical structure. To go back to our grammar analogy, we can all understand the sentence "I don't like peas" without a formal discussion of English grammar, but there

are some sentences in modern literature that are so long and complex that you will get lost unless you diagram them. To give a more mathematical comparison, think about when you learned algebra. You might have initially thought it was silly because you could solve easy equations like $2x + 1 = 13$ in your head by inspection. But the rules of algebra help when things become so complicated that you can no longer "see" the answer or hold all the details in your head.

One final comment before we start. Every book for a proof course has to have a chapter like this one, but most are dry because, by design, we have to step back from the theme of "I have some interesting math I want to communicate" and have to instead focus on "How can I say this simple thing correctly?" We do want to make sure that the focus of our discussion is on the logic, and so we choose examples that do not require substantial mathematical background to understand. Some of these examples will cover very simple mathematics, like the properties of even and odd integers. Some will discuss mathematical situations you may not have thought about before, but the emphasis will always be on thinking clearly through a logical structure, not learning specific new mathematical content. We will also try to intersperse non-mathematical examples, likely a bit quirky because, what's life without whimsy[1]?

3.1 The Basics

3.1.1 Statements and Open Sentences

One of the main themes of this chapter is that sometimes we use words in mathematics in different or more precise ways than we do in common English. In common English, a *statement* is a declarative sentence, for example "Of all the tree-dwelling animals of the tropical rainforest, none is as intriguing as the sloth ". In math, a *statement* must have a *truth value*, meaning that it is either true or it is false (but not both). The following are statements:

The integer 3 is odd.
The integer 7 is even.
Some sloths live in Central America.
Sloths eat tropical birds and wild cats.

Hopefully you recognize the first statement as true and the second as false. The third and fourth sentences are also statements, even if you personally don't know enough about sloths to know whether they are true or false. It turns out that some sloths *do* live in Central America, but they do not eat tropical birds or wild cats.

Consider next these sentences:

The integer n is greater than 100.
The number of toes on the sloth's front foot is 3.

[1]So in other words, this chapter may feel more boring than other chapters in this book, but hopefully it will be less boring than its counterparts in similar textbooks.

If we were to ask you whether these are true or false, you'd probably refuse to answer. In the first case, you would say that we need to specify what n is – some integers are more than 100 but some are less. For the second, you would probably have a similar objection, requiring us to specify which sloth we are discussing (as perhaps you recall that some sloths have 3 toes on each front foot and others have 2, not to mention the case of Quatro, the sloth with a rare genetic condition resulting in four toes per foot, or other outliers). You might think the issue is simply that we were vague, but it is better to think of each of these sentences the way you think of a function, with a domain. Once we specify an element of the domain, you obtain a specific statement with a truth value. We call such objects *open sentences*. For the first, the domain is clearly intended to be the integers. So for each integer n, we obtain the statement

$$P(n)\text{: The integer } n \text{ is greater than 100.}$$

Then $P(243)$ is true because 243 is greater than 100 but $P(-18)$ is false because -18 is not greater than 100. For the second statement, the domain is intended to be all sloths in the world. For each individual sloth, you can similarly count the toes on one of the sloth's front feet and determine the truth value for this particular sloth.

3.1.2 New Statements from Old; Truth Tables

Some of the propositions we want to prove really are just simple statements like the ones we discussed in the previous subsection. For example, in one of the projects at the end of this chapter, you will have the opportunity to prove the statement

$$\sqrt{2} \text{ is irrational.}$$

However, many propositions are more complicated combinations of statements. For example, you may be familiar with the Mean Value Theorem from first semester calculus:

Let $f\colon [a,b] \to \mathbb{R}$ be a function. If f is continuous on $[a,b]$ and differentiable on (a,b), then there exists $c \in (a,b)$ such that $f'(c) = \frac{f(b)-f(a)}{b-a}$.

This theorem involves three open sentences:

$$p_1(f)\text{: } f \text{ is continuous on } [a,b].$$
$$p_2(f)\text{: } f \text{ is differentiable on } (a,b).$$
$$q(f)\text{: There exists } c \in (a,b) \text{ such that } f'(c) = \tfrac{f(b)-f(a)}{b-a}.$$

The theorem states that if f is a function for which $p_1(f)$ and $p_2(f)$ are true, then $q(f)$ is also true. We need to be able to work with more complicated logical structures like these.

Moving forward, we will often use the term "statement" to mean either a statement or an open sentence because the definitions we make work the same for both. We simply need to keep in mind that open sentences have domains and we can't assign a truth value until we specify an element of the domain. Right now it might sound as if we're being sloppy, but we promise that no confusion will arise.

We start by defining several ways to make new statements from old. As with the example of the Mean Value Theorem, many statements that we work with in mathematics are really

built from two or more simpler statements. We want to understand how multiple statements can be combined to form a single larger statement and understand how determine the truth value of the larger statement. In fact, when we build a statement from several smaller statements, the truth value of the "big" statement depends on the truth values of the "small" statements. We can describe this relationship in words, but this can become unwieldy. It is often better to capture such dependencies through a visual aid called a *truth table*. A truth table records the truth values of the "small" parts of a complex statement and then displays how these truth values impact the truth of the "big" statement.

It will be helpful to illustrate these new logical structures with examples. We'll define four statements or open sentences to use throughout this section. For the first two, we take the domain to be all real numbers x.

$$p(x)\text{: } x < -2.$$
$$q(x)\text{: } x^2 > 4.$$

The second set of examples take us back to the wonderful world of tree-dwelling creatures:

$$P\text{: I am a sloth.}$$
$$Q\text{: I live in a tree.}$$

We first define the *negation* of a statement.

Definition 3.1.1. If p is a statement, its **negation** is the statement $\neg p$ with the opposite truth value.

So, the negation $\neg p$ is false if p is true and true if p is false. Although the definition might sound abstract, when we are working with a specific statement p, it is usually straight-forward to come up with its negation $\neg p$. For example, if statement p is "The card you are thinking of is the 3 of spades", then the negation $\neg p$ is the statement "The card you are thinking of is not the 3 of spades". The negation of a statement always concerns the same "subject" as the original statement. We would not say that the negation is "You are not wearing purple socks" even if there were some complete but mysterious synchronicity between your sock color and your thoughts about the 3 of spades.

Let's illustrate the process of building a truth table by summarizing the relationship between p and $\neg p$. In the first column of the table, we list the possible truth value for p. In the second column, we give the truth values for the negation given the truth value for p.

p	$\neg p$
T	F
F	T

You probably didn't need a fancy table to understand this definition, but such tables will become increasingly helpful as the statements become increasingly complex. Going back to the analogy with learning grammar, making truth tables is similar to diagramming English sentences; it seems unnecessary for simple sentences but can be quite helpful for complicated ones.

To make the discussion of negation more concrete, for our sample statements above, we have the negations

$\neg p(x)$: $x \geq -2$.
$\neg P$: I am not a sloth.[2]

Hopefully after a few examples, the idea of negation feels natural. However, forming negations is such a foundational skill that it is worth emphasizing. As you try to prove or refute conjectures, you will often need to understand the negation of a statement, and it will be helpful to think very carefully about what the negation really says. For any statement p, we can always form $\neg p$ by saying "It is not the case that p occurs", but such a formulation doesn't always help us think carefully about $\neg p$. For instance, in our example above, in which $p(x)$ is "$x < -2$", we did not just say $\neg p(x)$ is "It is not the case that $x < -2$". Instead, we stated $\neg p(x)$ as "$x \geq -2$" because this formulation gets at the property satisfied by those real numbers x for which $p(x)$ is false.

When you encounter logical statements in the wild, you will similarly want to directly understand their negations. Focus on what property *will* be satisfied when the statement is not true.

Exercise 3.1.1. For each statement or open sentence, formulate the negation. (If you do not remember the definition of some term used, feel free to look it up.)

(a) The triangle ABC is isosceles.

(b) The positive integer n is prime.

(c) On every single Tuesday, it rains.

(d) Somewhere in the world, there exists a purple sloth.

Often, as we saw with the Mean Value Theorem above, we wish to prove propositions in which we assume that multiple conditions hold simultaneously. We therefore study ways in which we can build complicated statements from multiple simpler ones.

Definition 3.1.2. Let p and q be statements. Their **conjunction** $p \wedge q$ (read "p and q") is the statement that is true if both p and q are true and is false otherwise. The **disjunction** $p \vee q$ (read "p or q ") is the statement that is true if at least one of p and q is true but is false if both are false.

Like with negations, for specific examples, we can talk ourselves through the conjunction or disjunction of two statements. But for *any* pair of statements, we can use the definition to make a truth table for the conjunction and disjunction. We see that now the truth table is probably easier to read than the definition. For the conjunction we have:

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

[2]Possibly attributed to Richard Nixon, November 1973.

and for the disjunction we have:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

.

Example 3.1.3. Let's explore the conjunction and disjunction for the sloth-related examples above. $P \wedge Q$ is only true if I am a sloth and I live in a tree. On the other hand, $P \vee Q$ is true in this case, but also if I am a sloth living in some different habitat (so that P is true but Q is false) or if I am not a sloth but I live in a tree – if, for example, I am a woodpecker or a squirrel.

Exercise 3.1.2. For the other sample open sentences $p(x)$ and $q(x)$ above about real numbers, form the conjunction $p(x) \wedge q(x)$ and disjunction $p(x) \vee q(x)$. Find values of x for which

(a) both $p(x) \wedge q(x)$ and $p(x) \vee q(x)$ are true,

(b) both $p(x) \wedge q(x)$ and $p(x) \vee q(x)$ are false,

(c) $p(x) \wedge q(x)$ is false but $p(x) \vee q(x)$ is true.

We can build more complicated statements by taking negations of conjunctions or disjunctions or by involving more statements. Such a thing can become arbitrarily complicated, but we illustrate with an example and ask you to explore further with some exercises.

Example 3.1.4. Suppose now we want to understand $\neg(p \wedge q)$. In a sense, we can do so easily; we can immediately make the truth table by negating the entries in the final column of the table for $p \wedge q$:

p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

.

Although this truth table is conclusive, we might ask for a deeper understanding. For example, when considering $\neg(p \wedge q)$, we might ask whether there is any sort of "distributive law" for the negation, as there would be if we had a negative 1 in front of a binomial. Is there a rule like $\neg(p \wedge q) = \neg p \wedge \neg q$?

Let's talk through this idea in the context of our sloth example. In that case, the conjunction is

I am a sloth and I live in a tree.

Under what conditions is the negation true? The negation is true if the statement itself is false, which can happen *either* if I am not a sloth *or* if I don't live in a tree. In other words, it appears that the negation is $\neg p \vee \neg q$. We can verify with an even larger truth table.

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

The previous example illustrates an important concept; just as two seemingly different algebraic expressions can turn out to be equal, two different logical statements can turn out to have the same truth table.

Definition 3.1.5. Two logical statements are said to be **logically equivalent** if they have identical truth tables. In this case, each is true if and only if the other is true – so, knowing the truth value of one automatically tells you the truth value of the other.

Right now, logical equivalence might seem to be a rather esoteric concept that we would seldom use. However, we will see that sometimes a thing we want to prove seems hard, but it is logically equivalent to a statement whose proof seems easier. We will explore this idea below.

Exercise 3.1.3. Let p and q be logical statements. Make a conjecture about a logical statement that is equivalent to $\neg(p \vee q)$ and prove it using a truth table.

Exercise 3.1.4. Let p, q, and r be logical statements. Make a truth table for each of the following and determine which, if any, are logically equivalent.

(a) $p \wedge (q \vee r)$

(b) $(p \vee q) \wedge (p \vee r)$

(c) $(p \wedge q) \vee (p \wedge r)$

(d) $p \vee (q \wedge r)$

In each case, note how many rows your truth table requires to illustrate the possible combinations of truth values for your "base" statements. In general, can you make a conjecture about the number of rows required for a truth table of a logical structure comprised of n statements?

As you continue in math, you probably won't make a formal truth table every time you need to consider a complex statement. But you *will* often need to think about such statements and parse out when they are true. Let's wrap up this discussion with another concrete example in which we need to keep multiple statements in mind at once. This situation has no mathematical prerequisites whatsoever. You may have encountered logic puzzles or deduction games in the past, where you are given some list of statements and rules about a situation and must draw some conclusion from these. With the language of formal logic, we can describe and verify the solutions of such puzzles. (Try this puzzle on your own before you read the solution. Hopefully it's fun!)

Example 3.1.6. At his quiet country estate, Baron Banach is hosting a mathematical dinner party. His guests discuss so many interesting proofs that they grow tired and are invited to stay overnight. Very early in the morning, Baron Banach is murdered![3] Because Baron Banach was a famous analyst and the crime happened very early, the detective on the case surmises that the murderer went to bed early (since early to bed, early to rise) and that the murderer does not enjoy analysis. The crime must have been committed by one of the dinner guests.[4] Here is a list of the guests, with some information about each.

1. **Professor Poincaré** does not enjoy graph theory and does not smoke.
2. **Herr Hilbert** does not enjoy graph theory and smokes.
3. **Doctor Dirac** enjoys graph theory and does not smoke.
4. **Mr. Mantel** enjoys graph theory and smokes.
5. **Lady L'Hospital** does not smoke and goes to bed early.

The detective also uncovers the following general facts:

1. All guests either enjoy analysis or enjoy graph theory, but none enjoy both.
2. Every guest who enjoys graph theory either smokes or drinks coffee.
3. Guests who drink coffee do not go to bed early.

Using these facts, can you determine which of the guests is the murderer?

When you tried this puzzle on your own, perhaps you just thought through the statements until you discovered how they connect to each other, or perhaps you wrote down some formal truth tables. There is nothing wrong with thinking through such an example in the less "formal" way; many mathematicians would approach this problem by thinking carefully and writing a few paragraphs, with no truth tables or logical symbols in sight. However, this example is also a good illustration of how the formalism can help if you are struggling to hold many statements in mind at once. Let's label some statements that appear in this example and see how they interact.

In the example, we are given some fairly complex statements, but they are all built from the following simpler ones:

a: The guest enjoys analysis.
g: The guest enjoys graph theory.
b: The guest goes to bed early.
c: The guest drinks coffee.
s: The guest smokes.

[3](In the conservatory with the candlestick.)

[4]Because the only other member of the household is the butler, and it would be too cliché for the butler to have done it.

So in logical symbols, we are told that $\neg a \wedge b$ is true about the murderer. So, we wish to determine for which (if any) of the guests this statement us true. But we are not directly told whether any of the guests enjoy analysis and we are only told that one of the guests goes to bed early. Unsurprisingly, we will have to use the interactions between the other statements to see whether the information we have about each guest is consistent with $\neg a \wedge b$.

Let's look at the general facts we are given and see what conclusions we can draw. The first fact is that all guests enjoy either analysis or graph theory, but not both. So, although we are not directly told how each guest feels about analysis, we can deduce how they feel about analysis by asking how they feel about graph theory. We can summarize this relationship in the following table:

g	a
T	F
F	T

Let's notice that this looks exactly like the truth table comparing g and $\neg g$ or comparing a and $\neg a$. In fact, we can augment this table with $\neg g$ and $\neg a$ to observe this relationship directly:

g	a	$\neg g$	$\neg a$
T	F	F	T
F	T	T	F

This means that g is logically equivalent to $\neg a$ and a is logically equivalent to $\neg g$. In other words, anyone who enjoys graph theory does not enjoy analysis. On the other hand, anyone who does not enjoy graph theory must enjoy analysis, so two suspects (Professor Poincaré and Herr Hilbert) are cleared because they do not enjoy graph theory. From now on, the condition "the murderer does not enjoy analysis" will be replaced with "the murderer enjoys graph theory", since that is what we are directly told about the guests.

Now, we must sort out the business with smoking, coffee, and going to bed early. Let's look at the second general fact we are given: Every guest who enjoys graph theory either smokes or drinks coffee. In logical symbols, the statement "This guest enjoys graph theory and either smokes or drinks coffee" is written $g \wedge (s \vee c)$. We have previously made truth tables for disjunctions; we know that $s \vee c$ is true if at least one of s or c is true, but $s \vee c$ is false if both s and c are false. Thus, for each guest who enjoys graph theory, if we know that s is false for that guest, then c must be true, and vice versa. (Of course, it is possible that both s and c are true for a guest.) Looking at the information we have, Doctor Dirac enjoys graph theory, so $g \wedge (s \vee c)$ is true of Doctor Dirac. So $s \vee c$ is true of Doctor Dirac, and we are told that s is not true, so c must be true of Doctor Dirac. We are told that both g and s are true of Mr. Mantel, so $g \wedge (s \vee c)$ is true of Mr. Mantel regardless of whether c is. So, we cannot conclude whether Mr. Mantel drinks coffee. Our last suspect is Lady L'Hospital. We are told that she does not smoke, but we are not told whether she enjoys graph theory or not. *If* we knew that Lady L'Hospital enjoys graph theory, then we would know that $g \wedge (s \vee c)$ is true of her, and make some conclusions. But if she does not enjoy grpah theory, then we do not know if $s \vee c$ is true of her. So we cannot conclude anything about Lady L'Hospital just yet.

We did not rule out any suspects with our second round of discussion, but we have made some progress. The last thing we are told is that guests who drink coffee do not go to bed early. So the statement $b \wedge c$ is false for every guest. Thus, the statement $\neg(b \wedge c)$ is *true* of every guest. As we previously verified with a truth table, $\neg(b \wedge c)$ is logically equivalent to $\neg b \vee \neg c$. We know that $\neg c$ is false for Doctor Dirac; we previously concluded that he must drink coffee. So $\neg b$ must be true of Doctor Dirac. Doctor Dirac does not go to bed early, so cannot be the murderer. Similarly, we know that $\neg b$ is false of Lady L'Hospital; we are directly told that she goes to bed early. So $\neg c$ must be true of Lady L'Hospital.

We are now down to only two suspects: Mr. Mantel and Lady L'Hospital. How can we choose between them? Let's revisit our rules with the full list of information we have deduced. We were previously stuck determining Lady L'Hospital's properties because we were not told whether she enjoys graph theory. But we now know that $\neg s$ and $\neg c$ are both true of Lady L'Hospital (since we are directly told that she does not smoke and just deduced that she does not drink coffee). We know that all guests who enjoy graph theory either smoke or drink coffee, so Lady L'Hospital does not enjoy graph theory. We know that the murderer does enjoy graph theory, so Lady L'Hospital cannot be the murderer. (In symbols, $g \wedge (s \vee c)$ is true of every guest who enjoys graph theory, but is not true of Lady L'Hospital.) By process of elimination, we suspect the murderer must be Mr. Mantel.

Let's note that the information we are given cannot allow us to *actually* conclude that $g \wedge b$ is true of Mr. Mantel. We know that g is true of Mr. Mantel, but there is no information establishing that b must be true of Mr. Mantel. The fact that s is true of Mr. Mantel doesn't mean that c *must* be false of him (and if c is true of Mr. Mantel, then he does not go to bed early). Indeed, even if we know that c is false of Mr. Mantel, there is no rule that every guest who does not drink coffee must go to bed early. So we are not able to reason from just these facts that Mr. Mantel must satisfy $g \wedge b$. However, we are told that one of the guests must be the murderer, and Mr. Mantel is the only guest among the five whose description is logically consistent with the murderer's attributes.[5]

3.1.3 Implications

You may have noticed that we have not yet formally discussed one of the most common logical structures we see in propositions. If you look again at the statement of the Mean Value Theorem, then you see that it has an "If. . . , then. . . " structure. Such a structure is called an *implication*.

Definition 3.1.7. Let p and q be statements or open sentences. The **implication** $p \implies q$ (read "p implies q") is the statement with the following truth table:

p	q	$p \implies q$
T	T	T
T	F	F
F	T	T
F	F	T

.

We call p the **hypothesis** and q the **conclusion**.

[5] Although one might note that butlers likely do not like analysis and do go to bed early.

In other words, an implication is only false if the hypothesis is true but the conclusion is false. Sometimes it strikes people as strange that we consider an implication to be true if the hypothesis is false. But such a convention is sensible; cases in which the hypothesis is false should have no bearing on the full statement. We illustrate with two examples, one mathematical, one not.

Example 3.1.8. For our open sentences $p(x)$ and $q(x)$ about real numbers, the implication is

$$p(x) \implies q(x)\text{: If } x < -2, \text{ then } x^2 > 4.$$

This implication is in fact true because for every real number that is less than -2, its square is indeed more than 4. If you were to say "But when $x = 1$, x^2 is not greater than 4", we'd say that this x does not give a counterexample because $x = 1$ does not satisfy the hypothesis. In a similar manner, we would consider it irrelevant to the matter at hand that when $x = 3$, $x^2 = 9$, which is greater than 4. The implication does not say anything about the squares of numbers that don't satisfy the hypothesis. They may or may not be greater than 4.

Example 3.1.9. Next consider the implication

If it is Sunday, then the bank is closed.

The hypothesis is

S: It is Sunday,

and the conclusion is

C: The bank is closed.

The implication $S \implies C$ is true if on every single Sunday, the bank is indeed closed. It is false if there is even one Sunday on which the bank is open. What happens on Saturday is irrelevant. If it is Saturday, the hypothesis is false, and so the implication makes no claim about the state of the bank.

Exercise 3.1.5. Make the truth tables for $\neg(p \implies q)$ and $p \wedge \neg q$ to show that they are logically equivalent. This result is important because it makes precise what we have already said about disproving implications: To show that $p \implies q$ is false, we must find an instance in which p is true but q is false.

Exercise 3.1.6. Consider the implication

If p is prime, then $p^2 - 2$ is prime.

Which of the following facts disprove the implication?

(i) 9 is not prime, but $9^2 - 2 = 79$ is prime.

(ii) 5 is prime, and $5^2 - 2 = 23$ is prime.

(iii) 11 is prime, but $11^2 - 2 = 119$ is not prime.

(iv) For any even number p, $p^2 - 2$ is even.

Next we want to talk about three other implications that are related to a given implication $p \implies q$. Although we could make the definition right now, we'd like to lead with an example to motivate some natural questions.

Example 3.1.10. In first-semester calculus, one of the first derivative rules you learn is that the derivative of a constant function is the zero function. Let's recast this fact in the language of this section. Consider

$$C\text{: } f : \mathbb{R} \to \mathbb{R} \text{ is a constant function.}$$
$$Z\text{: } f : \mathbb{R} \to \mathbb{R} \text{ is differentiable, with derivative the zero function.}$$

The fact stated above is $C \implies Z$. Later, after we have the Mean Value Theorem, we encounter a proposition that says

If $f : \mathbb{R} \to \mathbb{R}$ is differentiable, with derivative the zero function, then f is constant.

Sometimes when students see this proposition they say "We already know this – constant functions have derivative equal to the zero function." Is this logic correct? No. The new proposition has the form $Z \implies C$ – the hypothesis and the conclusion have been switched. We have not yet discussed whether a true implication can be rearranged in this way to give another true implication.

If you would rather not think about calculus, think about sloths.

Example 3.1.11. Let's compare the two related implications

If I am a sloth, then I live in a tree.
If I live in a tree, then I am a sloth.

Hopefully it is clear to you that these statements are not logically equivalent. The first is only false if there is a sloth who does not live in a tree. The second is false if there is a tree-dwelling animal that is not a sloth. According to our sloth facts, the first implication is true, but the second is clearly false – many birds, squirrels, and bugs also live in trees.[6]

We define the standard terms used for this and similar situations.

Definition 3.1.12. Consider the implication $p \implies q$. Its **converse** is the implication $q \implies p$. Its **inverse** is the implication $\neg p \implies \neg q$. Its **contrapositive** is the implication $\neg q \implies \neg p$.

Our last example illustrates that an implication and its converse are not logically equivalent. The conclusive proof, of course, comes by comparing truth tables. We can make a single truth table that shows all four related implications and immediately shows which, if any, are logically equivalent.

[6]And of course, this is not an exhaustive list. So we could not even say "If I live in a tree, then I am a sloth, a bird, a squirrel, or a bug". I could be a tree frog, a human living in a treehouse, a tree-dwelling Pokémon, etc.

p	q	$\neg p$	$\neg q$	$p \implies q$	$q \implies p$	$\neg p \implies \neg q$	$\neg q \implies \neg q$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

No surprise, the implication and its converse are not logically equivalent. However, there are a few surprises; the implication *is* logically equivalent to its contrapositive, and the converse and the inverse are logically equivalent. This discovery is important enough to be summarized in a proposition.

Proposition 3.1.13. *For logical statements p and q,*

(a) $p \implies q$ and $\neg q \implies \neg p$ are logically equivalent.

(b) $q \implies p$ and $\neg p \implies \neg q$ are logically equivalent.

Exercise 3.1.7. Proposition 3.1.13 is redundant. Explain how part (b) follows from part (a).

Proofs involving truth tables can seem easy, but there's a sense in which they are almost too slick; we can prove logical equivalences or inequivalences without really internalizing what all of it means in practice. We need to keep our feet on the ground by doing examples.

Example 3.1.14. We start with the implication

$$\text{If } x < -2, \text{ then } x^2 > 4.$$

This implication is true. We form the three related implications:

$$\begin{gathered}\text{converse: If } x^2 > 4, \text{ then } x < -2.\\ \text{inverse: If } x \geq -2, \text{ then } x^2 \leq 4.\\ \text{contrapositive: If } x^2 \leq 4, \text{ then } x \geq -2.\end{gathered}$$

In this case, the converse is certainly false; $x = 3$ is a real number with $x^2 > 4$ that does not satisfy the conclusion $x < -2$. The inverse must also be false. Indeed, $x = 3$ again gives an example of a real number that satisfies the hypothesis, $x \geq -2$, for which the conclusion $x^2 \leq 4$ fails. Finally, the contrapositive is true; if $x^2 \leq 4$, then $-2 \leq x \leq 2$, and so it is indeed the case that $x \geq -2$.

If the big truth table and long example seem dizzying, here are two important takeaways: First, just because an implication is true does not mean that its converse is true. This logical error is extremely common and easy to make, even for those who are looking out for it. Second, there *are* some logical equivalences, in particular between an implication and its contrapositive. Such an equivalence means that there are actually two valid ways to prove an implication – one in which we start with p and try to show that q follows, the other where we start with $\neg q$ and try to show that $\neg p$ follows. Sometimes it is much easier to wrap our minds around what it means for $\neg q$ to be true than it is to wrap our minds around p. See the subsection below on proof by contraposition.

One final remark: There are times when the implications in both directions are true. Such is the case above in our calculus example; It is true that if a function is constant on $\mathbb{R}$, then it is differentiable and its derivative is the zero function. It is also true that if a function is differentiable on $\mathbb{R}$ and its derivative is the zero function, then it is constant. Thus, both $C \implies Z$ and $Z \implies C$ are true. We call such an object a *biconditional statement*. We write $C \iff Z$, read "C if and only if Z".

Exercise 3.1.8. Consider the implication about an integer n:

If n is a multiple of 6, then n is a multiple of 3.

(a) Identify the open sentences that make up this implication.

(b) Is the implication true? Prove it or disprove it.

(c) Form the converse of the implication and determine whether it is true. Either prove or disprove the converse.

(d) Now form the inverse and the contrapositive of the original implication. Determine whether each is true or false.

3.1.4 Quantified Statements and Their Negations

To *quantify* a statement is to say to how many things it applies. In logic, there are two flavors of quantified statements – those that say "For all objects of this type, . . . " and those that say 11There exists an object of this type, . . . " In this subsection we study such statements in great detail and focus especially on how to negate them.

In order to make our discussion as precise as possible, we introduce the idea of a *set*. In English, the word "set" has a very large number of possible definitions. Sometimes it is a noun[7] and sometimes it is a verb.[8] In math, a set is a collection of objects. Many sets are defined by some rule – for example, the collection of all integers that are divisible by 2, better known as the *set of even integers*. Some commonly-used sets have special names and notation. For example, $\mathbb{R}$ denotes the set of all real numbers (the natural domain for most calculus functions), $\mathbb{Z}$ denotes the set of all integers, and $\mathbb{N}$ denotes the set of all natural numbers (i.e., positive integers). Hopefully, listing a few examples here makes you realize that you think about sets all the time. In Chapter 4, we will formally discuss many aspects of sets and mathematical questions that arise when we describe sets formally. But for now, all you need to know is what a set is. If we have a set S in front of us, we can figure out whether some object x is in S by looking through all of the objects in S and seeing whether x is there. If it is, we say x is an *element* of the set S (or just x is *in* S), and write $x \in S$. If it is not, we write $x \notin S$ and say that x is not an element of (or not in) S.

Already many times in this book, we have considered conjectures that claim that some property holds for all elements of some set. Examples of such conjectures include:

[7] You can win a *set* in tennis

[8] You can *set* something on a table or watch the sun *set* from your villa on the Italian Riviera, etc.

For all odd integers n and m, $n + m$ is even.
For every positive integer n, $11^n - 6$ is divisible by 5.
For every real number x, x^2 is positive.

In fact, the entire chapter on induction focused on proving that every statement in an infinite list of statements indexed by the positive integers is true, which we could think of as a claim about all members of a set. In our section in Chapter 1 on making and refuting conjectures, we informally explored the logic of the negation of such a statement, pointing out that in order to refute a claim that some property holds for every object of a certain type, it suffices to show that there exists some object of that type for which it does not hold. In this subsection, we develop some terminology and notation for such *quantified statements*. As with other sections of this chapter, the point is *not* that these are unfamiliar logical structures or that we have been thinking about them incorrectly. The point is that, as the situations we care about become more complicated, it will sometimes be helpful to lean on the additional structure we develop here.

To begin our more formal discussion, let's consider several simple statements without too much math content:

Every right triangle has exactly two acute angles.
All penguins eat fish.
Everyone in this class has taken calculus.

Such statements are said to be *universally quantified* because they give a property that holds universally. Each statement specifies an underlying set and contains an open sentence with that set as its domain. For example, for the first statement, the underlying set is the set R of all right triangles. The open sentence with this domain is

$A(T)$: Triangle T has exactly two acute angles.

Thus, a more formal version of the full first statement reads

For all triangles T in the set R of right triangles, $A(T)$ is true.

With some fancy notation, we can make the statement even shorter. We use the symbol $\forall$ in place of the words "for all" and use $T \in R$ to mean that T is in the set of right triangles, so our statement becomes

$$\forall T \in R, A(T).$$

Exercise 3.1.9. Give more precise restatements of the second and third universally quantified statements above. In each case, start by clearly defining the underlying set and relevant open sentence with this set as its domain. Then give both a more precise English formulation and a reformulation in terms of logical symbols.

There is a second flavor of quantified statement. Think about what each of the following statements has in common.

There exists a prime number that is not odd.
There is a penguin who does not eat fish.
Someone on this train is responsible for the death of Mr. Ratchett.

Each of these statements claims that some property holds for *at least one* element of some set. In other words, each is about the existence of an element of a set with a certain property. Such statements are said to be *existentially quantified*. Note that in order for an existentially quantified statement to be true, there needs to be *at least one* element in the given set with the stated property. There is no claim that there is *only one*. For the first statement above, it happens to be the case that there is precisely one even prime (namely 2). But the last statement would be considered to be true even if multiple people on a certain train were jointly responsible for Mr. Ratchett's death.[9]

We have a similar formalism for existentially quantified statements. Again there is some underlying set and an open sentence with this set as its domain. The quantified statement claims that there is at least one element of the set for which the associated statement is true. Consider the penguin example. Let P be the set of all penguins. Consider the open sentence

$$N(p)\text{: Penguin } p \text{ does not eat fish.}$$

Then our existentially quantified statement says

$$\text{There exists a penguin } p \text{ in the set } P \text{ of all penguins such that } N(p) \text{ is true.}$$

Symbolically, we use $\exists$ for the phrase "there exists" and write

$$\exists p \in P, N(p).$$

Exercise 3.1.10. Give more precise restatements of the first and third existentially quantified statements above (about primes that are not odd and Mr. Ratchett's murderer.) In each case, carefully define an appropriate set and open sentence. Give both a restatement as a more precise English sentence and a restatement using logical symbols, as above.

Let's return to the discussion we had at the beginning of the section about refuting conjectures. You might have noticed that we now have a precise logical formulation for the process. Each conjecture we mentioned is a universally quantified statement. To show that it is false means to show that its negation is true, and in each case, it is therefore necessary to prove an existentially quantified statement. This pattern always holds; the negation of a universally quantified statement is an existentially quantified one, and vice versa. Of course, in the specific examples we've talked through before this chapter, we were able to logically think through the statements each time and come to this conclusion. But moving forward, it will be convenient to know that this rule always applies, so we won't have to think through the logic anew each time.

To summarize the rules, if S is a set and $p(x)$ is an open sentence with S as domain, the negation of

$$\forall x \in S, p(x)$$

is

$$\exists x \in S, \neg p(x).$$

On the other hand, the negation of

$$\exists x \in S, p(x)$$

[9]The reference is to *Murder on the Orient Express*. Do you know how many murderers there were?

is

$$\forall x \in S, \neg p(x).$$

We solidify these ideas with a concrete example and some exercises.

Example 3.1.15. The negation of the universally quantified statement

For all positive integers n, $7^n - 1$ is a multiple of 6

is the existentially quantified statement

There exists a positive integer n such that $7^n - 1$ is not a multiple of 6.

For a non-mathematical example, the negation of the existentially quantified statement

There is a child who likes broccoli

is the universally quantified statement

All children dislike broccoli.

Exercise 3.1.11. Negate each of the following statements.

(a) There exists a prime number that is not odd.

(b) Every function that is differentiable on (a, b) is continuous on (a, b).

(c) Everyone loves moms and apple pie.

(d) Every critical point of a differentiable function is the location of either a local maximum or a local minimum.

(e) There is a positive integer less than 100 that is divisible by 2, 3, 5, and 7.

If quantified statements didn't get any more complicated than the ones above, we might not feel a need for much formalism surrounding them. After all, we have already worked with such statements (correctly) in previous chapters. However, some definitions and propositions involve complicated statements with nested quantifiers. Consider the following examples:

A: For every positive integer n, there exists a non-negative integer k such that $2^k \leq n < 2^{k+1}$.

B: (Definition of boundedness for a sequence) There exists a real number $M > 0$ such that, for all positive integers n, $|a_n| \leq M$.

We can write either of these more formally if we like. For example, for statement B, we might use $\mathbb{R}^+$ to denote the set of positive real numbers and $\mathbb{N}$ to denote the set of positive integers (also called natural numbers). Our restatement reads

$$\text{There exists } M \in \mathbb{R}^+ \text{ such that, for all } n \in \mathbb{N}, |a_n| \leq M,$$

or, even more tersely,

$$\exists M \in \mathbb{R}^+, \forall n \in \mathbb{N}, |a_n| \leq M.$$

Suppose we want to negate this statement. Your first question might be "Why?" Examples like this done out of context can feel artificial. It is possible to develop the skills and not know why you would ever care to use them. To understand the motivation, let's think about how we would use the definition of boundedness. Definitions don't just tell us what words mean – they tell us how to prove that certain mathematical objects have certain properties. Suppose you are investigating a sequence you've picked up along the side of the road. Suppose first that you suspect it is bounded. How would you prove it? Well, the definition tells you exactly what to do. Because a bounded sequence is one for which there exists a positive real number M with the property that for all positive integers n, $|a_n| \leq M$, your job would be to conjecture a positive real M with this property and then demonstrate, presumably using facts you know about inequalities, that $|a_n|$ never exceeds that M. Suppose, on the other hand, that the longer you interrogate the sequence, the more you suspect that it is not bounded. How would you prove this statement? It would not be enough to say "It doesn't seem to satisfy the definition." You would need to prove that it satisfies the *negation* of that definition, and to do so, you would need to know exactly what the negation says.

If you are systematic, negating a statement with nested quantifiers is easy because you work from left to right in layers. So to negate

$$\text{There exists } M \in \mathbb{R}^+ \text{ such that for all } n \in \mathbb{N}, |a_n| \leq M,$$

you first think of it as just having the form "There exists $M \in \mathbb{R}^+$ such that $P(M)$ is true" and negate it to form "For all $M \in \mathbb{R}^+$, the negation of $P(M)$ is true." The added complexity comes from the fact that the statement $P(M)$ we are now negating is itself a quantified statement, namely "For all $n \in \mathbb{N}, |a_n| \leq M$". Its negation is "There exists $n \in \mathbb{N}$ such that $|a_n| > M$." Putting it all together, $\neg B$ reads

$$\text{For all } M \in \mathbb{R}^+, \text{ there exists } n \in \mathbb{N} \text{ such that } |a_n| > M.$$

So if we believe we have an unbounded sequence, we need to begin with an arbitrary positive real number M and find at least one positive integer n for which $|a_n|$ exceeds M.

Exercise 3.1.12. Each of the statements below involves nested quantifiers. Write each statement with formal logical symbols, negate it, and then write its negation in English without logical symbols. In each case, decide which is true – the original statement or the negation? (You do not need to prove your assertions.)

(a) For every real number x in the interval $(0, 1)$, there exists a natural number n such that $x^n < \frac{1}{100}$.

(b) There exists a real number x in the interval $(0, 1)$ such that, for all natural numbers n, $x^n > x$.

Exercise 3.1.13. Each of the following non-mathematical statements involves nested quantifiers (though in some cases, sneakily). Negate each statement.

(a) Everyone in this class knows someone who has visited India.

(b) There is one day each year when all the Whos in Whoville eat roast beast and sing carols.

(c) Everybody loves somebody.

We end this section with a brief comment on notation. Some students become rather enamored with the logical symbols such as $\forall$ and $\exists$ and start using them liberally throughout their mathematical writing. We discourage this practice. We think of these symbols like shorthand – they are helpful when taking notes or writing quickly on the whiteboard or on scratch paper. But when you are really trying to communicate with a reader, even a mathematically advanced one, these symbols produce writing that is too terse. It takes a reader too long to parse such tight statements and it disrupts the flow of the writing. We will avoid using such symbols in our writing. We have introduced them here because not all writers will be as considerate. But please be one of these considerate writers. Your goal when you write is to make your writing as clear as possible, not as short as possible. The goal is always to help the reader understand and never to impress others.

3.2 Standard Proof Techniques

Although logic puzzles can be interesting in their own right, we aren't studying formal logic to divide pirate gold or to prepare us for detective work. We want to write proofs of mathematical propositions. We have something interesting or important we want to communicate, and we want to do so as effectively as possible, following patterns of communication others will recognize and understand. Thus, in this section, we discuss some of the most common logical structures for proofs.

3.2.1 Direct Proof

The most common proof technique is the direct proof. In a direct proof of an implication, we begin by assuming that the hypothesis is satisfied and show that the conclusion necessarily follows. If this all sounds familiar, it should. Many of the proofs we have given in this book so far are direct proofs. For example, consider Proposition 1.2.2 in Chapter 1, which claims that the sum of two odd integers is even. If we want to phrase the proposition formally as an implication, it would be "If n and m are odd integers, then $n + m$ is even." We proved the proposition by starting with n and m and invoking the definition of an odd integer to write them as $n = 2j + 1$ and $m = 2k + 1$. We then added the two and did some algebra to show that their sum has the defining property of an even integer, namely that it can be written as 2 times an integer.

Although it is customary and helpful to the reader to start a proof by indicating whether some special technique is being used, most mathematicians would not, in fact, begin a direct proof by saying "We give a direct proof". It is not wrong to do so, but generally unnecessary; in the absence of any comment, a reader will assume a direct approach because it is, well, direct.

Although we have used direct proof often, we give several exercises that allow you to practice, this time, being particularly mindful to the logical structure of the statements and how they guide how the proof proceeds.

Exercise 3.2.1. Give a direct proof of each of the following true statements about integers.

(a) If n is an odd integer, then n^2 is odd.

(b) If three integers are consecutive, then their sum is a multiple of 3.

(c) If five integers are consecutive, then their sum is a multiple of 5.

Exercise 3.2.2. Okay – we have to ask. If four integers are consecutive, then is their sum a multiple of 4? If so, give a direct proof of this fact. If not, prove that it is not. (Make sure you clearly understand the logical structure of this claim. What form does a refutation of this claim take?)

3.2.2 Proof by Contraposition

Mathematical research would be much easier (but, alas, also much less rewarding) if every true mathematical statement had a simple direct proof. But sometimes, direct proofs are hard to find. Fortunately, our understanding of logic opens up the possibility of other *indirect* proof methods, including proof by contraposition and proof by contradiction, the topics of the next two subsections. But before we present them, let's talk a little about how you should think about the topic of proof techniques.

Think of a proposition you are trying to prove as a sticky door in an old house that won't close properly. You've decided you want to tackle it, so you get out your toolbox. Your tool box probably has a hammer, a screwdriver, and a saw. One way to deal with the door is to start by whacking it with the hammer to see if it will close. If that fails, you might get out the screwdriver and tighten the hinges in case the door has sagged. If that too fails, it might be time to get serious and get our the saw and shave a bit off the top or the side that is rubbing. By this approach you might get the door to close. However, might you have approached the door in a more savvy way? Perhaps it would have been better to start by carefully studying the door to diagnose the problem. Is the door rubbing? Is the latch broken? Is it no longer aligned properly with the doorjamb? Diagnosing the problem will help you choose the right tool to smartly fix the door.

Similarly, when you try to prove a proposition, you could just get out each proof technique one by one and see if it works. But it might be preferable to consider first which technique is likely to work. To be clear, we are *not* suggesting here a sort of template-matching approach. What we are suggesting is that certain kinds of statements lend themselves better to one technique than another, and you can learn to recognize these features.

For example, consider the following three simple statements about an integer n:

If n^2 is odd, then n is odd.
If $n^2 - 4n + 6$ is even, then n is even.
If $n^2 + n + 15$ is not a multiple of 3, then n is not a multiple of 3.

Before reading on, try to give a direct proof of each statement. See if you can articulate why it is difficult.

Just to further extend your likely frustration,[10] let's consider the second statement. If we were to attempt a direct proof, we would have as our starting point an integer n for which n^2-4n+6 is even. It would follow that there is an integer k such that $n^2-4n+6=2k$. Our goal is to conclude that n is even. How? You might imagine solving for n. The equation is quadratic in n and doesn't factor. You could use the quadratic formula. Doing so certainly gives an expression for n in terms of k and the other various coefficients, but it isn't even obvious that it is ever an integer, let alone an even one.

You could get one step further down the road by noting that, with k as above,

$$n^2 = 2k + 4n - 6 = 2(k + 2n - 3),$$

and so we can conclude that n^2 is even. But we still have the same problem as above. In fact, proving that if n^2 is even, then n is even seems almost certainly to be the same level of difficulty as the first statement on the list above.

It's time to put away the hammer and get out a different tool. Recall that in our section on implications, we proved that a statement and its contrapositive are logically equivalent. Thus they are either both true or both false. Perhaps it is easier to prove that our current statement is true by instead proving that its contrapositive is true. In this case, noting that if an integer is not even, then it is odd, the contrapositive states

$$\text{If } n \text{ is odd, then } n^2 - 4n + 6 \text{ is odd.}$$

Now, this statement seems like one we could enjoy proving; we have done proofs like this before in which the hypothesis is that an integer has a certain parity (even or odd) and the goal is to conclude that some other object has a certain parity. We were good at doing such proofs in Chapter 1. We have now hit on our strategy. We show how we formally write such a proof.

Proposition 3.2.1. *Let n be an integer. If $n^2 - 4n + 6$ is even, then n is even.*

Proof. The proof is by contraposition. Thus, we seek to prove the contrapositive, namely that if n is odd, then $n^2 - 4n + 6$ is odd.

Suppose, then, that n is odd. Then there exists an integer k such that $n = 2k + 1$. Observe,

$$\begin{aligned} n^2 - 4n + 6 &= (2k + 1)^2 - 4(2k + 1) + 6 \\ &= (4k^2 + 4k + 1) - (8k + 4) + 6 \\ &= 4k^2 - 4k + 3 \\ &= 2(2k^2 - 2k + 1) + 1. \end{aligned}$$

This last expression is of the form $2j + 1$ for the integer $j = 2k^2 - 2k + 1$. Thus $n^2 - 4n + 6$ is indeed odd, proving the proposition. □

[10] We promise it gives us no pleasure.

Note that in the above proof, we explicitly told our reader that we were giving a proof by contraposition. Although the exact wording of this first paragraph will vary depending on the writer and the audience, it is customary and courteous to start by indicating the proof method when it is not direct. Although it is not strictly necessary to explicitly write down the contrapositive of the original implication at this point, when you are new to proof-writing it will probably help you think and write more clearly.

Exercise 3.2.3. The first and third statements at the beginning of the subsection can also be proved by contraposition. Do so.

Exercise 3.2.4. The **parity** of an integer refers to its classification as either even or odd. Thus, two integers have the same parity if either both are even or both are odd. Consider the implication

Let m and n be integers. If $m + n$ is even, then n and m have the same parity.

(a) Give a direct proof of this proposition. (As a hint, you might consider within the proof two separate cases – one in which n is even and one in which n is odd.)

(b) Now give a proof by contraposition. (It might seem like you need cases here too. See if you can write a sentence or two in the proof explaining why there is in fact really only one case.)

3.2.3 Proof by Contradiction

Proofs by contraposition rely on the fact that the statement $p \implies q$ is logically equivalent to the statement $\neg q \implies \neg p$. Our next indirect proof technique similarly begins by leveraging the formal logic we have been developing. We know that the statement "p is true" is equivalent to the statement "$\neg p$ is false". In other words, exactly one of p and $\neg p$ is true – and sometimes, it is easier to show that $\neg p$ *is not* true than to argue that p *is* true.

The "contradiction" part of proof by contradiction describes the strategy we typically use to argue that $\neg p$ is false. If we can show that $\neg p$ implies some false statement q, then it must be that $\neg p$ is itself false (because the implication $\neg p \implies q$ is false if the hypothesis is true but the conclusion is false). These ideas are summarized in the following logical structure:

1. Exactly one of p and $\neg p$ is true. So our goal is to show that $\neg p$ is false.

2. Assume *for the sake of a contradiction* that $\neg p$ is actually true, and see what consequences follow from this assumption.

3. Eventually observe that $\neg p \implies q$ for some statement q that you know to be false. The fact that q is false *contradicts* the assumption that $\neg p$ is true.

4. Because we conclude that $\neg p$ is in fact false, it follows that p is true.

If the formal logic make this process seem complicated, you might firm up your understanding by thinking about the idea in a non-mathematical context first. Suppose a friend

makes a statement you do not think is true – for instance, "The earth is flat."[11] How would you go about challenging this statement? The proof by contradiction structure might very naturally come to mind! You might say something like "Let's suppose you're right and the earth *is* flat. Hopefully we agree that the earth is finite in size. Doesn't that mean the earth must have an "edge" somewhere? Wouldn't people have found the edge? What about the boats and planes that claim to have circumnavigated the earth? If the earth had an edge, wouldn't they have fallen off it?"

This imagined dialog exactly follows the proof by contradiction structure. You "pretend" that the statement we want to disprove is really true and show that this assumption leads to an impossible conclusion. Even without the framework of formal logic, we understand that if a premise logically leads to a false conclusion, then the premise is not sound. In fact, this structure of argument is so flexible and useful that it is routinely used outside mathematics, in contexts ranging from ancient philosophy to thought experiments in physics. In a non-mathematical context, the argument style is often called *reductio ad absurdum*, Latin for "reduction to absurdity". Thought about this way, you probably have made contradiction-style arguments in your own life!

For a non-mathematical debate, say about the flat earth, it is possible to imagine a disagreement about a contradiction-style argument. The two sides may not be working with the same set of starting assumptions and "facts", and may not agree about whether a contradiction has been reached. One great thing about math is that we basically all start with some agreed-upon axioms, so mathematical proofs by contradiction are no more controversial than any other type of mathematical argument. Ideally, in a mathematical proof by contradiction, the absurd conclusion will be a simple statement that we have no difficulty agreeing is false – for example, "1 is smaller than 0", or "The positive integer n is simultaneously odd and even".

Now that we've discussed the structure of a proof by contradiction, let's illustrate the idea with some examples. We'll start with a simple claim about divisibility that you likely believe already. In addition to being a good first example of our technique, this claim will come up later in the section, so it's worth understanding it now. Recall, we say that an integer p *divides* another integer n if we can write $n = pk$ for some integer k.

Proposition 3.2.2. *Let p and n be positive integers, with $p > 1$. Then p does not divide both n and $n + 1$.*

Of course, we will give a proof by contradiction to establish Proposition 3.2.2. Before doing so, let's make sure we really understand the statement we are trying to prove and its negation. Many mathematicians would find the phrasing we've written above the most "natural", but to find the statement's negation, we might find it easier to rewrite the claim with explicit logical quantifiers. Proposition 3.2.2 is saying,

For all positive integers n, for all positive integers $p > 1$, the statement "p divides n and p divides $n + 1$" is false.

The negation of this claim is then

[11] With apologies to any flat earthers reading this text.

There exist positive integers n and $p > 1$ for which the statement "p divides n and p divides $n + 1$" is true.

With this negation in hand, we can prove Proposition 3.2.2 by contradiction. Before reading our proof, pause and see if you can find a proof (either using contradiction, or some other strategy) for Proposition 3.2.2.

Proof of Proposition 3.2.2. For the sake of a contradiction, suppose the claim is false. Then there exist positive integers n and p, with $p > 1$, such that p divides n and p divides $n + 1$. Thus, we can find integers k and m such that $n = pk$ and $n + 1 = pm$. Observe,

$$1 = (n + 1) - n = pm - pk = p(m - k),$$

so p divides 1. However, the only positive integer that divides 1 is 1, so we must have $p = 1$. This is a contradiction, because we assumed that $p > 1$; it is impossible to have $p = 1$ and $p > 1$ simultaneously. We conclude that the negation of the claim is false, so Proposition 3.2.2 is true. □

Let's note a few things about the above proof. We started with the very specific sentence, "For the sake of a contradiction, suppose the claim is false." We then immediately explained what it means for the claim to be false. While we do not ever want you to think of proof-writing as formulaic, this beginning structure is the best practice for any proof by contradiction. Proof by contradiction is a slightly twisty idea – you are starting out by assuming a statement that you expect to be false, and your goal is to reach a false conclusion. If you do not alert your reader to this strategy and make sure they understand your set-up, a proof by contradiction can be really confusing to read.

After reading this argument, you hopefully see (if you did not already) the key idea for why Proposition 3.2.2 is true. If p divides two distinct positive integers n_1 and n_2 (say with $n_1 > n_2$), then p also divide $n_1 - n_2$ (which is also a positive integer). So, since p can only divide positive integers at least as big as itself, if p divides n_1 and n_2, then $n_1 - n_2$ cannot equal 1. In fact, if p divides n_1 and n_2, then we must have $p \leq n_1 - n_2$.

The last paragraph might make you pause. In some sense, we have just rephrased the argument from our proof of Proposition 3.2.2, omitting some details because they were justified in the previous proof. But this rephrasing has a somewhat different form – it actually indicates a *direct* proof of Proposition 3.2.2. Is this a problem? Well, in some sense, no. We gave a valid, short, and hopefully clear proof of Proposition 3.2.2 using our new technique. However, let's return to the analogy of fixing a stuck door. If you have a mysterious carpentry situation in your house, you would probably prefer to solve it by the simplest means possible. If you can fix your door by getting out a screwdriver and tightening the hinges, then you would probably rather just do that than rent some complicated machine from the hardware store in order to re-size your door. Similarly, it is often the case that multiple proof techniques will work to justify a statement. Sometimes, when proof by contradiction works, the contradiction you find indicates a direct explanation for the truth of your claim. At that point, it is up to you as the mathematical communicator to choose which proof style explains the proposition best. But a general rule of thumb is that the simpler argument will usually be the most illuminating and easiest for your reader to

follow. So in this instance, you might choose not to prove Proposition 3.2.2 by contradiction if you worry that the added complexity of the proof structure will bog down your reader.

In our first example, the proof by contradiction and direct proof were arguably about the same level of complexity. However, proof by contradiction exists because for some problems, the contradiction style of argument is much easier to think about and explain than a direct proof. Our next example will illustrate such a case. Think about the following claim before reading on, so that you are certain you understand what it is saying.

Proposition 3.2.3. *Suppose n people attend a party, where $n > 1$ is an integer. Some (but perhaps not all) pairs of people shake hands. Each pair of people shakes hands at most once. No matter how these people decide to shake hands (or not), there exist two people at the party who participated in exactly the same number of handshakes.*

While this example may not be as obviously connected to formal math as problems about divisibility, it is certainly a quantitative statement that we could attempt to logically justify.[12] Let's start by writing our claim using logical quantifiers. This will help us understand the statement itself better, and help us understand the statement's negation. We will say a *handshake pattern* among n people is a list of handshakes that the people choose to perform; as in the statement of Proposition 3.2.3, we assume that any two people shake hands either 0 times or 1 time. A person's *handshake number* is the number of handshakes they perform in a certain handshake pattern. Then Proposition 3.2.3 says:

For all natural numbers $n > 1$, for all handshake patterns among n people, there exist two people with the same handshake number.

So, its negation is:

There exists a natural number $n > 1$ for which there exists a handshake pattern among n people such that no two people have the same handshake number.

To prove Proposition 3.2.3, we will proceed by contradiction. That is, we assume for the sake of contradiction that the negation of Proposition 3.2.3 holds, and then derive some false conclusion.

Proof of Proposition 3.2.3. Suppose, by way of contradiction, that there exists a natural number $n > 1$ for which there exists a handshake pattern among n people such that no two people have the same handshake number. We consider the possible handshake numbers that each person could have. There are n total people, so any person can shake hands at most $n - 1$ times (because a person cannot shake hands with themselves). And clearly, each person shakes hands at least 0 times. So the possible handshake numbers in this group are

$$0, 1, 2, \ldots, n - 1.$$

Note that there are exactly n possible handshake numbers. Because there are n people and n different handshake numbers, if no two people share a handshake number, then each of

[12] In fact, this strange property of handshakes is secretly a foundational statement in a branch of math called *graph theory*; see Chapter 13 for an introduction to the field.

the numbers on the list above is some person's handshake number.[13] But that means there is one person in the group with a handshake number of 0 and one person in the group with a handshake number of $n-1$. Here we have reached a contradiction because, a person with handshake number $n-1$ must shake hands with every other member of the group, and so would need to shake hands with the person whose handshake number is 0. We conclude that the negation of Proposition 3.2.3 is false, so Proposition 3.2.3 is true. □

We now give another example of proof by contradiction that is one of the most famous basic applications of the technique. It is actually a really important fact about the set of primes. As with many of our examples involving divisibility, we assume that you have a basic familiarity with what prime numbers are and accept basic facts about prime factorization. In this case, we take it as known that every number that is not prime has a prime divisor (which is certainly implied by the fact that every number has a prime factorization).

Theorem 3.2.4. *There are infinitely many prime numbers.*

Proof. Suppose for the sake of contradiction that there are finitely many prime numbers, $p_1, p_2, \ldots, p_k$. We will achieve a contradiction by exhibiting a number p_{k+1} that is not on our list but is prime. Let

$$n = p_1 p_2 \cdots p_k$$

and consider $p_{k+1} = n + 1$. We claim that p_{k+1} is a prime that is not on our list. First, note that $p_{k+1} > p_i$ for any i between 1 and k, so p_{k+1} is indeed different from all numbers on our list of primes. Next, we apply Proposition 3.2.2 with $p = p_i$ and n as above. Because p_i clearly divides n, p_i does not divide $n + 1 = p_{k+1}$. This conclusion applies to any i. Because we assumed that the numbers p_i are the only primes, this means that p_{k+1} does not have a prime divisor. But every number that is not prime has a prime divisor, so p_{k+1} is itself prime. This is a contradiction, because if p_{k+1} were prime, it should have been on our exhaustive list of primes. We conclude that we cannot make a finite exhaustive list of primes, so there are infinitely many. □

Now that you have seen proof by contradiction done in a few settings, hopefully you feel ready to try it yourself. The next exercise makes a statement that is somewhat similar-sounding to Theorem 3.2.4, but is actually substantially more straightforward to prove. As you can guess, the best strategy is a proof by contradiction.

Exercise 3.2.5. Show that there does not exist a smallest strictly positive real number.

Let's make some further remarks about the ways that contradictions themselves may look. In some examples, our contradiction is the equivalent of a bright flashing neon sign – a statement that is clearly nonsensical. For example, in Proposition 3.2.2, we get to a point where we exhibit a number p that simultaneously satisfies $p = 1$ and $p > 1$. It is quite clear that such a number doesn't exist. Many proofs by contradiction do end this way – you either conclude a false equality, like $1 = 0$, or you conclude that a number you

[13] Some people consider this statement to be an application of an observation called the *pigeonhole principle*, which we will discuss in detail in Chapter 4. We trust that, in this specific instance, you can justify the statement to yourself without invoking anything fancy.

are working with must satisfy two contradictory inequalities, or something along those lines. However, some contradictions are a bit more subtle. In the proof of Theorem 3.2.4, how does the contradiction really work? We first claimed that we had an exhaustive list of primes, and then we found an "extra" prime that was not on our exhaustive list. There is nothing inherently nonsensical or logically inconsistent about defining an integer and showing that integer to be prime. The contradiction came from our set-up – we assumed that we had the biggest possible list of primes and then found another prime to add to our list, so our list was in fact *not* exhaustive.

A proof-by-contradiction strategy with a somewhat similar flavor is the "minimal counterexample" (or, more colorfully, "minimal criminal") approach. Here is the basic logical structure. Suppose P is an open sentence whose domain is the positive integers. Thus, for each positive integer n, you get a statement $P(n)$ with a truth value. Say your goal is to show that $P(n)$ is true for all positive integers n, and you can directly show that $P(1)$ is true. For the sake of a contradiction, suppose $P(n)$ is not always true. So $P(n)$ is actually false for one or more positive integers n. Let k be the smallest positive integer for which $P(k)$ is false.[14] This is the "minimal criminal". The goal is to achieve a contradiction to the minimality of k by showing that $P(m)$ is false for some smaller positive integer m – so k is *not* the smallest possible counterexample.

The next proposition is a nice application of this proof technique.

Proposition 3.2.5. *Let d be a natural number. Then for every non-zero polynomial p, $(1+x)^d p(x)$ has at least $d+1$ terms.*

Proof. Consider first $d = 1$. Then if p is a non-zero polynomial, the polynomial $(1+x)p(x)$ has at least two terms. Indeed, if we take 1 times the lowest-degree term of p, we get a term whose degree equals the lowest degree appearing in p. On the other hand, if we take x times the highest-degree term in p, we get a term whose degree is one more than the degree of any term in p. Because these terms must be of different degrees, $(1+x)p(x)$ has at least $d+1 = 2$ terms.

Suppose, by way of contradiction, that the result does not hold for all natural numbers d. Then there exists $d \geq 2$ for which it doesn't hold. Let d_0 be the smallest such natural number and let p_0 be a polynomial for which $(1+x)^{d_0}p_0(x)$ has at most d_0 terms.[15] We may assume without loss of generality that p_0 has a non-zero constant term, for otherwise we could factor a power of x out of it and replace it with a new polynomial P_0 with a non-zero constant term for which $(1+x)^{d_0}P_0(x)$ would have exactly the same number of terms as $(1+x)^{d_0}p_0(x)$.

Let $q_0(x) = (1+x)^{d_0}p_0(x)$. We will consider its derivative. Note that q_0 can not be a constant polynomial, so its derivative is not 0. Furthermore, because it has a non-zero constant term (1 times the constant term of p_0), the number of terms in q_0' is one less than the number of terms in q_0. Thus it has at most $d_0 - 1$ terms. Now,

$$\begin{aligned} q_0'(x) &= d_0(1+x)^{d_0-1}p_0(x) + (1+x)^{d_0}p_0'(x) \\ &= (1+x)^{d_0-1}[d_0p_0(x) + (1+x)p_0'(x)]. \end{aligned}$$

[14] It is a fact that every non-empty set of positive integers has a smallest one. This fact is called the *Well-Ordering Principle*. We (and probably you) will take this as intuitively clear.

[15] We have found our minimal criminal. Let's put him on trial.

The expression in square brackets is a polynomial. It can not be the zero polynomial because the entire right-hand side does not equal the zero polynomial. So we now have a non-zero polynomial p_1 such that $(1+x)^{d_0-1}p_1(x)$ has at most d_0-1 terms. But then the proposition also fails for d_0-1, contradicting our assumption that d_0 was the smallest natural number for which it fails. Because we have reached a contradiction, the proposition must in fact hold for all natural numbers d.

□

If you want more practice with proof by contradiction, see the project at the end of the chapter on rational and irrational numbers. The Problems also contain a fun but some-what challenging situation in which you can write a minimal criminal argument.

3.2.4 Proof by Cases

Take a moment to consider how you might prove the following (true) proposition:

Proposition 3.2.6. *For any integer n, n^2-n+6 is even.*

This proposition seems rather similar to some we have considered in previous sections. In fact, it seems related to Proposition 3.2.1, especially in light of the fact that we proved that proposition by proving the contrapositive, that if n is odd, then n^2-4n+6 is odd. At this point we are quite good at such proofs. However, the current proposition is different because it doesn't make a hypothesis on the parity of n – we are simply assuming that n is an integer. We seem to have no place to start.

One way to proceed is to pattern our proof after the kinds we've done before by doing a *proof by cases*. We could separately consider the cases in which n is even and n is odd and see if in each case we can get to the conclusion. Because every integer is either even or odd, we would obtain a complete proof in this way. Here is such a proof.

Proof. We consider two cases, depending on the parity of n. Suppose, first, that n is even, so that $n=2k$ for some integer k. Then

$$\begin{aligned} n^2-n+6 &= (2k)^2-(2k)+6 \\ &= 4k^2-2k+6 \\ &= 2(2k^2-k+3). \end{aligned}$$

This last expression exhibits n^2-n+6 as $2j$ for the integer $j=2k^2-k+3$, proving that n is even in this case.

For the second case, suppose n is odd, so that $n=2k+1$ for some integer k. Then,

$$\begin{aligned} n^2-n+6 &= (2k+1)^2-(2k+1)+6 \\ &= 4k^2+4k+1-2k-1+6 \\ &= 4k^2+2k+6 \\ &= 2(2k^2+k+3), \end{aligned}$$

which again exhibits n^2-n+6 as twice an integer, proving that it is even. Thus, in all cases, n^2+n+6 is an even integer. □

We give a second example of a proposition whose proof has a similar flavor. We will walk you through the cases you must consider and then ask you to fill in the details.

Proposition 3.2.7. *Let n be an integer. If n is not a multiple of 3, then n^2 is not a multiple of 3.*

Pause and think for a minute about what it means to say that n is not a multiple of 3. How can you write such n?

To motivate the answer, consider a corresponding proposition involving divisibility by 2:

If n is not a multiple of 2, then n^2 is not a multiple of 2.

(A more concise and natural way to state the same proposition would be "If n is odd, then n^2 is odd.") We have used many times the fact that if n is not a multiple of 2, then it can be written in the form $n = 2k + 1$ for some integer k. We then know exactly how to proceed. But what does it mean to say that n is not a multiple of 3? Multiples of 3 have the form $n = 3k$, which can be rephrased by saying that 3 divides evenly into n, leaving a remainder of 0. If 3 does not divide evenly into n, the remainder after such a division is *either 1 or 2*. Thus, if n is not a multiple of 3, then *either* $n = 3k + 1$ for some integer k *or* $n = 3k + 2$ for some integer k. We see that it is natural, then, to consider two cases depending on which form n takes. Our goal in each case is to show that n^2 is also not a multiple of 3, meaning it takes one of the forms $3\ell + 1$ or $3\ell + 2$ for some integer ℓ.

Exercise 3.2.6. Write a proof of Proposition 3.2.7 by considering the two cases described in the preceding paragraph.

Exercise 3.2.7. We discussed prime factorization in Section 1.3. If you are comfortable with the idea of prime factorization, and willing to believe without a formal proof that every positive integer has a unique prime factorization, you can use this fact to prove Proposition 3.2.7 directly. Can you write such a proof?

There is another kind of proposition commonly treated by proof by cases. These are propositions involving the absolute value function. Recall that if x is a real number,

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}.$$

Proofs of propositions about absolute value often involve cases because the definition of the function itself does! The next proposition includes several elementary statements that can be proved by this method.

Proposition 3.2.8. *Let x and y be real numbers.*

(a) $|x|^2 = x^2$.

(b) $x \leq |x|$.

(c) $|xy| = |x||y|$.

Proof. We give the (shockingly tedious) proof of part (c) and leave the proofs of parts (a) and (b) as the next exercise. The proof requires three cases. In each case, we will allow ourselves to use basic facts about how algebra with inequalities works. Each one of these smaller steps could be proved from what are called the *ordered field axioms* for the real numbers. Some transition to proof books develop these ideas in great detail; such an approach here would violate our perspective that every proof has to assume that the reader knows certain things and these are the kinds of things we will assume are familiar to our reader.

Suppose first that $x \geq 0$ and $y \geq 0$. Then $|x| = x$ and $|y| = y$. Furthermore, $xy \geq 0$, so $|xy| = xy$. Thus

$$|xy| = xy = |x||y|,$$

as claimed.

Suppose next that $x < 0$ and $y < 0$. Then $|x| = -x$ and $|y| = -y$. In this case, $xy > 0$, so $|xy| = xy$. Thus,

$$|xy| = xy = (-1)^2 xy = (-x)(-y) = |x||y|,$$

as claimed.

Finally, suppose one of the real numbers is non-negative and the other is negative. We may assume without loss of generality that $x \geq 0$ and $y < 0$. Then $|x| = x$ and $|y| = -y$, and because $xy \leq 0$, $|xy| = -xy$. (Note that this last statement is even true if $xy = 0$.) Thus,

$$|xy| = -xy = x(-y) = |x||y|,$$

as claimed. Thus, in all cases, $|xy| = |x||y|$. □

Exercise 3.2.8. Prove parts (a) and (b) of Proposition 3.2.8. (These proofs should be much shorter and less tedious than the proof of (c).)

Proof by cases is, in a sense, the "workhorse" of proof techniques. It isn't a fancy tool and doesn't usually lead to what one would call elegant arguments. But sometimes you just need to plow the field, and this horse does the job.

If, however, you write a proof by cases and feel that it doesn't communicate as clearly or efficiently as you had hoped, it may be worth the time to look for another argument. For our last example in this section, we present a famous inequality about real numbers that is used often. Many books also use this proposition as a classic example suggesting a proof by cases. We will ask you to give this proof by cases but will then present an alternative proof that uses Proposition 3.2.8.

Proposition 3.2.9 (Triangle Inequality)**.** *For all real numbers x and y,*

$$|x + y| \leq |x| + |y|. \tag{3.1}$$

Exercise 3.2.9. Prove Proposition 3.2.9 by cases. The proof will resemble the proof of part (c) of Proposition 3.2.8. Study your proof to determine under what conditions equality holds.

Proof. We give a different proof using Proposition 3.2.8. Consider $|x+y|^2$.

$$\begin{aligned}
|x+y|^2 &= (x+y)^2 && \text{by part (a)}\\
&= x^2+2xy+y^2 &&\\
&= |x|^2+2xy+|y|^2 && \text{by part (a)}\\
&\le |x|^2+2|xy|+|y|^2 && \text{by part (b)}\\
&= |x|^2+2|x||y|+|y|^2 && \text{by part (c)}\\
&= (|x|+|y|)^2. &&
\end{aligned}$$

Because $|x+y|$ and $|x|+|y|$ are non-negative, it follows from $|x+y|^2 \le (|x|+|y|)^2$ that $|x+y| \le |x|+|y|$. □

This example is a good lead-in to the next section in which we discuss the (possibly unsettling) truth that there are often multiple valid ways to prove a proposition. If you do find this truth unsettling because you have come to like the "one right answer" feeling of math, we hope to persuade you that the real situation is even more satisfying. It *is* still true that every proposition is either true or not true and it is still true that for many problems posed, there is a single correct answer. The idea that may be new to you is that there can be many valid ways to establish the truth of a claim or to reason to a correct answer, and this feature of mathematics is part of what makes it flexible, creative, and rewarding.

3.3 Examples and Counterexamples

Before reading on, think about the following two conjectures. In each case, decide whether the conjecture is true or false and think about how to prove your assertion.

Conjecture 3.3.1. *There exists a positive real number x such that $x^2 < x$.*

Conjecture 3.3.2. *For every polynomial p, the number of real zeros of p' is less than the number of real zeros of p.*

Hopefully you decided that Conjecture 3.3.1 is true and that Conjecture 3.3.2 is false. If so, you probably also understand why. Here are our arguments.

Proof of Conjecture 3.3.1. Consider $x = 0.5$. Then $x^2 = 0.25$ and so $x^2 < x$, proving the conjecture. □

Disproof of Conjecture 3.3.2. Consider $p(x) = x^2+1$. Then p has no real zeros because there are no real solutions to $x^2+1=0$. On the other hand, $p'(x) = 2x$, which has a real zero at $x = 0$. Because we have exhibited a polynomial p whose derivative has more real zeros than p, the conjecture is false. □

Although the two conjectures pertained to different situations, do you see what the proofs have in common (aside from the fact that they are delightfully short)? For both, it was enough for us to provide a single example.

"But wait!" you say. "I thought one of our mantras in this book is that single examples don't prove theorems." Not exactly. What we have pointed out several times is that if we

want to prove that some property holds for all elements of a specific set, it is not enough to check that it holds for a few. In other words, you can not prove a universally-quantified statement by checking it only for some subset of things in the universal set. For example, you can't prove that for every integer n, $n^2 + n$ is even by checking it for $n = 1, 2, 3, \ldots, 10$ or even by checking it for all positive integers or all odd integers. You have to show it is true for every single integer. On the other hand, to prove an existentially-quantified statement, it *does* suffice to give a single example because you are simply trying to show that some such object exists and are making no claims about how common they are. Our first proof was of this sort. We simply needed to find a real number that exceeds its square. We found one, and thus had a proof of the conjecture. It doesn't matter whether there are others. We were not asked for a full characterization of those real numbers x for which $x^2 < x$. That would be a different conjecture with a different proof.

What about our second conjecture? It is a universally-quantified statement. But because we decided it was false, we were required to prove its negation, which is existentially quantified. Thus, again, our (dis)proof required us to simply produce a single example. So we see that existence proofs and disproofs of universally quantified statements have a similar flavor.

Take a moment to practice with some more elementary examples.

Exercise 3.3.1. Prove each of the following existentially quantified statements.

(a) There exist positive integers a, b, and c satisfying $a^2 + b^2 = c^2$.

(b) There exists a rational number q satisfying $\sqrt{2} < q < \sqrt{3}$.

(c) There is a natural number less than 10,000 that is divisible by 2, 3, 5, 7, and 11.

(d) There is a real solution to the equation $x^5 - 2x + 1 = 0$.

Exercise 3.3.2. Give a counterexample to each of the following false statements.

(a) Every product of 4 consecutive integers is positive.

(b) Every increasing sequence $\{a_n\}$ of real numbers is unbounded.

(c) Every linear system of the form

$$\begin{aligned} ax + by &= 0 \\ cx + dy &= 0 \end{aligned}$$

has exactly one solution.

At this point, you may be thinking that doing existence proofs and finding examples are pretty easy. Alas, finding examples and counterexamples can be just as hard as doing any other kind of proof and, in fact, some "big theorems" are exactly this sort. For example, each of the following true statements can be proved by giving an example.[16] However, finding these examples and proving that they work is in each case highly non-trivial:

[16] Don't worry if you don't know some of these words.

There exists an everywhere continuous, nowhere differentiable real-valued function on $\mathbb{R}$.

There exists a real number x which is not a root of any non-zero polynomial with integer coefficients. (Such numbers are called *transcendental*.)

There exists a non-measurable subset of $\mathbb{R}$.

The above statements are known results, so although it may not be simple to find examples of each, someone did. It may not surprise you to learn that some existence questions are still open (that is, unsolved). In fact, some of the most famous questions in modern mathematics could be settled with the right example – and yet, we are uncertain whether examples exist.

The story of the solution to one long-standing existence question in math made it into popular culture within our lifetimes, and so you may have heard about it too. Fermat's Last Theorem is a claim first made in the 1600's by the mathematician Pierre de Fermat, which states that the equation

$$a^n + b^n = c^n$$

has no positive integer solutions (that is, there *do not* exist positive integers a, b, c satisfying this equation) for any integer $n > 2$. Despite being very simple to state, mathematicians spent centuries failing to prove (or disprove) this claim. Fermat's Last Theorem was not proved until 1995, and its resolution is widely regarded as one of the biggest single results of the last century. Some of the biggest unsolved problems in math today concern the existence (or non-existence) of certain objects. For instance, the Riemann Hypothesis asks whether the Riemann-Zeta function has any non-trivial zeroes off of a certain vertical line in the complex plane. You might be surprised to hear that such an existence claim is so difficult to resolve – after all, how hard can it be to decide if a function is 0 somewhere? It turns out, the answer is *very hard*. In fact, the Riemann Hypothesis is one of the seven Millennium Prize Problems designated by the Clay Institute in 2000. These problems were chosen as among the most important and difficult open questions at the turn of the millennium, and solving any one of them would win you a million dollar prize. (As of this writing, only one of the seven Millennium Prize Problems has been resolved.)

So, when working with existence statements – either finding examples to show that certain objects exist, or disproving universally quantified statements by finding counterexamples, you should not feel that you are doing "easy" math. It is not always easy to find examples and counterexamples. There is an art to knowing how to look for likely examples and counterexamples. There is also an art, once you have found them, to writing them down well. We end this section by discussing what makes a good example or counterexample. Let's start with a simple situation. Consider the following true statement and see if you can find your own proof before reading further.

There exists a natural number n for which $3^n - 1$ is a multiple of 4.

Here is a possible proof.

Proof. Consider $n = 12$. Then $3^{12} - 1 = 531,440 = 4 \cdot 132,860$, proving the claim. □

Is this proof correct? Yes. Is it good? Well. . . that's debatable. There are certainly mathematicians who would argue that this proof is fine because it does what it needed to

do – it exhibited a natural number with the given property. As you probably know by now, our perspective is that a proof is more than just a logical argument that a conjecture is true – it is written with the goal of communicating with the reader. What did this proof communicate to you when you read it? Did you wonder if it is necessary to work with numbers this large? Were you annoyed that you would probably want to get out a calculator to check the arithmetic? If you came up with your own proof before reading ours, did ours make you wonder if you had missed something in your own work?

Although such an approach is not strictly required, we propose that when providing examples and counterexamples, you look for the simplest examples you can find. If it's easy to find examples of the thing you are looking for, you will confuse your reader by presenting a complicated one. "But," you might say, "531,440 is my very favorite integer. I was delighted to discover that I could use it as an example." Of course math can be fun and we don't want to discourage you from finding joy in your work, but do try to keep your reader in mind.

3.4 Problems

1. Consider the open sentences

$$p(n)\text{: The integer } n \text{ is divisible by 2.}$$
$$q(n)\text{: The integer } n \text{ is divisible by 3.}$$
$$r(n)\text{: The integer } n \text{ is divisible by 4.}$$

   Find the four smallest positive integers for which each of the following is true. In each case, say in words the properties held by integers for which the statements are true.

   (a) $p(n) \wedge (q(n) \vee r(n))$

   (b) $p(n) \wedge \neg(q(n) \vee r(n))$

   (c) $(p(n) \wedge q(n)) \wedge \neg r(n)$

2. (You do not need any knowledge of series to do this problem.) The Integral Test for convergence of infinite series says

   Suppose $f\colon [0, \infty) \to \mathbb{R}$ is positive, continuous, and decreasing. Let $a_n = f(n)$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\int_0^{\infty} f(x)\,dx$ converges.

   Note that the hypothesis is that f is a function for which three statements are true:

$$p(f)\text{: } f \text{ is positive.}$$
$$q(f)\text{: } f \text{ is continuous.}$$
$$r(f)\text{: } f \text{ is decreasing.}$$

   (a) Give an example of a function f for which $p \wedge q \wedge r$ is true.

   (b) Next consider $\neg(p \wedge q \wedge r)$.

i. Your classmate says that the only f for which this negation is true are discontinuous, negative, and increasing. Explain your classmate's logical error.

ii. Give three different examples of functions f illustrating three distinct ways that $\neg(p \wedge q \wedge r)$ can be true.

3. One of the best-known definitions in undergraduate mathematics involving nested quantifiers is the definition of convergence for a sequence $\{a_n\}$ of real numbers. We say that $\{a_n\}$ **converges** if there exists a real number L such that, for every $\varepsilon > 0$, there exists a natural number N such that, for all $n \geq N$, $|a_n - L| < \varepsilon$. (Whew!) The number L is called the **limit** of the sequence.

   (a) Write this definition more tersely using logical symbols (but also reread our warning about such restatements).

   (b) Consider the specific sequence whose n-th term is $a_n = 2 + \frac{1}{n}$. We claim that $\{a_n\}$ converges to $L = 2$. Generate or plot the first 20 terms of the sequence and explain in your own words why this conjecture is plausible.

   (c) Consider again the sequence in part (b). Take $\varepsilon = 0.01$ and find N with the property that, for all $n \geq N$, $|a_n - 2| < 0.01$. Have you proved that $\{a_n\}$ converges to 2? Explain.

   (d) Consider now the sequence $\{b_n\}$ with $b_n = (-1)^n$. Generate or plot the first 20 terms of the sequence and make a conjecture about the convergence of the sequence. Justify your conjecture as carefully as you can.

   (e) In order to prove that a sequence $\{a_n\}$ does not converge, we would have to prove that it satisfies the negation of the definition of convergence above. Form this logical negation.

4. In a statement with nested quantifiers, the order of the quantifiers matters. Consider

   A: For all real numbers x, there exists an integer n such that $n \leq x < n + 1$.
   B: There exists an integer n such that, for all real numbers x, $n \leq x < n + 1$.

   (a) Write both statements more tersely using logical symbols.

   (b) Explain as clearly as you can the difference between statements A and B. Is either true? Both? You do not need to prove your claims, but explain.

   (c) Construct your own example of a pair of statements with nested quantifiers for which the only difference is the order of the quantifiers. Make your example non-mathematical, and make it so that one statement is true and the other is false.

5. Consider the statement

   For all natural numbers n, $3^n - 1$ is a multiple of 4.

   (a) The statement is false. Give a counterexample.

(b) For even natural numbers n, the statement is true. Give a proof. Several correct proofs are possible.

6. Use a "minimal criminal" argument to prove: If p is a real polynomial of degree at least 1, then p has at most n real roots. In your argument, you will probably want to apply Rolle's Theorem, which is almost certainly proved in your calculus book. It says that if a function p is continuous on some interval $[a, b]$ and differentiable on (a, b) and if $p(a) = p(b)$, then there exists some c between a and b such that $p'(c) = 0$.

7. You may have learned a "trick" in grade school for deciding whether a number is divisible by 3: Add the digits in the base 10 representation of the number. If this sum is divisible by 3, so is the original number.

   (a) Start with a three-digit number N. If its digits are a, b, and c, then

   $$N = a \cdot 10^2 + b \cdot 10 + c.$$

   Prove that N is divisible by 3 if and only if $a + b + c$ is.

   (b) There is an analogous test for divisibility by 9. State it and indicate how your work above proves it for three-digit numbers.

   (c) (Challenging) Prove the above result for natural numbers of any size. As a hint, if you are doing it right, at some point you should need to prove a lemma that for every natural number n, $10^n - 1$ is a multiple of 3.

8. In Chapter 1, you were encouraged to make a conjecture about a factor shared by all numbers of the form $k^3 - k$. At that time, you likely conjectured that all such numbers are multiples of 6. This proposition is true. It is also an example of a proposition with many proofs. Give the following two:

   (a) First prove it by induction for all non-negative integers k. Then explain how this result can be used to prove that it also holds for negative k.

   (b) First prove two lemmas:

      i. Any product of three consecutive integers is even. (Consider two cases, depending on the parity of the middle number).

      ii. Any product of three consecutive integers is a multiple of 3. (Consider three cases according to whether the middle number has a remainder of 0, 1, or 2 when divided by 3.) Put these lemmas together to prove the result.

3.5 Suggested Projects

3.5.1 Rational and Irrational Numbers

Every real number can be classified as either *rational* or *irrational*. Rational numbers are those that can be written in the form $\frac{n}{m}$ for integers n and m with $m \neq 0$. The set of rational numbers is denoted by $\mathbb{Q}$ (because these numbers are quotients of integers). Of course $\frac{1}{2}$

and $-\frac{16}{17}$ are rational. So are the numbers with decimal representations 0.25 and $0.\overline{3}$, even though these are not displayed as ratios of integers. Real numbers that are not rational are called irrational and we will denote the set of all irrational numbers by $\mathbb{I}$. In several senses that are hard to make precise with the ideas we have developed so far, it turns out that there are "many more" irrational numbers than rational ones. Nevertheless, in practice it is often a bit hard to prove that a given number is irrational. Because the defining characteristic of an irrational number is that it can *not* be written in the form $\frac{n}{m}$ for integers n and m with $m \neq 0$, perhas not surprisingly, the most common way to prove that a number is irrational is by contradiction, supposing for the sake of contradiction that it is rational.

In this project, you will have the opportunity to do two things. First, you will be asked to write several proofs that specific numbers are irrational. Second, you will be asked to explore what happens when you add or multiply various combinations of rational and irrational numbers. More specifically, for your project, complete the exercises below and then write an essay that reads like a subsection of this book in which you present the definitions, some basic examples, and the propositions you have proved about these numbers.

The first thing you need to know is that there are, in fact, irrational numbers. Do at least one of the following:

1. Prove that $\log_2 3$ is irrational.
2. Prove that $\sqrt{2}$ is irrational. (Hint: Suppose $\sqrt{2} = \frac{n}{m}$ with the fraction in lowest terms.)
3. Prove that $\sqrt[3]{2}$ is irrational.

Next, we explore what happens when we combine rational and irrational numbers.

4. Must the sum of two rational numbers be rational? Must the product of two rational numbers be rational?
5. Must the sum of two irrational numbers be irrational? Must the product of two irrational numbers be irrational?
6. What happens if you add a rational number to an irrational number?
7. What happens if you multiply an irrational number by a rational number? Does the type of number you get depend on which rational number you multiply by?

If you find this project interesting and want an extension, you might notice that we omitted some famous irrational numbers from the problems above. The numbers π and e are both fundamentally important mathematical constants which are well known to be irrational. However, it is not actually so simple to prove that π and e are irrational (although if you have seen decimal representations of π and e, you might guess that their decimals do not repeat or terminate in the way that those for rational numbers do). Discovering proofs that e and π are irrational would require considerable ingenuity. However, understanding proofs found by famous mathematicians in history is itself a worthwhile endeavor – and, as long as you are comfortable with single variable integral calculus, you actually already have all of the background knowledge needed to understand several of these arguments.

If you are interested in turning this exploration of irrational numbers into a longer-form project, or even the basis of a capstone paper, here are some suggestions for where to start. For each suggested task, the idea would be not just to read the recommended source but to then make a paper or other product which communicates the ideas of the argument in your own words.

1. Learn why e is irrational by working through a proof due to Fourier which begins with the fact that e can be expressed as the infinite sum $\sum_{n=0}^{\infty} \frac{1}{n!}$. (This proof was originally discovered by Joseph Fourier; you can find modern treatments of it, so do not need to go back to the original source. The proof should be accessible to you so long as you remember some basic facts about working with series.)

2. Learn why π is irrational. There are several proofs of the irrationality of π, some substantially more complicated than others. We recommend that you find the proof by Ivan Niven (originally published in the Bulletin of the American Mathematical Society, 1946). If you read the original article (which is only one page long!) you will find a number of calculus details left to the reader. Because you are the reader, start by filling them in! More modern treatments of this proof also exist, which show more details than the original article.

3. Another way that one can understand the irrationality of both π and e is through continued fractions. In fact, the original proofs that π and e are irrational (due to Lambert and Euler, respectively) worked by proving that these numbers have some special continued fraction expansion. For a very substantial project, you could learn what continued fraction expansion is all about, and then as an application learn these two arguments which use the idea. Please be aware that this last suggestion is probably at the level of a capstone project, not a 1 to 2 week essay.

3.5.2 Equivalent Theorems

To be written.

Chapter 4

Sets, Functions, and Counting

We have already discussed the idea of a *set*: the mathematical word we use to describe a collection of objects. In fact, we have now spent a lot of time working with sets in various ways, both concretely (e.g., proving that all elements n in the set of positive integers have the property that $11^n - 6$ is divisible by 5) and abstractly (e.g., discussing the formal definitions of universally and existentially quantified statements). You may feel like you already know a lot about sets – and you do! But there are some big ideas about sets that we haven't yet touched on formally, and these are actually so foundational that we will spend an entire chapter exploring them.

Some natural questions about sets (the kind we have focused on up to now) stem from the mathematical properties of the objects inside the set. When we prove that all positive integers n have the property that $11^n - 6$ is divisible by 5, we are using the definition of our set (that its elements are all positive integers) and inferring some additional property that these elements must all share by virtue of this definition. But other natural questions might focus more on a set itself than the properties of its elements. To give you an example of how compelling such questions might be, let's pose a problem to start this chapter.

The following scenario has delighted math students for centuries. Suppose n mathematicians attend a fancy restaurant, where an attendant takes their coats before dinner. However, this attendant is new and sort of bad at his job, so he completely neglects to write down which coat belonged to which mathematician. At the end of the night, he just hands back the coats randomly, hoping that they miraculously return to their correct owners. In how many ways total could the attendant return the coats to the mathematicians? Of this total number of ways, how many result in the best possible outcome – everyone receives their own coat? How many result in the worst possible outcome – nobody receives their own coat? While we have not phrased this problem in the language of sets, there are natural sets under the surface here. We could make a set T_n (T for *total*) whose elements are all the different ways that the n coats can be handed back to n people.[1] We could then consider the set B_n (B for *best possible*) containing all the distributions where each person receives

[1]It is important to remember that sets can contain *any* type of object; they need not just be collections of real numbers. There are multiple ways we could cleanly encode a way that coats get handed back. Probably the easiest would be to make an ordered list of n numbers, where the position of each number on the list represents the mathematician and the number in that position represents which coat is handed back. So for instance, seeing a 3 in position 1 on the list would say that mathematician 1 received coat 3.

their own coat, and the set W_n (W for *worst possible*) where nobody receives their own coat. Phrased like this, we are asking the seemingly simple question: *how many elements are in each of these sets*? We are not necessarily that interested in any further properties of these elements themselves; all we care about is how "big" each set is. Asking about these sizes leads us in a number of fascinating directions. Are there clean formulas in terms of n that will let us predict how big T_n, B_n, and W_n will be? Can we use the sizes of these sets to quantify how much more often we see a "worst outcome" than a "best outcome", or quantify the probability that a "worst outcome" will occur?

There is an entire branch of math, called *enumerative combinatorics*, devoted to counting interesting things. This might not sound like a very challenging discipline, at first. We all know how to count, after all. But just think about the coat check example. If n is a very small number (say 2, 3, or 4), you could write down all of the elements of T_n by hand. But you might find it illustrative to generate T_5 by hand. It is possible, but takes a long time, and it's very easy to miss elements or write them down twice. For even moderately large n, it becomes prohibitively difficult to generate and count manually; for instance, T_{10} contains $3,628,800$ elements. You could use a computer to help you systematically generate and count things for a while, but soon even a computer will not be able to keep up with the sheer number of objects involved. And of course, any finite list of examples can at best give you evidence towards a general understanding for the sizes of these sets when n is arbitrary. Counting can in fact be *extremely* difficult, and in order to understand it, we might need to know a lot about how to systematically generate and compare sets.

Counting becomes an even thornier question when our sets become even bigger. For example, how many elements does the set of positive integers contain? You might immediately say "well, infinitely many – that's obvious!" But how precisely should we understand what it means for a set to be "infinitely big"? Can we compare sizes of infinite sets? For instance, are there fewer even positive integers than there are positive integers? Are there more real numbers than there are rational numbers? When we are discussing sets that contain infinitely many elements, do the words "more" and "bigger" still have precise meanings? Especially when we graduate to thinking about infinite sets, the question of what size even means is an interesting one in its own right.

This chapter will build towards the tools you need to understand counting, both for finite sets like in the coat check problem and for infinite sets like the positive integers. In order to lay the groundwork, we'll begin with some basics: notation, operations, and vocabulary that we use all the time when working with sets. As with some of the early sections of Chapter 3, you might not immediately see why these foundational skills are particularly interesting. But know that after mastering these fundamentals, you will have a good toolkit to address many beautiful problems involving sets, including both of the motivating examples discussed here.[2]

[2]See Section 4.4 for discussion of measuring infinite sets, and Project 4.6.5 if you would like to try the coat check problem for yourself.

4.1 Sets and Set Operations

4.1.1 Definitions, Notation, and Operations

We know that a *set* is a collection of objects. These objects can be anything – numbers, words, functions, lists, vectors, matrices, or any other type of object you can imagine. While this might make it seem like "anything goes" with a set, there some basic rules underlying how we think about and denote sets that we must discuss.

If we have a set S and x is an element of S, we'll write $x \in S$. If x is is *not* an element of S, we similarly write $x \notin S$. One obvious follow-up question is: How do we know if x is an element of S? This is a very important question, because if we can't describe what objects a set contains, we have little hope of counting the elements of that set or working with it in any other manner. It is sometimes easy to understand a set by listing its elements directly (e.g., "the set whose elements are the words 'dog', 'cat', and 'bat' "). But clearly, for sets with more than a few elements, it would become tedious (or impossible) to specify a set by listing all of its elements one by one. Fortunately, we can avoid making an exhaustive list if we instead explain what rule or property defines membership in our set. We could do so verbally, and often have in the past. When we say "Let E be the set of all even integers", we are making it completely unambiguous what E is, even though we have not actually listed a single element of E explicitly. Similarly, expanding on our animal set above, we could say "Let T be the set of all three-letter English words that denote mammals". When defining sets, we will often take this second approach, explaining the rules for membership in the set rather than listing all elements. The only drawback is that some sets are defined in pretty complicated ways, to the point that explanation in words becomes cumbersome and confusing. Just as we introduced notation to streamline our descriptions of complex logical statements, we often want a more streamlined way to denote complicated sets. The following notation, called *set-builder notation*, will recur throughout this chapter (and throughout your mathematical career).

To denote a set, we use curly braces (which look like { }). The curly braces suggest the idea of "collecting" objects inside themselves; the objects inside the braces are the elements of the set. So, one way to describe a set is to make a list of all the set's elements contained in braces. For instance

$$A = \{\text{dog, cat, bat}\}$$

is the first set described in words above. This is already a clearer and more succinct way of describing the set A than listing its elements verbally. The main idea of set-builder notation is to help define more complicated sets by putting all of the relevant information into our braces. We'll use the basic layout

$$\{\text{element notation } : \text{ defining properties}\}.$$

In this context, the colon is read "such that", and it naturally separates the two pieces that we need to explain how our set works. To the left of the colon, we show the notation we use to denote our elements. This could just be an "x", where we specify later what type of object x is. But depending on our set, perhaps every element has a specific form that is easy to write down. To the right of the colon, we say in words or symbols what rule(s) define

membership in our set. So, for example,

$$T = \{x : x \text{ is a 3 letter English word denoting a mammal}\}$$

describes exactly the set T we verbally described above. This first example may not seem that much more efficient than a verbal description, but let's look at a couple other scenarios to get familiar with the notation and see how it can help us out.

Example 4.1.1. Define the set

$$M = \left\{ \begin{bmatrix} a & b \\ 0 & 2a \end{bmatrix} : a, b \text{ are integers} \right\}$$

Here, M is a set of matrices[3] that all have two rows and two columns. A matrix in M must follow several rules at once: All of its entries must be integers, its lower left entry must specifically be 0, and its lower right entry must be twice its upper left entry. Of course we can say all of these rules in words, but it is clearly a hassle. Set-builder notation allows us to clearly display the common form that the elements of M share and specify all of the conditions on elements of M in very few words.

Exercise 4.1.1. Below, the set S is described in words, and the set T is described in set-builder notation. Write a description of S in set-builder notation. Then, write one or more English sentences to describe how the set T is defined.

(a) S is the set of all points in the plane whose x and y coordinates are both non-negative and whose x coordinate is exactly 4 times its y coordinate.

(b) $T = \{\frac{x}{2y} : x \text{ is an odd integer}, y \text{ is an even integer not equal to } 0\}$.

As a final comment about defining sets, there are some special sets that arise so often that we find it convenient to have even briefer descriptions than set-builder notation. We give these special sets designated names, which we will later use without any further description. Depending on what branch of math you are working with, there may be a variety of special named sets that you encounter and work with. Here, we only introduce notation for several very common sets that you already have plenty of experience with. We write

$$\mathbb{N} = \{x : x \text{ is a natural number}\}$$

$$\mathbb{Z} = \{x : x \text{ is an integer}\}$$

$$\mathbb{Q} = \{x : x \text{ is a rational number}\}$$

$$\mathbb{R} = \{x : x \text{ is a real number}\},$$

$\emptyset$ is the set that contains no elements, called the *empty set*.

[3]For any readers who have not taken linear algebra, a matrix is just a rectangular array of numbers; we can define algebraic rules for how to work with matrices logically, and efficiently store all kinds of information inside them. For this example, you don't need to know how to do anything with them, though, so it's fine to just view this as an array of numbers formatted in a particular way.

It might seem strange that we can have a set with nothing in it. But just as it is algebraically important to have the number 0, it is convenient to allow for the possibility of a set with no element.

With our set notation sorted out, let's pause to note a couple important properties that we'll want sets to possess. You may already implicitly understand these features, but we'll need to make them explicit before we start counting sets. Firstly, we do not allow a set to contain an object multiple times. For instance, if we say

$$S = \{x : x \text{ is an integer with } 1 \leq x \leq 5\},$$

then written out explicitly, $S = \{1, 2, 3, 4, 5\}$. We do *not* want to allow S to be something like $\{1, 1, 1, 1, 1, 2, 3, 4, 5\}$, even though all the individual numbers listed in these braces do follow the rule defining S.[4] Perhaps you can already think of some practical reasons why it makes sense to only allow a set to contain each object one time. From a mathematical perspective, we usually want to think about sets as answering the question "Which objects obey a certain rule?" From this viewpoint, it certainly doesn't make sense to include an element in a set more than once; that element either obeys the rule (is in the set) or does not obey the rule (is not in the set), and nothing extra could be communicated by making additional copies of the same element. The other big property of sets is that they are *unordered*. So our set S above could equally well be written as $S = \{3, 1, 4, 5, 2\}$ (or in many other orders). The thing that matters is which objects are contained in the set, not the order in which we choose to list them. If you think about the way we are defining sets, hopefully it is clear why taking sets to be unordered is reasonable. For a set like M in Example 4.1.1, there is not an obvious "canonical" way to list the elements in order, and even for the set $S = \{1, 2, 3, 4, 5\}$, the natural order we used is just a matter of convention. In either case, what we care about is which specific objects are in the set, not how we happen to list them.

Because sets are unordered, there is not always a single way that a set will be displayed, even if it is written out explicitly. It is also sometimes possible to describe the same set in two different ways. For instance, we could describe the set $\{1, 2, 3, 4, 5, 6\}$ by directly listing its elements, we could say that this set is the set of all positive integers smaller than 7, we could say that it is the set of numbers which appear on a standard 6-sided die, and so on. No matter what description we use, the underlying set is the same, and we want to be able to recognize this. We say that two sets A and B are *equal* if they contain exactly the same elements. We will talk much more in Section 4.1.2 about how to recognize and prove that two sets are equal, but in specific examples, you can already think about the notion of equality. Just think about deciding whether two different descriptions result in sets which really have exactly the same set of elements.

With the ability to formally describe sets, it will be much easier to write down a lot of sets that we might want to work with. It will also be easier to describe a variety of *set operations*. As with arithmetic in the real numbers, where we can add, multiply, and so on, there are a variety of basic ways we can combine sets to make new sets. These operations are not all directly analogous to the standard arithmetic operations, but you might think about learning these set operations as similar to learning how to add, multiply, or divide. The

[4] In some more advanced combinatorics courses, you might work with *multisets*, which are sets for which repetition of elements is allowed. But we will not consider multisets at all in this book.

operations themselves may seem like basic procedures to master, but it is very important that you are fluent with them if you want to solve more complicated problems involving sets.

Our first operations will be familiar to readers who have worked with Venn diagrams in the past.

Definition 4.1.2. Given two sets S and T, the **union** $S \cup T$ is the set

$$S \cup T = \{x : x \in S \text{ or } x \in T\}.$$

The **intersection** $S \cap T$ is the set

$$S \cap T = \{x : x \in S \text{ and } x \in T\}.$$

In words, $S \cup T$ is the set whose elements consist of every object in *at least one* of S and T.[5] And in words, $S \cap T$ is the set whose elements consist of every object in *both* S and T.[6] Unions and intersections are often extremely natural operations. For instance, they are ubiquitous in discrete probability theory, where a union of two sets can represent outcomes satisfying at least one of two conditions, and an intersection can represent outcomes satisfying two conditions at the same time.

Example 4.1.3. Let

$$A = \{\text{red, orange, yellow, green, blue, indigo, violet}\}, B = \{\text{cyan, magenta, yellow, black}\}$$

(You may recognize A as the set of colors in the rainbow and B as the set of colors used for printer ink.) Then we can take a union:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

$$= \{\text{red, orange, yellow, green, blue, indigo, violet, cyan, magenta, black}\}.$$

Note that we do *not* list yellow twice, even though yellow is common to both sets. Like with any other set, a union of two sets contains each element only once. We can also take an intersection:

$$A \cap B = \{x : x \in A \text{ and } x \in B\} = \{\text{yellow}\}$$

because yellow is the only element which appears in both sets.

When we use set-builder notation to describe sets, sometimes we must take a moment to "decode" the notation before understanding a union or intersection. Our next example will illustrate.

Example 4.1.4. Let

$$A = \{2k : k \text{ is an integer}\},\ B = \{2k + 1 : k \text{ is an integer}\},\ C = \{4k : k \text{ is an integer}\}.$$

[5] Note the similarity of the union symbol $\cup$ to the logical "or" symbol $\vee$. This is not coincidental; the two ideas are quite similar.

[6] Again, note the similarity of the intersection symbol $\cap$ to the logical "and" symbol $\wedge$.

We will find the unions and intersections of each pair of sets. First, let's look at unions. $A \cup B$ is the set whose elements are all the objects in at least one of A and B. Without thinking, we can write a clunky description in set builder notation:

$$A \cup B = \{x : x = 2k \text{ for some integer } k \text{ or } x = 2k + 1 \text{ for some integer } k\}.$$

However, if we pause to think about it for a moment, we'll recognize that A is the set of even integers and B is the set of odd integers. Every integer is in at least one of A, B (in fact, every integer is in *exactly* one, although this fact isn't important). So $A \cup B$ contains all integers, and clearly doesn't contain any element that isn't an integer. So we can much more clearly describe our union as

$$A \cup B = \mathbb{Z}.$$

(Recall, we use $\mathbb{Z}$ to denote the set of all integers.)

Next, let's consider $A \cup C$. The elements of $A \cup C$ are the objects that are in at least one of A and C. Again, we could express this idea in set-builder notation without pausing for thought, but this time, let's consider our two sets first. The elements of A are the even integers and the elements of C are the integers that are multiples of 4. Note that any element of C can be written as $4k = 2(2k) = 2m$ for some integer k, and the integer $m = 2k$, so every element of C is also an element of A. Thus, $A \cup C$ equals A, because every object that is in at least one of the two must already be in A.[7] Finally, $B \cup C$ does not have as nice of a description as $A \cup B$ or $A \cup C$ did; we have

$$B \cup C = \{x : x = 2k + 1 \text{ for some integer } k \text{ or } x = 4k \text{ for some integer } k\}.$$

In words, $B \cup C$ is the set of integers that are either odd or a multiple of 4.

We now consider intersections. Firstly, $A \cap B$ is the set of all objects that are in *both* A and B. We have already noted that A is the set of even integers and B is the set of odd integers. We know[8] that no integer is both even and odd, so no integer is in both A and B. Thus, there are *no elements* in $A \cap B$, so $A \cap B = \emptyset$. By very similar reasoning, because every element of C is an even integer (even though there are many even integers that are not in C), we also have $B \cap C = \emptyset$. Finally, consider $A \cap C$. Because every element of C is in both A and C, every element of C is in $A \cap C$. There can't be any elements of $A \cap C$ beyond the elements of C, because any object that isn't in C can't be in any intersection involving C. So $A \cap C = C$. In words, A is the set of even integers and C is the set of integers that are multiples of 4. The intersection of these sets is the set of every integer that is both even and a multiple of 4; since every multiple of 4 is automatically even, we can save time and say that $A \cap C$ is the set of all multiples of 4.

Unions and intersections take two sets and somehow combine them. Our next operations are based on comparing two sets and identifying where they differ.

Definition 4.1.5. Given two sets A and B, the **difference** $A \setminus B$ is the set

$$A \setminus B = \{x : x \in A \text{ and } x \notin B\}.$$

[7] We say that the set A *contains* the set C. We will talk much more about set containment in Section 4.1.2.

[8] Refer back to Proposition 1.2.3 if you need a refresher on this fact.

In words, $A \setminus B$ is the set whose elements consist of every object in A *but not* in B. Set differences are natural analogues of subtraction, and like subtraction, set difference is not a *commutative* operation – that is, $A \setminus B$ and $B \setminus A$ are typically not the same set. By contrast, for any two sets, $A \cup B = B \cup A$ and $A \cap B = B \cap A$. We illustrate both the basic premise of set differences, and the potentially non-commutative nature of set differences, with an example.

Example 4.1.6. We consider the same sets as in the previous example (though with more compact notation). Let

$$A = \{2k : k \in \mathbb{Z}\},\ B = \{2k+1 : k \in \mathbb{Z}\},\ C = \{4k : k \in \mathbb{Z}\}.$$

We find all possible differences of pairs of sets. There are *six* possible differences to consider because, as we have said, differences are not commutative. Firstly, we consider $A \setminus B$. We know that A is the set of even integers and B is the set of odd integers, and have observed that there is no element in both A and B. So, $A \setminus B$ is the same as A, because every object that is in A is also not in B. Similarly, $B \setminus A$ is just B. Next, we consider $A \setminus C$ and $C \setminus A$. C is the set of all integers that are multiples of 4, so

$$A \setminus C = \{x : x \text{ is an even integer and } x \text{ is not a multiple of } 4\}.$$

Note that there are plenty of even integers that are not multiples of 4 (for instance, 2, 6, and -14), so $A \setminus C$ is a non-empty set.[9] On the other hand, let's think about $C \setminus A$. if we just wrote out the definition of $C \setminus A$ in set-builder notation, we'd get

$$C \setminus A = \{x : x \text{ is a multiple of } 4 \text{ and } x \text{ is not even}\}.$$

This is technically a correct description of $C \setminus A$, but it obscures what $C \setminus A$ actually is. Every multiple of 4 is also even. Thus, there does not exist an integer x that is both a multiple of 4 and is not even. Thus, $C \setminus A = \emptyset$.

Finally, we consider $B \setminus C$ and $C \setminus B$. Very similar to $A \setminus B$ and $B \setminus A$, note that B and C share no elements in common. So $B \setminus C$ is just B and $C \setminus B$ is just C.

Thinking through the previous example, you may have conjectured one of the following facts. See if you can prove both of the following basic properties.

Exercise 4.1.2. Let A and B be sets.

(a) Show that $A \setminus B$ is exactly A if and only if $A \cap B = \emptyset$.

(b) Show that if $A \setminus B$ is not $\emptyset$, then $A \setminus B$ and $B \setminus A$ are different sets (that is, one contains some element that the other does not).

[9] If you are willing to accept some basic facts about prime factorization and divisibility, as we sometimes have in earlier chapters, you could more succinctly describe $A \setminus C$ as the set of all integers of the form $2k$ for k an *odd* integer.

There is a type of set difference that deserves special attention. Often, it makes sense to think of a set A as living within some "universe" of objects U. Here, the word "universe" suggests a larger set of all possible elements that a set like A might contain. The *complement* of A is $A^C = U \setminus A$. In words, A^C is the set of all elements of the universe U that are not in the set A.

Before we illustrate set complements, it is worth reflecting on this definition a bit. In some sense, taking a set complement isn't mechanically any harder than taking a set difference. Indeed, once we know A and U, the thought process for finding A^C is the same as for finding the difference of any two given sets. However, there are sometimes subtleties to understanding what U itself should be and how A^C should be interpreted. There is not a single set U that we always use as our universe. We might imagine making a set U containing every single mathematical object we can think of (all numbers of all different types, all matrices, all functions, and so on). But not only are there practical limitations to defining such a set,[10] it is famously impossible to have a set containing everything.[11] Even if it were possible, such a massive universal set would not result in complements that are useful for anything. We typically will find A^C in the context of a specific problem, where A is defined as a set containing objects of *one specific type* that have *some specific property*. In such a context, the sensible thing is to say that our universe U is the set of *all objects of the type that A contains*, whether or not these objects satisfy the specific property defining membership in A. If this discussion sounds a bit abstract, a few examples should help.

Example 4.1.7. At a certain Japanese restaurant, you can order onigiri[12] in two sizes: a "personal-sized" order of 2 or a "sharing-sized" order of 7. The restaurant will not sell partial orders, so if a customer wants a specific number of these delicious rice balls, they must achieve this number by purchasing some combination of the two options. Let A be the set of impossible order sizes. What is A? In the context of this problem, what universe U does it make sense to think of A as "living in"? With this universe U, what is A^C?

Let's begin by finding A. You may remember doing very similar problems in Chapter 2. In this case, we have chosen the numbers so that we can see the answer pretty quickly and don't really need a formal inductive argument. Clearly, all even numbers can be ordered (because we can place any number of orders of size 2). We can directly order 7 onigiri, and we can achieve an order of any odd size larger than 7 by making one order of size 7 and then an appropriate number of orders of size 2. However, there is no way to order an odd number smaller than 7. So $A = \{1, 3, 5\}$.

Now we turn to our universe U. We want U to be all objects of the type that A contains. If you just see the set $\{1, 3, 5\}$ in isolation and are asked "What kind of objects does A contain?", you could have many different answers. You could say that the elements are all integers. You could say they are actually all odd integers. You could be more general and say that they are all real numbers. But in the context of this problem, we don't have to consider the set A in isolation. The problem is about sizes of onigiri orders. A is the set of order sizes that are impossible given the restaurant menu's restrictions. In other words,

[10] Depending on your perspective, it is either exciting or frightening to think about how many mathematical objects there are and how few of them any one human personally knows about.

[11] Read about Russell's paradox.

[12] (Seaweed wrapped rice balls that can be filled with a variety of delicious ingredients.)

the type of object that A contains is "potential order sizes" – that is, natural numbers. A is the set of potential order sizes *with a special extra property* – that is, that the potential order sizes in A are actually not possible to achieve. So in the context of this problem, it makes sense to say $U = \mathbb{N}$ and $A^C = \mathbb{N} \setminus \{1, 3, 5\}$, the set of all natural numbers that are not equal to 1,3, or 5. Note that A^C has a very nice interpretation – because $\mathbb{N}$ is the set of all potential order sizes and A is the set of all order sizes that are not achievable, then A^C is the set of all order sizes that *are* achievable.

As we see from this last example, if you are not explicitly told what the universe U is, it may take a bit of care to determine what makes sense for U. However, a similar thought process to what we modeled will usually help you make a good choice. The next exercise will give you a chance to try it for yourself.

Exercise 4.1.3. Let $f(x) = x^3 - 9x^2 + 23x - 15$. Here, we are thinking of f in the standard way that you do in calculus or pre-calculus, as a function that can take any real number as its input. Let A be the set of real roots of f, that is, real numbers x for which $f(x) = 0$. Note that we can factor $f(x) = (x - 1)(x - 3)(x - 5)$, so $A = \{1, 3, 5\}$, exactly the same set as in Example 4.1.7. In the context of this problem, what universe U makes sense to define? With this universe U, what is A^C? Can you interpret the set A^C in any nice way?

Finally, we define a type of product of sets, called the *Cartesian product*. While the word "product" might make us think of multiplication, we will not be multiplying elements together. (Indeed, because sets may contain many types of objects for which multiplication is not naturally defined, it wouldn't make sense to define a general type of set product that multiplies the elements of the sets themselves.) Rather than trying to multiply them, we will combine elements of our two sets into ordered pairs. Formally, we have the following definition.

Definition 4.1.8. For sets A and B, the **Cartesian product** $A \times B$ is

$$A \times B = \{(x, y) : x \in A \text{ and } y \in B\}.$$

While the Cartesian product may not immediately seem natural, it is actually a very important operation that you have seen before. The term "Cartesian product" probably makes you think of Cartesian coordinates, and indeed, the points in the Cartesian plane are exactly the elements of $\mathbb{R} \times \mathbb{R}$. Cartesian products come up in many other contexts in higher mathematics as well, and will recur throughout this book.

Example 4.1.9. The Cartesian product $\mathbb{Z} \times \mathbb{Z}$ is the set $\{(x, y) : x \in \mathbb{Z}, y \in \mathbb{Z}\}$. In words, $\mathbb{Z} \times \mathbb{Z}$ consists of all ordered pairs where both coordinates are integers. If you graph the points of $\mathbb{Z} \times \mathbb{Z}$, you will see that the points are evenly spaced in a grid-like pattern. For this reason, $\mathbb{Z} \times \mathbb{Z}$ is often called the *integer lattice* (and often denoted $\mathbb{Z}^2$).

Example 4.1.10. We can, of course, take Cartesian products of two different sets. For instance, consider

$$A = \{\text{Anna, Jen, Phoebe}\},\ B = \{\text{red, orange, yellow, green, blue, purple}\}.$$

Then $A \times B$ is a set with many elements, all of which look like an ordered pair whose first entry is one of the three names in A and whose second entry is one of the six colors in B. For example, it contains (Anna, red), (Anna, orange), (Anna, yellow), and so on, and it contains similar elements with "Jen" and "Phoebe" in the first coordinate instead of "Anna". Without context, it might seem like the elements of $A \times B$ do not have much meaning. However, depending on the interpretation of A and B, the set $A \times B$ could have a very natural interpretation. For instance, if A is a set of interesting people and B is a set of potential favorite colors, then the Cartesian product $A \times B$ is the set of all possible ways that a person could be associated with a favorite color. If we actually polled the three people in set A and asked them their favorite colors, the resulting pairs of people and colors would be taken from the universe $A \times B$. (We will talk much more about the idea of "associating" elements of one set with elements of another in Section 4.2 and in the subsection in the next chapter on relations.)

We end with some exercises that allow you to explore Cartesian products yourself.

Exercise 4.1.4. Let $A = \{\text{cat, rat, bat}\}$ and $B = \{\text{hat, mat}\}$. Explicitly write out all elements of $A \times B$. Can you explain how many elements $A \times B$ has, and why? (We will mostly wait until Section 4.3 to describe techniques for counting sets. But it is likely that you can figure out an explanation for the size of $A \times B$ without any formal introduction.) Predict (without writing all elements down explicitly) how many elements are in the Cartesian product from Example 4.1.10?

Exercise 4.1.5. Describe $\mathbb{Z} \times \mathbb{R}$ and $\mathbb{R} \times \mathbb{Z}$. Can you graph or visualize these two Cartesian products? Are they the same set, or does the order in which we take the product matter?

4.1.2 Set Containment and Equality

We have already talked about the basic idea of set equality: Two sets A and B are equal if they contain exactly the same elements, even if A and B are described in different ways. For small sets, you can often determine equality just by inspection; if you write out the elements of A and B and can see directly that the two have exactly the same elements, then there is nothing more needed to show that $A = B$. However, for larger sets that may be defined more abstractly, we will need a different approach. In Examples 4.1.4 and 4.1.6, we found some specific infinite sets that we argued were actually equal to a more simply-described set. For example, we said that

$$\{x : x = 2k \text{ for some integer } k \text{ or } x = 2k + 1 \text{ for some integer } k\} = \mathbb{Z}.$$

To establish this equality, we of course did not list all elements of each set and compare them. Rather, we thought through a general description of what elements the first set contains, and convinced ourselves that the second set contains exactly the same things. We'll want to do something similar in general when establishing set equality. But we will be aided by a somewhat more systematic approach, relying on the following important set relationship.

Given sets A and B, we say that A *contains* B (or B *is contained in* A), and write $B \subseteq A$, if every element of B is an element of A. Note the horizontal bar at the bottom of

the containment symbol. This bar is meant to evoke the bars in an equals sign $=$. While the word "contains" might make you think of one set being smaller than the other, if A and B are actually equal sets, then it is in fact true that every element of B is an element of A, so A contains B. But in this situation it is equally true that every element of A is an element of B, so B contains A. In fact, we see that the statement

$$A \subseteq B \text{ and } B \subseteq A$$

is the same as the statement $A = B$. Typically, to prove that two sets are equal, we take this approach: Instead of directly comparing sets or just talking through the equality, show that each set contains the other. Moreover, to show that A is contained in B, we take an arbitrary element of A and argue that it is also an element of B. Similarly, to show that B is contained in A, we take an arbitrary element of B and show that it is also in A. To show that A is *not* contained in B, it suffices to display a single element of A that is not in B, and vice versa to show that B is not contained in A. Because we have fairly straightforward strategies for proving that containments hold (or do not hold), the set containment approach to showing equality is very nice to use. It makes it very clear to the reader (and to you) what must be demonstrated to establish the equality, and gives a systematic way to approach the proof.

We begin with an example with some small sets so that we can directly compare and verify equality that way. Still, we will focus on showing two containments when writing our arguments.

Example 4.1.11. Let

$$A = \{k : k \text{ is an odd integer with } 3 \leq k \leq 7\},$$

$$B = \{p : p \text{ is a prime number with } p < 11\},$$

and

$$C = \{m : m \text{ is an integer for which } m + 3 \text{ is even and } 6 \leq m + 3 \leq 10\}.$$

Which, if any, of these sets are equal?

Firstly, we ask if A and B could be equal. Is A contained in B? We know that the odd integers between 3 and 7 are 3, 5, and 7, each of which happens to be prime and less than 11. So every element of A satisfies the condition for membership in B, meaning $A \subseteq B$. Next, is B a subset of A? We recall that 2 is prime, but 2 is not an odd integer, so B contains an element that A does not. This single element suffices to show that B is *not* contained in A. Thus, we conclude that $A \neq B$.

We can similarly compare B and C. We have already noted that 2 is an element of B, and $2 + 3 = 5$ is not an even integer between 6 and 10. So B contains an element that is not in C, meaning B is not contained in C. Now, if we are curious, we can check whether C is contained in B. But if we only want to determine whether B and C are equal, we do not need to go any farther. For two sets to be equal, each must be contained in the other, so the fact that B is not contained in C means $B \neq C$.

Finally, we compare A and C. Let k be an element of A. So $3 \leq k \leq 7$ and k is odd. We can add 3 to all parts of this inequality to obtain another valid inequality, so we observe

that $3+3 \leq k+3 \leq 7+3$, i.e., $6 \leq k+3 \leq 10$. Note also that because k and 3 are odd, $k+3$ is even,[13] so k is an integer such that $k+3$ is even and $6 \leq k+3 \leq 10$. Thus, k satisfies all the properties that define membership in C, meaning that k is in C. So $A \subseteq C$. On the other hand, let m be an element of C. Then $m+3$ is even and $6 \leq m+3 \leq 10$. Subtracting 3 from the inequality gives $3 \leq m \leq 7$. Moreover, because $m+3$ is even and 3 is odd, $m = (m+3)-3$ is odd.[14] So m is an odd integer and $3 \leq m \leq 7$, meaning that m is in A. So $C \subseteq A$. Because both $A \subseteq C$ and $C \subseteq A$, we conclude that $A = C$.

In this example, of course, we could have just written out the sets explicitly:

$$A = \{3, 5, 7\},\ B = \{2, 3, 5, 7\},\ C = \{3, 5, 7\}$$

and immediately identified which are equal. But the purpose of the example is to demonstrate the general method for verifying set equality.

Example 4.1.12. Let $S = \{x^2 + 5 : x \in \mathbb{R}\}$ and $T = \{r : r \in \mathbb{R} \text{ and } r \geq 5\}$. How are S and T related?

If we look at these sets for a moment, we may convince ourselves that they are equal. These sets both contain infinitely many elements, so we cannot write out all of their elements explicitly to compare them. However, we can prove equality by showing two containments. Firstly we take an arbitrary element of S, which looks like $x^2 + 5$ for some $x \in \mathbb{R}$. For all real numbers x, we know that $x^2 + 5$ is a real number and that $x^2 \geq 0$, so $x^2 + 5 \geq 5$. Thus, this element of S is also an element of T, and we conclude that $S \subseteq T$. Next we take an arbitrary element r of T. We know that r is a real number and $r \geq 5$. Thus, $r - 5 \geq 0$, and we can write

$$r = (r-5) + 5 = \left(\sqrt{r-5}\right)^2 + 5.$$

Because $r - 5 \geq 0$, we know that $\sqrt{r-5}$ is a real number. Thus, we have displayed r in the form 5 plus the square of a real number, which means r is an element of S. We conclude that $T \subseteq S$. Because both set containments hold, the two sets are equal.

If you think back to notation from calculus, you might recall a more compact way to describe the set T in Example 4.1.12. Given two real numbers a and b, with $a < b$, the *closed interval* $[a, b]$ is the set $\{x : x \in \mathbb{R} \text{ and } a \leq x \leq b\}$. The *open interval* (a, b) is the set $\{x : x \in \mathbb{R} \text{ and } a < x < b\}$. We can similarly define "half-open" intervals with endpoints a, b, denoted by $(a, b]$ or $[a, b)$, by including only one of the two endpoints in the set. While infinity isn't a number and so we do not want to think of a set of real numbers including infinity, we can define open or half-open intervals with ∞ or $-\infty$ as an "endpoint". So $[a, \infty) = \{x : x \in \mathbb{R} \text{ and } a \leq x < \infty\}$, and so on. In this interval notation, the set T from Example 4.1.12 is simply $[5, \infty)$.

Like almost all notation, interval notation should be used when it makes your writing clearer. If there is a situation where it seems better to use set-builder notation, you should feel free to choose set-builder notation instead. However, often interval notation *is* easier to read, and you should be prepared to use it when appropriate.

Exercise 4.1.6. Let $A = \{x^3 : x \in \mathbb{R} \text{ and } -2 \leq x \leq -1 \text{ or } 1 \leq x \leq 2\}$. Find an interval or union of intervals equal to A, and prove this equality.

[13] See Proposition 1.2.2 if you would like to review this fact.

[14] This is not exactly Proposition 1.2.2; we are now claiming that an even integer minus an odd integer is an odd integer. If you recall the proof of Proposition 1.2.2, a similar argument proves this claim.

4.1.3 Indexed Collections and De Morgan's Laws

Thus far, we have focused on situations where we work with only two sets at a time. However, some of the set operations we defined in Section 4.1.1 can be extended to collections of more than two sets. Now that we understand the operations for pairs of sets, it will feel more natural to extend these ideas to larger collections. We will highlight through examples a few of the situations where such operations are useful.

When we work with a large number of sets at once, we require some notation to make sure we know how many sets we have and how to refer to each of these sets. The typical solution is to assign each set a label, called its *index*, from an "index set" α. Each element of α corresponds to exactly one of the sets. We thus might denote a *collection*[15] of sets as

$$\{S_i : i \in \alpha\}.$$

We note that the index set can be anything at all – a finite set, the natural numbers, the real numbers, etc. It doesn't even need to be a set of numbers. So when you see an index i attached to a set, don't just assume it is a natural number or that you can list the sets in an indexed collection in some systematic way.

When we define a collection of sets, we will of course need to clearly specify α and describe how each S_i is defined. While the general description of an indexed collection may seem abstract, in practice it is usually clear how to think about it as long as you are comfortable with set-builder notation. We illustrate the idea with an example that will recur throughout the section.

Example 4.1.13. We recall the "coat check" problem from this chapter's introduction. A group of n mathematicians give their coats to an attendant at a coat check. The attendant fails to note who owns each coat, so at the end of the night, he passes back the coats at random. We will encode the pattern in which the attendant passes the coats back as an ordered list of n numbers. The i-th number of the list will be the coat passed back to mathematician i, so if j appears in position i on the list, then mathematician i has received mathematician j's coat. Call each such list a *length-n coat list*.

Let $\alpha = \{k : k \in \mathbb{Z},\ 1 \leq k \leq n\}$. For $i \in \alpha$, let A_i be the set of length-n coat lists in which mathematician i receives their own coat. We then have constructed an indexed collection,

$$\{A_i : i \in \alpha\}.$$

After we construct an indexed collection, we can perform set operations on its elements. The collection constructed in Example 4.1.13 is actually a very useful collection to work with as we start towards an understanding of the coat check problem. The next exercise indicates why; try it before reading on to check your understanding of the problem and the notation we have established.

Exercise 4.1.7. Take the collection $\{A_i : i \in \alpha\}$ established in Example 4.1.13. Describe in words the set $A_1 \cap A_2$. More generally, let i and j be two distinct integers in α. Describe in words the set $A_i \cap A_j$. What is $(A_i \cap A_j)^C$?

[15] Looking at the notation, you might say we are making a set of sets. This is true, and it is fine to refer to a collection as a set. But many mathematicians refer to a set that contains other sets as a collection. For one thing, calling it a collection helps keep it distinct from the sets that it contains.

After thinking through Exercise 4.1.7, perhaps you can imagine why we might want to extend intersections to more than two sets at a time. Just as $A_i \cap A_j$ has a specific interpretation, could we define a "triple intersection" $A_i \cap A_j \cap A_k$ that would tell us something else about the problem? Indeed, the ideas of union and intersection naturally generalize to any indexed collection of sets.

Definition 4.1.14. Let $\{S_i : i \in \alpha\}$ be an indexed collection of sets. Then we define the union

$$\bigcup_{i\in\alpha} S_i = \{x : \exists i \in \alpha \text{ such that } x \in S_i\}$$

and the intersection

$$\bigcap_{i\in\alpha} S_i = \{x : \forall i \in \alpha, x \in S_i\}.$$

Our definition allows the index set α to be any set we like, either finite or infinite. However, it is common to see $\alpha = \{k : k \in \mathbb{Z}, 1 \leq k \leq n\}$ for some n or $\alpha = \mathbb{N}$. In these cases, rather than separately defining α and writing $\bigcup_{i\in\alpha} S_i$, many authors write $\bigcup_{i=1}^{n} S_i$ (if the index set is the natural numbers up to n) or $\bigcup_{i=1}^{\infty} S_i$ (if the index set is $\mathbb{N}$). In these cases, of course, intersections can be similarly denoted.

Example 4.1.15. Let $\alpha = \mathbb{Z}$ and for each $k \in \alpha$, let $S_k = \{p(x) = ax^2 + bx + c : a, b, c \in \mathbb{R} \text{ and } p(k) = 0\}$. In words, what are the sets

$$\bigcup_{k=1}^{3} S_k \text{ and } \bigcap_{k=1}^{3} S_k?$$

What about

$$\bigcup_{k\in\alpha} S_k \text{ and } \bigcap_{k\in\alpha} S_k?$$

We begin with the union and intersection involving only three sets. In set-builder notation,

$$\bigcup_{k=1}^{3} S_k = \{p(x) = ax^2 + bx + c : \exists k \in \{1, 2, 3\} \text{ such that } p(x) \in S_k\}.$$

While this set-builder notation is very compact, it may not be the most clear way to explain what the union means. Each S_k is the set of polynomials of degree at most 2 with real coefficients for which the integer k is a root. (Note that we do *not* say S_k is a set of quadratic polynomials because the coefficient a could be 0.) The union above is the set of all polynomials in *at least one* of S_1, S_2, or S_3. In words, it is the set of polynomials of degree at most 2 with real coefficients such that at least one of 1, 2, or 3 is a root.

Similarly,

$$\bigcap_{k=1}^{3} S_k = \{p(x) = ax^2 + bx + c : \forall k \in \{1, 2, 3\}, p(x) \in S_k\}.$$

But again, it might be clearer to say in words that this intersection is the set of polynomials of degree at most 2 with real coefficients such that 1, 2 and 3 are all roots.[16]

Having examined the union and intersection involving three sets, we turn to the union and intersection over all of α. While the index set is now larger, the thought process is the same. The union

$$\bigcup_{k\in\alpha} S_k = \{p(x) = ax^2 + bx + c : \exists k \in \alpha \text{ such that } p(x) \in S_k\}$$

is, in words, the set of polynomials of degree at most 2 with real coefficients for which some integer k is a root. The set

$$\bigcap_{k\in\alpha} S_k = \{p(x) = ax^2 + bx + c : \forall k \in \alpha, p(x) \in S_k\}$$

is the set of polynomials of degree at most 2 with real coefficients for which every integer k is a root.

For our next example, we return to the indexed collection defined in Example 4.1.13. Refer back to Example 4.1.13 before reading to remind yourself of the notation and terminology we set up.

Example 4.1.16. Fix an integer $n \geq 3$. As in Example 4.1.13, let $\alpha = \{k : k \in \mathbb{Z}, 1 \leq k \leq n\}$ and for $i \in \alpha$, let A_i be the set of length-n coat lists in which mathematician i receives their own coat. Explain in words the following sets:

(a) $\displaystyle\bigcup_{i=1}^{3} A_i$

(b) $\displaystyle\bigcap_{i=1}^{n} A_i$

(c) $\displaystyle\left(\bigcup_{i=1}^{3} A_i\right)^C$

(d) $\displaystyle\left(\bigcap_{i=1}^{n} A_i\right)^C$

[16] You likely recall from previous courses that if k is a root of a non-zero polynomial, then $(x - k)$ is a factor of that polynomial. If you are comfortable using this fact, then you could very quickly determine that this intersection contains only the zero polynomial, since a non-zero, degree ≤ 2 polynomial cannot be the product of three (or more) linear factors. Because we have not yet proved how factoring of polynomials works, we'll leave the exercise here. We comment that given any three points, there is a unique polynomial of the form $ax^2 + bx + c$ that goes through these three points, and the coefficients a, b, c can be found with high-school algebra. So, if you want to pursue the point, there are means available that do not appeal to additional theory.

For (a), we are describing the set of all length-n coat lists in at least one of A_1, A_2, or A_3. Thus, $\bigcup_{i=1}^{3} A_i$ is the set of length-n coat lists in which *at least one* of mathematicians 1, 2, or 3 receives their own coat. Thinking about how length-n coat lists are defined, the lists in $\bigcup_{i=1}^{3} A_i$ are precisely the length-n coat lists that have a 1 in entry 1, a 2 in entry 2, or a 3 in entry 3. (As always, we use the inclusive "or"; it is perfectly fine if more than one of the first three mathematicians receive their own coat.)

For (b), we are describing the set of all length-n coat lists in *all* of the sets A_i. A list is in A_i if mathematician i receives their own coat back, so a length-n coat list is in all of the A_i if *every* mathematician receives their own coat back.[17] There is a single list in this set: $\bigcap_{i=1}^{n} A_i$ contains the single list $1, 2, 3, \ldots, n$, where the i-th entry is equal to i for all i.

For (c), we now must describe the complement of the set we described in (a). The sensible universe U here is the set of all length-n coat lists. By definition,

$$\left(\bigcup_{i=1}^{3} A_i\right)^C = U \setminus \bigcup_{i=1}^{3} A_i.$$

To describe this complement, we simply negate the feature of length-n coat lists that defines membership in $\bigcup_{i=1}^{3} A_i$. In words, this complement is the set of all length-n coat lists in which *none* of mathematicians 1, 2, or 3 receive their own coat back. These are the length-n coat lists for which the first entry is not 1, the second entry is not 2, and the third entry is not 3.

Finally, for (d), we are again taking the complement of a set we have already thought about. By definition,

$$\left(\bigcap_{i=1}^{n} A_i\right)^C = U \setminus \bigcap_{i=1}^{n} A_i.$$

In words, this is the set of all length-n coat lists in which *at least one* mathematician does not receive their own coat.

After several examples, you hopefully are becoming convinced that taking unions and intersections across large indexed collections can be very useful. Indeed, many interesting sets are most naturally described as the union or intersection of some collection. Try the next exercises on your own to make sure you understand the ideas.

Exercise 4.1.8. Using the same definitions and notation as Example 4.1.13, explain in words what each of the following sets means:

(a) $\bigcup_{i=1}^{n} A_i$

[17] In the language we used in the introduction to this chapter, $\bigcap_{i=1}^{n} A_i$ is the set B_n of "best possible" outcomes.

(b) $\left(\bigcup_{i=1}^{n} A_i\right)^C$

Exercise 4.1.9. For each $k \in \mathbb{Z}$, let $A_k = \{(x, y) \in \mathbb{R} \times \mathbb{R} : y = kx\}$.[18] Describe in words

$$\bigcup_{k\in\mathbb{Z}} A_k \text{ and } \bigcap_{k\in\mathbb{Z}} A_k.$$

Our final topic in the realm of set operations concerns the relationship between unions, intersections, and complements. In the context of the coat check problem, we have already seen that the complement of a union or the complement of an intersection may be an interesting set in its own right. While thinking through those specific examples, you might have noticed that there seems to be a specific logic to describing such complements.[19] We now note a general way to describe the complement of a union or intersection.

Theorem 4.1.17 (De Morgan's Laws)**.** *Let $\{A_i : i \in \alpha\}$ be an indexed collection of sets, all of which are contained in some universe U. Then*

(a) $\left(\bigcup_{i\in\alpha} A_i\right)^C = \bigcap_{i\in\alpha} A_i^C$

(b) $\left(\bigcap_{i\in\alpha} A_i\right)^C = \bigcup_{i\in\alpha} A_i^C$

Proof. We first prove the equality in (a) by proving two containments. We have $x \in \left(\bigcup_{i\in\alpha} A_i\right)^C$ if and only if $x \notin \bigcup_{i\in\alpha} A_i$. Thus, $x \notin A_i$ for any $i \in \alpha$, so $x \in A_i^C$ for all $i \in \alpha$ We conclude that $x \in \bigcap_{i\in\alpha} A_i^C$. We conclude

$$\left(\bigcup_{i\in\alpha} A_i\right)^C \subseteq \bigcap_{i\in\alpha} A_i^C.$$

On the other hand, suppose $x \in \bigcap_{i\in\alpha} A_i^C$. Then $x \in A_i^C$ for all $i \in \alpha$, so $x \notin A_i$ for any $i \in \alpha$. Thus, $x \notin \bigcup_{i\in\alpha} A_i$, so $x \in \left(\bigcup_{i\in\alpha} A_i\right)^C$. We conclude

$$\bigcap_{i\in\alpha} A_i^C \subseteq \left(\bigcup_{i\in\alpha} A_i\right)^C.$$

[18] Recall that $\mathbb{R} \times \mathbb{R}$ is the set of ordered pairs (x, y). Because ordered pairs and open intervals look the same, we want to make it clear how we are using the notation (x, y) here.

[19] Returning to the analogy of set operations and logical symbols, if unions are similar to the logical "and" and intersections are similar to the logical "or", then taking a complement is similar to negation. Just as we built up an understanding of a statement like $\neg(p \vee q \vee r)$, we can build a systematic understanding of a set like $(P \cup Q \cup R)^C$.

Thus, the two sets are equal, proving (a).

The proof of (b) is similar. First, suppose $x \in \left(\bigcap_{i\in\alpha} A_i\right)^C$. Then $x \notin \bigcap_{i\in\alpha} A_i$, so for some $i \in \alpha$, $x \notin A_i$. Thus, for some $i \in \alpha$, $x \in A_i^C$, so $x \in \bigcup_{i\in\alpha} A_i^C$. We conclude

$$\left(\bigcap_{i\in\alpha} A_i\right)^C \subseteq \bigcup_{i\in\alpha} A_i^C.$$

On the other hand, suppose $x \in \bigcup_{i\in\alpha} A_i^C$. Then for some $i \in \alpha$, $x \in A_i^C$, so $x \notin A_i$. Thus, $x \notin \bigcap_{i\in\alpha} A_i$, so $x \in \left(\bigcap_{i\in\alpha} A_i\right)^C$. We conclude

$$\bigcup_{i\in\alpha} A_i^C \subseteq \left(\bigcap_{i\in\alpha} A_i\right)^C.$$

Thus, the two sets are equal, proving (b). □

Exercise 4.1.10. Using the same definitions and notation as Example 4.1.13, consider

$$\left(\bigcup_{i=1}^{n} A_i\right)^C \text{ and } \left(\bigcap_{i=1}^{n} A_i\right)^C.$$

We have thought about these sets already, in Example 4.1.16 and Exercise 4.1.8. Use De Morgan's Laws to write each set as a union or intersection of complements. Reflect upon the interpretation of each set. Do you think one way of writing the set makes it easier to understand what elements the set contains?

4.2 Functions Between Sets

We now have a very thorough understanding of sets – how to build them, combine them, and decide when two are equal. As our next step along the road to developing a rigorous theory of counting,[20] we will study a powerful way of creating an association between objects in two sets. Our new concept, called a *function*, will be crucial in our discussion of counting but is also ubiquitous in mathematics more generally.

4.2.1 Definitions and Examples

Reflect upon your last few math classes. Most likely, you recently took one or more semesters' worth of calculus. If so, it won't be hard to convince you that functions play a huge role

[20] Yes, we know it still seems hard to believe that counting is a hard problem.

in mathematics. However, your calculus experience has likely skewed your thinking about functions so that you probably almost always think about real-valued functions of a real variable and their graphs. Although these functions are very important to many branches of mathematics and its applications, the concept of a function is much more general. In this section, we look at functions from this more general perspective. We will certainly use familiar functions from calculus as examples, but we will also study functions with an eye to one of our intended applications to the counting of finite and infinite sets.

The following definition is likely quite similar to that which you encountered in a pre-calculus course.

Definition 4.2.1. Let A and B be non-empty sets. A **function** $f\colon A \to B$ is a rule that assigns to each element a of the set A exactly one element $f(a)$ of the set B. We call A the **domain** and B the **codomain**.

When $f(a) = b$, we sometimes informally refer to a as the "input" and b as the "output". We also say that a is "mapped to" b.

When we say that a function is a rule for assigning to each element of the domain an element of the codomain, we do not mean to suggest that the rule needs to be described by a formula. We give a number of examples and non-examples, some familiar, some strange, to start to give you a feeling for the range of possibilities.

Example 4.2.2. Let's start with something familiar. Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ given by the formula $f(x) = x^2$. Thus, the domain and codomain are both the set of real numbers. It is indeed true that to each element x of the domain we have associated exactly one element of the codomain, namely the real number x^2. Now, it happens to be the case that not every element of the codomain is assigned to something in the domain. For example, there is no x for which $f(x) = -1$, because there is no real number whose square is negative. There is no problem here; the definition requires every element of the domain to be used but does not require every element of the codomain to be. We will later have a special term for functions that indeed "hit" every element of the codomain.

On the other hand, consider the rule $g(x) = \sqrt{x}$. Does g define a function from $\mathbb{R}$ to $\mathbb{R}$? No, because if x is a negative real number, $\sqrt{x}$ is undefined. We could, however, talk about the function $g\colon [0, \infty) \to \mathbb{R}$ that assigns to each non-negative x the real number $g(x) = \sqrt{x}$.

Our next example takes us out of the familiar context of calculus functions and illustrates that the domain and codomain can be any kind of set.

Example 4.2.3. Let

$$A = \{\text{Jen, Anna, Phoebe}\}$$

and let

$$B = \{\text{red, orange, yellow, green, blue, purple}\}.$$

Define $C\colon A \to B$ by

$$C(\text{Jen}) = \text{purple}, \quad C(\text{Anna}) = \text{red}, \quad C(\text{Phoebe}) = \text{blue}.$$

This function doesn't have a formula, and you may have no idea whether it describes anything in the world.[21] Still, it meets the definition of assigning to each element of the domain exactly one element of the codomain.

Example 4.2.4. Let $A = \{a, b, c, d\}$ and let $B = \{0, 1\}$. Consider the rule

$$p(a) = 0, \quad p(b) = 0, \quad p(c) = 0, \quad p(d) = 1.$$

Is p a function from A to B? Is it a problem that it assigns the value 0 to three different elements of the domain?

Yes, p is a function from A to B and there is no problem here. The definition says that we must assign an element of the codomain to every element of the domain, *but it doesn't say we can't repeat elements of* B. Again, there is a special property a function has when the outputs are all distinct, but this property is not part of being a function.

On the other hand, consider a different rule for the same sets:

$$q(a) = 0, \quad q(a) = 1, \quad q(b) = 0, \quad q(c) = 1.$$

This rule has two "personal problems" that disqualify it from being a function. First, d is an element of the domain, but there is no element of the codomain assigned to it. Second, there is a problem with a. A function must assign exactly one element of the codomain to each element of the domain, but we have assigned both 0 and 1 to the element a.

Exercise 4.2.1. Each of the items below describes a possible relationship between two sets A and B. Determine whether the relationship is a function with domain A and codomain B.

(a) Let $A = [0, \infty)$ and let $B = \mathbb{R}$. For each $x \in A$, let $f(x)$ be an element of B whose square is x.

(b) Let $A = [0, \infty)$ and let $B = [0, \infty)$. For each $x \in A$, let $f(x)$ be an element of B whose square is x. (Parts (a) and (b) together show that the same rule may or may not define a function depending on the domain and codomain we specify.)

Exercise 4.2.2. Let $A = B = \mathbb{N}$. Determine whether each rule defines a function from A to B.

(a) $f(n) = 2n$

(b) $g(n) = \frac{1}{2}n$

(c) $h(n)$ is a prime factor of n

(d) $p(n)$ is the number of divisors of n

[21] If you guessed that it describes the favorite colors of the people in set A, you would be correct.

We make one final definition that is a bit subtle but will be quite important for us later. If A and B are sets between which we are trying to define a function, it may be the case that the rule we give leads to the assignment of different elements of the codomain depending on the particular representation we pick for the element of the domain. When we encounter this problem, we say that our rule does not yield a *well-defined function*. Let's look at an example.

Example 4.2.5. Let $\mathbb{Q}$ denote the rational numbers. Thus every element of $\mathbb{Q}$ can be written in the form $\frac{m}{n}$ for integers m and n with $n \neq 0$. As you know from your study of fractions, many different fractions can represent the same rational number. For example, all of the following represent the rational number with decimal expansion 0.5:

$$\frac{1}{2}, \quad \frac{5}{10}, \quad \frac{-25}{-50}, \quad \frac{3000}{6000}.$$

Suppose we forget about this fact and try to define a function $f\colon \mathbb{Q} \to \mathbb{Q}$ according to the rule that, for $x \in \mathbb{Q}$, $f(x)$ is "the" denominator of "the" fraction representing x. We would say that this rule is not a well-defined function because the output depends on the particular representation we choose for x. On the other hand, if we agreed to represent every rational number by a fraction in lowest terms with a positive denominator, then the same rule *would* yield a well-defined function.

4.2.2 Images and Pre-Images

In the previous section, we mainly considered what functions do to individual elements of the domain. Sometimes, we want to step back and look at how a function behaves on a subset of the domain or determine all the elements of the domain mapped to a specific part of the codomain. If you are dubious that such things would ever be interesting, consider the next example.

Example 4.2.6. Let $T(n)$ denote the high temperature in Provo, Utah (in degrees Fahrenheit) on the n-th day of 2025. Thus $T(1)$ is the high temperature on January 1, 2025, T(2) is the high temperature on January 2, 2025, and so on. The domain of this function is

$$A = \{n \in \mathbb{N} : 1 \leq n \leq 365\}$$

and we can take the codomain to be $\mathbb{Z}$.[22] Let's interpret each of the sets:

(a) $D = \{T(n) : 1 \leq n \leq 7\}$

(b) $C = \{n : T(n) \geq 100\}$.

Set D is a subset of the codomain. It consists of all the high temperatures achieved in Provo over the first seven days of the year. Its largest element is the highest temperature reached during that week.

Set C, on the other hand, is a subset of the domain. It contains the numbers of all of the days in 2025 when the high temperature in Provo was at least $100°F$. Could such a set be empty? Yes! If the temperature never reached $100°F$ on any day in 2025, then C would be empty. Alas, a web search indicates that C has 11 elements.

[22] With hopes that many, many of these temperatures are never realized. . .

Set D above is an example of an *image* of a set under a function and set C is an example of a *pre-image*. Although the abstract definitions below may look esoteric, as our example illustrates, when the function has a concrete interpretation, such sets answer real questions about the behavior of a function.

Definition 4.2.7. Let A and B be non-empty sets and let $f\colon A \to B$. If $C \subseteq A$, we define its **image** under f by

$$f(C) = \{b \in B : b = f(c) \text{ for some } c \in C\}.$$

If $D \subseteq B$, we define its **pre-image** by

$$f^{-1}(D) = \{a \in A : f(a) \in D\}.$$

In words, the image of a subset C of the domain is the set of all elements of the codomain that are outputs of the function for some c in C. The pre-image of a subset D of the codomain is the set of all elements a of the domain for which $f(a)$ is in D.

To solidify these definitions, we consider an example with a more "traditional" function.

Example 4.2.8. Consider $f\colon \mathbb{Z} \to \mathbb{Z}$ given by $f(n) = n^2 + 1$. Let

$$C = \{-1, 0, 1, 2\},\ D = \{1, 2, 5\},\ E = \{-2, -1, 0\}.$$

We find $f(C)$, $f^{-1}(D)$ and $f^{-1}(E)$.

First we find $f(C)$. This set consists of all outputs of f that can be obtained using inputs from C. Note that

$$f(-1) = 2,\ f(0) = 1,\ f(1) = 2,\ f(2) = 5.$$

Thus, (because we do not repeat elements when we write sets)

$$f(C) = \{1, 2, 5\}.$$

Next, we consider $f^{-1}(D)$. This set consists of all elements of the domain that lead to the output 1, 2, or 5. Now,

$$\begin{aligned} f(n) = n^2 + 1 = 1 \quad &\text{if and only if} \quad n = 0, \\ f(n) = n^2 + 1 = 2 \quad &\text{if and only if} \quad n = -1, 1, \text{ and} \\ f(n) = n^2 + 1 = 5 \quad &\text{if and only if} \quad n = -2, 2. \end{aligned}$$

Thus, $f^{-1}(D) = \{-2, -1, 0, 1, 2\}$.

Finally, consider $f^{-1}(E)$. Because $f(n) = n^2 + 1 \geq 1$ for all n, f never takes the values -2, -1, or 0. Thus, $f^{-1}(E) = \emptyset$.

Notice something interesting in the previous example; we found that

$$f(C) = \{1, 2, 5\} = D,$$

but

$$f^{-1}(f(C)) = f^{-1}(D) = \{-2, -1, 0, 1, 2\} \neq C.$$

If you just look at the symbols and don't think about exactly what they mean, the fact that these two expressions are different may seem strange – after all, shouldn't the f^{-1} and f "cancel out"? Well, no. Although this notation resembles the notation we use for certain kinds of "inverses", in this case, it means something a little different different.[23] The set $f^{-1}(f(C))$ is the set of all things in the domain that are sent to the same place where C is sent. Obviously C is part of this set, but because there is no reason to think that an output can come from only one input, it is entirely possible that elements outside of C are mapped to the same place as elements of C. The next exercise asks you to thoroughly explore such relationships between images and pre-images.

Exercise 4.2.3. As in Definition 4.2.7, let $f\colon A \to B$ and let $C \subseteq A$ and $D \subseteq B$.

(a) Prove that $C \subseteq f^{-1}(f(C))$.

(b) Prove that $f(f^{-1}(D)) \subseteq D$.

(c) In general, neither of the above set containments need be an equality. For about the simplest possible example, consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$. *By taking sets C and D with only a single element*, give simple examples where equality fails in parts (b) and (c).

In Example 4.2.8, the sets we worked with were small enough that we could easily identify the elements in our image and pre-images – there was no lack of rigor. For more complicated functions or bigger sets, it can take a bit of work to rigorously show that an image or pre-image is what we say it is. Because we are working with sets in such problems, a rigorous proof about these objects uses the principles laid out in the subsection on set containment and set equality. The next example illustrates such an argument.

Example 4.2.9. Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = \frac{1}{x^2+1}$. Let's find $f(\mathbb{R})$. In such a case, where we are finding the image of the entire domain, we refer to the set as simply *the image of f*.

Before we can write a proof, we need a conjecture. A simple graph made using your favorite piece of technology shows a curve with a bell-like shape,[24] with a maximum value of 1 when $x = 0$ and the x-axis as a horizontal asymptote. We would therefore conjecture that

$$f(\mathbb{R}) = \{y \in \mathbb{R} : 0 < y \leq 1\} = (0, 1].$$

Now, in the modern world, we have come to trust our technology in such matters and so you likely feel pretty confident that we have identified the correct image for f. However, you probably have seen demonstrations that illustrate various pitfalls of computer graphing software. One of the most annoying is that it shows you exactly what you ask it to. If

[23] Though some authors do call f^{-1} the "inverse image" of D. Furthermore, when f has an honest-to-goodness inverse function, images under the inverse and pre-images under the function are the same. See Subsection 4.2.4.

[24] The function whose graph is most often called "the bell curve" has equation $f(x) = e^{-\pi x^2}$.

you only ask it to graph the function for x-values between -10 and 10, that's all it does, and so if it has unusual behavior far away from the origin, you would never see it. You can also obscure what is going on if the range of outputs is large because the scale needed for the vertical axis to show these large values may result in "hiding" wiggly behavior for other parts of the domain where the function is of moderate size. In any event. we need to prove our conjecture. In doing so, we will also illustrate the general method, which applies even when the domain and codomain are not sets of real numbers and so drawing a graph doesn't make sense.

Our conjecture is that the set $f(\mathbb{R})$ is equal to the set $(0, 1]$, so we prove two containments. Note first that, for all real numbers x, $x^2 + 1 \geq 1$. Also, the quotient of two strictly positive real numbers is positive. Thus,

$$0 < \frac{1}{x^2 + 1} \leq 1.$$

In other words, for all $x \in \mathbb{R}$, $f(x) \in (0, 1]$, proving that $f(\mathbb{R}) \subseteq (0, 1]$.

For the reverse containment, take $y \in (0, 1]$. To show that $y \in f(\mathbb{R})$, we must show that there exists a real number x such that $f(x) = y$, i.e., we must find a solution to the equation $\frac{1}{x^2+1} = y$. Because $y \neq 0$, x is a solution if and only if

$$x^2 + 1 = \frac{1}{y},$$

or, equivalently, if and only if

$$x^2 = \frac{1}{y} - 1.$$

Now, because $y \leq 1$, $\frac{1}{y} \geq 1$ and so $\frac{1}{y} - 1$ is a non-negative real number. It thus has a real square root. Thus,

$$x = \sqrt{\frac{1}{y} - 1}$$

(or its opposite) is a real number satisfying $f(x) = y$. Thus, $(0, 1] \subseteq f(\mathbb{R})$. Because we have proved two containments, the set equality follows.

Some mathematicians (and, more to the point, some professors teaching a course like this one) have strong opinions about the way in which one writes the second half of the proof, the proof of the containment $(0, 1] \subseteq f(\mathbb{R})$. They would say that given y, what is logically required is to exhibit a real number x for which $f(x) = y$. (True.) They would argue that one shouldn't show the work of solving the equation $\frac{1}{x^2+1} = y$ because, when one first writes it down, one *does not know* that it has a solution, so it is squirrely to start manipulating it. They would instead advocate for an approach in which one begins with $x = \sqrt{\frac{1}{y} - 1}$, argues that it is real, and then verifies that $f(x) = y$. We will present such a proof in a moment. Their argument is valid; all that is required is to produce an x and show that $f(x) = y$, and, furthermore, there are some pitfalls with the other approach.

However, there is a counterargument to be made. The entire endeavor of algebra is to determine whether equations have solutions and to find them. The logic of solving an equation is: Suppose I have an x that satisfies some equation. Then I can use legal algebraic operations to replace that equation with an equivalent one, and I can keep doing so until I

arrive at an equivalent equation of the form $x =$ something. As long as the right-hand side of this last is a valid object (that is, living in the set in which I hope to find a solution), then, because all my equations are equivalent, I have a solution to the first. Of course it is a great idea to check one's solution by substituting it back into the original equation, but if all equations are equivalent and we have not made an error, there is no doubt that things will check.

There's a second argument for writing things as we did; one of our main themes in this book is that proof-writing is first and foremost an act of communication. A proof that presents an x and then verifies that it works often begs the question: Where did it come from? Did it come from scratch work? Was it a lucky guess? Was it generated by a computer algebra system or AI (which, incidentally, does many of these problems using those same computer algebra systems)? Was it purchased in a dark alley for ten bucks and a Slim Jim? We want to not only show you correct proofs but help you learn how to find them by illustrating our own thought process, and so we will tend to write arguments that show how we got our solution.

For those of you who would like to write arguments in the other style, here is such an argument showing that if $y \in (0, 1]$, then there exists a real number x such that $f(x) = y$.

Proof. Consider $x = \sqrt{\frac{1}{y} - 1}$. Because $y > 0$, $\frac{1}{y}$ is defined. Furthermore, because $y \leq 1$, $\frac{1}{y} \geq 1$, and so $\frac{1}{y} - 1 \geq 0$. It therefore has a non-negative square root in the real numbers. In other words, the x we have defined is a real number. Observe,

$$\begin{aligned} f(x) &= \frac{1}{\left(\sqrt{\frac{1}{y} - 1}\right)^2 + 1} \\ &= \frac{1}{\left(\frac{1}{y} - 1\right) + 1} \\ &= \frac{1}{\frac{1}{y}} = y. \end{aligned}$$

Thus for every $y \in (0, 1]$, there exists a real number x such that $f(x) = y$. □

Exercise 4.2.4. Consider $f : [0, \infty) \to \mathbb{R}$ given by $f(x) = \sqrt{x} + 2$. Find $f([0, \infty))$, the image of f, and prove your conjectured set equality by proving two containments.

4.2.3 Injections, Surjections, and Bijections

The concept of a function is extremely powerful, in part because of how general it is and how many varied situations it covers – it is simply an association between two sets in which every element of the domain is paired with exactly one element of the codomain. The downside of this generality is that, for some applications, we have not required enough for the concept to be useful. Consider one of our intended applications, to counting. When we count, we are making an association between a set we want to understand (e.g., the set {Jen, Anna, Phoebe, . . . } of people in a family) and a subset of natural numbers. So, for example, Jen gets paired with 1, Anna gets paired with 2, Phoebe with 3, and so on. But

the general definition of function does not force us to use every element of the codomain and allows us to use elements more than once. If a child were to pair Jen with 1 and Anna with 1 and then skipped and paired Phoebe with 4, we would correct the child and tell her to start over. The child was defining a fine function between the set of people and a set of natural numbers, but the function doesn't have the properties we have come to expect of the ones we use when we count. In this subsection, we discuss additional properties a function might have that address this situation.

To begin with, we make a special definition for functions that don't repeat outputs. As usual, for all our functions, the domain A and the codomain B are non-empty sets.

Definition 4.2.10. Let $f : A \to B$. f is **injective** if, for all $a_1, a_2 \in A$, if $f(a_1) = f(a_2)$, then $a_1 = a_2$.

You might find it easier to think of the definition in terms of its contrapositive: For all $a_1, a_2 \in A$, if $a_1 \neq a_2$, then $f(a_1) \neq f(a_2)$. Although this definition has more intuitive appeal (distinct inputs produce distinct outputs), the actual definition puts us in a better position to *prove* injectivity because we start with the equality $f(a_1) = f(a_2)$ and show that $a_1 = a_2$.

Let's illustrate how proofs and disproofs of injectivity work.

Example 4.2.11. Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x + 1$. Then f is injective.

Proof. Let $x_1, x_2 \in \mathbb{R}$ with $f(x_1) = f(x_2)$. Thus,

$$3x_1 + 1 = 3x_2 + 1.$$

Subtracting 1 from both sides gives the equivalent equality

$$3x_1 = 3x_2,$$

or, equivalently, $x_1 = x_2$. Thus, f is injective. □

Example 4.2.12. Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$. Then f is *not* injective; $-2 \neq 2$, but $f(-2) = 4 = f(2)$.

On the other hand, consider $g : [0, \infty) \to \mathbb{R}$ given by $g(x) = x^2$. Because the domain consists of *non-negative* real numbers, it is now the case that $x_1^2 = x_2^2$ does imply that $x_1 = x_2$.

In addition to showing how to demonstrate that a function is not injective, the previous example makes an important point that will come up over and over; the exact properties of a function depend not only on the rule or formula – they also depend heavily on the choice of domain and codomain. The same rule may define an injective function on one domain but may fail to do so on another.

Exercise 4.2.5. Consider the function $f : (1, \infty) \to \mathbb{R}$ given by $f(x) = \frac{1}{x-1}$. Prove that f is injective.

Exercise 4.2.6. Consider the rule $g(x) = \frac{1}{x^2+1}$.

(a) Give a choice of domain A and codomain B for which $g : A \to B$ is an injective function. Prove injectivity on this domain.

(b) Give a choice of domain A and codomain B for which $g : A \to B$ is not an injective function. Prove your claim.

We next define those types of functions that hit everything in the codomain.

Definition 4.2.13. Let $f : A \to B$. f is **surjective** if, for every $b \in B$, there exists $a \in A$ such that $f(a) = b$.

As usual, the definition tells us exactly what we must check in order to show that a function is surjective; we must take an arbitrary element out of the codomain and show that there is at least one element of the domain that is mapped to it. For very simple functions, it may be easy to guess the form of the element of the domain mapped to it. For more complicated functions, the style of proof may resemble our argument in the previous subsection about the image of the function with rule $f(x) = \frac{1}{x^2+1}$, and the same discussion applies. We look at several examples of varying difficulties.

Example 4.2.14. We start with one for which we can all do the algebra in our heads. Consider $f : \mathbb{Z} \to \mathbb{Z}$ given by $f(n) = n + 5$. Then f is surjective.

Proof. Let $m \in \mathbb{Z}$. We must show that there exists $n \in \mathbb{Z}$ such that $f(n) = m$. Consider $n = m - 5$, which is clearly an integer. Then

$$f(n) = f(m - 5) = (m - 5) + 5 = m.$$

Thus f is surjective. □

Example 4.2.15. The function $f : \mathbb{R} \to \mathbb{R}$ with rule $f(x) = 3x + 1$ that we considered above is also surjective. Before reading the proof, see if, for an arbitrary y in the codomain, you can write down the element x of the domain that is mapped to y. Then read our proof.

Proof. Let $y \in \mathbb{R}$ and consider $x = \frac{y-1}{3}$, which is clearly a real number. Observe,

$$f(x) = f\left(\frac{y-1}{3}\right) = 3\left(\frac{y-1}{3}\right) + 1 = (y - 1) + 1 = y.$$

Thus f is surjective. □

It can also be easy at times to prove that a function is not surjective. The next example not only illustrates such an argument but further emphasizes our point that the properties of a function depend not only on the rule but on the domain and codomain.

Example 4.2.16. Consider $h : \mathbb{Z} \to \mathbb{Z}$ given by $h(n) = 3n + 1$. Then h is not surjective. Indeed, we claim that there is no element of the domain mapped to $m = 2$ in the codomain. If there were, we would have

$$3n + 1 = 2,$$

or, equivalently, $3n = 1$. There is a *real number* satisfying this equation, namely $n = \frac{1}{3}$, but no integer because 1 is not a multiple of 3.

Exercise 4.2.7. Consider $f : (1, \infty) \to (0, \infty)$ given by $f(x) = \frac{1}{x-1}$. Prove that f is surjective.

Exercise 4.2.8. Consider the rule $g(x) = x^2 + 1$.

(a) Give a choice of domain A and codomain B for which $g : A \to B$ is not surjective. Prove your claim.

(b) Give a choice of domain A and codomain B for which $g : A \to B$ is surjective. Prove the surjectivity of this function.

It is likely that you have seen the concepts in this subsection in a previous course, though it is also likely that you used different terminology. Injective functions are sometimes said to be*one-to-one* and surjective functions are said to be *onto*. It is fine to use these terms, though the ones we have introduced here are perhaps more common in abstract math.

We make one final definition for a function that has both of the properties we've discussed.

Definition 4.2.17. Let $f : A \to B$. f is **bijective** if it is both injective and surjective.

A bijective function is also sometimes called a *one-to-one correspondence*. Such functions are, in a sense, the "gold standard" for functions, satisfying the most properties. Returning to our motivation at the beginning of the subsection when we discussed counting, these are the associations we want to make between two sets. For example, if we claim a mystery set has 17 elements, we want to not only make an association between the mystery set and the integers from 1 to 17, we want the function to be injective so that no two elements are paired with the same number and we want the function to be surjective so that every number from 1 to 17 has something associate with it (otherwise, we shouldn't say the set has 17 elements). We will discuss these ideas in great detail in the subsections on counting and cardinality.

The good news is that you already know how to prove that a function is bijective – you give two proofs. One proof shows that it is injective and the other shows that it is surjective.

Example 4.2.18. The function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 3x + 1$ is both injective and surjective, hence it is bijective.

Exercise 4.2.9. Let $A = B = \mathbb{R} \setminus \{0\}$. Consider $f : A \to B$ given by $f(x) = \frac{1}{x}$. Prove that f is a bijection.

4.2.4 Compositions and Inverses

Function composition gives a way of combining two functions to make a new function. Because the general definition seems somewhat abstract, we start with a concrete example reminding you why such an operation may be useful.

Example 4.2.19. A scientist is interested in conditions in the ocean at different depths and sends down a probe. The probe measures the temperatures T in degrees Fahrenheit at different depths d in feet below the surface. The depth of the probe is also a function of

time t in seconds. The *composition* of the two functions gives the temperature as a function of time. More specifically, $(T \circ d)(t) = T(d(t))$ expressed the dependence of temperature on time.

In order to define a composition like in the above example, we need two functions that "fit together" so that the image of the "inside" function is contained in the domain of the "outside" function. The definition that follows guarantees this sort of compatibility.

Definition 4.2.20. Let $f : A \to B$ and $g : B \to C$ be functions. Their **composition** is the function $g \circ f : A \to C$ given by $(g \circ f)(a) = g(f(a))$.

No doubt you have some familiarity with compositions of functions from previous courses. One important fact that we stress here is that, even when the domains and codomains are such that the composition in either order makes sense, $g \circ f$ and $f \circ g$ are most likely not the same function. Our first example illustrates this fact.

Example 4.2.21. Consider $f : \mathbb{Z} \to \mathbb{Z}$ and $g : \mathbb{Z} \to \mathbb{Z}$ given by $f(n) = n - 1$ and $g(n) = n^2$. Both compositions make sense and define function from $\mathbb{Z}$ to $\mathbb{Z}$ with rules

$$(g \circ f)(n) = g(f(n)) = (n-1)^2$$
$$(f \circ g)(n) = f(g(n)) = n^2 - 1.$$

These two functions are not the same. To see this, it is not enough to just say that the formulas are different; sometimes two algebraic expressions are equivalent even though they are presented in different ways. We need to find an n for which the two functions have different outputs. Take $n = 2$. Then $(g \circ f)(2) = (2-1)^2 = 1$ whereas $(f \circ g)(2) = 2^2 - 1 = 3$.

One of our most important goals in this subsection is to understand which properties of the functions f and g are "inherited" by the composition $g \circ f$. The next proposition summarizes some important relationships.

Proposition 4.2.22. *Let $f : A \to B$ and $g : B \to C$.*

(a) If f and g are injective, then $g \circ f$ is injective.

(b) If f and g are surjective, then $g \circ f$ is surjective.

(c) If f and g are bijective, then $g \circ f$ is bijective.

Proof. We prove part (a) and leave the proofs of parts (b) and (c) as exercises.

Suppose f and g are injective and suppose for some $a_1, a_2 \in A$, $(g \circ f)(a_1) = (g \circ f)(a_2)$. Then, by definition,

$$g(f(a_1)) = g(f(a_2)).$$

By the definition of injectivity for g, it follows that

$$f(a_1) = f(a_2).$$

Then, by the definition of injectivity for f, we conclude that $a_1 = a_2$. Thus $g \circ f$ is injective. □

Exercise 4.2.10. Prove parts (b) and (c) of Proposition 4.2.22. (Once you prove part (b), the proof of part (c) should be quick.)

Before we leave our discussion of functions, we discuss *inverse functions*. Perhaps the first question we should ask is, "Why do we want them?" One answer is practical; even if we don't conceptualize it this way, we use inverse functions every time we solve an equation. For example, suppose we want to solve

$$(x-1)^3 = 27.$$

You know what to do – take the cube root of both sides. But why? One way to think about this equation is that it is saying that some object $(x-1)$ put into the "cubing function" gives 27. Thus, to solve for x, we should first "undo" the cubing function. We do so by applying the inverse function, the "cube root function" to both sides to obtain

$$\sqrt[3]{(x-1)^3} = \sqrt[3]{27}.$$

Simplifying yields

$$x - 1 = 3,$$

and so $x = 4$.

Are there situations in which such a strategy doesn't work? Well, there are certainly situations in which care must be taken. Suppose, for example, we want to solve the very similar equation

$$(x-1)^2 = 16.$$

We think of this as saying that the squaring function evaluated at $x-1$ gives 16, and so, again, we try to isolate $x-1$ by applying the inverse function. The problem is that there is not a unique real number whose square is some given real number such as 16 or $(x-1)^2$. When such a number exists, there are usually two such, and there are some real numbers (the negatives) for which there is no real number with that square. So our strategy runs into an issue.

The difference between these two situations has to do with the different properties possessed by the functions $f, g : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3$ and $g(x) = x^2$. The first is a bijection, so that every real number is the image of a unique element of the domain. In this case, it is possible to go "backwards" and say for each y in the codomain, what element x of the domain satisfies $x^3 = y$. We call this number the cube root of y and denote it by $\sqrt[3]{y}$. On the other hand, the function g is not bijective. In fact, it is neither surjective nor injective. Thus given a real number y, in general we can not find a unique x in the domain such that $x^2 = y$. If $y < 0$, no such x exists, and if $y > 0$, two distinct values of x exist.

With this discussion, the next definition should seem unsurprising.

Definition 4.2.23. Suppose $f : A \to B$ is bijective. Its **inverse function** is the function $f^{-1} : B \to A$ defined by the rule $f^{-1}(b) = a$ if and only if $f(a) = b$.

It follows immediately that $f^{-1}(f(a)) = a$ for every $a \in A$ and $f(f^{-1}(b)) = b$ for every $b \in B$.

At this point, you may be fretting about notation. If $f : A \to B$ is bijective and $D \subseteq B$, we now have two possible meanings for $f^{-1}(D)$. First, we could think of it as the pre-image of the set D under the function f – that is, as the set of all elements of the domain of f that are mapped to the set D in the codomain. We now could also think of it as the image of the set D under the inverse function f^{-1}. Thankfully, these sets agree!

The next exercise is basic, but gives you an opportunity to think about the process of finding an inverse function.

Exercise 4.2.11. Consider $f : [0, 1] \to [3, 5]$ given by $f(x) = 2x + 3$.

(a) Show that f is bijective, so that f^{-1} exists.

(b) What is the formula for $f^{-1}(y)$?

As we indicated above when we were talking about finding all real solutions to $(x-1)^2 = 16$, not all functions have inverses. For example, the "squaring function", viewed as a function from $\mathbb{R}$ to $\mathbb{R}$, is not bijective, hence not invertible. But, then what's up with the "square root" button on every calculator?

There's a standard work-around when we are dealing with a function that is not invertible. Refusing to be thwarted, we restrict the domain to a set on which the function *is* injective and then restrict the codomain if necessary so that the function is also surjective. Then we form the inverse of the new restricted function.

Example 4.2.24. We illustrate the idea of restricting a function to obtain something invertible. As we have noted, $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^2$ is not bijective. However, consider the function $g : [0, \infty) \to [0, \infty)$ given by the rule $g(x) = x^2$. This function *is* bijective, and its inverse is given by the rule $g^{-1}(y) = \sqrt{y}$.

When we have a function $f : A \to B$ and we want to restrict it to a subset C of A, as above, we sometimes use the notation $f|_C$.

Exercise 4.2.12. Consider $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = x^3 - x$.

(a) Prove that f is not injective, hence not invertible.

(b) By using the same rule but changing the domain and codomain, describe a function that appears to be bijective. (You probably can't prove your conjecture without using some calculus and other facts about continuous functions, so you needn't try to write a proof. However, explain your thinking.)

Exercise 4.2.13. Let $f : A \to B$ be a bijection. Prove that $f^{-1} : B \to A$ is a bijection.

4.3 Counting

4.3.1 Set Sizes and Basic Principles

We now have a good overview of the basics about sets: how they are defined and notated, how to perform operations with sets, and how to define functions between sets. We will

now illustrate the use of these tools through the application which we used to motivate the introduction of sets as an independent object of study: counting.

It is clear that sets will appear when we pose counting problems, as we have noted already. If we say "how many ways can a coat check attendant return 10 coats so that no customer receives their own coat", then we are implicitly defining a set of interest, containing all possible length-10 coat lists in which nobody receives their own coat, and asking how many elements this set contains. The big question is, given a set, how can we determine the number of elements in this set?

Let us begin with the formal definitions, and then make some comments about them. For readability, both here and throughout the rest of the section, we will write $\{1, 2, \ldots, n\}$ to indicate the set $\{k : k \in \mathbb{Z}, 1 \leq k \leq n\}$.[25]

Definition 4.3.1. We say that a set S is **finite** if either $S = \emptyset$ or there exists some positive integer n such that there is a bijection

$$f : S \to \{1, 2, \ldots, n\}.$$

The **size**, or **cardinality**, of S, denoted $|S|$, describes how many elements are in S. If $S = \emptyset$, then we say $|S| = 0$. If $f : S \to \{1, 2, \ldots, n\}$ is a bijection, then we say $|S| = n$.

If you think about how you would naturally decide how many elements a set contains, you might not imagine defining a function between sets. Perhaps you visualize your set S as consisting of a "list" or "bucket" of objects, and imagine pointing to these elements one by one as you count "one, two, three, ...". But indeed, such a counting procedure really *is* defining a bijection f between S and some set of natural numbers $\{1, 2, \ldots, n\}$. When you point to an element x and say "one", you are determining that $f(x) = 1$, and so on. So, while it might be phrased in more abstract language, the definition above in some sense is not describing any more complicated procedure than the way you have thought about counting since childhood.

It is worth noting that there *are* subtleties about this definition, though we will not explore all of these subtleties in this text. If you read our definition critically, you might have some follow-up questions. What if $S \neq \emptyset$ and there is not a bijection from S to any set of the form $\{1, 2, \ldots, n\}$? Could it happen that there are bijections $f : S \to \{1, 2, \ldots, n\}$ and $g : S \to \{1, 2, \ldots, m\}$ for two different integers n and m? If these situations are possible, how do we understand $|S|$ when they happen?

It certainly *is* possible that $S \neq \emptyset$ and there is no integer n for which there exists a bijection $f : S \to \{1, 2, \ldots, n\}$. In this case, by our definition, f is not finite. We say that any set which is not finite is **infinite**, and we will discuss infinite sets and their cardinalities in Section 4.4. For now, we will only be thinking about the sizes of finite sets.

The second question we posed is beyond the scope of this text to address. Our definition of cardinality for finite sets essentially claims that we can define a function (which we are denoting $|S|$) which takes a finite set and maps it to a non-negative integer. But have we actually defined a function at all here? There are many possible ways that we could map

[25] Many mathematicians dislike notation which includes "...", since it can potentially be ambiguous. However, in this case, you should understand what we mean, and it is visually easier to parse this notation than the explicit set-builder formulation.

the elements of a set S to different sets of integers. What if there is some ingenious way that we could find a bijection between some set S and *both* $\{1, 2\}$ and $\{1, 2, 3\}$? If this were possible, then our definition of cardinality would not be usable, since a single input could have multiple outputs. The situation would be dire.

It is probably intuitively clear to you that cardinality *should* be a proper function; it might surprise you to learn that formally proving this fact is a bit annoying. Depending on the courses you take in the future, you may see cardinality treated more formally later on. For now, we will accept that every finite set has a single cardinality. If you believe that a pile of coins on the table cannot simultaneously contain both exactly 8 and exactly 9 coins, then you hopefully will be happy enough to accept this similar (though more general) claim about the size of finite sets.

When working with small sets, we can of course find cardinalities easily by making explicit bijections. We illustrate with an example; however, you'll hopefully feel that nothing particularly difficult is transpiring.

Example 4.3.2. Let

$$A = \{\text{win, lose, draw}\}$$

and

$$B = \{\text{Blübaum, Caruana, Esipenko, Giri, Nakamura, Praggnanandhaa, Sindarov, Wei}\}.^{26}$$

We claim that $|A| = 3$ and $|B| = 8$. To formally verify this using our definition, we would define bijections. For instance,

$$f : A \to \{1, 2, 3\}$$

given by $f(\text{win}) = 1$, $f(\text{lose}) = 2$, $f(\text{draw}) = 3$ is a bijection from A to $\{1, 2, 3\}$, so $|A| = 3$. But of course, if we simply count the elements of the set "by hand", we see by inspection that there are three elements in A. Similarly, to use our formal definition, we could write a bijection $g : B \to \{1, 2, \ldots, 8\}$ which formally assigns a number between 1 and 8 to each name. We will not do so here.

If all enumeration simply consisted of generating finite sets, lining their elements up with those in $\{1, 2, \ldots, n\}$, and formally establishing a bijection, then this clearly would not be an interesting type of mathematical question to study. The problem of counting really arises when we have a set consisting of interesting objects and it is not straightforward to list them all by hand (either because the set has a very large number of elements or because we are asking about a more abstract scenario, like the number of length-n coat lists, where the elements in the set will depend upon some parameter which may not be fixed).

Fortunately, there are a variety of techniques and ideas which can help us determine the sizes of sets without explicitly listing their elements. The first two principles which we will discuss are likely familiar to you, even if you have not heard them formally stated before.

Proposition 4.3.3 (The Addition Principle)**.** *Suppose A, B are two finite sets such that $A \cap B = \emptyset$. Then $|A \cup B| = |A| + |B|$.*

[26] You may recognize A as the set of possible outcomes of a chess game and B as the set of family names of the participants in the 2026 Candidates Tournament, which was held to determine the next challenger for the World Chess Championship.

While the Addition Principle may seem intuitively obvious, we give a proof for completeness, and to illustrate again the definition of cardinality.

Proof. Say $|A| = n$ and $|B| = m$. We can then find bijections $f : A \to \{1, 2, \ldots, n\}$ and $g : B \to \{1, 2, \ldots, m\}$. Define a new function $h : A \cup B \to \{1, 2, \ldots, n + m\}$ by the rule

$$h(x) = \begin{cases} f(x) & \text{if } x \in A \\ g(x) + n & \text{if } x \in B \end{cases}.$$

We claim that h is a bijection. (Note that because $A \cap B = \emptyset$, each element in $A \cup B$ is in exactly one of A, B. So h is a proper function.)

Firstly, we show that h is injective. Take distinct elements $x, y \in A \cup B$. If both x, y are in A, then $h(x) = f(x)$ and $h(y) = f(y)$. Because f is an injection from A to $\{1, 2, \ldots, n\}$, we know that $f(x) \neq f(y)$, so $h(x) \neq h(y)$. Similarly, if x, y are both in B, then $h(x) = g(x) + n$ and $h(y) = g(y) + n$. Since g is an injection from B to $\{1, 2, \ldots, m\}$, we have that $g(x) \neq g(y)$, so $g(x) + n \neq g(y) + n$. Finally, suppose $x \in A$ and $y \in B$. Then $h(x) = f(x)$ and $h(y) = g(y) + n$. We know that $f(x) \leq n$ and $g(y) + n \geq 1 + n > n$, so $h(x) \neq h(y)$. Thus, if x, y are distinct elements of $A \cup B$, then $h(x) \neq h(y)$, so h is injective.

Next, we show that h is surjective. Let $k \in \{1, 2, \ldots, n + m\}$. If $k \in \{1, 2, \ldots, n\}$, then because f is a surjective function from A to $\{1, 2, \ldots, n\}$, there exists $x \in A$ such that $f(x) = k$. So (since $f(x) = h(x)$) there is an element of $A \cup B$ that h maps to k. Next, if $k \notin \{1, 2, \ldots, n\}$, then we can write $k = \ell + n$ for some $\ell \in \{1, 2, \ldots, m\}$. Since g is a surjective function from A to $\{1, 2, \ldots, m\}$, there exists $x \in B$ such that $g(x) = \ell$, so $h(x) = \ell + n = k$. So there is an element of $A \cup B$ that h maps to k. We conclude that h is a surjection. Since h is both surjective and injective, h is a bijection from $A \cup B$ to $\{1, 2, \ldots, n + m\}$. Thus, $|A \cup B| = n + m$. □

We comment that the Addition Principle can be generalized to finite unions of sets. We state this generalization here, though do not prove it; if desired, you can prove it yourself using induction.

Proposition 4.3.4 (Generalized Addition Principle)**.** *Let $A_1, A_2, \ldots, A_n$ be a collection of sets such that, for any $i \neq j$, we have $A_i \cap A_j = \emptyset$. Then*

$$\left| \bigcup_{k=1}^{n} A_k \right| = \sum_{k=1}^{n} |A_k|.$$

You likely use the Addition Principle mentally all the time; for instance, if you have a bowl containing three apples, six peaches, and four bunches of grapes, then the total number of pieces of fruit in your bowl is $3 + 6 + 4 = 13$. In set theory language, we have defined three sets ("apples", "peaches", "grapes") and are asking about the size of their union. But note that in this example, it is important that we know that all pairs of sets have an empty intersection (since a given piece of fruit cannot be both an apple and a bunch of grapes). We say that two sets A, B are **disjoint** if $A \cap B = \emptyset$; otherwise, we say that $A \cap B$ **intersect**. When sets intersect, we must be a little more careful in determining the size of their union.

Example 4.3.5. Suppose A, B are two sets which may or may not intersect. Find a formula for $|A \cup B|$.

We know that if A and B are disjoint, then $|A \cup B| = |A| + |B|$. If A and B intersect, however, then $|A \cup B|$ will not equal $|A| + |B|$. Intuitively, this is because the elements in $A \cap B$ are "counted twice" by $|A| + |B|$. We claim that we can fix this over-counting in all cases with the following modification:

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

To justify this claimed formula, observe that

$$B = (B \cap A) \cup (B \setminus A),$$

since $B \cap A$ contains every element of B which is in A and $B \setminus A$ contains every element of B which is not in A. Moreover, from their definitions, it is clear that $B \cap A$ and $B \setminus A$ are disjoint. Thus, by the Addition Principle, we have

$$|B| = |B \cap A| + |B \setminus A|,$$

which we can rewrite as

$$|B \setminus A| = |B| - |B \cap A|.$$

Breaking down B in this way is relevant because we can also observe that

$$A \cup B = A \cup (B \cap A) \cup (B \setminus A) = A \cup (B \setminus A)$$

because $B \cap A \subseteq A$ and so $A \cup (B \cap A)$ is just equal to A. Since $B \setminus A$ is clearly disjoint from A, we can use the Addition Principle again to write

$$|A \cup B| = |A \cup (B \setminus A)| = |A| + |B \setminus A| = |A| + |B| - |A \cap B|$$

by the above work.

There is a systematic way to determine $\left|\bigcup_{k=1}^{n} A_k\right|$ when some or all of the sets A_k have intersections with one another. The most general formula to find the size of such a union is called the *Principle of Inclusion-Exclusion*,[27] and can be explored in Project 4.6.5. (This is also the project where we fully resolve our questions about the coat check problem, which should give you a sense that the principle under discussion here really is applicable to interesting questions.) We will not state the full Principle of Inclusion-Exclusion here, but give you a chance to think about a slightly larger union and see if you can figure out how to treat it.

Exercise 4.3.1. Suppose A, B, C are three sets, each pair of which may or may not intersect. Find a formula for $|A \cup B \cup C|$ in terms of the sizes of A, B, C and intersections of these sets. (If you are not immediately sure what such a formula should look like, we suggest starting with a specific example, and perhaps drawing a Venn diagram to help you visualize what is going on.)

[27] Or "PIE" for short.

We now turn to a second fundamental principle, called the Multiplication Principle. Again, it is likely that you have thought about this principle before, even if you have not formally written it out.

Proposition 4.3.6 (The Multiplication Principle)**.** *Suppose two decisions are to be made, with n distinct choices for the first decision and m distinct choices for the second decision. Then the total number of ways to choose a pair of outcomes is $n \cdot m$.*

We phrase the Multiplication Principle in terms of "making choices" because this is how it typically is used in practice; from a different perspective, however, it can be phrased as a statement about Cartesian products, and we will use this phrasing to indicate a proof.

Example 4.3.7. Rephrase the Multiplication Principle as a statement about the size of a certain Cartesian product.

Suppose we have a situation where there are two decisions to be made, with n choices for the first and m for the second. Let A be the set of choices for the first decision and B the set of choices for the second decision. Then a pair of choices for the two decisions is precisely an element of $A \times B$. Thus, the Multiplication Principle is essentially claiming that for finite sets A, B, we have

$$|A \times B| = |A| \cdot |B|.$$

Exercise 4.3.2. Prove the Multiplication Principle, interpreted as the statement that

$$|A \times B| = |A| \cdot |B|,$$

by using the Generalized Addition Principle. (Hint: break up elements of $A \times B$ according to their first coordinate. How many sets do you break the Cartesian product into, and what can you say about the sizes of these sets?)

Just as the Addition Principle can be generalized, the Multiplication Principle can. The following form is frequently useful.

Proposition 4.3.8 (Generalized Multiplication Principle)**.** *Suppose k decisions are to be made, with n_i distinct choices for the ith decision. Then the total number of ways to choose the outcomes for this set of decisions is*

$$n_1 \cdot n_2 \cdot \cdots \cdot n_k.$$

This next example will illustrate the Multiplication Principle, and also will remind us of a mathematical object which will recur throughout this section.

Example 4.3.9. Recall, a *length-n binary string* is an ordered list of n numbers, each of which is either 0 or 1. We can think of any length-n binary string as corresponding to the *binary representation* of some number n – that is, the (unique) way of writing n as a sum of distinct integer powers of 2. The ith "digit" in a binary string represents how many copies of 2^i are in the binary representation corresponding to the given string. So for instance, 1011 corresponds to $1 \cdot 2^3 + 0 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = 8 + 2 + 1 = 11$. By convention, the "digits" of a binary string are often called its *bits*. How many length-3 binary strings are there?

Let S be the set of length-3 binary strings. One approach is to write out all the elements of S and count them directly. The only way that this could go wrong is if we miss a string, or write down the same string twice. We can avoid both problems with a little care. Our set is

$$S = \{111, 110, 101, 100, 011, 010, 001, 000\}.$$

We can count and see that S has 8 elements. Now that we have S explicitly written out, we could also define a bijection $f : S \to \{1, 2, \ldots, 8\}$ as in Example 4.3.2. However, even if the approach above is fairly fast, there is probably a *better* approach. A reader might wonder: have we really listed every element of S exactly once? Is the number 8 just a random number, or is there a some real reason why S has exactly 8 elements? Let's note that to generate a length-3 binary string, we have three choices: we choose its first bit, its second bit, and its third bit. For each bit, there are two options (0 or 1). So by the Multiplication Principle, there are $2 \cdot 2 \cdot 2 = 8$ length-3 binary strings.

Let us emphasize that using the Multiplication Principle gives us *dramatically* more insight into Example 4.3.9 than generating all length-3 binary strings and counting them by hand does. In practice, we think of counting as an interesting question in cases where it is not simple to just write down all the elements of a set and directly count them. But whether or not we are faced with a set where we *could* directly count its elements, it is worth thinking about whether there are other potential approaches. As mathematical communicators, we want to choose the *best* explanation we know to help a reader understand both how big the set is and, if possible, why the set has the size that it does.

Choosing a strategy with explanatory power is especially helpful in a situations where your set can be naturally generalized. For instance, an obvious follow-up question to this exercise is, "How many length-n binary strings are there?" Generating and counting the length-3 strings by hand gives little insight into the more general question, but if we have the idea to use the multiplication principle, it is clear how to answer the general case. There are 2^n length-n binary strings, because each length-n binary string is the result of n independent choices, each with two possible options.

Exercise 4.3.3. It might occur to you that there is another natural way to count the length-n binary strings. In fact, we reminded you in Example 4.3.9 that there is a specific relationship between binary strings and certain integers. Use this relationship to define a bijection f between the set of length-3 binary strings and $\{1, 2, \ldots, 8\}$ which follows a consistent rule (that is, you could describe in one or two sentences how to compute $f(x)$ for *any* length-3 binary string). You may use the facts that every non-negative integer has a binary representation, and when written as a binary string, this representation is unique up to adding extra zeros to the front of the string (so 2 can correspond to 10, 010, 0010, and so on).

If you wanted to generalize this idea to define an explicit bijection between the set of length-n binary strings and $\{1, 2, \ldots, 2^n\}$, would the same idea work? Are there any extra claims or facts you would need in order to prove that your function is a bijection, regardless of the value of n?

Finally, while the Addition and Multiplication Principles will recur in future subsections, it is worthwhile to practice with them now. Try the next exercise before going on to ensure

that you can use these fundamental ideas before we graduate to more complex ideas in counting.

Exercise 4.3.4. A passcode for a certain brand of safe must be six digits long. When generating such a code, there may be some additional rules imposed. For each scenario below, determine how many passcodes fit the stated restrictions:

(a) there are no restrictions on the digits used (i.e., each digit can be any integer between 0 and 9, inclusive);

(b) the first digit used cannot be 0;

(c) the last digit used must be 0;

(d) either the first digit used is not zero or the last digit used is not zero.

4.3.2 Foundations of Counting

In this subsection, we will go beyond the Addition and Multiplication Principles to discuss some of the basic objects and ideas of enumerative combinatorics. There are entire books on the subject of enumeration, and so the introduction given here will certainly not cover all the possible ground in this direction. However, even just a few foundational concepts can equip you to answer a fairly broad range of counting questions. Because we are discussing counting, there is still some sense in which bijections between sets are occurring in the background. However, now that we have the Addition and Multiplication Principles, we will in practice be applying these primarily and not explicitly constructing bijections.

We begin with a fundamental question which recurs both within and outside of combinatorics: given a set of n objects, how many ways can we order these? Imagine you have a bucket of n objects and want to arrange them in a line, so that one object is assigned position 1, one is assigned position 2, and so on. Such an ordered arrangement of n distinct objects is called a **permutation** of these objects.

Proposition 4.3.10. *Let S be a set with $|S| = n$. Then there are $n!$ permutations of the elements of S.*

Proof. We apply the Multiplication Principle. We can think of building up a permutation of the elements of S in n steps. First, we will select an element of S to appear in position 1. Then we will select an element of S to appear in position 2, and so on. Because we cannot repeat elements of S in our permutation, each time we select an element of S to appear in a given position, we will remove that element from consideration for subsequent steps. Thus, there are n choices for which element will appear in position 1, $n-1$ choices for which element will appear in position 2, and so on. Applying the Multiplication Principle, the total number of ways in which we can determine a permutation of the elements in S is

$$n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot 2 \cdot 1 = n!.$$

□

Permutations in their own right come up fairly frequently; in fact, for example, the set of length-n coat lists which we have discussed throughout this chapter is just the set of all permutations of $\{1, 2, \ldots, n\}$. Counting the number of permutations of a fixed set has a simple formula, and so it might not be clear what else there is to do with the concept. However, many natural counting questions involve modifying the idea of permutations in some way. For instance, perhaps we have a large set S and want to form a permutation of the elements from some subset of S.

Example 4.3.11. Let S be a set with $|S| = n$, and fix an integer k with $1 \leq k \leq n$. How many k-tuples (that is, ordered lists of k objects) can we form whose entries are distinct elements of S?

Note that if $k = n$, then this is really just asking for how many permutations of the elements of S are possible; we have seen that the answer is $n!$. But the same logic that we used in proving Proposition 4.3.10 applies for any value of k. We think of building a k-tuple in k steps. We will first select an element of S to be the first entry in the k-tuple, then select a distinct element of S to be the second entry, and so on. We have n choices for the first entry, $(n-1)$ for the second, and so on, for a total of

$$n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot (n-k+1)$$

total k-tuples.

Let's note that we could express the answer to Example 4.3.11 in a somewhat more compact form. Observe that

$$n \cdot (n-1) \cdot (n-2) \cdot \cdots \cdot (n-k+1) = \frac{n!}{(n-k)!},$$

which is substantially shorter to write. In fact, not only do we obtain a nice form for our answer here, but this form might suggest another way to think about the process of building k-tuples. One big theme of counting is that often, thinking of multiple explanations will help you think more deeply about the problem. Let's revisit Example 4.3.11 and give an alternative argument; the ideas of this argument will also foreshadow our second big idea in enumeration.

Example 4.3.12. Show that the number of k-tuples whose entries consist of distinct elements from a set S with $|S| = n$ is $\frac{n!}{(n-k)!}$. Do not use the same argument as Example 4.3.11.

Written like this, the formula $\frac{n!}{(n-k)!}$ makes us think about the formula for the total number of permutations of the elements of S. We are dividing this number by a second specific number, $(n-k)!$. Let us observe that every k-tuple of the desired type can be constructed from a permutation of the elements of S as follows. Firstly, we construct a permutation of all elements of S. Secondly, we take as our k-tuple the first k elements of our chosen permutation (in order). We know that there are $n!$ ways in which to construct a permutation of all n elements of S. However, there are not $n!$ different k-tuples of the desired type, because many different permutations of the elements of S can correspond to the same k-tuple. Any time two different permutations agree in the first k positions, they correspond to the same k-tuple, even if they differ in later positions.

We are in a situation of *over-counting*. To figure out how many k-tuples actually exist, we need to figure out how many distinct permutations of the elements of S correspond to the same k-tuple. Fortunately, we can do this. Fix a specific k-tuple T. The permutations corresponding to the tuple T have their first k elements fixed (since they must match exactly the elements and order observed in T). There are $n-k$ elements of S which are not elements of T; in a permutation corresponding to T, these appear in positions $k+1, k+2, \ldots n$. If we are building a permutation corresponding to T, there is thus one choice for how to fill in the first k entries, $n-k$ choices for entry $k+1$, $n-k-1$ choices for entry $k+2$, and so on. Thus, the total number of permutations corresponding to T is

$$1 \cdot (n-k) \cdot (n-k-1) \cdot \cdots \cdot 2 \cdot 1 = (n-k)!$$

Now, let P be the set of all permutations of the elements of S. Intuitively, we have $n!$ objects in P and P includes $(n-k)!$ objects for every one in the set of k-tuples that we wish to count, so we should be able to find the desired number of k-tuples by dividing $n!$ by $(n-k)!$. Indeed, this matches what we found in Example 4.3.11.

While it clearly lines up with the results of Example 4.3.11, the argument in Example 4.3.12 may feel a little different than what we have been doing up to this point. Dividing by our "over-counting factor" may seem intuitively legitimate, but we haven't appealed to the definition of cardinality or the Addition or Multiplication Principles to justify that it really works this way. If you are concerned, let us note that you could formally tie this idea of over-counting and then dividing to the ideas developed through Section 4.3.1; we will not supply all details below, but filling them in for yourself is another good exercise.

Once we have defined the set P of all permutations of the elements of S, we can **partition** P into disjoint subsets P_T, each corresponding to a different k-tuple T of the desired type. (A partition of a set S is a decomposition of the set into two or more subsets $S_1, \ldots, S_k$, with the property that every element of S is in exactly one of the subsets S_i. Convince yourself that the collection of described subsets P_T forms a partition of P.) Our goal is to determine how many k-tuples T there are; this is clearly the same as the number of subsets P_T we split P into. Because the subsets P_T are disjoint, we can apply the Generalized Addition Principle to say that

$$|P| = \left| \bigcup_{T \text{ a } k\text{-tuple}} P_T \right| = \sum_{T \text{ a } k\text{-tuple}} |P_T|.$$

We know that $|P| = n!$ and that $|P_T| = (n-k)!$ for any T, so this reduces down to

$$n! = \sum_{T \text{ a } k\text{-tuple}} (n-k)!.$$

Because we are summing a constant, the right-hand side is just equal to $(n-k)!$ times the number of k-tuples we have, so we can algebraically solve for the number of k-tuples from here.

While it can be tied back to our basic principles, we typically discuss this process of over-counting and dividing without explicitly going through such an argument. It is a very common idea in counting, and we will need it to discuss our second main type of counting problem.

Proposition 4.3.13. *Let $1 \leq k \leq n$ be integers. The number of k-element subsets of a set S with $|S| = n$ is $\frac{n!}{k!(n-k)!}$.*

Proof. Note that this is different from Example 4.3.12 because a k-element subset of S is not ordered. However, we can adapt the results of Example 4.3.12 to this situation. There are $\frac{n!}{(n-k)!}$ k-tuples whose entries are distinct elements of S. Each such k-tuple corresponds naturally to a k-element subset of S (namely, the subset whose elements are those used as entries in the chosen tuple). Note that each k-element subset of S corresponds to $k!$ distinct k-tuples, so to find the number of k-element subsets of S, we divide the number of k-tuples by $k!$. Thus, the number of k-element subsets from S is

$$\frac{1}{k!} \cdot \frac{n!}{(n-k)!} = \frac{n!}{k!(n-k)!}.$$

□

We note that, in the above propositions and examples, we assume $k \geq 1$. It is in fact possible to discuss ordered or unordered selections of 0 elements from a set of size n; however, since this is an exceptional case, we treat it separately here so as not to draw focus from the combinatorial arguments above. We say there is one way to select 0 objects (either ordered or unordered) from a set of n objects, namely by taking the empty set (in the unordered case) or the empty "0-tuple" (in the ordered case). In order to make our formulas consistent, we adopt the convention that $0! = 1$.

Given a set S of size n, an unordered selection of k elements from S is sometimes called a **combination** of k elements, or a k-**combination**, taken from a set of size n. Combinations are so common that we want specific notation to describe how many there are. We define the **binomial coefficient** $\binom{n}{k}$ (read "n choose k") to be the number of ways to choose a subset of size k from a set of size n. Given Proposition 4.3.13 and our discussion of the case $k = 0$, it follows that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

if n, k are positive integers with $0 \leq k \leq n$. If n, k are integers but do not satisfy $0 \leq k \leq n$, we say that $\binom{n}{k} = 0$. (This should make sense as, for instance, there is no way to select a subset of 11 objects from a set of 9 objects.) We do not consider binomial coefficients whose entries are not integers.

If you completed Project 2.4.2, you will remember working with binomial coefficients. Indeed, given the formula $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, you can learn a fair amount about binomial coefficients by just algebraically manipulating them. However, keeping in mind the "counting" interpretation of $\binom{n}{k}$ is a good idea for several reasons. When we are faced with counting problems which involve choosing unordered sets of objects, we can now use the binomial coefficients to address these without going back through the argument of Proposition 4.3.13 again. We can also learn more about the behavior of binomial coefficients themselves by considering this interpretation; there are many properties and identities involving binomial coefficients which are best proved by interpreting $\binom{n}{k}$ as making a statement about the number of ways that we can construct a certain kind of object. Several applications and properties of binomial coefficients will be highlighted in the problems for this chapter, and in Project 4.6.1. We also give some preliminary examples here.

Example 4.3.14. How many length-10 binary strings are there in which exactly four bits are 1? How many length-10 binary strings have at most four bits equal to 1?

Let B_i be the set of length-10 binary strings with exactly i bits equal to 1. Our first goal is to find $|B_4|$. We have previously thought about building a binary string bit-by-bit, starting from one end and working down the line. However, this doesn't seem like an appropriate approach here, because if we fix bits one at a time, it is difficult to guarantee that we will have chosen exactly four to equal 1 by the end of the string. We instead note that, because every bit in a binary string must either be 0 or 1, we can think of choosing the locations of our four 1's all at once; every bit which we do not choose to equal 1 must automatically equal 0. Thus, there are as many strings in B_4 as there are ways to select a set of four bits to equal 1. It does not matter in what order we select these four bits, only the set of bits we select. So, $|B_4| = \binom{10}{4}$.

Next, we wish to determine how many length-10 binary strings contain *at most* four bits equal to 1. This is similar to our first question, but now, there are multiple numbers of bits which could equal 1 in a valid string. In fact, the length-10 binary strings with at most four bits equal to 1 are exactly the strings in

$$\bigcup_{i=0}^{4} B_i.$$

Clearly the sets B_i are pairwise disjoint, so we have

$$\left|\bigcup_{i=0}^{4} B_i\right| = \sum_{i=0}^{4} |B_i| = \sum_{i=0}^{4} \binom{10}{i},$$

since by a very similar argument to that above, $|B_i| = \binom{10}{i}$ for any i.

We pause here to give you the chance to apply these same principles yourself.

Exercise 4.3.5. Similar ideas can be applied to count strings where there are more than two options for each bit. For instance, a **length-n ternary string** is an ordered list of n numbers (still called bits), each of which equals either 0, 1, or 2. How many length-10 ternary strings have exactly 5 bits equal to 0?

This next exercise may sound quite different from the previous, but can be approached with the same techniques we have been developing.

Exercise 4.3.6. The **integer lattice** is the set of all points in $\mathbb{Z} \times \mathbb{Z}$. (You may recall discussing the integer lattice in Example 4.1.9.) We often visualize the integer lattice as not just the points in the lattice, but a grid of lines "connecting" the points of the lattice into a grid. We can think of taking a "step" between points on the integer lattice by starting at a chosen point (i, j) and moving either "up" (so, adding 1 to j to move to the point $(i, j+1)$), "down" (subtracting 1 from j), "right" (adding 1 to i), or "left" (subtracting 1 from i). We visualize each such "step" as moving along a line segment connecting two distance-1 points in the integer lattice, and say that a **walk** in the integer lattice is a sequence of such steps. Given a positive integer n, how many ways are there to walk from $(0, 0)$ to (n, n) using only "right" and "up" steps?

Our next example will return us to the realm of binary strings, but perhaps not in the way you expect. Indeed, this problem asks us to count a set which you may already feel you understand completely. However, the whole point will be to exploit the multiple ways we now know to think about binary strings: the total number of length-n binary strings can be found using the multiplication principle, but we can also count specific types of binary strings by using binomial coefficients. When we think about counting objects in different ways, we can find connections between different formulas, as this example will show.

Example 4.3.15. Find a formula for the number of length-n binary strings in terms of binomial coefficients. What property of binomial coefficients does this imply?

Note that, very similar to Example 4.3.14, there are $\binom{n}{k}$ length-n binary strings in which exactly k bits are equal to 1. We let B be the set of all length-n binary strings and B_k be the set of length-n binary strings in which exactly k bits are equal to 1. Then the collection $\{B_i\}_{i=0}^n$ partitions B, because clearly, for any fixed length-n binary string s, there is exactly one number k such that s has exactly k bits equal to 1. Thus, we can apply the Generalized Addition Principle to say that

$$|B| = \left|\bigcup_{k=0}^{n} B_k\right| = \sum_{k=0}^{n} |B_k| = \sum_{k=0}^{n} \binom{n}{k}.$$

Thus, the number of length-n binary strings is exactly $\displaystyle\sum_{k=0}^{n} \binom{n}{k}$. Note that we already know another formula for the number of length-n binary strings: by the Multiplication Principle, there are exactly 2^n of them. We now have two very different-looking formulas for the same count; however, this is not necessarily a problem! We have justified that both formulas are correct, so they must in fact be *equal*, even though they look different. We have thus proved the beautiful identity:[28]

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Example 4.3.15 illustrates a common strategy for establishing identities, often referred to as **counting two ways**. We have a single set – in this case, the set B of all length-n binary strings. We can think of two distinct methods to count it – in this case, either directly applying the Multiplication Principle or partitioning the set based on the number of 1's contained in each string. If both counting methods are valid, then the resulting formulas we get from each for the size of our set must be equal to one another.

There are many combinatorial identities, and like any topic in math, some are more straightforward to show than others. Here are two famous identities which follow quickly from counting the appropriate set in the correct two ways. They can also be verified algebraically, and in fact you may recognize part (b) from Project 2.4.2. However, we strongly recommend that you attempt combinatorial (that is, counting-based) justifications.

Exercise 4.3.7. Prove the given identities by counting the indicated set in two ways.

[28] A *combinatorial identity* is just an equality involving two expressions derived from some sort of counting process. These frequently involve binomial coefficients, and there are many, many of them!

(a) Let $0 \leq k \leq n$. Show that $\binom{n}{k} = \binom{n}{n-k}$ by counting the set of length-n binary strings containing exactly k bits equal to 1 in two ways.

(b) Let $n \geq 1$ and $0 \leq k \leq n-1$. Show that $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ by counting the collection of k-element subsets from an n-element set in two ways.

We close with one more example to tie together a number of the techniques and ideas thus far encountered in this section. While we have by no means given an exhaustive introduction to the subject of counting, the ideas we have developed are already enough to address some reasonably complex problems.

Example 4.3.16. As in Example 4.3.2, let

$$A = \{\text{win, lose, draw}\}$$

and

$$B = \{\text{Blübaum, Caruana, Esipenko, Giri, Nakamura, Praggnanandhaa, Sindarov, Wei}\}.$$

As we mentioned when we introduced these sets, A is the set of possible outcomes of a chess game and B is the set of grandmasters who competed in the 2026 Candidates Tournament. In a single round of a chess tournament, participants are paired off to play one another. The order in which pairs are selected does not matter. However, the order of players within a pair *does* matter, because in a chess game, one player receives the white pieces and the other receives the black pieces; the two sides are not symmetric, because white has the advantage of going first. Once players are paired off and assigned pieces, all pairs will play a game and arrive at an outcome. In a tournament consisting of the players from set B, in how many ways can a single round of the tournament transpire?

We have all the tools to answer this question, but there is a lot going on here. We will describe one way to think about this problem, but there are multiple equally valid approaches to the count. Let's first begin by sorting out the number of ways to select pairs of players who will compete in the round. We know that four players will be assigned white pieces and four players will be assigned black pieces. We begin by selecting the four players who will be assigned white. At this point, this is an unordered selection; since the order in which pairs of players are listed does not matter, we just need the set of four players who will get white pieces, not a specific order for them. Thus, there are $\binom{8}{4}$ ways to select which players receive the white pieces. Now, we will apply the Multiplication Principle. To make a list of pairs who will play, we can first select the players with white pieces and then select an opponent for each player with white pieces (and these opponents are automatically assigned the black pieces). When we assign opponents, order *does* matter; for example, if Blübaum, Caruana, Esipenko, and Giri are playing with the white pieces, it is a different outcome to say that Wei plays Blübaum and Sindarov plays Caruana than to say that Sindarov plays Blübaum and Wei plays Caruana. We can thus choose some order of the players with white pieces arbitrarily, and choose a permutation of the four players who will receive black pieces in order to line up pairs of opponents. There are thus 4! ways to assign opponents to the players with white pieces. By the Multiplication Principle, there are thus $\binom{8}{4} \cdot 4! = \frac{8!}{4! \cdot 4!} \cdot 4! = \frac{8!}{4!}$ ways in which to select pairs who will play in the round.[29]

[29] Can you find another justification for this formula?

Now, the outcome of a round in this tournament is not solely determined by what pairs of players are assigned. There are three possible results for each pair of players, as summarized in set A: either white wins, white loses (so black wins), or the game is a draw. An outcome of a tournament round consists of a list of pairs who are playing (we already know how many ways we can generate such a list) *and* a list of the outcome arrived at by each pair. We will now use the Multiplication Principle again. Since there are three potential outcomes for each pair, and four pairs total, there are $3 \cdot 3 \cdot 3 \cdot 3 = 3^4$ possible ways that the four pairs can finish their games. Applying the Multiplication Principle a final time, the total number of outcomes possible in a round is

$$\text{(number of possible assignments into four pairs)} \cdot \text{(number of ways four games can end)}$$

$$= \frac{8!}{4!} \cdot 3^4 = 136,080.$$

Hopefully you now see the benefit of counting systematically! Not only do we obtain a lot more explanatory power from our justification than we would by brute-force enumeration, but you could never generate a list of all outcomes in this example without computer assistance. We hope that this final example also illustrates how many of the basic concepts we have touched on thus far can be used together. From the individual elements of the Addition and Multiplication Principles, permutations, and combinations, a wide variety of questions become tractable.

4.3.3 Sets With Equal Size

The previous subsections may give you the impression that all counting problems will be addressed by either finding an explicit bijection between a set A and some set $\{1, 2, \ldots, n\}$ or by applying some combination of basic techniques (Addition Principle, Multiplication Principle, permutations, combinations) to directly find a formula for the number of elements in A. But often, it is actually very hard to directly determine the size of a set through such methods alone. In this subsection, we will return to the idea of using bijections to describe size, but in a slightly different way than we have so far. We begin with a definition.

Definition 4.3.17. We say that set A has the same cardinality as set B if there exists a bijection $f : A \to B$.

This definition requires several comments. A first sensible question, before we even discuss the interpretation of this definition, is whether this notion is symmetric. That is, if A has the same cardinality as B, does B have the same cardinality as A? It would be distressing if it didn't work this way. Thankfully, it does. If there is a bijection $f : A \to B$, then f is invertible and the inverse function $f^{-1} : B \to A$ is also a bijection. This fact allows us to be "lazy" and just say that sets A and B have the same cardinality, without worrying about which set comes first. If A and B have the same cardinality, we can write $|A| = |B|$ or $|B| = |A|$ interchangeably.

The other big question is how this definition fits in with the ideas about counting which we've built up already. We have discussed cardinality before, but our discussion in Section 4.3.1 looked a little different. Previously, we said that a set A has cardinality n if there

exists a bijection $f : A \to \{1, 2, \ldots, n\}$. Is this new definition using the word "cardinality" in a different way? Actually, no. Cardinality here still has exactly the same interpretation as before: we are talking about how many elements the sets A and B contain. But here, we are not directly directly determining what the sizes of A and B actually are. Instead, Definition 4.3.17 asserts that we can *compare* the sizes of A and B (whether we know exactly what these sizes are or not) by asking whether there is a bijection between A and B. If there is, then we say that A and B have the same size.

Definition 4.3.17 actually applies whether A and B are finite or infinite sets. Again, we will wait until Section 4.4 to start discussing infinite sets in general. If A and B are finite sets, let's verify that Definition 4.3.17 squares with our previous understanding of how to determine the cardinality of a set. Let n be a positive integer and suppose A is a set with $|A| = n$. Then there exists a bijection $g : A \to \{1, 2, \ldots, n\}$. If B is another set such that B has the same cardinality as A, then there is a bijection $f : B \to A$. Then observe that $g \circ f : B \to \{1, 2, \ldots, n\}$ is a bijection[30] between B and $\{1, 2, \ldots, n\}$, so $|B| = n$. Thus, indeed, if A and B are finite sets and there is a bijection between A and B, it follows from our original procedure for determining cardinality that $|A| = |B|$.

Why might we want to show that $|A| = |B|$ by finding a bijection between A and B? Could we not just separately determine the cardinalities of A and B and then observe that they are the same? In Section 4.4, this will be an incredibly important concept, because when we work with infinite sets, there is not a single number we can assign to describe the size of an infinite set. But even for finite sets, there are many reasons why finding a bijection between A and B might be nicer than finding $|A|$ and $|B|$ separately. Sometimes, it is difficult to directly find a bijection between a set A and $\{1, 2, \ldots, n\}$, but there is a different set B which seems more naturally related to A and which we find it easier to understand. Finding a bijection between two different-looking sets A and B can also give us deeper insight into the sets A and B themselves. In addition to telling us that the sets have the same size, the bijection could help us understand some structural similarity between the elements of A and B. An example will demonstrate.

Example 4.3.18. Let A_n be the set of walks from $(0, 0)$ to (n, n) in the integer lattice which use only "right" and "up" steps. (Refer to Exercise 4.3.6 if you need to recall the vocabulary here.) Let B_n be the set of length-$2n$ binary strings in which exactly n bits are equal to 1. Show that $|A_n| = |B_n|$.

We have discussed binary strings at length, and if you completed Exercise 4.3.6, you will have thought about the elements of A_n on your own. If you were thinking about the two problems separately, and just cared about finding $|A_n|$ and $|B_n|$, you might not have mentally connected the elements of these two sets as similar. However, from a certain perspective, a walk from $(0, 0)$ to (n, n) using only "right" and "up" steps is a very similar object to a certain kind of binary string.

Let us first discuss how to think of walks from $(0, 0)$ to (n, n). Recall that in such a walk, we are always taking steps between points on the integer lattice. A "right" step moves us from an initial point (i, j) to the point $(i + 1, j)$, and an "up" step moves us from (i, j) to $(i, j + 1)$. The key observation is that the two types of steps we can use are in some sense independent of one another – "right" steps only impact the x-coordinate of our point, and

[30] Refer to Proposition 4.2.22.

"up" steps only impact the y-coordinate. Any walk which begins at $(0,0)$ and uses exactly k "right" steps and exactly ℓ "up" steps will end at the point (k,ℓ), no matter what order the steps occur in, because on exactly k of the $k+\ell$ total steps, our x-coordinate will increase by 1 (with the y-coordinate staying fixed) and on exactly ℓ of the $k+\ell$ total steps, our y-coordinate will increase by 1 (with the x-coordinate remaining fixed). In particular, to walk from $(0,0)$ to (n,n), we must take exactly n "right" steps and n "up" steps in the course of our walk, for a walk consisting of $2n$ total steps.

Now, there are still different ways in which we could generate a walk from $(0,0)$ to (n,n). This is because we could combine n "right" steps and n "up" steps into different patterns. Thought of this way, it is natural to associate each walk from $(0,0)$ to (n,n) to a string of $2n$ characters, each of which is either R or U. We will read each string from left to right, and correspond a string to a walk by taking a "right" step each time we read an R and an "up" step each time we read a U. Now, define a function $f : A_n \to B_n$ as follows. For a walk W in A_n, let $f(W)$ be the string obtained by first representing W as a length-$2n$ string consisting of R's and U's, as described above, and then replacing each R with a 0 and each U with a 1. Observe, $f(W)$ is then a length-$2n$ binary string. By the above comments, for any $W \in A_n$, the $R-U$ string corresponding to W contains exactly n bits equal to U, so $f(W)$ contains exactly n bits equal to 1, and is thus in fact an element of B_n. Two distinct walks $W_1, W_2 \in A_n$ must correspond to different $R-U$ strings, and so in particular must differ in some bit, say the kth bit. Then one of $f(W_1), f(w_2)$ equals 1 in the kth bit and the other equals 0 in the kth bit, so if $W_1 \neq W_2$, then $f(W_1) \neq f(W_2)$. Thus, f is injective. Moreover, for any binary string $s \in B_n$, we can identify the pre-image of s by simply "reversing" our rule, replacing all 1's in s with U's and all 0's in s with R's. The resulting string then describes a walk with exactly n "right" steps and exactly n "up" steps, which by the previous comments must be a walk in A_n. Thus, f is surjective. We conclude that f is a bijection, so $|A_n| = |B_n|$.

In this example, we already had a fairly straightforward way to determine both $|A_n|$ and $|B_n|$ directly. As the logic of our solution has hopefully reminded you, both A_n and B_n have size $\binom{2n}{n}$ (thinking of building a walk as choosing the positions of the n "up" steps and building a string as choosing the positions of the n bits equal to 1). However, even when we have a good idea of how to find the sizes of the sets individually, finding a logical bijection between them might give us deeper insight into *why* the two sets have the same size. Moreover, suppose you left the last subsection with a good understanding of binary strings but without having tried Exercise 4.3.6. If you noticed the bijection displayed above, you would not need to go through the extra work of writing down an argument to find $|A_n|$. It would suffice to say that there is a bijection between A_n and B_n and that we know $|B_n| = \binom{2n}{n}$; these facts tell us automatically that $|A_n| = \binom{2n}{n}$. It is sometimes the case that a complicated set is hard to count or understand directly, but can be bijectively related to a more familiar set that you already know how to count. Thus, the strategy of finding bijections between sets really can be practically useful.

The next exercise introduces some sets which cannot be quickly counted using only the techniques of Sections 4.3.1 and 4.3.2. The sizes of these sets *can* in fact be described with relatively simple formulas, but it is extremely hard to guess what these formulas should be without a deeper study of combinatorics. However, you definitely have the ability to

experiment with all three sets. See if you notice any patterns or interesting features.

Exercise 4.3.8. Three sets are defined below, each in terms of n (which will be a non-negative integer). For $n = 0, 1, 2, 3$ and (if feeling a little ambitious) 4, find all elements of the described sets.

(a) A_n is the set of length-$2n$ strings with exactly n bits equal to 1 and exactly n bits equal to -1, with the additional property that each partial sum of bits (read left to right) is non-negative. For example, $11-11-1-1$ is in A_3, but $1-1-111-1$ is not because the third partial sum, $1+(-1)+(-1) = -1$, is negative.

(b) B_n is the set of all valid parenthesizations using n total sets of parentheses. A parenthesization is just a way of arranging open and closed parentheses in a way which could correspond to an algebraically sensible order of operations. The general rule is that each open parenthesis must be matched to a closing parenthesis that lies to its right. For instance, (()(())) is a parenthesization in B_4, but ())()()(is not, because we cannot end with an open parenthesis that does not "match" with a corresponding closed parenthesis, and neither can we see two closed parentheses with only one open parenthesis to their left.

(c) C_n is the set of all triangulations of a regular $(n+2)$-gon with labeled vertices. A triangulation is a way of drawing straight lines between vertices of a polygon, so that no two lines cross and the lines result in the inside of the polygon being broken up into triangles. Some examples and non-examples are shown below.

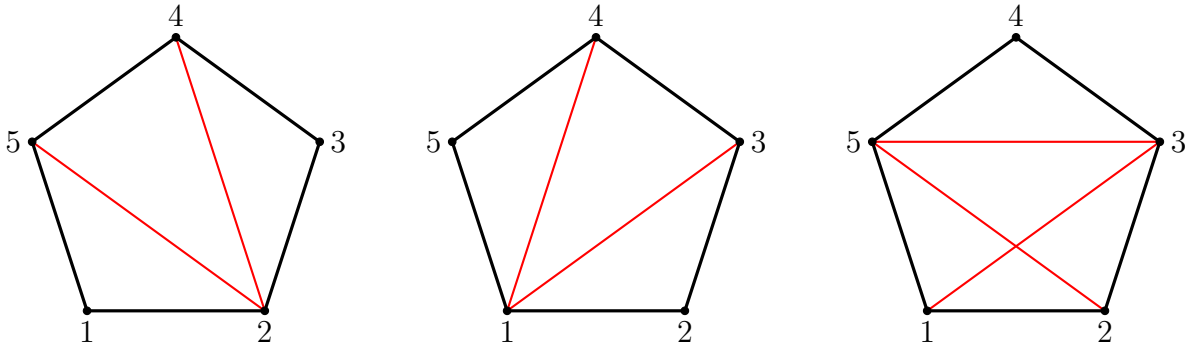


Figure 4.1: In (a) and (b), we see two different triangulations of the 5-gon with labeled vertices. (Although the two pictures could be "rotated" to look the same, we consider them different because the red lines go between vertices with different labels). In (c), the arrangement of lines is *not* a triangulation of the 5-gon. Although it looks like we have broken up the region inside the 5-gon into regions shaped like triangles, some of the red lines cross each other, which is not allowed in a triangulation.

After working with the sets in Exercise 4.3.8, you hopefully have some conjectures about how the sizes of these sets relate. But again, it is likely not clear what $|A_n|, |B_n|$, and $|C_n|$

will be for general n. This is one of the situations where finding bijections between sets is potentially a very useful idea. Even if you have no idea what a general formula for $|A_n|$ would look like, you could at least show something about $|A_n|$ – that is, that it equals the size of some other interesting set. In fact, it *is* possible to find bijections between each pair of sets in $\{A_n, B_n, C_n\}$, for any natural number n. All three of these sets are counted by the **Catalan numbers**, a famous sequence which counts many seemingly different objects. Perhaps, as you were constructing sets in Exercise 4.3.8, you even got a sense of some similar procedures that you could use to build the objects of each type.

Just because it is possible to find a bijection between two sets does not necessarily mean that the bijection is obvious. It might take a lot of thought, for instance, to discover a systematic way to associate the strings in A_n with the triangulations in C_n. Indeed, of the hundreds of types of objects counted by the Catalan numbers, some pairs sets can be very naturally bijected and other pairs require very substantial work before a direct correspondence can be understood. Below, we indicate the pair of sets from Exercise 4.3.8 between which a bijection is most natural. See if you can understand why the two have the same size.

Exercise 4.3.9. Fix n a non-negative integer and let A_n and B_n be the sets described in parts (a) and (b) of Exercise 4.3.8. Describe a bijection $f : A_n \to B_n$, and justify why f is in fact a bijection.

4.4 Cardinality and Infinite Sets

When we told you that one of our main goals for this chapter would be to understand the sizes of sets, you may have thought it couldn't be that deep. After all, you have known how to count since you were a young child. Presumably now you see that, even for finite sets, things can get pretty complicated. Just think how much more fun things could get once we consider infinite sets![31] Several questions naturally arise:

1. If a set is not finite, is there even a way to talk about its size?

2. If so, are there different "sizes" for infinite sets?

3. How can we determine whether two infinite sets have the same size?

We address all of these questions in the subsections to follow and, along the way, will encounter some examples that we guarantee will stretch your thinking. You will likely discover that most of the intuition you have about set sizes comes from your experience with finite sets, and so you will have to be extra careful not to assume that statements that are obviously true for finite sets are still true for infinite ones. We will point out some of these common misconceptions as we go along.

[31] You might think we are being sarcastic. A bit. But it might interest you to know that, for both of the authors, questions about the size of certain infinite sets were the math topic that really sparked our interest in studying advanced mathematics.

4.4.1 Definition and First Examples

Just as bijections played a central role in our discussion of cardinality for finite sets, they are important when studying infinite sets. Recall, we say that sets A and B have the same cardinality if there exists a bijection $f : A \to B$. This definition applies whether A and B are finite or infinite sets. There is a big caution here, though. If the sets are infinite, $|A|$ and $|B|$ are not numbers, and so $|A| = |B|$ is not an equation we can manipulate – it is a description of a relationship.

With this definition, we can do some examples. As we suggested in the introduction to the section, these examples may go against some of the intuition you have about set size.

Example 4.4.1. Let $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. We prove that $\mathbb{N}_0$ and $\mathbb{N}$ have the same cardinality. This result may be surprising; the set $\mathbb{N}$ is a proper subset of $\mathbb{N}_0$. For finite sets, it is not possible for a set to have the same cardinality as a proper subset. This example illustrates that with infinite sets, it is possible.

To prove the claim, we need to make a bijection $f : \mathbb{N}_0 \to \mathbb{N}$. The most natural choice is the function with rule $f(n) = n + 1$. If n is an integer with $n \geq 0$, then $n + 1 \geq 1$ and so the function indeed maps $\mathbb{N}_0$ into $\mathbb{N}$. To see that it is injective, suppose $f(n_1) = f(n_2)$. By definition, $n_1 + 1 = n_2 + 1$. Subtracting 1 from both sides of the equation shows that $n_1 = n_2$. Thus f is injective.

To see that f is surjective, take $m \in \mathbb{N}$, so that $m \geq 1$. Take $n = m - 1$. Then $n \geq 0$ and $n \in \mathbb{N}_0$. Furthermore,

$$f(n) = f(m - 1) = (m - 1) + 1 = m.$$

Thus f is surjective. Because we have established a bijection between $\mathbb{N}_0$ and $\mathbb{N}$, the two sets have the same cardinality.

Our next example gives an even more dramatic display of the fact that we can have a bijection between a set and a proper subset.

Example 4.4.2. We show that the open interval $(-1, 1)$ has the same cardinality as $\mathbb{R}$! To establish this fact, we construct a bijection between the two sets. Define $f : (-1, 1) \to \mathbb{R}$ by

$$f(x) = \frac{x}{1 - x^2}.$$

The denominator is only zero at $x = \pm 1$, so this function is indeed defined at every point of the domain.

First we show that it is injective. Indeed, suppose $f(x_1) = f(x_2)$ for $x_1, x_2 \in (-1, 1)$. Thus,

$$\frac{x_1}{1 - x_1^2} = \frac{x_2}{1 - x_2^2},$$

or, equivalently,

$$x_1 - x_1 x_2^2 = x_2 - x_1^2 x_2.$$

Rearranging gives the equivalent equation

$$x_1 - x_2 = x_1 x_2^2 - x_1^2 x_2 = -x_1 x_2 (x_1 - x_2).$$

Of course this solution will be satisfied is $x_1 = x_2$ (which is what we hope is true). But we have to rule out the possibility that it is satisfied for some other choices of x_1 and x_2. Now, if $x_1 \neq x_2$, then our equation is equivalent to

$$1 = -x_1 x_2.$$

Here's where we can use the fact that $x_1, x_2 \in (-1, 1)$. In other words, $|x_1|, |x_2| < 1$ and so $|x_1 x_2| < 1$. It is therefore not possible for $x_1 x_2$ to equal -1. We conclude that the only solution to the equation is $x_1 = x_2$.

Next, we must argue that f is surjective. We actually won't give the complete argument here. If we had access to all the theorems in calculus, we could argue as follows: f is continuous, tends to $-\infty$ as x tends to -1 from the right, and tends to ∞ as x tends to 1 from the left, so by the Intermediate Value Theorem, its image contains all real numbers. We've probably convinced you, but we broke the rules by invoking theory we have not developed. A rigorous proof using the techniques we have developed so far involves some non-trivial algebra. We start with an arbitrary y in the codomain, which in this case just means that y is an arbitrary real number. We need to show that there exists $x \in (-1, 1)$ such that $f(x) = \frac{x}{1-x^2} = y$. It is clear that $f(0) = 0$, so we may assume that $y \neq 0$. We thus need an $x \in (-1, 1)$ satisfying $x = y(1 - x^2)$, or, equivalently,

$$yx^2 + x - y = 0.$$

This equation has two real solutions,

$$x = \frac{-1 \pm \sqrt{1 + 4y^2}}{2y}$$

We are not finished, because we need to show that one of these is between -1 and 1. We leave the details to the next exercise.

Exercise 4.4.1. Show that if $y \neq 0$, one of the real numbers $\frac{-1 \pm \sqrt{1+4y^2}}{2y}$ is in the open interval $(-1, 1)$. The proof of this claim completes the proof that $(-1, 1)$ and $\mathbb{R}$ have the same cardinality.

Let's pause to reflect on how amazing this last example is! Remember, "cardinality" is our fancy word for the size of a set. For finite sets, it is the number of things in the set. So as we discuss infinite sets, we still think of cardinality as answering the question, "How big is this set?" When we did Example 4.4.1, it might have initially seemed counterintuitive that the non-negative integers and the positive integers "have the same size", but then if you thought about it for a minute, you might have decided that maybe it is intuitive after all that if a set is already infinite, you can throw one more thing into it and not change its size. But Example 4.4.2 seems to be a whole different level of counter-intuitive. How can an interval two units long have the same size as the entire real line? One has a finite length and the other doesn't. Part of the issue here is that this last bit of reasoning conflates two different notions of size – cardinality and length. If I were to say "Andy is bigger than Bob; Andy is 6 feet tall and Bob is only 5 feet 10 inches," you might say "No, Bob is bigger than

Andy; Bob weighs 190 pounds and Andy only weighs 175." We would agree that our issue was that we were working with two different definitions of size.

With sets, we have several different ways we could imagine comparing them to answer the question, "Which is bigger?" First, at times one is contained in the other, and that seems to give a sort of answer. Next, we could talk about cardinality, as we are in this chapter. Finally, if both live inside some space in which we have a notion of length or area (what in analysis we call *measure*) we might have a third way. They may not give the same answer. In these cases, the problem is really with the question. Back to the question of "Who is bigger, Andy or Bob?" if this were posed to you, instead of answering it, you should ask for clarification. "Are you asking who is taller or who is heavier?" Similarly for sets, unless it is already obvious from the context (for example, you are looking at exercises in a textbook on the cardinalities of finite sets) you should ask the clarifying question, "What is your definition of size?" Cardinality is a good, general way to talk about size because two sets can have the same cardinality even if their elements don't come from the same universe[32] and doesn't require us to be living in some highly-structured other set with a notion of measure.

Exercise 4.4.2. Recall that, for any natural number $k \geq 2$, $k\mathbb{Z}$ is the set of all integers that are multiples of k. Prove that $k\mathbb{Z}$ and $\mathbb{Z}$ have the same cardinality. As a consequence, the set of even integers and the set of integers hae the same cardinality, even though it might seem as if there are only "half as many" even integers as even integers.

Exercise 4.4.3. In this exercise, let $a, b, c, d \in \mathbb{R}$ with $a < b$ and $c < d$.

(a) Prove that the closed intervals $[0, 1]$ and $[a, b]$ have the same cardinality.

(b) Prove that $[a, b]$ and $[c, d]$ have the same cardinality. (Hint: Perhaps the most elegant argument uses part (a) instead of explicitly writing down a bijection between the two, though it would not be incorrect to do so.)

4.4.2 Countably Infinite Sets

In the previous subsection, e showed some fun examples of infinite sets that have the same cardinality. But we still don't know whether all infinite sets have the same cardinality or whether there are different "flavors" of infinite sets. How could we move towards answering this question? Return to our problem of counting finite sets. Although at some point we did make the definition of what it means for two finite sets to have the same cardinality and remarked that we could therefore compare sizes of sets without first determining their cardinality, this was not how we *started* the section. We actually started by taking some "special" sets – the natural numbers $\{1, 2, \ldots, n\}$ – and trying to specifically establish bijections from mystery sets into one of these. In this section, we take a similar approach. What would be our prototypical infinite set? The natural numbers, of course. Thus, this section is devoted to determining which sets have the same cardinality as the natural numbers. Here is the definition:

Definition 4.4.3. A set A is **countably infinite** if it has the same cardinality as $\mathbb{N}$, that is, if there exists a bijection $f : A \to \mathbb{N}$.

[32] So we *can* compare apples to oranges.

We can therefore reinterpret Example 4.4.1 as saying that the non-negative integers form a countably infinite set. We look at one more specific example and then prove some general results.

Example 4.4.4. We prove that $\mathbb{Z}$ is countably infinite. We need to construct a bijection from $\mathbb{Z}$ to $\mathbb{N}$. Before reading on, see if you can come up with such a function.

One standard approach is to define $f : \mathbb{Z} \to \mathbb{N}$ by

$$f(n) = \begin{cases} 2n+1 & n \geq 0 \\ -2n & n < 0. \end{cases}$$

In words, we send the non-negative integers to the odd natural numbers and the negative integers to the even natural numbers.

We must show that f is bijective. For injectivity, suppose $f(n_1) = f(n_2)$. If this common natural number is odd, then both n_1 and n_2 are non-negative integers with $2n_1+1 = 2n_2+1$. It follows that $n_1 = n_2$. On the other hand, if the common function value is even, then n_1 and n_2 are both negative integers with $-2n_1 = -2n_2$, which again implies $n_1 = n_2$. Thus f is injective.

For surjectivity, let $m \in \mathbb{N}$. If m is an odd natural number, then $m-1$ is a non-negative even integer, hence $n = \frac{m-1}{2}$ is a non-negative integer with

$$f(n) = f\left(\frac{m-1}{2}\right) = 2\left(\frac{m-1}{2}\right) + 1 = (m-1) + 1 = m.$$

On the other hand, if m is an even natural number, then $n = -\frac{m}{2}$ is a negative integer and

$$f(n) = f\left(-\frac{m}{2}\right) = -2\left(-\frac{m}{2}\right) = m.$$

Thus f is surjective.

Try to use a similar argument in the following exercise.

Exercise 4.4.4. Let $A = \{0, 1\} \times \mathbb{N}$. Prove that A is countably infinite by explicitly constructing a bijection from A to $\mathbb{N}$.

As fun as it is to construct bijections,[33] it can be of tremendous value to prove general propositions that can help us decide more quickly whether a given set is countably infinite. The next proposition is a very helpful tool.

Proposition 4.4.5. *Let A be an infinite set. Suppose there exists an injection $f : A \to \mathbb{N}$. Then A is countably infinite.*

To summarize, this proposition states that if we know our set A is infinite (which we usually allow ourselves to discern by inspection), then in order to show that it is countably infinite, we only need to construct an injection to the natural numbers. This doesn't just save us half the work in a proof by no longer requiring us to check surjectivity – it allows us to save work at the front end when we construct the function and not even attempt to make it surjective. Before we prove the proposition, let's use it in an example to see what an amazing result it is.

[33] Sarcasm?

Example 4.4.6. Let's show that $\mathbb{N} \times \mathbb{N}$ is countably infinite. By Proposition 4.4.5, we need only construct an injection $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$. Recall that the elements of $\mathbb{N} \times \mathbb{N}$ have the form (n, m) for natural numbers n and m. Define

$$f(n, m) = 2^n 3^m.$$

Suppose $f(n_1, m_1) = f(n_2, m_2)$, so that $2^{n_1}3^{m_1} = 2^{n_2}3^{m_2}$. We must show that $n_1 = n_2$ *and* $m_1 = m_2$. If we accept uniqueness of prime factorization for natural numbers of at least 2, we are finished.

If we want to give more details, we can. Suppose, first, that $n_1 \neq n_2$ and assume without loss of generality that $n_2 > n_1$. Then

$$2^{n_1}3^{m_1} = 2^{n_2}3^{m_2} \text{ is equivalent to } 3^{m_1} = 2^{n_2-n_1}3^{m_2}.$$

Thus the right-hand side is even, but the left-hand side is not (being a power of an odd number). It follows that $n_1 = n_2$. But then

$$3^{m_1} = 3^{m_2}.$$

It follows that $m_1 = m_2$. Thus f is injective, and, by the proposition, $\mathbb{N} \times \mathbb{N}$ is countably infinite.

Wasn't that delightfully easy? There are methods for writing down an explicit bijection, but most are rather involved (and hands-down harder than what we did). So we definitely want the proposition in our tool box. We will prove it now. It uses a property of the natural numbers called the *Well-Ordering Principle*. This principle states that every non-empty subset of $\mathbb{N}$ has a smallest element. It certainly seems non-controversial, and we accept it as an axiom. If doing so makes you squirm a bit, it may or may not help you to know that the Well-Ordering Principle is, in fact, equivalent to the Principle of Mathematical Induction, meaning that if you accept either, you can use it to prove the other.

Proof of Proposition 4.4.5. Suppose $f : A \to \mathbb{N}$ is an injection. We must construct a bijection. Let $B_1 = f(A)$, the image of A under f. Because A is infinite and f is injective, B_1 is infinite and, in fact, the function from A to B_1 is a bijection. We now define a function $g : B_1 \to \mathbb{N}$. This function will be a bijection, and then the composition $g \circ f : A \to \mathbb{N}$ will be the bijection we are looking for.

Here's how we construct g. Because $B_1 = f(A)$ is a non-empty subset of $\mathbb{N}$, by the well-ordering principle, it has a smallest element, which we denote b_1. Now, $B_2 = f(A) \setminus \{b_1\}$ is still an infinite subset of $\mathbb{N}$, so it has a smallest element b_2 that is different from b_1. Proceed inductively, so that at step $k \geq 2$, we define $B_k = f(A) \setminus \{b_1, \ldots, b_{k-1}\}$. Because $f(A)$ is infinite but to form B_k, we remove only finitely many elements, B_k is still an infinite subset of $\mathbb{N}$. Thus, it has a smallest element, b_k, distinct from $b_1, \ldots, b_{k-1}$. We claim that every element of B_1 is eventually chosen through this process. Indeed, let n be an arbitrary element of B_1. Then, thought of as an element of $\mathbb{N}$, n has $n-1$ natural numbers that precede it. In other words, there are at most $n-1$ elements of B_1 less than n. Thus, there is some $k \leq n$ for which n is the smallest element in $B_1 \setminus \{b_1, \ldots, b_{k-1}\}$. In simple English, after at most n steps of the process, n is selected to be one of the b_k.

We define $g : B_1 \to \mathbb{N}$ by $g(b_k) = k$. Because the b_k are distinct, g is injective. It is also surjective, because for every $k \in \mathbb{N}$, we were indeed able to find an element in $f(A) \setminus \{b_1, \ldots, b_{k-1}\}$ to assign to b_k.

Because $g \circ f : A \to \mathbb{N}$ is a composition of a bijection from A to $f(A) = B_1$ with a bijection from B_1 to $\mathbb{N}$, it is a bijection. Thus A is countably infinite. □

We make a comment about the above proof; once we identified $b_1, b_2, \ldots, b_k, \ldots$, it almost seemed unnecessary to form the function $g : B_1 \to \mathbb{N}$ given by $g(b_k) = k$. In fact, we do sometimes skip this step. Why? When we make the list $b_1, b_2, \ldots, b_k, \ldots$ of elements in B_1, *we are defining a one-to-one correspondence between* B_1 *and* $\mathbb{N}$ – we are pairing 1 with b_1, 2 with b_2, and, in general, k with b_k. We have guaranteed that the b_k are distinct and that each element of B_1 is on the list, so we have guaranteed that we have constructed a bijection. A bijection with $\mathbb{N}$ displayed in this way is called an *enumeration* of a set. Once you have obtained an enumeration of a set, you have shown that it is countably infinite.

Our next few proofs show how we can use this idea that a set is countably infinite if and only if there is an enumeration of it to prove other propositions about countably infinite sets.

Proposition 4.4.7. *Let A and B be countably infinite sets. Suppose A and B are disjoint. Then $A \cup B$ is countably infinite.*

A similar proposition holds even if the sets are not disjoint. See the exercises.

We give two proofs that are essentially the same idea, one entirely using the language of bijections and one using the language of enumerations.

First Proof: Because A and B are countably infinite, there are bijections $f : A \to \mathbb{N}$ and $g : B \to \mathbb{N}$. Define $h : A \cup B \to \mathbb{N}$ by

$$h(x) = \begin{cases} 2f(x) - 1 & x \in A \\ 2g(x) & x \in B. \end{cases}$$

Because A and B are disjoint, h returns a single output for each x in $A \cup B$ and so is indeed a function. We must show that h is a bijection.

Suppose that $h(x_1) = h(x_2)$. This number is either odd or even. If it is odd, then both x_1 and x_2 are in A and

$$2f(x_1) - 1 = h(x_1) = h(x_2) = 2f(x_2) - 1.$$

Thus $f(x_1) = f(x_2)$, and because f is injective, $x_1 = x_2$. A similar argument treats the case in which $h(x_1) = h(x_2)$ is even. Thus h is injective.

For surjectivity, let $n \in \mathbb{N}$. Suppose first that n is odd. Thus $n = 2k + 1$ for some integer $k \geq 0$. Equivalently, $n = 2(k + 1) - 1$ for $k + 1 \geq 1$. Because f is surjective, there exists $x \in A$ such that $f(x) = k + 1$ and so

$$h(x) = 2f(x) - 1 = 2(k + 1) - 1 = n.$$

A similar argument deals with the case in which n is even. Thus h is surjective.

We have shown that h is a bijection and so $A \cup B$ is countably infinite. □

Second Proof: Because A and B are countably infinite, each has an enumeration. Thus let $a_1, a_2, \ldots$ be an enumeration of A and let $b_1, b_2, \ldots$ be an enumeration of B. Because A and B are disjoint, all these elements are distinct. We would like our enumeration of $A \cup B$ to be

$$a_1, b_1, a_2, b_2, \ldots.$$

To be more explicit, letting c_n denote the n-th element in the enumeration of $A \cup B$, we have the rule

$$c_n = \begin{cases} a_{\frac{n+1}{2}} & n \text{ odd} \\ b_{\frac{n}{2}} & n \text{ even.} \end{cases}$$

□

These exercises lead you to the stronger version of the proposition that does not require A and B to be disjoint.

Exercise 4.4.5. Suppose A is finite and B is countably infinite. Suppose that A and B are disjoint. Show that $A \cup B$ is countalby infinite.

Exercise 4.4.6. Let A and B be countably infinite. Prove that $A \cup B$ is countably infinite. (Hint: Write $A \cup B = A \cup (B \setminus A)$. Can you be sure that $B \setminus A$ is countably infinite?)

With the concept of enumeration and with Proposition 4.4.5, you can prove the following useful proposition. In simple language, it says that there is no "size" for infinite sets smaller than "countably infinite".

Proposition 4.4.8. *Let A be a countably infinite set. If $B \subseteq A$ and if B is infinite, then B is countably infinite.*

Exercise 4.4.7. Prove Proposition 4.4.8.

We can also use enumerations to prove that the Cartesian product of two countably infinite sets is countably infinite following closely our proof that $\mathbb{N} \times \mathbb{N}$ is countably infinite.

Proposition 4.4.9. *Let A and B be countably infinite sets. Then $A \times B$ is countably infinite.*

Proof. Because A and B are countably infinite, there is an enumeration $a_1, a_2, \ldots$ of A and an enumeration $b_1, b_2, \ldots$ of B. W construct an injection from $A \times B$ to $\mathbb{N}$ as follows: Let $(x, y) \in A \times B$ so that $x \in A$ and $y \in B$. Because x is in A and $a_1, a_2, \ldots$ is an enumeration of A, there exists a unique $j \in \mathbb{N}$ such that $x = a_j$. Similarly, there exists a unique $k \in \mathbb{N}$ such that $y = b_k$. Define $f : A \times B \to \mathbb{N}$ by

$$f(a_j, b_k) = 2^j 3^k.$$

The same argument as above shows that this function is injective and so, by Proposition 4.4.5, $A \times B$ is countably infinite. □

Our next situation initially sounds different from the above, but we will see that the argument is startlingly familiar.

Proposition 4.4.10. *Let $\{A_j : j \in \mathbb{N}\}$ be a countably infinite collection of sets in some universe U. Suppose that each A_j is countably infinite and that the sets are pairwise disjoint, so that if $j \neq k$, $A_j \cap A_k = \emptyset$. Then $\bigcup_{j=1}^{\infty} A_j$ is countably infinite.*

This proposition might seem scary. The point of the proof that follows is that, if we set up good notation, it will follow quickly in a manner similar to previous proofs.

Proof. Because A_j is countably infinite, there is an enumeration of its elements, $a_{j,1}, a_{j,2}, \ldots, a_{j,k}, \ldots$. Consider the set $S = \{a_{j,k} : j, k \in \mathbb{N}\}$. Every element of any one of the A_j is in this set, and all the $a_{j,k}$ are distinct because the sets are pairwise disjoint. S is therefore the union of the A_j, and the function $f : S \to \mathbb{N}$ given by

$$f(a_{j,k}) = 2^j 3^k$$

is an injection. Thus $S = \bigcup_{j=1}^{\infty} A_j$ is countably infinite. □

As lovely as this proposition is, we would again like to relax some of the hypotheses. We would like it to apply to an arbitrary finite or countably infinite collection of finite or countably infinite sets, and we would like it not to require the sets to be disjoint. The next exercise guides you through these modifications. First, we make a definition.

Definition 4.4.11. Set A is **countable** if it is either finite or countably infinite.

Don't get tripped up on the definition. It is easy to forget that "countable" and "countably infinite" are *not* synonyms.

Exercise 4.4.8. There are two ways we would like to generalize Proposition 4.4.10. The first way allows either the sets themselves or the collection to be countable rather than just countably infinite. In this exercise, suppose $\{A_j : j \in \mathbb{N}\}$ is a collection of pairwise disjoint, non-empty, countable sets.

(a) If each set is finite, explain why a finite union $\bigcup_{j=1}^{n} A_j$ is finite. (You do not need to give a new proof. It is a fact we already know, by a different name.)

(b) Prove that if at least one of the sets involved is countably infinite, then $\bigcup_{j=1}^{n} A_j$ is countably infinite.

(c) Prove that $\bigcup_{j=1}^{\infty} A_j$ is countably infinite.

As a consequence of the previous exercise, we know that *if the sets are pairwise disjoint*, the countable untion of countable sets is countable.

Exercise 4.4.9. The next way we could generalize the result is to get rid of the hypothesis that the sets are pairwise disjoint. Let $\{A_j\}$ be a countable collection of countable, non-empty sets. Define a new collection of sets that *are* pairwise disjoint but have the same union. Explain how doing so leads to a proof that the countable union of countable sets is countable, even if the sets are not pairwise disjoint.

For ease of reference, we summarize the previous exercises in the following proposition:

Proposition 4.4.12. *A countable union of countable sets is countable.*

We end this section by proving one more counter-intuitive result. Recall that $\mathbb{Q}$, the set of rational numbers, consists of ratios $\frac{n}{m}$ of integers in which $m \neq 0$. As we have discussed previously, one thing that makes working with $\mathbb{Q}$ a challenge is that any given rational number can be represented in infinitely many different ways as a ratio of integers. Our goal now is to determine the cardinality of $\mathbb{Q}$. In our argument, we will want to be careful about this point.

Proposition 4.4.13. $\mathbb{Q}$ *is countably infinite.*

Does this proposition seem surprising? It might. It may have already stretched your thinking when we proved that $\mathbb{Z}$ is countably infinite because it almost seems as if there are "twice as many" integers as natural numbers. Proposition 4.4.13 may seem even more extreme; you probably visualize $\mathbb{Z}$ as a set of discrete dots spaced at unit intervals along the real line. If you know anything about rational numbers from calculus, you might have heard that every open interval, no matter how small, contains infinitely many rational numbers! This fact is, in itself, mind boggling. Still the proposition is true, as we now prove.

Proof. For each rational number x, we can express x uniquely as a ration $\frac{n}{m}$ of integers in which $m > 0$ and n and m have no common factors. We say such a fraction is in *lowest terms*. We define $f : \mathbb{Q} \to \mathbb{Z} \times \mathbb{N}$ by the rule $f(x) = (n, m)$ where n and m are the numerator and denominator in this canonical representation. The function f is injective, for if $f(x_1) = f(x_2)$, then this image is precisely the same pair (n, m), meaning that $x_1 = \frac{n}{m} = x_2$. Now, $f(\mathbb{Q})$ is an infinite subset of $\mathbb{Z} \times \mathbb{N}$ because it contains all pairs of the form $(n, 1)$, which is clearly in one-to-one correspondence with the integers. Because $\mathbb{Z} \times \mathbb{N}$ is the Cartesian product of countably infinite sets, it is countably infinite. Thus $f(\mathbb{Q})$ is countably infinite. Finally, because f defines a bijection from $\mathbb{Q}$ to $f(\mathbb{Q})$, $\mathbb{Q}$ is itself countably infinite. □

4.4.3 Uncountable Sets

We have now thoroughly studied one flavor of infinite sets – the countably infinite ones. We saw that many sets that we think of as very large (like the rational numbers) turn out to be countably infinite. This counter-intuitive result might make us wonder if there is really, then, only a single size of infinite set. The answer is no. Notably absent from the previous section were intervals of the real line. In this section, we start by exhibiting an example of a set this is infinite but not of the same cardinality as $\mathbb{N}$. We then prove some consequences.

Proposition 4.4.14. *Let S be the set of all infinite sequences of zeros and ones. Then S is not countably infinite.*

Although the proof involves a clever idea at some point, you should be able to decide on a proof technique. Think about this question for a minute or two before reading on.

In order to show that any infinite set S is not countably infinite, we have to argue that there is *not* a bijection from S to $\mathbb{N}$. Think about other situations in which we wanted to show that something is impossible – say, that it is impossible to write $\sqrt{2}$ as a ratio of integers or that it is not possible to find a largest prime or a smallest positive real number. These were classic proof by contradiction situations. Often the best way to show that something is impossible is to suppose it *is* and show that this assumption leads to a contradiction. That will be our strategy here.

Proof. Assume, by way of contradiction, that S is countably infinite. Thus, there exists an enumeration $s_1, s_2, \ldots$ of S. Our goal is to obtain a contradiction by showing that some element of S is not, in fact, on this list, meaning that we did not have an enumeration.

Now, each one of the s_j in the enumeration is a sequence of zeros and ones. Letting $s_{j,k}$ denote the k-th term in the sequence s_j, we can display our enumeration as follow:

$$\begin{aligned} s_1 &= \{s_{1,1}, s_{1,2}, s_{1,3}, \ldots\} \\ s_2 &= \{s_{2,1}, s_{2,2}, s_{2,3}, \ldots\} \\ s_3 &= \{s_{3,1}, s_{3,2}, s_{3,3}, \ldots\} \\ &\vdots \\ s_j &= \{s_{j,1}, s_{j,2}, s_{j,3}, \ldots\} \\ &\vdots \end{aligned}$$

We construct a sequence $t = \{t_1, t_2, t_3, \ldots\}$ that is not on the list by making sure it differs from sequence s_j in entry j. Thus we define

$$t_j = \begin{cases} 1 & \text{if } s_{j,j} = 0 \\ 0 & \text{if } s_{j,j} = 1. \end{cases}$$

Because for all j, the sequence t is different from the sequence s_j in entry j, sequence t is not on our list, contradicting the supposition that our list was in fact an enumeration of S. Thus no such enumeration exists, and S is not countably infinite. □

This method of listing the infinitely many sequences and then making a new sequence that disagrees with sequence s_j in entry j is called the *Cantor diagonalization argument* because we're obtaining t by switching the diagonal elements $s_{j,j}$ of our array.

Inspired by the example, we make a definition:

Definition 4.4.15. A set A is **uncountable** if it is infinite but not countably infinite.

This definition makes it sound like there are two flavors of infinite sets – the countably infinite ones and all the others. Although it is true that uncoutable sets are the infinite ones that do not have the same cardinality as $\mathbb{N}$, there is no reason to think they all have the same cardinality (and, in fact, they do not). We could ask many more questions about cardinalities of infinite sets. How many different classes of infinite sets are there? Once we

have an infinite set with a certain identified cardinality, is there a systematic way to make one with a larger cardinality? We don't address most of these questions here. We simply pose a couple of exercises that allow you to practice with the style of argument above and then let you explore what happens when we do set operations with countable and uncountable sets. The problems give you an opportunity to explore a little more.

Exercise 4.4.10. Let S be the set of all infinite strings of letters from the 26-letter English alphabet. Prove that S is uncountable.

Exercise 4.4.11. Let A be countably infinite and let B be uncountable.

(a) What can you say about $A \cap B$? Prove your claim.

(b) What can you say about $B \setminus A$? Prove your claim.

(c) What can you say about $A \cup B$? Prove your claim.

4.5 Problems

1. For a natural number n, let $n\mathbb{Z}$ denote the set of all integers that are multiples of n. (So $2\mathbb{Z}$ is the set of even numbers, $3\mathbb{Z}$ is the set of all multiples of 3, and so on.)

   (a) What is the relationship between $3\mathbb{Z}$ and $6\mathbb{Z}$? Prove your claim.

   (b) Is $4\mathbb{Z} \subseteq 5\mathbb{Z}$? Is $5\mathbb{Z} \subseteq 4\mathbb{Z}$?

   (c) Describe in words the elements of the set $3\mathbb{Z} \setminus 2\mathbb{Z}$.

   (d) Let $S = \{6k+3 : k \in \mathbb{Z}\}$. What is the relationship between S and $3\mathbb{Z} \setminus 2\mathbb{Z}$? Prove your assertion.

2. There are many identities relating various combinations of sets. Not surprisingly, one standard way to prove them is by proving two containments. Let A, B, and C be sets.

   (a) Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

   (b) Find and prove a similar identity involving $A \cup (B \cap C)$.

3. Let a and b be real constants. Consider the function $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = ax + b$. For which choices of a and b is f injective? Surjective? Bijective?

4. Compare this problem with the previous problem. It illustrates how much the choice of domain and codomain matters when discussing functions.

   (a) There are many choices of real constants a and b for which the rule $g(n) = an + b$ doe *not* define a function from $\mathbb{Z}$ to $\mathbb{Z}$. Give two specific examples and explain carefully what goes wrong.

   (b) Now fix *integers* a and b. Because sums and products of integers are integers, now the rule $g(n) = an + b$ *does* define a function from $\mathbb{Z}$ to $\mathbb{Z}$. For which integers a and b is g injective? Surjective? Bijective?

5. Any quadratic function $q\colon \mathbb{R} \to \mathbb{R}$ can be written in the form

$$q(x) = a(x-h)^2 + k$$

for real numbers a, h, and k with $a \neq 0$. Prove that q is not injective and not surjective.

The next few problems ask you to consider a union or intersection of an indexed collection of intervals of real numbers. For these problems, you may assume several facts about $\mathbb{R}$. First, for every real number $x > 0$, there exists a natural number n such that $n > x$. Second, for every real number $x > 0$, there exists a natural number n such that $\frac{1}{n} < x$. These two statements are actually equivalent and are called the *Axiom of Archimedes*. We do not prove these statements. However, cite them when you use them.

6. Prove that $\bigcup_{k=1}^{\infty}(-k,k) = \mathbb{R}$.

7. Express $\bigcap_{k=1}^{\infty}(-k,k)$ in a simple form and prove your claim.

8. Express $\bigcap_{k=1}^{\infty}\left(-\frac{1}{k},\frac{1}{k}\right)$ in a simple form and prove your claim.

9. For each $k \in \mathbb{N}$, let $A_k = \left[\frac{1}{k}, 1+\frac{1}{k}\right]$.

   (a) By hand, find A_k for $k = 1, 2, 3$. Using this, explicitly describe each of

$$\bigcup_{k=1}^{3} A_k \text{ and } \bigcap_{k=1}^{3} A_k$$

   as an interval or union of intervals.

   (b) We claim that

$$\bigcup_{k=1}^{\infty} A_k = (0,2].$$

   Prove this set equality. (Can you explain why even though we are taking the union of closed intervals, the result is a half-open interval?)

10. Let $f : A \to B$ be a function. It is important to understand how images and pre-images interact with unions and intersections. Let C_1 and C_2 be subesets of A and let D_1 and D_2 be subsets of B.

    (a) Prove that $f(C_1 \cup C_2) = f(C_1) \cup f(C_2)$.

    (b) Prove that $f^{-1}(D_1 \cap D_2) = f^{-1}(D_1) \cap f^{-1}(D_2)$.

    (c) Prove that $f^{-1}(D_1 \cup D_2) = f^{-1}(D_1) \cup f^{-1}(D_2)$.

    (d) How are $f(C_1 \cap C_2)$ and $f(C_1) \cap f(C_2)$ related? Prove a containment and illustrate with an example that we need not have equality.

11. At a sandwich shop, customers have four options of bread (white, wheat, rye, sourdough), three options of meat (turkey, ham, beef), three options of cheese (cheddar, Swiss, American), and five options of vegetable (lettuce, tomato, onion, pepper, mushroom). Find the number of distinct sandwiches possible if

(a) customers must select exactly one item from each category;

(b) customers must select one bread option, and may select either zero or one items from the other three categories;

(c) customers must select one bread option and one cheese option, and may select either one meat option or one vegetable option (but cannot have both a meat and vegetable);

(d) customers must select one bread option and one cheese option, and may select either one meat option or one vegetable option (with the potential for having both a meat and a vegetable).

12. Consider a set of n people (labeled by the numbers $1, 2, \ldots, n$). Some of these people choose to shake hands with some others. Let $h(i)$ be the number of handshakes that person i participates in, and let H be the total number of handshakes which take place. Show by counting two ways that

$$\sum_{i=1}^{n} h(i) = 2H.$$

13. Let $k \leq n$ be positive integers. A k-**composition** of n is a k-tuple whose entries are positive integers which add up to n. For example, $(3, 1, 1), (1, 1, 3)$, and $(2, 2, 1)$ are all 3-compositions of 5.

(a) Find all 3-compositions of 5 by hand.

(b) The number of 3-compositions of 5 is equal to $\binom{4}{2}$. Can you explain why? (Hint: visually represent the number 5 as a row of 5 objects. How can you represent "breaking these up" into a 3-composition?)

(c) Find and prove a general formula for the number of k-compositions of n.

14. At a dinner party, 10 guests will be seated around a circular table. The chairs of this table are labeled 1 through 10, so all seats are distinguishable. Of the 10 guests, 5 belong to the same book club and the other 5 are enrolled in the same combinatorics class. To ensure that everyone gets to know each other, the host of the party wants to assign seats so that members of the book club are not seated next to one another, and members of the combinatorics class are not seated next to one another. How many such seating assignments are possible?

15. In the game of bridge, a full deck of 52 cards is dealt to four players (who are labeled as North, South, East, and West) so that each player receives exactly 13 cards. How many distinct bridge deals are possible? (The order that a player receives cards in does not meaningfully change the deal, so do not take this order into account. However, the player to which each set of 13 cards is assigned *does* matter, so you should factor this in.)

16. Let n be an odd positive integer. Let E_n be the subsets of $\{1, 2, \ldots, n\}$ with even size, and O_n the subsets of $\{1, 2, \ldots, n\}$ with odd size. Conjecture and prove a relationship between $|E_n|$ and $|O_n|$.

The next few problems concern the *power set* os a set A. If A is any set, its *power set*, denoted $\mathcal{P}(A)$, is the set of all subsets of A. (Recall that $\emptyset$ is considered to be a subset of every set.)

17. Let A be a finite set.

    (a) If $|A| = 0, 1, 2, 3, 4$, determine $|\mathcal{P}(A)|$. (Suggestion: The names of the elements in A don't matter, so for definiteness, you could suppose $A = \{1, 2, \ldots, n\}$ for various choices of n and systematically list all the subsets. The purpose of this question is to explore and to help you make a conjecture. below.)

    (b) Suppose $|A| = n$ for some non-negative integer n. Show that there is a bijection between $\mathcal{P}(A)$ and length n binary strings.

    (c) Suppose $|A| = n$ for some non-negative integer n. Make and prove a conjecture about $|\mathcal{P}(A)|$.

18. Show that there is a bijection between $\mathcal{P}(\mathbb{N})$ and infinite sequences of zeros and ones. What does this say about the cardinality of $\mathcal{P}(\mathbb{N})$? (Comment: This problem illustrates a more general phenomenon that the power set of a set and the set itself never have the same cardinality. This fact gives a nice way of obtaining sets of different cardinalities.)

4.6 Suggested Projects

4.6.1 Binomial Coefficients, Part II

(1◷)

4.6.2 The Poisoned Rum Puzzle

(2◷) A pirate captain has ten excellent bottles of rum in his cabin. This pirate captain has many enemies who wish him dead, one of whom sneaks into the cabin and poisons exactly one of the ten bottles. The poison used is undetectable, except by drinking it and subsequently dying. It has no immediate effect when it is consumed, but anyone who drinks even a little will die within one day. There is no way to dilute the poison to a safe level or otherwise make a poisoned bottle safe to drink from.

The pirate captain's parrot saw everything and alerts him to the situation, but the parrot does not remember which bottle was poisoned[34]. The captain realizes that the only way to test a bottle for poison is to give a small sip to someone and wait a day to see if they expire. In shortly more than one day, he is holding a party for other pirate captains at which he must serve the other nine bottles of rum, so he only has one day in which to test the rum. The captain looks around for test subjects, and can only find four rats to use.

[34]Well, you can't expect a parrot to do everything for you.

Find a strategy by which the captain can determine, using only four rats and one day, which of the ten bottles is poisoned. (Note that you can find a successful strategy which does not depend on some clever timing of the administration of the doses. We promise the poison will take effect *within* one day, but imagine the timing of its impact is somewhat variable. So you cannot infer based on the moment of death exactly when the poison was consumed.) Write a careful description of this strategy and a discussion which includes

1. A proof that your strategy will always succeed in identifying the poisoned bottle.
2. A discussion of the dependence between the number of bottles and number of rats. What is the largest number of bottles that you could differentiate between if you have four rats and only one day to test the bottles?
3. A discussion of the same problem performed with n bottles of rum for some positive integer n. Can you adapt your strategy to this general situation? How many rats would you need to find one poisoned bottle among n bottles of rum in just one day?

For a longer project, after you complete the analysis above, consider this related question. A different pirate captain has a very similar problem. He has 200 bottles of excellent rum, and knows that exactly one is poisoned using this same undetectable but deadly substance. This captain is having a very large party in shortly more than two days, and needs to determine which 199 bottles are safe to drink. The captain has only 5 rats in total to test the bottles on.

Explain why it is impossible to determine in one day only which bottle is poisoned under these conditions. Find and describe a strategy which *will* determine the poisoned bottle within two days. (Again, this strategy should not rely on some clever timing of administration of doses.) Prove that your strategy always works. Is your strategy the same as the one day strategy? If so, why does the same strategy work in both situations? If not, how do they differ?

4.6.3 Intersecting Sets

$(1-2\clock)$

4.6.4 Catalan Bijections

$(1-3\clock)$

4.6.5 The Principle of Inclusion-Exclusion

$(2-3\clock)$

4.6.6 The Size of the Cantor Set

$(2\clock)$

Chapter 5

Relations, Equivalence, and Modular Arithmetic

A big part of what we do in math is to determine how various mathematical objects are related. In arithmetic, we compare sizes of numbers. In algebra and trigonometry, we determine whether two expressions or equations are equivalent. In set theory, we determine whether one set is contained in another or whether they are equal. We define functions between sets, which are special ways of associating an element of one set with an element of another. In combinatorics, we compare cardinalities. The list goes on.

In this chapter, we abstract these ideas. We first talk about general relations between two sets and discuss the kinds of properties a relation can have. There are many ways to take two sets and define a rule by which some elements of one are "similar to" or can be "compared to" some elements of another. While the notion of relations will capture a very wide range of examples, we will discover certain properties that are nice to have in a relation; these will abstract the ways in which well-known relations, like equality and inequality of real numbers, behave. Once we understand how to describe the "important" features of a relationship like equality in an abstract way, we will be able to define special kinds of relations that mimic familiar ways of comparing objects.

In particular, we will be able to define and study in detail a very special kind of relation called an *equivalence relation*. As the name suggests, equivalence relations somehow describe when two objects are "equivalent" with respect to some property. You might immediately think of algebraic equality – which, perhaps, does not sound all that exciting. But as we shall see, there are many ways in which it might make sense to consider two objects meaningfully "the same", even if they are not actually equal in every respect. The notion of equivalence is in fact ubiquitous in mathematics, and we will show a variety of examples to give you a sense of how broadly the ideas of this chapter can be applied. However, we have one special kind of equivalence relation we study in detail, called congruence modulo n.

With congruence modulo n, we consider two integers to be "equivalent" if they have the same remainder when divided by n. While this might sound like a somewhat arbitrary rule at first, there are in fact some natural situations where you think about congruence modulo n already, though not in these words. For example, you might consider every January 1st or December 25th to be "equivalent" in the sense that these are dates which fall at the same place in the calendar and (if you celebrate New Year's Day and Christmas) likely involve

similar activities every time they arrive. When thinking about the calendar in this "cyclic" way, we are really thinking about congruence modulo 365.[1] Similarly, if you wake up at 7 am every day, then you in some sense consider many different times to be "equivalent" modulo 24. Pretty much anything that has a regular, cyclical nature like this could be modeled by thinking about congruence modulo n for some n. Thus, when we discuss congruence modulo n, we hope that you will have some existing intuition and think back to these examples for initial motivation. However, as we shall see, the idea of congruence modulo n is actually a lot deeper than just "clock arithmetic" and the like. In fact, studying this topic in detail is a great way to prepare for future classes in number theory and abstract algebra. There are plenty of other places in higher mathematics where congruence modulo n arises as well; for instance, many methods in cryptography rely on using this notion of equivalence in clever ways.

5.1 Relations

5.1.1 Basic Definition

The definition of a relation is extremely general:

Definition 5.1.1. Let A and B be non-empty sets. A **relation** is a subset R of $A \times B$. We write either $(a, b) \in R$ or aRb if (a, b) is in the relation. If $A = B$, we call R a **relation on** A.

The notation aRb most likely looks strange to you. But R is just the name of this general relation. In specific contexts, we use other symbols, many of which will look more familiar to you.

It's hard to form much intuition from such a general definition so we consider a variety of examples.

Example 5.1.2. Let $A = \{$Jen, Anna, Phoebe$\}$ and let $B = \{$c,r, g,p$\}$ (where the letters stand for the names of the foods chocolate, raspberries, goat cheese, and pickles.) We define a relation

$$R = \{\text{(Jen, c), (Jen, r), (Anna, g), (Phoebe, c), (Phoebe, p)}\}.$$

As it stands, this set is indeed a relation between A and B – it is a subset of their Cartesian product, and this is a valid way to describe it.

We could describe it in other ways, especially if we had some idea what the underlying rule is. It turns out that this relation is about the food preferences of three specific people. We could write it as

$$R = \{(x, y) \in A \times B : \text{Person } x \text{ likes food } y\}.$$

So, Jen likes chocolate and raspberries, Anna likes goat cheese, and Phoebe likes chocolate and pickles.

[1] Ignoring leap years.

Example 5.1.3. Our next example will be mathematical. Let $A = B = \mathbb{N}$ and define a relation $<$ on $\mathbb{N}$ by $n < m$ is n is strictly less than m. We could write this relation in set notation as

$$<= \{(n, m) \in \mathbb{N} \times \mathbb{N} : n \text{ is less than } m\},$$

but whereas the first notation seems standard, this set notation seems strange; you don't think of $<$ as the name of a set, you actually already think of it as a relation between objects.

Example 5.1.4. Let $\mathcal{P}(A)$ be the set of all subsets of some fixed set A. (This object is called the *power set* of A and was introduced in the Problems at the end of Chapter 4.) Then $\subseteq$ is a relation on $\mathcal{P}(A)$, with $S \subseteq T$ if every element of S is also an element of T. This relation is also very familiar to you.

Example 5.1.5. Let $A = B = \mathbb{Z}$. Fix a natural number $n \geq 2$ and define a relation $\equiv$ on $\mathbb{Z}$ by $j \equiv k$ if $j - k$ is a multiple of n. For example, if $n = 3$, $7 \equiv 1$ because $7 - 1 = 3 \cdot 2$ and $11 \equiv -1$ because $11 - (-1) = 12 = 3 \cdot 4$.

These examples show how varied relations can be. Without any more definitions, there isn't much more we could say about relations. But if you think about the examples, you will notice that some of the relations have special properties not possessed by others. For instance, in Example 5.1.4, a set S is "related to" itself, that is, $S \subseteq S$, but in Example 5.1.3, there is no natural number for which $n < n$ is true.

We now define a variety of properties a relation could satisfy. Later, we will see that various combinations of them come up often enough that we give them a special name.

Definition 5.1.6. Let R be a relation on a non-empty set A.

- R is **reflexive** if for all $a \in A$, aRa.
- R is **symmetric** if aRb implies bRa.
- R is **transitive** if aRb and bRc implies aRc.
- R is **anti-symmetric** if aRb and bRa implies $a = b$.

We continue two of our examples above. For Example 5.1.4, we see that the relation is reflexive because for all $S \in \mathcal{P}(A)$, $S \subseteq S$. The relation is anti-symmetric, for we have already discussed extensively that $S \subseteq T$ and $T \subseteq S$ imply that $S = T$. Finally, the relation is transitive, for if $S \subseteq T$ and $T \subseteq U$, then every element of S is in T, hence in U, meaning that $S \subseteq U$. A relation that is reflexive, anti-symmetric, and transitive is called a *partial order*.

For Example 5.1.5, let's focus again specifically on $n = 3$ (other cases are similar). The relation is reflexive, because for all integers j, $j - j = 0$ is a multiple of 3, meaning that $j \equiv j$. The relation is symmetric, because if $j \equiv k$, there is an integer d for which $j - k = 3d$. Then $k - j = 3(-d)$, and so $k \equiv j$. Finally, the relation is transitive. Indeed, if $j \equiv k$ and $k \equiv \ell$, then there are integers d_1, d_2 such that $j - k = 3d_1$ and $k - \ell = 3d_2$.. Then

$$j - \ell = (j - k) + (k - \ell) = 3d_1 + 3d_2 = 3(d_1 + d_2),$$

and $j \equiv \ell$. A relation that is reflexive, symmetric, and transitive is called an *equivalence relation*. Such relations have an entire section devoted to them.

The next few exercises are simply designed to give you the opportunity to explore different relations and their properties. Later sections will develop these ideas much more systematically.

Exercise 5.1.1. Consider the relation in Example 5.1.3. Which, if any, of the four properties above does this relation satisfy?

Exercise 5.1.2. Let $A = B = \mathbb{R}$ and consider the usual relation $\leq$ on $\mathbb{R}$. Which of the above properties does this relation satisfy?

Exercise 5.1.3. Let P be the set of all people, living or deceased. Define a relation $\sim$ on P by $p \sim q$ if p is a direct-line ancestor of q. What properties does the relation satisfy?

5.1.2 Functions as Relations

In Chapter 4, we studied functions extensively. At that time, we defined a function to be a rule that assigns to each element of a set A exactly one element of a set B. We can now give a defferent definition in terms of relations.

Definition 5.1.7. Let A and B be non-empty sets. A function $f : A \to B$ is a relation containing exactly one ordered pair (a, b) for each element a of A. When f is a function, instead of writing afb, we write $f(a) = b$. We call A the **domain** and B the **codomain**.

Is this definition the same as our previous one? Yes! They use different words, but both involve two sets (a domain A and a codomain B) and both require that each a in the domain is "paired" with exactly one element of the codomain.

Example 5.1.8. Let's return to some sets we discussed in Chapter 4. Let

$$A = \{\text{Jen, Anna, Phoebe}\}$$
$$B = \{\text{red, orange, yellow, green, blue, purple}\}.$$

We defined a function $C : A \to B$ giving each person's favorite color. In our new notation:

$$C = \{\text{(Jen, purple), (Anna, red), (Phoebe, blue)}\}.$$

As we see, there is exactly one ordered pair in the set for each element of the domain. On the other hand,

$$S = \{\text{(Jen, red), (Anna, blue), (Phoebe, orange), (Phoebe, yellow), (Phoebe, purple)}\}$$

is not a function because there are three elements of the set with first coordinate "Phoebe". Of course S is a relation between A and B. It doesn't matter, but the rule here is that the two words in the ordered pairs have the same number of letters.

Do we gain anything new by thinking of functions as a special kind of relation? Let's look at one more example.

Example 5.1.9. Consider

$$R = \{(x, y) \in \mathbb{R}^2 : x = y^2\}.$$

R is a perfectly good relation on $\mathbb{R}$, being a subset of $\mathbb{R}^2$. It is not a function because $(4, 2)$ and $(4, -2)$ are both in R. If you visualize it as a curve in the plane, it is a sideways-opening parabola that does not pass the vertical line test and so does not define y as a function of x. If, however, we were to swap the variables, we would have a new relation

$$R' = \{(y, x) \in \mathbb{R}^2 : x = y^2\}.$$

This new relation R' is a function because every first coordinate y in $\mathbb{R}$ is paired with exactly one real number, namely y^2. Even though one is a function and one is not, the two sets are clearly very closely related.Many kinds of curves in the plane are defined by relations like these that may or may not be functions. Regardless, they are objects of study in geometry and analysis.

Exercise 5.1.4. Thinking of a function as a set of ordered pairs in which every element of the domain appears exactly once, describe carefully the additional properties of this set if (a) the function is injective and (b) the function is surjective.

5.2 Equivalence Relations

In the previous section, we not only saw many examples of familiar relations, but we also explored various properties these relations satisfy. We saw that several combinations of properties occur in multiple examples, and so we gave those special kinds of relations their own names. Here, we see the real power of abstract mathematics – instead of studying each specific example in isolation, trying to understand how it works, if we form a larger class to which it belongs, we can study that class, in effect studying all examples known, unknown, and not-yet-created at once.

In this section, we explore one of these classes in detail – the equivalence relation. Although we already gave a definition, we state it again here formally. We'll also pick a standard notation to use whenever we are talking about a general equivalence relation.

Definition 5.2.1. A relation $\sim$ on a non-empty set A is an **equivalence relation** if it is reflexive, symmetric, and transitive.

Equivalence relations abstract the concept of "equality". Familiar examples include $=$ on any set and $\cong$ for geometric objects like angles, triangles, and so forth. We also saw that congruence modulo 3 is an equivalence relation. In the next subsection, we work through several more involved examples, some familiar, some not.

Equivalence relations are significant because (as the name suggests), when we have two elements of a set that are equivalent, it often means they function the same way or are in some sense "interchangeable". We explore this idea in great detail in our section on equivalence classes.

5.2.1 Examples

Our first example is not surprising but will be significant throughout this chapter. It is so important that it deserves its own formal definition.

Definition 5.2.2. Let $n \in \mathbb{N}$ with $n \geq 2$. We define a relation on $\mathbb{Z}$ called **congruence modulo** n by $j \equiv k \mod n$ if $j - k = nd$ for some integer d.

We showed above that congruence modulo 3 is an equivalence relation. You can follow an almost identical proof to show that, in fact, we have an equivalence relation for any $n \geq 2$.

Exercise 5.2.1. Let $n \in \mathbb{N}$ with $n \geq 2$. Prove that congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Congruence may be a new equivalence relation to you, but there is propbably at least one place in elementary school math where you heard the word "equivalent" a lot, and that is in the context of fractions. It is familiar to you by now that every rational number can be represented by (infinitely) many equivalent fractions. Is it just coincidental that we use the word "equivalent" here, or is there an underlying equivalence relation? It is the latter, and the next example begins our formalization of our discussion about what rational numbers really are.

Example 5.2.3. We won't be working with fractions. Instead, we will be working with pairs of integers. More formally, let $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$. So the elements of A are ordered pairs (a, b) of integers in which the second coordinate is not zero. (Do you see the connection yet with fractions?) We define a relation $\sim$ on A by

$$(a, b) \sim (c, d) \text{ if and only if } ad = cb.$$

(Think about "cross multiplying.") We claim that $\sim$ is an equivalence relation on A.

Proof. To check reflexivity, consider $(a, b) \in A$. Because $ab = ab$, $(a, b) \sim (a, b)$ and hence $\sim$ is reflexive.

To check symmetry, suppose $(a, b) \sim (c, d)$. Thus, by definition, $ad = cb$. Now, because equality of integers is a symmetric relation, $cb = ad$ and so by definition $(c, d) \sim (a, b)$. Thus the relation is symmetric.

To check transitivity, suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then, by definition, $ad = cb$ and $cf = ed$. Then,

$$adcf = cbed.$$

Because $d \neq 0$, we have (rearranging letters a bit)

$$caf = ceb.$$

If $c \neq 0$, we immediately find $af = eb$, If $c = 0$, then necessarily $a = e = 0$, and again $af = eb$. Thus $(a, b) \sim (e, f)$, proving that the relation is transitive. □

Exercise 5.2.2. The purpose of this exercise is to help you bridge the gap between the previous example and the way you are used to thinking about fractions.

(a) In the usual notation of fractions, list three fractions that are equivalent to $\frac{1}{2}$. Then Write down the general form of a fraction equivalent to $\frac{1}{2}$ with numerator a.

(b) Now, in the notation of the equivalence relation above, write down all pairs with $(a, b) \sim (1, 2)$.

We now turn to an example that is likely new to you. It is a foundational example from the field of *graph theory*, which is the abstract mathematical study of "network-like" structures. As well as forming a major area of modern mathematical study, graph theory has applications to many other fields, because many structures in the world are "network-like". Before beginning this example, we will require a few definitions. Fortunately, the objects of interest here can be naturally visualized, which will help you understand how they work.

A *graph* G is an object made of two sets.[2] The first set $V(G)$ is the set of **vertices**. $V(G)$ can consist of any type of elements we desire, and typically we visualize the elements of $V(G)$ as labeled points drawn in the plane. The second set needed to define G is denoted $E(G)$, and called the **edge set** of G. The elements of $E(G)$ are some (perhaps all) unordered pairs $\{x, y\}$ such that $x, y \in V(G)$. We typically visualize an element $\{x, y\} \in E(G)$ as a line or curve in the plane with endpoints x and y. For example, the graph G with

$$V(G) = \{1, 2, 3, 4\}, E(G) = \{\{1, 2\}, \{1, 3\}, \{2, 4\}\}$$

could be visualized as a drawing consisting of four labeled points with three specific pairs of points connected by edges. We illustrate the idea in Figure 5.1

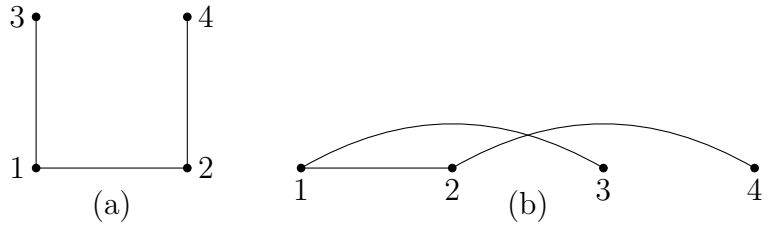


Figure 5.1: Two different drawings of the graph G defined above. Note that while the two look visually different, they contain exactly the same labeled points (although in different locations) and exactly the same set of connections. So we consider both (a) and (b) to represent the same graph G.

Note, as in the caption to Figure 5.1, that there are many different ways to draw the same graph G. This is why we say that the graph G is comprised of its vertex and edge sets, instead of defining G as a type of drawing. Graphs do not have the fixed spatial nature that you might associate with something like a road network – all they record are objects and the connections between objects. Nonetheless, using drawings to help us visualize and understand graphs can be very useful.

Let's define another graph. Say D is a graph with

$$V(D) = \{B, H, L, N\}, E(D) = \{\{B, N\}, \{H, L\}, \{L, N\}\}.$$

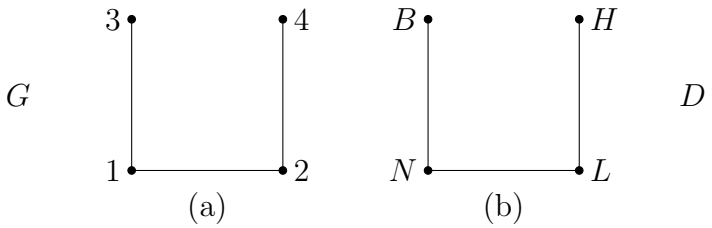


Figure 5.2: In (a), the graph G from Figure 5.1; in (b), the newly defined graph D.

We illustrate D in Figure 5.2

Now, the two graphs depicted in Figure 5.2 are *not* the same. One has vertex set $\{1, 2, 3, 4\}$ and one has vertex set $\{B, H, L, N\}$. Two graphs are only considered equal if their vertex sets are equal and their edge sets are equal. There are sensible reasons to be so specific about this. For instance, suppose the vertices $1, 2, 3, 4$ in G represent four people, and edges represent pairs of people who are friends. Then "mixing up" the labels on the vertices of G would clearly result in different information being conveyed. Completely replacing the labels on G – for instance, by calling the vertices B, H, L, and N – could lose the entire interpretation of the graph. Perhaps the graph D has vertices corresponding to Buffalo, New York City, Los Angeles, and Honolulu, with edges representing pairs of cities connected by direct flights. The graph D would then store completely different information than G.

However, if you look at G and D in Figure 5.2, you likely feel that in *some* sense, these graphs should be considered "the same". We could make the picture of G look exactly like the picture of D if we "re-labeled" its vertices, changing 3 to B, 1 to N, 2 to L, and 4 to H. The edges of G would then also be "re-labeled"; for instance, $\{1, 3\}$ would become $\{B, N\}$. Importantly, when we re-label, the vertices that were connected by an edge in G receive the labels of connected vertices in D, and vertices not connected by an edge in G receive the labels of vertices in D which are also not connected. This re-labeling procedure is a formal way to verify that, while their vertices might be different objects and edges might convey different information about how those vertices are related, the underlying structure of G and D look exactly the same. We have naturally discovered a notion of equivalence between graphs. This notion is, in fact, one of the first foundational ideas you learn when beginning the study of graph theory. We formalize it in the next example.

Example 5.2.4. Given two graphs G and H, we say that G and H are **isomorphic**, and write $G \cong H$, if there exists a bijection $\varphi : V(G) \to V(H)$ with the property that $\{x, y\} \in E(G)$ if and only if $\{\varphi(x), \varphi(y)\} \in E(H)$. Show that graph isomorphism is an equivalence relation.

Proof. We must verify that graph isomorphism is reflexive, symmetric, and transitive. Firstly, we show that it is reflexive. Let G be a graph, and define $\varphi : V(G) \to V(G)$ by $\varphi(x) = x$ for all $x \in V(G)$.[3] Then it is clear that φ is a bijection from$V(G)$ to $V(G)$, and because

[2]Of course, we are talking about a different sense of the word "graph" than you are used to. This notion of the word "graph" has nothing to do with visually representing a function in the Cartesian plane.

[3]That is, φ is the *identity* map on $V(G)$; it maps every element of $V(G)$ to itself without changing anything.

$\{\varphi(x), \varphi(y)\} = \{x, y\}$, it is also immediate that $\{x, y\} \in E(G)$ if and only if $\{\varphi(x), \varphi(y)\} \in E(G)$. Thus, $G \cong G$.

Next, we show that graph isomorphism is symmetric. Suppose G and H are graphs with $G \cong H$, so there exists a bijection $\varphi : V(G) \to V(H)$ with the property that $\{x, y\} \in E(G)$ if and only if $\{\varphi(x), \varphi(y)\} \in E(H)$. We must show that $H \cong G$, so, we must find a bijection from $V(H)$ to $V(G)$ that interacts with $E(H)$ and $E(G)$ in the desired way. Recall, every bijective function has an inverse function and[4] the inverse of a bijection is itself a bijection. So, we consider the bijection $\varphi^{-1} : V(H) \to V(G)$. We claim that $\{u, v\} \in E(H)$ if and only if $\{\varphi^{-1}(u), \varphi^{-1}(v)\} \in E(G)$. Fix distinct vertices $u, v \in V(H)$. Since φ is a bijection from $V(G)$ to $V(H)$, there exist unique, distinct vertices $x, y \in V(G)$ such that $u = \varphi(x)$ and $v = \varphi(y)$. Because $\varphi : V(G) \to V(H)$ satisfies the graph isomorphism property, we know that $\{x, y\} \in E(G)$ if and only if $\{\varphi(x), \varphi(y)\} \in E(H)$. Observe, we have said that $\{\varphi(x), \varphi(y)\}$ is really just $\{u, v\}$, so $\{\varphi^{-1}(u), \varphi^{-1}(v)\} = \{\varphi^{-1}(\varphi(x)), \varphi^{-1}(\varphi(y))\} = \{x, y\}$. Using these equalities to replace $\{x, y\}$ with $\{\varphi^{-1}(u), \varphi^{-1}(v)\}$ and $\{\varphi(x), \varphi(y)\}$ with $\{u, v\}$, the statement

$$\{x, y\} \in E(G) \iff \{\varphi(x), \varphi(y)\} \in E(H)$$

is in fact equivalent to

$$\{u, v\} \in E(H) \iff \{\varphi^{-1}(u), \varphi^{-1}(v)\} \in E(G).$$

Thus, $H \cong G$, so graph isomorphism is symmetric.

Finally, we argue that graph isomorphism is transitive. Suppose G, H, K are graphs such that $G \cong H$ and $H \cong K$. So there exist bijections $\varphi : V(G) \to V(H)$ and $\psi : V(H) \to V(K)$ which have the correct interactions with $E(G), E(H)$, and $E(K)$. We must find an appropriate bijection from $V(G)$ to $V(K)$. The natural thing to do is take the composition $\psi \circ \varphi : V(G) \to V(H)$. We have seen[5] that the composition of bijections is a bijection, so $\psi \circ \varphi$ is a bijection between $V(G)$ and $V(K)$. Now, consider distinct vertices $x, y \in V(G)$. We know (because φ demonstrates a graph isomorphism from G to H) that $\{x, y\} \in E(G)$ if and only if $\{\varphi(x), \varphi(y)\} \in E(H)$. Moreover (because ψ demonstrates a graph isomorphism from H to k), we have $\{\varphi(x), \varphi(y)\} \in E(H)$ if and only if $\{\psi(\varphi(x)), \psi(\varphi(y))\} \in E(K)$. Thus, combining the two statements, we have that $\{x, y\} \in E(G)$ if and only if $\{\psi(\varphi(x)), \psi(\varphi(y))\} \in E(K)$. We conclude that the composition $\psi \circ \varphi$ correctly interacts with $E(G)$ and $E(K)$, so we have established a graph isomorphism $G \cong K$. We conclude that graph isomorphism is transitive. □

This last example might seem a bit heavy on the notation. Indeed, to describe even natural ideas like "relabeling a picture" with mathematical precision, we sometimes need a surprising number of tools! Please note how important it was that we have some basic facts about compositions and inverses ready to hand. While those topics may not have been the most immediately interesting in Chapter 4, they crop up in many places.

[4]See Exercise 4.2.13

[5]See Proposition 4.2.22

5.2.2 Equivalence Classes

If you remember our motivation when we began discussing equivalence relations, the idea is that if we have an equivalence relation $\sim$ on a set A, then it is supposed to identify elements of the set that function equivalently in some manner. If you try this idea out with some of the familiar equivalence relations we have already discussed, this statement should seem accurate. For example, when you work with the geometric equivalence relation of congruence on the set of all triangles, you think of any two congruent triangles as being effectively "the same", with the only difference being exactly where they are situated. All of them have the same three angle measures and the same three side lengths. Similarly, you think of equivalent fractions as functioning equivalently in an arithmetic problem; you know that if you take $\frac{1}{2} \cdot \frac{2}{3}$, you will get the same rational number as if you take $\frac{2}{4} \cdot \frac{4}{6}$. In this case, the actual fractions you get are different, but you know that in both cases, they represent the same rational number.

The purpose of this subsection is to formalize this idea that, when we have an equivalence relation, it "divides up" our set into distinct subsets of elements that all "function" the same way. Some of the notation and terminology will seem abstract, but if you keep some of these familiar examples in mind, it will help.

Definition 5.2.5. Let A be a non-empty set and let $\sim$ be an equivalence relation on A. For $a \in A$, Let

$$[a] = \{x \in A : x \sim a\}.$$

We call $[a]$ the **equivalence class** determined by a and call the elements of $[a]$ **representatives** of the class.

So, in the example of the equivalence relation of congruence on the set of all triangles, $[\Delta ABC]$ consists of all triangles that are congruent to a given triangle ΔABC. So, for example, $[\Delta ABC]$ could consist of all right triangles with side lengths 3, 4, and 5 no matter how they might be situated, but a right triangle with other side lengths would be in a different equivalence class.

The next proposition will seem intuitive given our motivation and discussion, but it still requires a short argument.

Proposition 5.2.6. *Let A be a non-empty set and let $\sim$ be an equivalence relation on A. Then any two equivalence classes $[a]$ and $[b]$ are either disjoint or equal.*

Proof. Let $a, b \in A$ and suppose that $[a] \cap [b]$ is not empty. Thus, there exists $c \in [a] \cap [b]$. By definition, $c \sim a$ and $c \sim b$. Because by symmetry, $a \sim c$, by transitivity, we conclude that $a \sim b$. It follows that, if x is an arbitrary element of $[a]$, then because $x \sim a$, by transitivity, $x \sim b$ and so $x \in [b]$. Thus $[a] \subseteq [b]$. A similar argument shows that $[b] \subseteq [a]$ and so $[a] = [b]$. □

Because, of course, every element of A is in *some* equivalence class, the above proposition implies that the equivalence relation *partitions* A, meaning that A can be written as a union of pairwise disjoint sets, each consisting of elements that are equivalent to one another.

Example 5.2.7. As an example, we consider $\mathbb{Z}$ with the equivalence relation congruence modulo 3. What are the equivalence classes?

When dealing with congruence modulo n, it is customary to denote the equivalence classes using a bar instead of square brackets. So, if k is an integer,

$$\overline{k} = \{j \in \mathbb{Z} : j \equiv k \mod 3\}.$$

Let's consider some values of k. When $k = 0$, $j \equiv 0 \mod 3$ means that there is an integer d such that $j = j - 0 = 3d$. Using our notation,

$$\overline{0} = \{3d : d \in \mathbb{Z}\}.$$

In other words, $\overline{0}$ contains all the integers that are a multiple of 3.

For $k = 1$, $j \equiv 1 \mod 3$ if there exists an integer d such that $j - 1 = 3d$, that is, if $j = 3d + 1$ for some integer d. In notation,

$$\overline{1} = \{3d + 1 : d \in \mathbb{Z}\}.$$

Thus $\overline{1}$ contains all the integers that are one more than a multiple of 3.

Next, consider $k = 2$. If $j \equiv 2 \mod 3$, then there is an integer d such that $j - 2 = 3d$, and so $j = 3d + 2$. In our set notation,

$$\overline{2} = \{3d + 2 : d \in \mathbb{Z}\}.$$

These are the three equivalence classes. We know we have them all, because the union of the ones we found is all of $\mathbb{Z}$, and so by Proposition 5.2.6, because it is not possible for another equivalence class to be disjoint from all of these, any other class must be equal to one of these. For example, if $k = 3$, then $j \equiv 3 \mod 3$ if $j - 3 = 3d$ for some integer d, in which case, $j = 3d + 3 = 3(d + 1)$. So we again obtain the multiples of 3, i.e., $\overline{0} = \overline{3}$. By a similar argument, $\overline{4} = \overline{1}$, $\overline{5} = \overline{2}$, and so on.

Exercise 5.2.3. To test your understanding of the above example, consider $\mathbb{Z}$ again, but now with congruence modulo 5 as the equivalence relation. Describe the equivalence classes.

For our next example, we return to the task of defining the rational number system.

Example 5.2.8. As in the previous subsection, let $A = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$ and define $\sim$ on A by $(a, b) \sim (c, d)$ if $ad = cb$. We'll use $[(a, b)]$ to denote the equivalence class containing (a, b). For example,

$$[(1, 4)] = \{(a, b) : (a, b) \sim (1, 4)\}.$$

The pairs in this set satisfy the condition $4a = b$. Now, b can not be 0, so neither can a. Thus

$$[(1, 4)] = \{(a, 4a) : a \in \mathbb{Z} \setminus \{0\}\}.$$

In light of this example, we can now actually rigorously define $\mathbb{Q}$, the set of rational numbers. It is actually the set of all equivalence classes of $\mathbb{Z} \times \mathbb{Z} \setminus \{0\}$ under the relation $\sim$. In other words, a "rational number" is not a single fraction, but, rather, a class of objects (a numerator and a non-zero denominator) that all function the same way when we do arithmetic. We like the following analogy: A "rational number" is a label on a drawer that has inside of it an infinite collection of equivalent fractions.

Exercise 5.2.4. Continuing the example above, describe the elements of $[(-4, 10)]$. In terms of rational numbers, what rational number do all these ordered pairs represent?

Example 5.2.9. Recall, from Example 5.2.4, we can define an equivalence relation on the set of graphs by saying that two graphs G, H with vertex sets $V(G), V(H)$ and edge sets $E(G), E(H)$ are isomorphic (written $G \cong H$) if there exists a bijection $\varphi : V(G) \to V(H)$ with the property that $\{x, y\} \in E(G)$ if and only if $\{\varphi(x), \varphi(y)\} \in E(H)$. Show that there are exactly four equivalence classes which contain 3-vertex graphs.

Proof. We begin with a simple observation which is nonetheless quite important. If $G \cong H$, then we must have $|V(G)| = |V(H)|$ and $|E(G)| = |E(H)|$. Indeed, if $G \cong H$, then there exists a bijection φ between $|V(G)|$ and $|V(H)|$, so by definition these sets have the same size. Moreover, we claim that $\psi : E(G) \to E(H)$ given by $\psi(\{x, y\}) = \{\varphi(x), \varphi(y)\}$ is a bijection between $E(G)$ and $E(H)$. The fact that if $\{x, y\} \in E(G)$, then $\{\varphi(x), \varphi(y)\}$ must be in $E(H)$ means that ψ is indeed a function from $E(G)$ to $E(H)$. We now must check that ψ is injective and surjective. We have that $\psi(\{x, y\}) = \psi(\{u, v\})$ if and only if $\{\varphi(x), \varphi(y)\} = \{\varphi(u), \varphi(v)\}$, meaning that either $\varphi(x) = \varphi(u)$ and $\varphi(y) = \varphi(v)$ or $\varphi(x) = \varphi(v)$ and $\varphi(y) = \varphi(u)$. Since φ is a bijection, these equalities are only possible if either $x = u, y = v$ or $x = v, y = u$. Either way, since our edges are unordered pairs, we have $\{x, y\} = \{u, v\}$, so ψ is injective. Moreover, say $\{z, w\} \in E(H)$. Because φ is a bijection from $V(G)$ to $V(H)$, we know that there are some vertices $x, y \in V(G)$ such that $\phi(x) = z$ and $\phi(y) = w$. Thus, $\psi(\{x, y\}) = \{\phi(x), \phi(y)\} = \{z, w\}$, so ψ is surjective. We conclude that ψ is a bijection, and thus $|E(G)| = |E(H)|$.

For any graph G, the equivalence class $[G]$ consists of all graphs H which are isomorphic to G. We thus have proved a fundamental fact about equivalence classes under graph isomorphism: all graphs in $[G]$ contain the same number of vertices as G and the same number of edges as G. Note, we have proved that having equal numbers of vertices and edges is a necessary condition for isomorphism; we have *not* proved that it is sufficient, and indeed, in general, it is not a sufficient condition. However, it will still be very helpful in this example. We are asking for the equivalence classes containing 3-vertex graphs. In a graph G with $|V(G)| = 3$, there are $\binom{3}{2} = 3$ possible unordered pairs taken from $V(G)$, so $|E(G)| \leq 3$ (since every edge of G is a different unordered pair from $V(G)$). Thus, there are four possible values for $|E(G)|$: 0, 1, 2, or 3. By the above work, these four possibilities immediately tell us that there are *at least* four equivalence classes containing 3-vertex graphs. We now must determine whether there are exactly four. To do so, we will consider in turn each possible number of edges, and argue that up to isomorphism, there is only one 3-vertex graph with the given number of edges.

Say G, H are two 3-vertex graphs with $|E(G)| = |E(H)|$. We claim that $G \cong H$. There are four cases to consider. In order to make the cases as simple as possible to write, we will preface them with one more observation. Because we have already shown that isomorphism is an equivalence relation, and equivalence relations are transitive, we can show that $G \cong H$ as long as we show that *some graph isomorphic to* G is isomorphic to *some graph isomorphic to* H. In particular, we can start by "renaming" the vertices of G and H so that they are labeled in some consistent way.[6] We may assume (relabeling vertices if needed) that

[6] As an exercise, you might find it interesting to formally write out a justification for the following

$V(G) = \{x, y, z\}$ with the added property that x is contained in at least as many edges of G as y and that y is contained in at least as many edges of G as z. Similarly, we assume that $V(H) = \{a, b, c\}$ with the added property that a is contained in at least as many edges of G as b and that b is contained in at least as many edges of G as c. We consider four cases, depending on the sizes of $|E(G)|$ and $|E(H)|$.

Case 1. $|E(G)| = |E(H)| = 0$. In this case, consider $\varphi : V(G) \to V(H)$ given by $\varphi(x) = a, \varphi(y) = b, \varphi(z) = c$. Then φ is clearly a bijection, and because $E(G)$ and $E(H)$ are empty, the condition on how φ must interact with these edge sets is trivially satisfied. (Explicitly, every pair from $\{x, y, z\}$ is not an edge, and this pair is mapped to a pair from $\{a, b, c\}$ which is also not an edge.)

Case 2. $|E(G)| = |E(H)| = 1$. In this case, exactly one of $\{x, y\}, \{x, z\}, \{y, z\}$ is an edge and exactly one of $\{a, b\}, \{a, c\}, \{b, c\}$ is an edge. Since x is contained in at least as many edges as y and y is contained in at least as many edges as z, it must be that $\{x, y\}$ is the single edge of G and z is contained in no edge. Similarly, it must be that $\{a, b\}$ is the single edge of H. We define $\varphi : V(G) \to V(H)$ by $\varphi(x) = a, \varphi(y) = b, \varphi(z) = c$. Then φ is clearly a bijection, and we see by inspection that φ correctly interacts with $E(G)$ and $E(H)$. So φ is an isomorphism.

Case 3. $|E(G)| = |E(H)| = 1$. In this case, exactly two of $\{x, y\}, \{x, z\}, \{y, z\}$ are edges and exactly two of $\{a, b\}, \{a, c\}, \{b, c\}$ are edges. Thus, exactly one of x, y, z is in two edges, and the other two vertices of G are in exactly one edge each. By our labeling of vertices, x must be in two edges, so $E(G) = \{\{x, y\}, \{x, z\}\}$. Similarly, we must have $E(H) = \{\{a, b\}, \{a, c\}\}$. We again define $\varphi : V(G) \to V(H)$ by $\varphi(x) = a, \varphi(y) = b, \varphi(z) = c$. Then φ is clearly a bijection, and we see by inspection that φ correctly interacts with $E(G)$ and $E(H)$. So φ is an isomorphism.

Case 4. $|E(G)| = |E(H)| = 0$. In this case, consider $\varphi : V(G) \to V(H)$ given by $\varphi(x) = a, \varphi(y) = b, \varphi(z) = c$. Then φ is clearly a bijection, and because $E(G)$ and $E(H)$ each contain all possible edges, the condition on how φ must interact with these edge sets is certainly satisfied. (Explicitly, every pair from $\{x, y, z\}$ is an edge, and this pair is mapped to a pair from $\{a, b, c\}$ which is also an edge.)

We conclude that there are exactly four equivalence classes of 3-vertex graphs. These are depicted in Figure 5.3. □

While the last example took some time to establish some basic, useful properties of isomorphism, it may not seem have seemed very complicated to determine isomorphism classes once we had those facts established. Essentially, we found that as long as we label our graphs in a consistent way so that we knew which vertices were in the most edges,

statement, which really formalizes the notion that graph isomorphism just allows us to "rename" vertices however we like. Let G be a graph with $|V(G)| = k$ and $\mathcal{L} = \{L_1, L_2, \ldots, L_k\}$ be any set of k distinct labels (possibly intersecting with $V(G)$). Define any bijection $\varphi : V(G) \to \mathcal{L}$. Then φ is an isomorphism between G and the graph $G_{\mathcal{L}}$ defined by $V(G_{\mathcal{L}}) = \mathcal{L}$ and with $\{L_i, L_j\} \in E(G_{\mathcal{L}})$ if and only if $\{\varphi^{-1}(L_i), \varphi^{-1}(L_j)\} \in E(G)$. You can visualize making $V(G)$ by just "erasing" each vertex label $v = \varphi^{-1}(L_i)$ in G and replacing it with $\varphi(v) = L_i$.

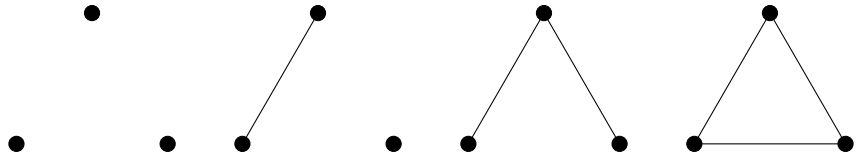


Figure 5.3: Depictions of the four equivalence classes of isomorphic 3-vertex graphs. Because graph isomorphism essentially amounts to "renaming" vertices, we often depict the equivalence classes of isomorphic graphs as *unlabeled*, meaning that vertices have no name associated to them and only the structure of the graph's connections is displayed. By assigning any desired labels to the vertices of an unlabeled graph, you could find a specific representative of the graph's equivalence class.

we could always deduce how many edges each vertex was in and quickly write down an isomorphism by sending a vertex in one graph to a vertex in another which appeared in the same number of edges. (Moreover, this function actually *was* always an isomorphism – that is, all vertices which appeared in the same number of edges looked "the same".)

Unfortunately, while this approach worked for 3-vertex graphs, in general, it is much more complicated to determine whether two n-vertex graphs are isomorphic. When the number of vertices is larger, there are many more possible configurations of edges, and so perhaps it is not surprising that isomorphism will depend on more than the number of vertices and edges in a graph. For a few small values of n, it is possible to determine the number of equivalence classes of n-vertex graphs by hand, but the numbers quickly become large and counting non-isomorphic graphs systematically becomes a serious theoretical question. Still, with a little patience, you are perfectly capable of determining the next case up.

Exercise 5.2.5. How many equivalence classes of 4-vertex graphs are there? Justify your answer as carefully as you can.

5.3 Modular Arithmetic

We have now seen that for any natural number $n \geq 2$, congruence modulo n is an equivalence relation on $\mathbb{Z}$ that partitions $\mathbb{Z}$ into equivalence classes. We worked out in great detail what these equivalence classes look like when $n = 3$ and an exercise asked you to describe them when $n = 5$. These examples suggest that there are, in general, n distinct equivalence classes. We summarize in the next proposition.

Proposition 5.3.1. *Let $n \in \mathbb{N}$ with $n \geq 2$. Congruence modulo n partitions $\mathbb{Z}$ into n distinct equivalence classes*

$$\overline{j} = \{nd + j : d \in \mathbb{Z}\}$$

for integers j with $0 \leq j \leq n - 1$.

The proof that the union of these sets is in fact $\mathbb{Z}$ requires the *division algorithm*, the statement that, given a non-zero integer n (the divisor), an arbitrary integer m can be written in the form

$$m = nd + r$$

for some *remainder* r with $0 \leq r \leq n-1$. We have not officially proved this result, but we have used it throughout the book many times. A future chapter on the basics of number theory includes a longer discussion of this and other propositions. For now, we take it as familiar. In light of this fact, Proposition 5.3.1 follows; the union of the sets is $\mathbb{Z}$, the sets are disjoint, and each one contains all the elements congruent to j modulo n.

Of course, as we discussed previously, in a sense there are infinitely many equivalence classes we could write down, one associated with each integer. But only n of them are *distinct*. In general, $\overline{n}$ is the same set as $\overline{0}$, $\overline{n+1}$ is the same as $\overline{1}$, and so on. So we *could* make the partition *look* different by labeling these equivalence classes with different representatives, but they would be the same n sets. The way we have labeled them, using the integers $0, 1, \ldots, n$, is the canonical way to do it. We define some notation.

Definition 5.3.2. Let $n \in \mathbb{N}$, $n \geq 2$. We denote by $\mathbb{Z}_n$ the set of equivalence classes of $\mathbb{Z}$ modulo n. We call $\mathbb{Z}_n$ the **integers modulo** n and label them canonically $\overline{j}$ for $0 \leq j \leq n-1$. We call $\overline{j}$ the **congruence class** of j.

If we were finished discussing this relation, this definition would seem like unnecessary notation. *Any* time we have an equivalence relation, it partitions the parent set, and we could always then give a notation to the set of all equivalence classes under that relation. Why are we so keenly interested in doing so for this example? The answer could be "Because we can", but that's a poor answer. We didn't just partition $\mathbb{Z}$ in some random meaningless way. The classes are supposed to consist of integers that function the same way in some sense, so we should now ask them to, well, function. What kinds of problems can be addressed more easily by thinking about congruence classes? Go back to the introduction to the chapter. We talked about the fact that certain cyclical phenomena can be modeled using integers modulo n. Let's do one very simple example.

Example 5.3.3. Suppose the math club at your university decides to do a Pi-Day marathon in which you broadcast people doing fun math problems for 100 hours.[7] If the marathon begins at noon on March 13, at what time does it end?[8]

We can do this problem without officially thinking about congruence; we figure out how many times 24 goes into 100. The answer is 4, with a remainder of 4. Thus, the marathon ends after 4 days and 4 hours, so at 4pm, which is hour 16 in the day.

Let's think about it again, more explicitly in the language of congruence classes. In this case, n is 24 because each day has 24 hours in it. In the United States we're odd[9] and use the same numbers for the hours before noon and the ones after noon with the "am" and "pm" designation, but it is probably more reasonable to think of the 24-hour clock in which 1pm is 13, 2pm is 14, etc. So we're working in $\mathbb{Z}_{24}$. Every single hour, past, present, and future, can be considered to be a representative from one of the 24 congruence classes. We actually don't know exactly *which* representative it is because we don't have a universally-agreed-upon "time 0", but it doesn't matter. For this problem, we may as well set time 0 as the beginning of March 13. So the marathon starts at noon, which is in $\overline{12}$. We want to

[7] Anna has been part of such Pi-Day marathons hosted at the University of Montana, though they "only" go for 24 hours. . .

[8] . . . with a trip to Denny's.

[9] Not even?

go ahead 100 hours, up to 112. To determine the time, we need to figure out what integer j with $0 \leq j \leq 23$ has 112 in its class. Division gives

$$112 = 4 \cdot 24 + 16.$$

That is, $112 \in \overline{16}$, so the marathon ends when the time is 16 (corresponding to 4pm).

It seems like maybe we should write our calculation as follows:

$$\overline{12} + \overline{100} = \overline{16}.$$

We come to the point: In order to solve problems like the above that really only care about the congruence class of the answer, we should develop some kind of good theory surrounding arithmetic with congruence classes. We call the process of doing operations on congruence classes *modular arithmetic*.

Definition 5.3.4. Let $n \in \mathbb{N}$ with $n \geq 2$. We define operations in $\mathbb{Z}_n$ by

$$\overline{j} + \overline{k} = \overline{j+k} \quad \text{and} \quad \overline{j} \cdot \overline{k} = \overline{jk}.$$

In words, to add two classes, we take something out of each (j and k), add them together, and report the name of the class in which their sum lives. The product of classes works similarly.

"Now wait a minute," you say, being your usual skeptical self. "It seems like I could possibly get different answers this way depending on which integer I take out of each congruence class to add or multiply."[10] There *is* something to worry about. In the example above, what would the answer have been if we had decided that noon on March 13 should be considered to be -12 (because time should really begin on Pi Day itself) and if we had decided that instead of adding 100 hours, we would recognize that 100 is in the same class as 4, so we'd just add 4? We would have then taken $-12 + 4 = -8$. What class is -8 in? It is in the same class as 16, because $16 - (-8) = 24$. So we would again conclude that the end time for the Pi-Day marathon is hour 16, meaning 4pm. Is it a coincidence that we got the same answer? No. Is there something to prove? Yes! We need to prove that the operations defined in Definition 5.3.4 are *well-defined*. Recall from Chapter 4 that a function is well-defined if the result does not depend on the particular choice we make of how to represent the input. The situation we are in right now, where we are defining the result of the addition or multiplication of classes by saying "Take something out of each class, do the operation, and then report the class of the result" is precisely the kind of situation in which we need to be really careful. Things work out here, as the next proposition shows.

Proposition 5.3.5. *Let $n \in \mathbb{N}$ with $n \geq 2$. Let $a, b, c, d \in \mathbb{Z}$.*

(a) If $a \equiv c \mod n$ and $b \equiv d \mod n$, then $a + b \equiv c + d \mod n$.

(b) If $a \equiv c \mod n$ and $b \equiv d \mod n$, then $ab \equiv cd \mod n$.

Thus addition and multiplication of congruence classes, as described in Definition 5.3.4, are well-defined.

[10] Great comment.

Proof. We prove part (b) and leave the proof of part (a) as the next exercise. By hypothesis, there exist integers j and k such that $a - c = nj$ and $b - d = nk$. Now,

$$\begin{aligned} ab - cd &= ab - cb + cb - cd \\ &= (a - c)b + c(b - d) \\ &= njb + nkc \\ &= n(jb + kc). \end{aligned}$$

Thus $ab \equiv cd \mod n$. □

Exercise 5.3.1. Prove part (a) of Proposition 5.3.5.

Modular arithmetic can be pretty fun. This next example is not very hard but does show the power of the results.

Example 5.3.6. Suppose in $\mathbb{Z}_7$ we want to find $\overline{701} \cdot \overline{52}$. One correct strategy is to take 701 and 52, multiply them together to get an obnoxiously big number, then divide that number by 7 to find the remainder. A savvy way to do it is to choose more convenient representatives from the same equivalence classes and do the arithmetic with them. For example, it is clear that $701 \equiv 1 \mod 7$ and $52 \equiv 3 \mod 7$, so $701 \cdot 52 \equiv 1 \cdot 3 \mod 7$. So

$$\overline{701} \cdot \overline{52} = \overline{3}.$$

It can be instructive to make the full addition and multiplication tables for $\mathbb{Z}_n$ for various values of n. We will do one as an example and then have you try several as exercises.

Example 5.3.7. We make the full addition and multiplication tables for $\mathbb{Z}_4$. For addition, we have:

$+$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$
$\overline{0}$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$
$\overline{1}$	$\overline{1}$	$\overline{2}$	$\overline{3}$	$\overline{0}$
$\overline{2}$	$\overline{2}$	$\overline{3}$	$\overline{0}$	$\overline{1}$
$\overline{3}$	$\overline{3}$	$\overline{0}$	$\overline{1}$	$\overline{2}$

For multiplication, we have:

$\cdot$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$
$\overline{0}$	$\overline{0}$	$\overline{0}$	$\overline{0}$	$\overline{0}$
$\overline{1}$	$\overline{0}$	$\overline{1}$	$\overline{2}$	$\overline{3}$
$\overline{2}$	$\overline{0}$	$\overline{2}$	$\overline{0}$	$\overline{2}$
$\overline{3}$	$\overline{0}$	$\overline{3}$	$\overline{2}$	$\overline{1}$

Exercise 5.3.2. Make the full addition and multiplication tables for $\mathbb{Z}_5$ and $\mathbb{Z}_6$. How are they similar? Are there any noticeable differences?

Exercise 5.3.3. When we solve algebraic equations about real numbers, we often use the fact that, if a and b are real numbers, the only way for ab to equal 0 is if either $a = 0$ or $b = 0$. Is this statement true about arithmetic in $\mathbb{Z}_n$?

You may have noticed that the operations on $\mathbb{Z}_n$ satisfy many of the same properties satisfied by operations on integers. For example, there is an *additive identity* – an element of the set that, when added to anything, leaves it unchanged. The additive identity is $\overline{0}$. There is also a *multiplicative identity*, namely $\overline{1}$. Furthermore, every element has an *additive inverse* that can be added to it to get $\overline{0}$. In a later chapter, we will explore in more detail sets with operations and the properties they satisfy. The branch of mathematics devoted to such topics is *abstract algebra*. Here, we have just given you the tiniest of tastes.

5.4 Problems

1. Let $A = \mathbb{R}^2 \setminus \{(0,0)\}$. Define a relation $\sim$ on A by $(a,b) \sim (c,d)$ if there exists a real number $r > 0$ such that $(c,d) = (ra, rb)$.

   (a) Show that $\sim$ is an equivalence relation on A.

   (a) Find the set of all elements of A that are equivalent to each of the following. Each of these sets is a subset of the plane. Make a rough sketch of each.

      i. $(1,0)$
      ii. $(1,1)$
      iii. $(-2,1)$

2. Let $A = \mathbb{R}^2$ and define a relation $\sim$ on A by $(a,b) \sim (c,d)$ if $ab = cd$.

   (a) Show that $\sim$ is an equivalence relation on A.

   (b) Let $\overline{(a,b)}$ denote the equivalence class containing (a,b). Find four distinct elements of A in $\overline{(1,1)}$. If you were to plot all the points in the plane that are in $\overline{(1,1)}$, what would the set look like?

   (c) Find each of the following:

      i. $\overline{(0,0)}$
      ii. $\overline{(1,-2)}$

5.5 Suggested Projects

Part II

A Mathematical Sampler

Chapter 6

Arguments in Analysis

Chapter 7

Number Theory

Chapter 8

Algebraic Structures

Chapter 9

Discrete Probability

Part III

Excursions and Explorations

Chapter 10

Recurrence Relations

1. Fibonacci, Lucas, Tribonacci
2. Catalan Numbers
3. Characteristic polynomial method – can students discover it themselves?
4. Other recursive objects, recursively defined polynomials

Chapter 11

Abstract Vector Spaces

Chapter 12

Complex Analysis

In first-year calculus, you studied real-valued functions of a single real variable. As we indicated earlier, the field of *real analysis* focuses on putting establishing the theoretical foundations for calculus and extending the ideas to functions of multiple real variables. Once can also extend calculus to *complex-valued* functions of a *complex* variable. Not surprisingly, the resulting field is called *complex analysis*. Although complex analysis is a huge field and we can only just brush the surface, we devote this chapter to some of the basic ideas and suggest some exploratory projects.

12.1 Algebra and Geometry of Complex Numbers

12.1.1 Motivation and definitions of operations

12.1.2 Polar form and geometry

12.2 Project: Roots of Unity

Now that we understand the basics about complex numbers and the operations with them, we can start to explore some beautiful topics. In this project, we explore *roots of unity*. A complex number z is an n-th root of unity if

$$z^n = 1.$$

In other words, an n-th root of unity is a complex number that, when raised to the n-th power, gives 1 – or "unity." For example, the fourth roots of unity are

$$1, i, -1, -i$$

because

$$1^4 = 1, \quad i^4 = (i^2)^2 = (-1)^2 = 1, \quad \text{etc.}$$

This simple example also allows us to illustrate an important concept. Some fourth roots of unity are *primitive*, meaning that not only is the fourth power equal to 1, but no lower positive power equals 1. Indeed,

$$i^1 = i, \quad i^2 = -1, \quad \text{and} \quad i^3 = i^2 \cdot i = -i.$$

On the other hand, -1 is a fourth root of unity that is not primitive, because we only actually need to square it to get 1. Because many of the exercises in this project will ask you to explore which roots of unity are primitive for a given n, we make the definition precise.

Definition 12.2.1. Let $n \in \mathbb{N}$, $n \geq 2$. The complex number ω is an n-th root of unity if $\omega^n = 1$. It is **primitive** if for all $1 \leq k \leq n-1$, $\omega^k \neq 1$.

12.2.1 Primitive roots of unity

Exercise 12.2.1. For each natural number n between 2 and 8, find all n-th roots of unity, plot them, and determine which are primitive.

Exercise 12.2.2. Based on the results of the previous exercise, find a condition that allows one to determine, for fixed n, which n-th roots of unity are primitive and which are not. Prove your proposition.

The next exercises require concepts from the chapter on algebraic structures.

Exercise 12.2.3. Show that for a fixed natural number n, the set of n-th roots of unity forms a group under multiplication. Show that the group is cyclic and determine its order. Which of the members of the group generate the group? Which generate subgroups of lower order?

Exercise 12.2.4. Recall that two groups are isomorphic if there is a bijection between them that preserves the group operation. The multiplicative group of n-th roots of unity is isomorphic to a group you met earlier in this book. What is it? Prove your claim by explicitly constructing an isomorphism.

12.2.2 Some fun sums

12.3 Project: Analytic Mappings of the Complex Plane

In this project, you will explore different ways to visualize a function whose domain and codomain are both $\mathbb{C}$. To do this project, you will need some knowledge of some plotting software. In order to do the later more theoretical parts, you will also need to be comfortable with some ideas from multivariable calculus, in particular, partial derivatives, gradients, level curves, and Jacobian matrices.

12.4 Project: Zeros of Polynomials

Chapter 13

Graph Theory